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Modulators of HSD17B13 expression

Abstract

Methods, compounds, and compositions useful for inhibiting HSD17B13 expression are provided. Such compounds, compositions, and methods are useful for treating, preventing, or ameliorating a disease associated with HSD 17B 13.

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Background/Summary

SEQUENCE LISTING

(1) The present application is being filed along with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled BIOL0350WOSEQ_ST25.txt created Dec. 13, 2019, which is 640 kb in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD

(2) The present embodiments provide methods, compounds, and compositions useful for inhibiting hydroxysteroid 17-beta dehydrogenase 13 (LOC345275; 17-beta hydroxysteroid dehydrogenase; HSD17B13; HSD17013) expression, and in certain instances, reducing the amount of HSD17B13 protein in a cell or animal, which can be useful for treating, preventing, or ameliorating a disease associated with HSD17B13.

BACKGROUND

(3) Non-alcoholic fatty liver disease (NAFLD) covers a spectrum of liver disease from steatosis to nonalcoholic steatohepatitis (NASH) and cirrhosis. NAFLD is defined as fat accumulation in the liver exceeding 5% by weight, in the absence of significant alcohol consumption, steatogenic medication, or hereditary disorders (Kotronen et al, *Arterioscler Thromb. Vasc. Biol.* 2008, 28: 27-38).

(4) Non-alcoholic steatohepatitis (NASH) is NAFLD with signs of inflammation and hepatic injury. NASH is defined histologically by macrovesicular steatosis, hepatocellular ballooning, and lobular inflammatory infiltrates (Sanyal, *Hepatol. Res.* 2011. 41: 670-4). NASH is estimated to affect 2-3% of the general population. In the presence of other pathologies, such as obesity or diabetes, the estimated prevalence increases to 7% and 62% respectively (Hashimoto et al, *J. Gastroenterol.* 2011. 46(1): 63-69).

SUMMARY

(5) Certain embodiments provided herein are compounds and methods for reducing the amount or activity of HSD17B13 mRNA, and in certain embodiments, reducing the amount or activity of HSD17B13 protein in a cell or animal. In certain embodiments, the animal has a liver disease. In certain embodiments, the disease is NASH. In certain embodiments, the disease is alcoholic steatohepatitis (ASH). In certain embodiments, the disease is NAFLD. In certain embodiments, the disease is hepatic steatosis. In certain embodiments, the disease is cirrhosis. In certain embodiments, the disease is hepatocellular carcinoma. In certain embodiments, the disease is alcoholic liver disease. In certain embodiments, the disease is HCV hepatitis. In certain embodiments, the disease is chronic hepatitis. In certain embodiments, the disease is hereditary hemochromatosis. In certain embodiments, the disease is primary sclerosing cholangitis. Certain compounds provided herein are directed to compounds and compositions that reduce liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an animal.

(6) Certain embodiments provided herein are directed to potent and tolerable compounds and compositions useful for inhibiting HSD17B13 expression, which can be useful for treating, preventing, ameliorating, or slowing progression of liver diseases. Certain embodiments provided herein are directed to compounds and compositions that are more potent or have greater therapeutic value than compounds publicly disclosed.

Description

DETAILED DESCRIPTION

(1) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the embodiments, as claimed. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of “or” means “and/or” unless stated otherwise. Furthermore, the use of the term “including” as well as other forms, such as “includes” and “included”, is not limiting.

(2) The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, treatises, and GenBank and NCBI reference sequence records are hereby expressly incorporated by reference for the portions of the document discussed herein, as well as in their entirety.

(3) It is understood that the sequence set forth in each SEQ ID NO in the examples contained herein is independent of any modification to a sugar moiety, an internucleoside linkage, or a nucleobase. As such, compounds defined by a SEQ ID NO may comprise, independently, one or more modifications to a sugar moiety, an internucleoside linkage, or a nucleobase. Compounds

described by ION number indicate a combination of nucleobase sequence, chemical modification, and motif.

Definitions

(4) Unless otherwise indicated, the following terms have the following meanings:

(5) “2'-deoxyfuranosyl sugar moiety” or “2'-deoxyfuranosyl sugar” means a furanosyl sugar moiety having two hydrogens at the 2'-position. 2'-deoxyfuranosyl sugar moieties may be unmodified or modified and may be substituted at positions other than the 2'-position or unsubstituted. A β -D-2'-deoxyribose sugar moiety in the context of an oligonucleotide is an unsubstituted, unmodified 2'-deoxyfuranosyl and is found in naturally occurring deoxyribonucleic acids (DNA).

(6) “2'-deoxynucleoside” means a nucleoside comprising 2'-H(H) furanosyl sugar moiety, as found in naturally occurring deoxyribonucleic acids (DNA). In certain embodiments, a 2'-deoxynucleoside may comprise a modified nucleobase or may comprise an RNA nucleobase (uracil).

(7) “2'-O-methoxyethyl” (also 2'-MOE) refers to a 2'-O(CH₂)₂—OCH₃ in the place of the 2'-OH group of a ribosyl ring. A 2'-O-methoxyethyl modified sugar is a modified sugar.

(8) “2'-MOE nucleoside” (also 2'-O-methoxyethyl nucleoside) means a nucleoside comprising a 2'-MOE modified sugar moiety.

(9) “2'-substituted nucleoside” or “2'-modified nucleoside” means a nucleoside comprising a 2'-substituted or 2'-modified sugar moiety. As used herein, “2'-substituted” or “2'-modified” in reference to a sugar moiety means a sugar moiety comprising at least one 2'-substituent group other than H or OH.

(10) “3' target site” refers to the nucleotide of a target nucleic acid which is complementary to the 3'-most nucleotide of a particular compound.

(11) “5' target site” refers to the nucleotide of a target nucleic acid which is complementary to the 5'-most nucleotide of a particular compound.

(12) “5-methylcytosine” means a cytosine with a methyl group attached to the 5 position.

(13) “About” means within 10% of a value. For example, if it is stated, “the compounds affected about 70% inhibition of HSD17B13”, it is implied that HSD17B13 levels are inhibited within a range of 60% and 80%.

(14) “Administration” or “administering” refers to routes of introducing a compound or composition provided herein to an individual to perform its intended function. An example of a route of administration that can be used includes, but is not limited to parenteral administration, such as subcutaneous, intravenous, or intramuscular injection or infusion.

(15) “Administered concomitantly” or “co-administration” means administration of two or more compounds in any manner in which the pharmacological effects of both are manifest in the patient. Concomitant administration does not require that both compounds be administered in a single pharmaceutical composition, in the same dosage form, by the same route of administration, or at the same time. The effects of both compounds need not manifest themselves at the same time. The effects need only be overlapping for a period of time and need not be coextensive. Concomitant administration or co-administration encompasses administration in parallel or sequentially.

(16) “Amelioration” refers to an improvement or lessening of at least one indicator, sign, or symptom of an associated disease, disorder, or condition. In certain embodiments, amelioration includes a delay or slowing in the progression or severity of one or more indicators of a condition or disease. The progression or severity of indicators may be determined by subjective or objective measures, which are known to those skilled in the art.

(17) “Animal” refers to a human or non-human animal, including, but not limited to, mice, rats, rabbits, dogs, cats, pigs, and non-human primates, including, but not limited to, monkeys and chimpanzees.

(18) “Antisense activity” means any detectable and/or measurable activity attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense

activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid compared to target nucleic acid levels or target protein levels in the absence of the antisense compound to the target.

(19) “Antisense compound” means a compound comprising an oligonucleotide and optionally one or more additional features, such as a conjugate group or terminal group. Examples of antisense compounds include single-stranded and double-stranded compounds, such as, oligonucleotides, ribozymes, siRNAs, shRNAs, ssRNAs, and occupancy-based compounds.

(20) “Antisense inhibition” means reduction of target nucleic acid levels in the presence of an antisense compound complementary to a target nucleic acid compared to target nucleic acid levels in the absence of the antisense compound.

(21) “Antisense mechanisms” are all those mechanisms involving hybridization of a compound with target nucleic acid, wherein the outcome or effect of the hybridization is either target degradation or target occupancy with concomitant stalling of the cellular machinery involving, for example, transcription or splicing.

(22) “Antisense oligonucleotide” means an oligonucleotide having a nucleobase sequence that is complementary to a target nucleic acid or region or segment thereof. In certain embodiments, an antisense oligonucleotide is specifically hybridizable to a target nucleic acid or region or segment thereof.

(23) “Bicyclic nucleoside” or “BNA” means a nucleoside comprising a bicyclic sugar moiety. “Bicyclic sugar” or “bicyclic sugar moiety” means a modified sugar moiety comprising two rings, wherein the second ring is formed via a bridge connecting two of the atoms in the first ring thereby forming a bicyclic structure. In certain embodiments, the first ring of the bicyclic sugar moiety is a furanosyl moiety. In certain embodiments, the bicyclic sugar moiety does not comprise a furanosyl moiety.

(24) “Branching group” means a group of atoms having at least 3 positions that are capable of forming covalent linkages to at least 3 groups. In certain embodiments, a branching group provides a plurality of reactive sites for connecting tethered ligands to an oligonucleotide via a conjugate linker and/or a cleavable moiety.

(25) “Cell-targeting moiety” means a conjugate group or portion of a conjugate group that is capable of binding to a particular cell type or particular cell types.

(26) “cEt” or “constrained ethyl” means a ribosyl bicyclic sugar moiety wherein the second ring of the bicyclic sugar is formed via a bridge connecting the 4'-carbon and the 2'-carbon, wherein the bridge has the formula: 4'-CH(CH₂sub.3)—O-2', and wherein the methyl group of the bridge is in the S configuration.

(27) “cEt nucleoside” means a nucleoside comprising a cEt modified sugar moiety.

(28) “Chemical modification” in a compound describes the substitutions or changes through chemical reaction, of any of the units in the compound relative to the original state of such unit.

“Modified nucleoside” means a nucleoside having, independently, a modified sugar moiety and/or modified nucleobase. “Modified oligonucleotide” means an oligonucleotide comprising at least one modified internucleoside linkage, a modified sugar, and/or a modified nucleobase.

(29) “Chemically distinct region” refers to a region of a compound that is in some way chemically different than another region of the same compound. For example, a region having 2'-O-methoxyethyl nucleotides is chemically distinct from a region having nucleotides without 2'-O-methoxyethyl modifications.

(30) “Chimeric antisense compounds” means antisense compounds that have at least 2 chemically distinct regions, each position having a plurality of subunits.

(31) “Cleavable bond” means any chemical bond capable of being split. In certain embodiments, a cleavable bond is selected from among: an amide, a polyamide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, a di-sulfide, or a peptide.

(32) “Cleavable moiety” means a bond or group of atoms that is cleaved under physiological

conditions, for example, inside a cell, an animal, or a human.

(33) “Complementary” in reference to an oligonucleotide means the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to the following pairs: adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (.sup.mC) and guanine (G) unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. By contrast, “fully complementary” or “100% complementary” in reference to oligonucleotides means that such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches.

(34) “Conjugate group” means a group of atoms that is attached to an oligonucleotide. Conjugate groups include a conjugate moiety and a conjugate linker that attaches the conjugate moiety to the oligonucleotide.

(35) “Conjugate linker” means a group of atoms comprising at least one bond that connects a conjugate moiety to an oligonucleotide.

(36) “Conjugate moiety” means a group of atoms that is attached to an oligonucleotide via a conjugate linker.

(37) “Contiguous” in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, “contiguous nucleobases” means nucleobases that are immediately adjacent to each other in a sequence.

(38) “Designing” or “Designed to” refer to the process of designing a compound that specifically hybridizes with a selected nucleic acid molecule.

(39) “Diluent” means an ingredient in a composition that lacks pharmacological activity, but is pharmaceutically necessary or desirable. For example, the diluent in an injected composition can be a liquid, e.g. saline solution.

(40) “Differently modified” means chemical modifications or chemical substituents that are different from one another, including absence of modifications. Thus, for example, a MOE nucleoside and an unmodified DNA nucleoside are “differently modified,” even though the DNA nucleoside is unmodified. Likewise, DNA and RNA are “differently modified,” even though both are naturally-occurring unmodified nucleosides.

(41) Nucleosides that are the same but for comprising different nucleobases are not differently modified. For example, a nucleoside comprising a 2'-OMe modified sugar and an unmodified adenine nucleobase and a nucleoside comprising a 2'-OMe modified sugar and an unmodified thymine nucleobase are not differently modified.

(42) “Dose” means a specified quantity of a compound or pharmaceutical agent provided in a single administration, or in a specified time period. In certain embodiments, a dose may be administered in two or more boluses, tablets, or injections. For example, in certain embodiments, where subcutaneous administration is desired, the desired dose may require a volume not easily accommodated by a single injection. In such embodiments, two or more injections may be used to achieve the desired dose. In certain embodiments, a dose may be administered in two or more injections to minimize injection site reaction in an individual. In other embodiments, the compound or pharmaceutical agent is administered by infusion over an extended period of time or continuously. Doses may be stated as the amount of pharmaceutical agent per hour, day, week or month.

(43) “Dosing regimen” is a combination of doses designed to achieve one or more desired effects.

(44) “Double-stranded antisense compound” means an antisense compound comprising two oligomeric compounds that are complementary to each other and form a duplex, and wherein one

of the two said oligomeric compounds comprises an oligonucleotide.

(45) “Effective amount” means the amount of compound sufficient to effectuate a desired physiological outcome in an individual in need of the compound. The effective amount may vary among individuals depending on the health and physical condition of the individual to be treated, the taxonomic group of the individuals to be treated, the formulation of the composition, assessment of the individual's medical condition, and other relevant factors.

(46) “Efficacy” means the ability to produce a desired effect.

(47) “Expression” includes all the functions by which a gene's coded information is converted into structures present and operating in a cell. Such structures include, but are not limited to, the products of transcription and translation.

(48) “Gapmer” means an oligonucleotide comprising an internal region having a plurality of nucleosides that support RNase H cleavage positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as the “gap” and the external regions may be referred to as the “wings.”

(49) “HSD17B13” means any nucleic acid or protein of HSD17B13. “HSD17B13 nucleic acid” means any nucleic acid encoding HSD17B13. For example, in certain embodiments, a HSD17B13 nucleic acid includes a DNA sequence encoding HSD17B13, an RNA sequence transcribed from DNA encoding HSD17B13 (including genomic DNA comprising introns and exons), and an mRNA sequence encoding HSD17B13. “HSD17B13 mRNA” means an mRNA encoding a HSD17B13 protein. The target may be referred to in either upper or lower case.

(50) “HSD17B13 specific inhibitor” refers to any agent capable of specifically inhibiting HSD17B13 RNA and/or HSD17B13 protein expression or activity at the molecular level. For example, HSD17B13 specific inhibitors include nucleic acids (including antisense compounds), peptides, antibodies, small molecules, and other agents capable of inhibiting the expression of HSD17B13 RNA and/or HSD17B13 protein.

(51) “Hybridization” means the annealing of oligonucleotides and/or nucleic acids. While not limited to a particular mechanism, the most common mechanism of hybridization involves hydrogen bonding, which may be Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an antisense compound and a nucleic acid target. In certain embodiments, complementary nucleic acid molecules include, but are not limited to, an oligonucleotide and a nucleic acid target.

(52) “Immediately adjacent” means there are no intervening elements between the immediately adjacent elements of the same kind (e.g. no intervening nucleobases between the immediately adjacent nucleobases).

(53) “Individual” means a human or non-human animal selected for treatment or therapy.

(54) “Inhibiting the expression or activity” refers to a reduction or blockade of the expression or activity relative to the expression of activity in an untreated or control sample and does not necessarily indicate a total elimination of expression or activity.

(55) “Internucleoside linkage” means a group or bond that forms a covalent linkage between adjacent nucleosides in an oligonucleotide. “Modified internucleoside linkage” means any internucleoside linkage other than a naturally occurring, phosphate internucleoside linkage. Non-phosphate linkages are referred to herein as modified internucleoside linkages.

(56) “Lengthened oligonucleotides” are those that have one or more additional nucleosides relative to an oligonucleotide disclosed herein, e.g. a parent oligonucleotide.

(57) “Linked nucleosides” means adjacent nucleosides linked together by an internucleoside linkage.

(58) “Linker-nucleoside” means a nucleoside that links an oligonucleotide to a conjugate moiety. Linker-nucleosides are located within the conjugate linker of a compound. Linker-nucleosides are

not considered part of the oligonucleotide portion of a compound even if they are contiguous with the oligonucleotide.

(59) “Mismatch” or “non-complementary” means a nucleobase of a first oligonucleotide that is not complementary to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned. For example, nucleobases including but not limited to a universal nucleobase, inosine, and hypoxanthine, are capable of hybridizing with at least one nucleobase but are still mismatched or non-complementary with respect to nucleobase to which it hybridized. As another example, a nucleobase of a first oligonucleotide that is not capable of hybridizing to the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotides are aligned is a mismatch or non-complementary nucleobase.

(60) “Modulating” refers to changing or adjusting a feature in a cell, tissue, organ or organism. For example, modulating HSD17B13 RNA can mean to increase or decrease the level of HSD17B13 RNA and/or HSD17B13 protein in a cell, tissue, organ or organism. A “modulator” effects the change in the cell, tissue, organ or organism. For example, a HSD17B13 compound can be a modulator that decreases the amount of HSD17B13 RNA and/or HSD17B13 protein in a cell, tissue, organ or organism.

(61) “MOE” means methoxyethyl.

(62) “Monomer” refers to a single unit of an oligomer. Monomers include, but are not limited to, nucleosides and nucleotides.

(63) “Motif” means the pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages, in an oligonucleotide.

(64) “Natural” or “naturally occurring” means found in nature.

(65) “Non-bicyclic modified sugar” or “non-bicyclic modified sugar moiety” means a modified sugar moiety that comprises a modification, such as a substituent, that does not form a bridge between two atoms of the sugar to form a second ring.

(66) “Nucleic acid” refers to molecules composed of monomeric nucleotides. A nucleic acid includes, but is not limited to, ribonucleic acids (RNA), deoxyribonucleic acids (DNA), single-stranded nucleic acids, and double-stranded nucleic acids.

(67) “Nucleobase” means a heterocyclic moiety capable of pairing with a base of another nucleic acid. As used herein a “naturally occurring nucleobase” is adenine (A), thymine (T), cytosine (C), uracil (U), and guanine (G). A “modified nucleobase” is a naturally occurring nucleobase that is chemically modified. A “universal base” or “universal nucleobase” is a nucleobase other than a naturally occurring nucleobase and modified nucleobase, and is capable of pairing with any nucleobase.

(68) “Nucleobase sequence” means the order of contiguous nucleobases in a nucleic acid or oligonucleotide independent of any sugar or internucleoside linkage.

(69) “Nucleoside” means a compound comprising a nucleobase and a sugar moiety. The nucleobase and sugar moiety are each, independently, unmodified or modified. “Modified nucleoside” means a nucleoside comprising a modified nucleobase and/or a modified sugar moiety. Modified nucleosides include abasic nucleosides, which lack a nucleobase.

(70) “Oligomeric compound” means a compound comprising a single oligonucleotide and, optionally, one or more additional features, such as a conjugate group or terminal group.

(71) “Oligonucleotide” means a polymer of linked nucleosides each of which can be modified or unmodified, independent one from another. Unless otherwise indicated, oligonucleotides consist of 8-80 linked nucleosides. “Modified oligonucleotide” means an oligonucleotide, wherein at least one sugar, nucleobase, or internucleoside linkage is modified. “Unmodified oligonucleotide” means an oligonucleotide that does not comprise any sugar, nucleobase, or internucleoside modification.

(72) “Parent oligonucleotide” means an oligonucleotide whose sequence is used as the basis of design for more oligonucleotides of similar sequence but with different lengths, motifs, and/or

chemistries. The newly designed oligonucleotides may have the same or overlapping sequence as the parent oligonucleotide.

(73) “Parenteral administration” means administration through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g. intrathecal or intracerebroventricular administration.

(74) “Pharmaceutically acceptable carrier or diluent” means any substance suitable for use in administering to an individual. For example, a pharmaceutically acceptable carrier can be a sterile aqueous solution, such as PBS or water-for-injection.

(75) “Pharmaceutically acceptable salts” means physiologically and pharmaceutically acceptable salts of compounds, such as oligomeric compounds or oligonucleotides, i.e., salts that retain the desired biological activity of the parent compound and do not impart undesired toxicological effects thereto.

(76) “Pharmaceutical agent” means a compound that provides a therapeutic benefit when administered to an individual.

(77) “Pharmaceutical composition” means a mixture of substances suitable for administering to an individual. For example, a pharmaceutical composition may comprise one or more compounds or salt thereof and a sterile aqueous solution.

(78) “Phosphorothioate linkage” means a modified phosphate linkage in which one of the non-bridging oxygen atoms is replaced with a sulfur atom. A phosphorothioate internucleoside linkage is a modified internucleoside linkage.

(79) “Phosphorus moiety” means a group of atoms comprising a phosphorus atom. In certain embodiments, a phosphorus moiety comprises a mono-, di-, or tri-phosphate, or phosphorothioate.

(80) “Portion” means a defined number of contiguous (i.e., linked) nucleobases of a nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of a target nucleic acid. In certain embodiments, a portion is a defined number of contiguous nucleobases of an oligomeric compound.

(81) “Prevent” refers to delaying or forestalling the onset, development or progression of a disease, disorder, or condition for a period of time from minutes to indefinitely.

(82) “Prodrug” means a compound in a form outside the body which, when administered to an individual, is metabolized to another form within the body or cells thereof. In certain embodiments, the metabolized form is the active, or more active, form of the compound (e.g., drug). Typically conversion of a prodrug within the body is facilitated by the action of an enzyme(s) (e.g., endogenous or viral enzyme) or chemical(s) present in cells or tissues, and/or by physiologic conditions.

(83) “Reduce” means to bring down to a smaller extent, size, amount, or number.

(84) “RefSeq No.” is a unique combination of letters and numbers assigned to a sequence to indicate the sequence is for a particular target transcript (e.g., target gene). Such sequence and information about the target gene (collectively, the gene record) can be found in a genetic sequence database. Genetic sequence-databases include the NCBI Reference Sequence database, GenBank, the European Nucleotide Archive, and the DNA Data Bank of Japan (the latter three forming the International Nucleotide Sequence Database Collaboration or INSDC).

(85) “Region” is defined as a portion of the target nucleic acid having at least one identifiable structure, function, or characteristic.

(86) “RNAi compound” means an antisense compound that acts, at least in part, through RISC or Ago2, but not through RNase H, to modulate a target nucleic acid and/or protein encoded by a target nucleic acid. RNAi compounds include, but are not limited to double-stranded siRNA, single-stranded RNA (ssRNA), and microRNA, including microRNA mimics.

(87) “Segments” are defined as smaller or sub-portions of regions within a nucleic acid.

(88) “Side effects” means physiological disease and/or conditions attributable to a treatment other

than the desired effects. In certain embodiments, side effects include injection site reactions, liver function test abnormalities, renal function abnormalities, liver toxicity, renal toxicity, central nervous system abnormalities, myopathies, and malaise. For example, increased aminotransferase levels in serum may indicate liver toxicity or liver function abnormality. For example, increased bilirubin may indicate liver toxicity or liver function abnormality.

(89) “Single-stranded” in reference to a compound means the compound has only one oligonucleotide.

(90) “Self-complementary” means an oligonucleotide that at least partially hybridizes to itself. A compound consisting of one oligonucleotide, wherein the oligonucleotide of the compound is self-complementary, is a single-stranded compound. A single-stranded compound may be capable of binding to a complementary compound to form a duplex.

(91) “Sites” are defined as unique nucleobase positions within a target nucleic acid.

(92) “Specifically hybridizable” refers to an oligonucleotide having a sufficient degree of complementarity between the oligonucleotide and a target nucleic acid to induce a desired effect, while exhibiting minimal or no effects on non-target nucleic acids. In certain embodiments, specific hybridization occurs under physiological conditions.

(93) “Specifically inhibit” with reference to a target nucleic acid means to reduce or block expression of the target nucleic acid while exhibiting fewer, minimal, or no effects on non-target nucleic acids. Reduction does not necessarily indicate a total elimination of the target nucleic acid's expression.

(94) “Standard cell assay” means assay(s) described in the Examples and reasonable variations thereof.

(95) “Standard in vivo experiment” means the procedure(s) described in the Example(s) and reasonable variations thereof.

(96) “Stereorandom chiral center” in the context of a population of molecules of identical molecular formula means a chiral center having a random stereochemical configuration. For example, in a population of molecules comprising a stereorandom chiral center, the number of molecules having the (S) configuration of the stereorandom chiral center may be but is not necessarily the same as the number of molecules having the (R) configuration of the stereorandom chiral center. The stereochemical configuration of a chiral center is considered random when it is the result of a synthetic method that is not designed to control the stereochemical configuration. In certain embodiments, a stereorandom chiral center is a stereorandom phosphorothioate internucleoside linkage.

(97) “Sugar moiety” means an unmodified sugar moiety or a modified sugar moiety. As used herein, “unmodified sugar moiety” means a β -D-ribosyl moiety, as found in naturally occurring RNA, or a β -D-2'-deoxyribosyl sugar moiety as found in naturally occurring DNA. As used herein, “modified sugar moiety” or “modified sugar” means a sugar surrogate or a furanosyl sugar moiety other than a β -D-ribosyl or a β -D-2'-deoxyribosyl. Modified furanosyl sugar moieties may be modified or substituted at a certain position(s) of the sugar moiety, substituted, or unsubstituted, and they may or may not have a stereoconfiguration other than β -D-ribosyl. Modified furanosyl sugar moieties include bicyclic sugars and non-bicyclic sugars.

(98) “Sugar surrogate” means a modified sugar moiety that does not comprise a furanosyl or tetrahydrofuran ring (is not a “furanosyl sugar moiety”) and that can link a nucleobase to another group, such as an internucleoside linkage, conjugate group, or terminal group in an oligonucleotide. Modified nucleosides comprising sugar surrogates can be incorporated into one or more positions within an oligonucleotide and such oligonucleotides are capable of hybridizing to complementary oligomeric compounds or nucleic acids.

(99) “Synergy” or “synergize” refers to an effect of a combination that is greater than additive of the effects of each component alone at the same doses.

(100) “Target gene” refers to a gene encoding a target.

(101) “Targeting” means the specific hybridization of a compound to a target nucleic acid in order to induce a desired effect.

(102) “Target nucleic acid,” “target RNA,” “target RNA transcript” and “nucleic acid target” all mean a nucleic acid capable of being targeted by compounds described herein.

(103) “Target region” means a portion of a target nucleic acid to which one or more compounds is targeted.

(104) “Target segment” means the sequence of nucleotides of a target nucleic acid to which a compound is targeted. “5′ target site” refers to the 5′-most nucleotide of a target segment. “3′ target site” refers to the 3′-most nucleotide of a target segment.

(105) “Terminal group” means a chemical group or group of atoms that is covalently linked to a terminus of an oligonucleotide.

(106) “Therapeutically effective amount” means an amount of a compound, pharmaceutical agent, or composition that provides a therapeutic benefit to an individual.

(107) “Treat” refers to administering a compound or pharmaceutical composition to an animal in order to effect an alteration or improvement of a disease, disorder, or condition in the animal.

Certain Embodiments

(108) Certain embodiments provide methods, compounds and compositions for inhibiting HSD17B13 expression.

(109) Certain embodiments provide compounds targeted to a HSD17B13 nucleic acid. In certain embodiments, the HSD17B13 nucleic acid has the sequence set forth in RefSeq or GENBANK Accession No. NM_178135.4 (incorporated by reference, disclosed herein as SEQ ID NO: 1), the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000 (incorporated by reference, disclosed herein as SEQ ID NO: 2), the sequence listed as SEQ ID NO: 3, or GENBANK Accession No. NM_001136230.2 (incorporated by reference, disclosed herein as SEQ ID NO: 4). In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded.

(110) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

(111) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 9 to 80 linked nucleosides and having a nucleobase sequence comprising at least 9 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

(112) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 10 to 80 linked nucleosides and having a nucleobase sequence comprising at least 10 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides.

(113) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 11 to 80 linked nucleosides and having a nucleobase sequence comprising at least 11 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain

embodiments, the modified oligonucleotide consists of 11 to 30 linked nucleosides.

(114) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 80 linked nucleosides and having a nucleobase sequence comprising at least 12 contiguous nucleobases of any of the nucleobase sequences of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 12 to 30 linked nucleosides.

(115) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

(116) Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of the nucleobase sequence of any one of 8-2896. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded.

(117) In certain embodiments, the compound is double-stranded.

(118) Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 3095-3369, 3371-7564, 7565-7672, 7673-8777, 8778-8909, 8910-10401, 10402-10508, 10509-12040, 12041-12178, 12179-15641, 15642-15758, 15759-20692, or 20693-22212 of SEQ ID NO: 2, wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2. In certain embodiments, the compound is an antisense compound or oligomeric compound. In certain embodiments, the compound is single-stranded. In certain embodiments, the compound is double-stranded. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

(119) In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

(120) In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

(121) In certain embodiments, compounds target within nucleobases 3095-3454, 3095-3140, 3152-

3236, 3168-3183, 3328-3328, 3312-3328, 3334-3339, 3348-3390, 3401-3429, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-

20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2.

(122) In certain embodiments, compounds have at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion complementary to an equal length portion within nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-

18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

(123) In certain embodiments, compounds have a nucleobase sequence complementary to nucleobases 1323-1338, 1600-1615, 1627-1642, or 1782-1797 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1.

(124) In certain embodiments, compounds have a nucleobase sequence comprising a portion of at least 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobases complementary within an equal length portion of nucleobases 1323-1338, 1600-1615, 1627-1642, or 1782-1797 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 1. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

(125) In certain embodiments, compounds have a nucleobase sequence complementary to nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2.

(126) In certain embodiments, compounds have a nucleobase sequence comprising a portion of at least 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobases complementary within an equal length portion of nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of an HSD17B13 nucleic acid having the nucleobase sequence of SEQ ID NO: 2. In certain embodiments, these compounds are antisense compounds, oligomeric compounds, or oligonucleotides. In certain embodiments, the modified oligonucleotide consists of 10 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides.

(127) In certain embodiments, any of the foregoing modified oligonucleotides has at least one modified internucleoside linkage, at least one modified sugar, and/or at least one modified nucleobase.

(128) In certain embodiments, at least one nucleoside of any of the foregoing modified oligonucleotides comprises a modified sugar. In certain embodiments, the modified sugar comprises a 2'-O-methoxyethyl group. In certain embodiments, the modified sugar is a bicyclic sugar, such as a 4'-CH(CH₃)-O-2' group, a 4'-CH₂-O-2' group, or a 4'-(CH₂).sub.2—O-2' group.

(129) In certain embodiments, at least one internucleoside linkage of the modified oligonucleotide comprises a modified internucleoside linkage, such as a phosphorothioate internucleoside linkage.

(130) In certain embodiments, at least one nucleobase of any of the foregoing modified oligonucleotides comprises a modified nucleobase, such as 5-methylcytosine.

(131) In certain embodiments, any of the foregoing modified oligonucleotides has: a gap segment

consisting of linked deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide consists of 12 to 30 linked nucleosides and has a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 8-2896. In certain embodiments, the modified oligonucleotide is 16 to 30 linked nucleosides in length having a nucleobase sequence comprising the nucleobase sequence recited in any one of SEQ ID NOs: 8-2896. In certain embodiments, the modified oligonucleotide is 16 linked nucleosides in length having a nucleobase sequence consisting of the sequence recited in any one of SEQ ID NOs: 8-2896.

(132) In certain embodiments, a compound comprises or consists of a modified oligonucleotide consisting of 16 to 80 linked nucleobases and having a nucleobase sequence comprising the nucleobase sequence recited in any one of 8-2896, wherein the modified oligonucleotide has: a gap segment consisting of linked 2'-deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides;

(133) wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide consists of 16 to 30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides

(134) In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, and 1595, wherein the modified oligonucleotide has: a gap segment consisting of ten linked 2'-deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of three linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

(135) In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 819, wherein the modified oligonucleotide has: a gap segment consisting of nine linked 2'-deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt sugar; wherein the sugar residues of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

(136) In certain embodiments, a compound comprises or consists of a modified oligonucleotide 12-30 linked nucleobases in length having a nucleobase sequence comprising the sequence recited in any one of SEQ ID NOs: 1595, wherein the modified oligonucleotide has: a gap segment consisting of ten linked nucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides;

wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight

deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide consists of 16-30 linked nucleosides. In certain embodiments, the modified oligonucleotide consists of 16 linked nucleosides.

(137) In any of the foregoing embodiments, the compound or oligonucleotide can be at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% complementary to a nucleic acid encoding HSD17B13.

(138) In any of the foregoing embodiments, the compound can be single-stranded. In certain embodiments, the compound comprises deoxyribonucleotides. In certain embodiments, the compound is double-stranded. In certain embodiments, the compound is double-stranded and comprises ribonucleotides. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

(139) In any of the foregoing embodiments, the compound can consist of 8 to 80, 10 to 30, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked nucleosides. In certain embodiments, the compound comprises or consists of an oligonucleotide.

(140) In certain embodiments, a compound comprises a modified oligonucleotide described herein and a conjugate group. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 5' end of the modified oligonucleotide. In certain embodiments, the conjugate group is linked to the modified oligonucleotide at the 3' end of the modified oligonucleotide. In certain embodiments, the conjugate group comprises at least one N-Acetylgalactosamine (GalNAc), at least two N-Acetylgalactosamines (GalNAcs), or at least three N-Acetylgalactosamines (GalNAcs).

(141) In certain embodiments, compounds or compositions provided herein comprise a pharmaceutically acceptable salt of the modified oligonucleotide. In certain embodiments, the salt is a sodium salt. In certain embodiments, the salt is a potassium salt.

(142) In certain embodiments, a compound provided herein is according to the following chemical structure:

(143) ##STR00001##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(144) In certain embodiments, a compound provided herein is according to the following chemical structure:

(145) ##STR00002##

(146) In certain embodiments, a compound provided herein is according to the following chemical structure:

(147) ##STR00003##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(148) In certain embodiments, a compound provided herein is according to the following chemical structure:

(149) ##STR00004##

(150) In certain embodiments, a compound provided herein is according to the following chemical structure:

(151) ##STR00005##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(152) In certain embodiments, a compound provided herein is according to the following chemical structure:

(153) ##STR00006##

(154) In certain embodiments, a compound provided herein is according to the following chemical structure:

(155) ##STR00007##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(156) In certain embodiments, a compound provided herein is according to the following chemical structure:

(157) ##STR00008##

(158) In certain embodiments, a compound provided herein is according to the following chemical structure:

(159) ##STR00009##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(160) In certain embodiments, a compound provided herein is according to the following chemical structure:

(161) ##STR00010##

(162) In certain embodiments, a compound provided herein is according to the following chemical structure:

(163) ##STR00011##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(164) In certain embodiments, a compound provided herein is according to the following chemical structure:

(165) ##STR00012##

(166) In certain embodiments, a compound provided herein is according to the following chemical structure:

(167) ##STR00013##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(168) In certain embodiments a compound provided herein is according to the following chemical structure:

(169) ##STR00014##

(170) In certain embodiments, a compound provided herein is according to the following chemical structure:

(171) ##STR00015##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(172) In certain embodiments, a compound provided herein is according to the following chemical structure:

(173) ##STR00016##

(174) In certain embodiments, a compound provided herein k according to the following chemical structure:

(175) ##STR00017##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(176) In certain embodiments, a compound provided herein is according to the following chemical structure:

(177) ##STR00018##

(178) In certain embodiments, a compound provided herein is according to the following chemical

structure:

(179) ##STR00019##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(180) In certain embodiments, a compound provided herein is according to the following chemical structure:

(181) ##STR00020##

(182) In certain embodiments, a compound provided herein is according to the following chemical structure:

(183) ##STR00021##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(184) In certain embodiments, a compound provided herein is according to the following chemical structure:

(185) ##STR00022##

(186) In certain embodiments, a compound provided herein is according to the following chemical structure:

(187) ##STR00023##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(188) In certain embodiments, a compound provided herein is according to the following chemical structure:

(189) ##STR00024##

(190) In certain embodiments, a compound provided herein is according to the following chemical structure:

(191) ##STR00025##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(192) In certain embodiments, a compound provided herein is according to the following chemical structure:

(193) ##STR00026##

(194) In certain embodiments, a compound provided herein is according to the following chemical structure:

(195) ##STR00027##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(196) In certain embodiments, a compound provided herein is according to the following chemical structure:

(197) ##STR00028##

(198) In certain embodiments, a compound provided herein is according to the following chemical structure:

(199) ##STR00029##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(200) In certain embodiments, a compound provided herein is according to the following chemical structure:

(201) ##STR00030##

(202) In certain embodiments, a compound provided herein is according to the following chemical structure:

(203) ##STR00031##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(204) In certain embodiments, a compound provided herein is according to the following chemical structure:

(205) ##STR00032##

(206) In certain embodiments, a compound provided herein is according to the following chemical structure:

(207) ##STR00033##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(208) In certain embodiments, a compound provided herein is according to the following chemical structure:

(209) ##STR00034##

(210) In certain embodiments, a compound provided herein is according to the following chemical structure:

(211) ##STR00035##

or a salt thereof. In certain embodiments, the compound is a sodium salt. In certain embodiments, the compound is a potassium salt.

(212) In certain embodiments, a compound provided herein is according to the following chemical structure:

(213) ##STR00036##

(214) Under certain conditions, certain compounds disclosed herein act as acids. Although such compounds may be drawn or described in protonated (free acid) form, or ionized and in association with a cation (salt) form, aqueous solutions of such compounds exist in equilibrium among such forms. For example, a phosphate linkage of an oligonucleotide in aqueous solution exists in equilibrium among free acid, anion and salt forms. Unless otherwise indicated, compounds described herein are intended to include all such forms. Moreover, certain oligonucleotides have several such linkages, each of which is in equilibrium. Thus, oligonucleotides in solution exist in an ensemble of forms at multiple positions all at equilibrium. The term "oligonucleotide" is intended to include all such forms. Drawn structures necessarily depict a single form. Nevertheless, unless otherwise indicated, such drawings are likewise intended to include corresponding forms. Herein, a structure depicting the free acid of a compound followed by the term "or a salt thereof" expressly includes all such forms that may be fully or partially protonated/de-protonated/in association with a cation. In certain instances, one or more specific cation is identified.

(215) In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with sodium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with potassium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in PBS. In certain embodiments, modified oligonucleotides or oligomeric compounds are in water. In certain such embodiments, the pH of the solution is adjusted with NaOH and/or HCl to achieve a desired pH. In certain embodiments, the compounds or compositions as described herein are active by virtue of having at least one of an in vitro IC₅₀ of less than 2 μ M, less than 1.5 μ M, less than 1 M, less than 0.9 μ M, less than 0.8 μ M, less than 0.7 M, less than 0.6 μ M, less than 0.5 M, less than 0.4 μ M, less than 0.3 μ M, less than 0.2 μ M, less than 0.1 μ M, less than 0.05 μ M, less than 0.04 μ M, less than 0.03 μ M, less than 0.02 μ M, or less than 0.01 μ M.

(216) In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having at least one of an increase in alanine transaminase (ALT) or aspartate transaminase (AST) value of no more than 4 fold, 3 fold, or 2 fold over control animals, or an increase in liver, spleen, or kidney weight of no more than 30%, 20%, 15%, 12%, 10%, 5%, or 2% compared to control animals. In certain embodiments, the compounds or compositions as

described herein are highly tolerable as demonstrated by having no increase of ALT or AST over control animals. In certain embodiments, the compounds or compositions as described herein are highly tolerable as demonstrated by having no increase in liver, spleen, or kidney weight over control animals.

(217) Certain embodiments provide a composition comprising the compound of any of the aforementioned embodiments or any pharmaceutically acceptable salt thereof and at least one of a pharmaceutically acceptable carrier or diluent. In certain embodiments, the composition has a viscosity less than about 40 centipoise (cP), less than about 30 centipoise (cP), less than about 20 centipoise (cP), less than about 15 centipoise (cP), or less than about 10 centipoise (cP). In certain embodiments, the composition having any of the aforementioned viscosities comprises a compound provided herein at a concentration of about 100 mg/mL, about 125 mg/mL, about 150 mg/mL, about 175 mg/mL, about 200 mg/mL, about 225 mg/mL, about 250 mg/mL, about 275 mg/mL, or about 300 mg/mL. In certain embodiments, the composition having any of the aforementioned viscosities and/or compound concentrations has a temperature of room temperature, or about 20° C., about 21° C., about 22° C., about 23° C., about 24° C., about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., or about 30° C.

(218) Non-Limiting Numbered Embodiments Include:

(219) Embodiment 1: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, or at least 12 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, or at least 12 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. Embodiment 2: Certain embodiments provide a compound comprising modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. Certain embodiments provide a compound comprising modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. Embodiment 3: Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NOs: 8-2896. Embodiment 4: Certain embodiments provide a compound comprising modified oligonucleotide consisting of 8 to 80 linked nucleosides, or consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases complementary to an equal length portion of nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-5319, 5395-5950, 5646-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5777, 5782-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565,

7550-7634, 7563-7699, 7653-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7789, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 85/c, at least 90/c, at least 95%, or 100% complementary to SEQ ID NO: 2. Embodiment 5: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence complementary within nucleobases 3095-3454, 3095-3140, 3152-3236, 3168-3183, 3248-3328, 3312-3328, 3334-3390, 3348-3390, 3401-3420, 3429-3451, 3429-3454, 3474-3719, 3474-3644, 3478-3494, 3505-3531, 3544-3559, 3568-3634, 3671-3736, 3676-3717, 3721-3799, 3741-3771, 3750-3770, 3782-3799, 3834-3849, 3834-4154, 3877-3897, 3902-3917, 3933-3967, 3933-3979, 3988-4005, 3988-4017, 4068-4091, 4103-4118, 4123-4155, 4133-4155, 4220-4240, 4220-4264, 4249-4264, 4222-4240, 4296-4311, 4360-4378, 4389-4427, 4389-4431, 4405-4449, 4476-4496, 4517-4588, 4517-4552, 4537-4552, 4572-4588, 4640-4678, 4640-4879, 4641-4667, 4862-4879, 4991-5006, 4991-5305, 5029-5098, 5046-5098, 5104-5120, 5121-5141, 5125-5141, 5156-5256, 5158-5217, 5223-5242, 5267-5319, 5268-

5319, 5395-5950, 5645-5695, 5655-5858, 5712-5752, 5712-5777, 5758-5797, 5782-5862, 5804-5862, 5870-5910, 5871-6005, 5870-5910, 5870-5911, 5916-6007, 5935-5950, 5974-6007, 6029-6048, 6029-6055, 6035-6164, 6084-6099, 6084-6104, 6134-6164, 6170-6248, 6170-6275, 6170-6277, 6191-6218, 6221-6243, 6246-6275, 6284-6350, 6284-6352, 6298-6372, 6357-6372, 6388-6406, 6422-6493, 6422-6497, 6426-6621, 6567-6758, 6567-6582, 6597-6653, 6626-6653, 6710-6779, 6784-6811, 6785-6800, 6795-6811, 6913-6930, 6914-6929, 6914-6930, 6915-6930, 7101-7138, 7101-7441, 7125-7145, 7426-7441, 7473-7494, 7473-7498, 7481-7498, 7526-7543, 7550-7565, 7550-7634, 7563-7699, 7643-7676, 7680-7698, 7680-7699, 7708-7723, 7751-7819, 7768-7796, 7828-7866, 7834-7969, 7938-7968, 7938-7969, 7973-7994, 8017-8032, 8039-8186, 8039-8065, 8040-8065, 8121-8156, 8125-8151, 8164-8185, 8164-8186, 8205-8297, 8205-8220, 8236-8252, 8236-8271, 8256-8294, 8279-8297, 8487-8502, 8487-8627, 8529-8579, 8587-8628, 8636-8732, 8643-8910, 8744-8762, 8786-8910, 8939-9027, 8939-8996, 9000-9027, 9057-9096, 9080-9230, 9110-9173, 9188-9240, 9256-9275, 9312-9365, 9372-9401, 9373-9394, 9410-9446, 9410-9491, 9476-9555, 9497-9555, 9588-9678, 9588-9681, 9712-9737, 9712-9742, 9751-9811, 9751-10040, 9887-9902, 9908-10017, 10025-10040, 10058-10079, 10112-10160, 10124-10160, 10138-10160, 10180-10227, 10180-10233, 10258-10279, 10292-10310, 10293-10310, 10335-10375, 10335-10538, 10380-10538, 10554-10569, 10584-10599, 10728-10745, 10831-10870, 10930-10946, 10931-10946, 10970-10993, 10971-11015, 1100-11015, 11030-11045, 11073-11094, 11074-11094, 11115-11132, 11117-11132, 11155-11171, 11210-11228, 11210-11276, 11235-11276, 11290-11323, 11308-11323, 11384-11407, 11385-11443, 11428-11443, 11468-11529, 11542-11605, 11621-11636, 11621-11638, 11621-11668, 11652-11668, 11742-11808, 11742-11811, 11853-11881, 11853-11909, 11892-11912, 11924-11940, 11925-12226, 11953-12027, 12038-12226, 12238-12270, 12337-12531, 12337-12368, 12410-12448, 12465-12531, 12598-12642, 12598-12742, 12647-12742, 12758-12832, 12761-12832, 12862-12904, 13548-13627, 13551-13654, 13638-13674, 13688-13798, 13704-13798, 13813-13932, 13949-14064, 13951-13969, 13975-14064, 14077-14107, 14089-14107, 14126-14147, 14165-14221, 14166-14221, 14243-14298, 14243-14299, 14315-14336, 14316-14336, 14362-14414, 14432-14454, 14461-14514, 14465-14514, 14541-14815, 14541-14636, 14724-14746, 1400-14815, 14905-14951, 14916-14951, 15019-15039, 15022-15039, 15170-15185, 15200-15232, 15211-15268, 15244-15268, 15279-15295, 15279-15399, 15312-15343, 15350-15399, 15405-15460, 15492-15526, 15570-15586, 15633-15808, 15641-15657, 15643-15711, 15716-15765, 15771-15802, 16116-16144, 16116-16149, 16189-16206, 16189-16342, 16218-16240, 16254-16310, 16327-16352, 16377-16397, 16377-16400, 16407-16439, 16407-16440, 16461-16476, 16533-16548, 16755-16770, 16895-16920, 16905-16920, 16956-16984, 16956-17092, 17014-17034, 17135-17159, 17041-17062, 17077-17092, 17135-17159, 17227-17254, 17672-17795, 17675-17872, 17802-17817, 17857-17872, 17909-17945, 17909-17971, 17953-17971, 17954-17971, 17984-18061, 17985-18048, 18075-18117, 18087-18115, 18138-18160, 18176-18193, 18505-18555, 18506-18536, 18585-18600, 18658-18712, 18662-18762, 18720-18763, 18798-18864, 18871-18888, 18901-18940, 18925-18940, 18958-18983, 18958-19013, 19388-19460, 19467-19513, 19474-19505, 19531-19626, 19533-19626, 19649-19717, 19649-19726, 19731-19801, 19731-19842, 19815-19875, 19860-19951, 19887-19951, 19970-19985, 19970-20054, 19999-20024, 20037-20061, 20087-20102, 20109-20153, 20087-20207, 20172-20212, 20356-20438, 20248-20437, 20248-20264, 20470-20508, 20481-20507, 20571-20623, 20571-20644, 20685-20700, 20706-21032, 20706-20772, 20781-20859, 20869-20984, 20990-21033, 21065-21102, 21065-21263, 21092-21632, 21276-21500, 21517-21632, 21989-22034, 22059-22087, or 22164-22209 of SEQ ID NO: 2, and wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2. Embodiment 6: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8 contiguous nucleobases 100% complementary to an equal length portion of nucleobases 1-46, 58-

142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of SEQ ID NO: 1, and wherein the nucleobase sequence of the modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 1. Embodiment 7: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 8 to 80 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence complementary within nucleobases 1-46, 58-142, 1-142, 74-89, 154-197, 154-234, 200-232, 218-234, 240-345, 254-269, 354-746, 354-830, 519-534, 565-580, 603-628, 693-716, 734-750, 734-830, 749-828, 835-862, 835-957, 966-1044, 1017-1043, 1054-1106, 1055-1070, 1075-1169, 1074-1132, 1136-1169, 1175-1218, 1180-1217, 1250-1292, 1251-1279, 1296-1146, 1296-1448, 1308-1448, 1461-1481, 1461-1519, 1461-1647, 1461-1646, 1483-1535, 1538-1646, 1544-1561, 1597-1645, 1654-1685, 1654-1817, 1702-1817, 1705-1741, 1748-2211, 2174-2219, 2240-2272, 2244-2272, 2349-2370, 2350-2371, or 2378-2394 of SEQ ID NO: 1, and wherein said modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 1. Embodiment 8: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising a portion of at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16 contiguous nucleobases 100% complementary to an equal length portion of nucleobases 3439-3454, 3552-3567, 3754-3769, 3963-3978, 4406-4421, 4123-4138, 4139-4154, 4991-5006, 5045-5060, 5662-5677, 6476-6491, 6478-6493, 17992-18007, 21138-21153, 21415-21430, 21442-21457, or 21597-21612 of SEQ ID NO: 2, and wherein the nucleobase sequence of the modified oligonucleotide is at least 85%, at least 90%, at least 95%, or 100% complementary to SEQ ID NO: 2. Embodiment 9: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence comprising any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819. Embodiment 10: Certain embodiments provide a compound comprising a modified oligonucleotide having a nucleobase sequence consisting of any one of SEQ ID NO: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819. Embodiment 11: Certain embodiments provide the compound of embodiments 1-6, wherein the oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to SEQ ID NO: 2 over the entire length of the oligonucleotide. Embodiment 12: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises at least one modification selected from at least one modified internucleoside linkage, at least one modified nucleoside comprising a modified sugar moiety, and at least one modified nucleoside comprising a modified nucleobase. Embodiment 13: Certain embodiments provide the compound of embodiment 12, wherein the modified internucleoside linkage is a phosphorothioate internucleoside linkage. Embodiment 14: Certain embodiments provide the compound of embodiments 12 or 13, wherein at least one nucleoside of the modified oligonucleotide is a 2'-substituted nucleoside or a bicyclic nucleoside. Embodiment 15: Certain embodiments provide the compound of embodiment 14, wherein the bicyclic sugar moiety of the bicyclic nucleoside is selected from the group consisting of: 4'-(CH₂)—O-2' (LNA); 4'-(CH₂)₂—O-2' (ENA); and 4'-CH(CH₂)—O-2' (cEt). Embodiment 16: Certain embodiments provide the compound of any one of embodiments 12-15, wherein the modified nucleoside is 2'-O-methoxyethyl nucleoside. Embodiment 17: Certain embodiments provide the compound of any one of embodiments 12-16, wherein the modified nucleobase is a 5-methylcytosine. Embodiment 18:

Certain embodiments provide the compound of any one of embodiments 1-17, wherein the modified oligonucleotide has: a gap segment consisting of linked 2'-deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. Embodiment 19: Certain embodiments provide the compound of any one of embodiment **1-18**, wherein the modified sugar moiety of each nucleoside of each wing segment is selected from 2'-O-Me, 2'-MOE, and cEt sugar moieties. Embodiment 20: Certain embodiments provide the compound of any of one embodiment **1-19**, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of three linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt nucleoside; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 21: Certain embodiments provide the compound of any of one embodiment **1-20**, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of three linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of each wing segment comprises a cEt nucleoside; wherein at least one internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 22: Certain embodiments provide the compound of any of the preceding embodiments, wherein the modified oligonucleotide comprises: a gap segment consisting of nine linked 2'-deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt nucleoside; wherein the sugar moieties of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 23: Certain embodiments provide the compound of any of the preceding embodiments, wherein the modified oligonucleotide comprises: a gap segment consisting of nine linked 2'-deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt nucleoside; wherein each sugar moiety of the nucleosides of the 3' wing segment are selected from cEt and 2'-MOE sugar moieties; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 24: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked nucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt nucleoside; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 25: Embodiment 24: Certain embodiments provide the compound of any one of the preceding embodiments, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked nucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the nucleosides of the gap segment are selected from deoxynucleosides, cEt nucleosides and 2'-MOE nucleosides, wherein each nucleoside of each of the wing segments comprises a cEt nucleoside; wherein each

internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 26: Certain embodiments provide the compound of any one of embodiments 1-25, wherein the compound is single-stranded. Embodiment 27: Certain embodiments provide the compound of any one of embodiments 1-25, wherein the modified oligonucleotide is hybridized to a second oligonucleotide to form a double-stranded antisense compound. Embodiment 28: Certain embodiments provide the compound of any one of embodiments 1-27, wherein at least one of the nucleotides of the modified oligonucleotide is a ribonucleotide. Embodiment 29: Certain embodiments provide the compound of any one of embodiments 1-27, wherein at least one of the nucleotides of the modified oligonucleotide is a deoxyribonucleotide. Embodiment 30: Certain embodiments provide the compound of any one of embodiments 1-23, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides. Embodiment 31: Certain embodiments provide the compound of any one of embodiments 1-30, wherein the modified oligonucleotide consists of 10 to 30 linked nucleosides. Embodiment 32: Certain embodiments provide the compound of any one of embodiments 1-30, wherein the modified oligonucleotide consists of 16 to 30 linked nucleosides. Embodiment 33: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence selected from any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, 1595, and 819, wherein the modified oligonucleotide comprises: a gap segment consisting of linked deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. Embodiment 34: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence selected from any one of SEQ ID NOs: 286, 817, 983, 1215, 747, 43, 355, 1602, 201, 734, 1249, 208, 513, 1449, 1448, and 1595, wherein the modified oligonucleotide comprises a gap segment consisting of ten linked deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of three linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein the 5' wing segment and the 3' wing segment comprise cEt sugars; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine. Embodiment 35: Certain embodiments provide a compound comprising a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence consisting of SEQ ID NO: 819, wherein the modified oligonucleotide comprises: a gap segment consisting of nine linked 2'-deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; (220) wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment, wherein each nucleoside of the 5' wing segment comprises a cEt sugar; wherein the sugar residues of the nucleosides of the 3' wing segment are 5'-cEt-cEt-cEt-MOE-3'; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 36: Certain embodiments provide a modified oligonucleotide consisting of 16 linked nucleosides, wherein the modified oligonucleotide has a nucleobase sequence consisting of SEQ ID NO: 1595, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked nucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of four linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein the gap segment consists of 5'-one deoxy, one 2'-O-methyl nucleoside, and eight deoxynucleosides-3', wherein each nucleoside of each of the wing segments comprises a cEt sugar; wherein each internucleoside linkage is a phosphorothioate linkage and wherein each cytosine is a 5-methylcytosine. Embodiment 37: Certain embodiments provide the compound of any preceding embodiment, comprising a conjugate group. Embodiment

38: Certain embodiments provide the compound of embodiment 37, wherein the conjugate group comprises a GalNAc cluster comprising 1-3 GalNAc ligands. Embodiment 39: Certain embodiments provide the compound of embodiment 37 or embodiment 38, wherein the conjugate group comprises a conjugate linker consisting of a single bond. Embodiment 40: Certain embodiments provide the compound of any one of embodiments 37-39, wherein the conjugate group comprises a cleavable linker. Embodiment 41: Certain embodiments provide the compound of any one of embodiments 37-40, wherein the conjugate group comprises a conjugate linker comprising 1-3 linker-nucleosides. Embodiment 42: Certain embodiments provide the compound of any one of embodiments 37-41, wherein the conjugate group is attached to the modified oligonucleotide at the 5'-end of the modified oligonucleotide. Embodiment 42: Certain embodiments provide the compound of any one of embodiments 37-41, wherein the conjugate group is attached to the modified oligonucleotide at the 3'-end of the modified oligonucleotide. Embodiment 43: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(221) ##STR00037## Embodiment 44: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(222) ##STR00038## Embodiment 45: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(223) ##STR00039## Embodiment 46: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(224) ##STR00040## Embodiment 47: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(225) ##STR00041## Embodiment 48: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(226) ##STR00042## Embodiment 49: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(227) ##STR00043## Embodiment 50: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(228) ##STR00044## Embodiment 51: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(229) ##STR00045## Embodiment 52: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(230) ##STR00046## Embodiment 53: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(231) ##STR00047## Embodiment 54: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(232) ##STR00048## Embodiment 55: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(233) ##STR00049## Embodiment 56: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(234) ##STR00050## Embodiment 57: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(235) ##STR00051## Embodiment 58: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(236) ##STR00052## Embodiment 59: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(237) ##STR00053## Embodiment 60: Certain embodiments provide the compound having the following chemical structure or salt thereof:

(238) ##STR00054## Embodiment 61: Certain embodiments provide the compound of any one of the embodiments 43-60, wherein the compound is a sodium salt. Embodiment 62: Certain

embodiments provide the compound of any one of the embodiments 43-60, wherein the compound is a potassium salt. Embodiment 63: Certain embodiments provide the chirally enriched population of the compound of any one of embodiments 1-62, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having a particular stereochemical configuration. Embodiment 64: Certain embodiments provide the chirally enriched population of embodiment 63, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having the (Sp) configuration. Embodiment 65: Certain embodiments provide the chirally enriched population of embodiment 63 or embodiment 64, wherein the population is enriched for compounds having a modified oligonucleotide comprising at least one particular phosphorothioate internucleoside linkage having the (Rp) configuration. Embodiment 66: Certain embodiments provide the chirally enriched population of embodiment **63**, wherein the population is enriched for compounds having a modified oligonucleotide having a particular, independently selected stereochemical configuration at each phosphorothioate internucleoside linkage. Embodiment 67: Certain embodiments provide the chirally enriched population of embodiment 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having the (Sp) configuration at each phosphorothioate internucleoside linkage. Embodiment 68: Certain embodiments provide the chirally enriched population of embodiment 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having the (Rp) configuration at each phosphorothioate internucleoside linkage. Embodiment 69: Certain embodiments provide the chirally enriched population of embodiment 63 or 66, wherein the population is enriched for oligomeric compounds having a modified oligonucleotide having at least 3 contiguous phosphorothioate internucleoside linkages in the Sp-Sp-Rp configurations, in the 5' to 3' direction. Embodiment 70: Certain embodiments provide a population of compounds having a modified oligonucleotide of any of embodiments 69, wherein all of the phosphorothioate internucleoside linkages of the modified oligonucleotide are stereorandom. Embodiment 71: Certain embodiments provide a composition comprising the compound of any one of embodiments 1-70 and a pharmaceutically acceptable carrier. Embodiment 72: Certain embodiments provide a composition comprising a compound of any preceding embodiment, for use in therapy. Embodiment 73: Certain embodiments provide a method of treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual comprising administering to the individual a compound complementary to HSD17B13, thereby treating, preventing, or ameliorating the disease. Embodiment 74: Certain embodiments provide a method of administering the compound of embodiments 1-70 or composition of embodiment 71 or embodiment 72 to an individual. Embodiment 75: Certain embodiments provide the method of embodiments 73 or 74, wherein the individual has a liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Embodiment 76: Certain embodiments provide the method of embodiment 73 or embodiment 74, wherein the compound is an antisense compound. Embodiment 77: Certain embodiments provide the method of any of embodiments 73-76, wherein administering the compound ameliorates liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in the individual. Embodiment 78: Certain embodiments provide a method of inhibiting expression of HSD17B13 in a cell comprising contacting the cell with a compound comprising a modified oligonucleotide complementary to HSD17B13, thereby inhibiting expression of HSD17B13 in the cell. Embodiment 79: Certain embodiments provide the method of embodiment 78, wherein the cell is in the liver of an individual. Embodiment 80: Certain embodiments provide the method of embodiment 79, wherein the individual has, or is at risk of having, liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing

cholangitis. Embodiment 81: Certain embodiments provide a method of ameliorating, reducing or inhibiting liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an individual, comprising administering a compound comprising a modified oligonucleotide complementary to HSD17B13 to the individual, thereby reducing or inhibiting liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in the individual. Embodiment 82: Certain embodiments provide the method of embodiment 81, wherein the individual has, or is at risk of having, liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Embodiment 83: Certain embodiments provide the method of any one of embodiments 78-82, wherein the compound is an antisense compound. Embodiment 84: Certain embodiments provide the method of any one of embodiments 78-83, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72. Embodiment 85: Certain embodiments provide the method of embodiment 83 or embodiment 84, wherein the compound or composition is administered parenterally. Embodiment 86: Certain embodiments provide the use of a compound comprising a modified oligonucleotide complementary to HSD17B13 for treating, preventing, or ameliorating a disease associated with HSD17B13. Embodiment 87: Certain embodiments provide the use of embodiment 86, wherein the disease is liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Embodiment 88: Certain embodiments provide the use of embodiments 86 or 87, wherein the compound is an antisense compound. Embodiment 89: Certain embodiments provide the use of any one of embodiments 86-88, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72. Embodiment 90: Certain embodiments provide the use of a compound comprising a modified oligonucleotide complementary to HSD17B13 in the manufacture of a medicament for treating, preventing, or ameliorating a disease associated with HSD17B13. Embodiment 91: Certain embodiments provide the use of embodiment 90, wherein the disease is liver disease, NAFLD, NASH, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Embodiment 92: Certain embodiments provide the use of embodiments 90 or 91, wherein the compound is an antisense compound. Embodiment 93: Certain embodiments provide the use of any one of embodiments 90-92, wherein the compound is the compound of any one of embodiments 1-70 or composition of embodiment 71 or embodiment 72.

Certain Indications

(239) Certain embodiments provided herein relate to methods of inhibiting HSD17B13 expression, which can be useful for treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual, by administration of a compound that targets HSD17B13. In certain embodiments, the compound can be a HSD17B13 specific inhibitor. In certain embodiments, the compound can be an antisense compound, an oligomeric compound, or an oligonucleotide targeted to HSD17B13.

(240) Examples of diseases associated with HSD17B13 treatable, preventable, and/or ameliorable with the methods provided herein include liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Certain compounds provided herein are directed to compounds and compositions that reduce liver damage, steatosis, liver fibrosis, liver inflammation, liver scarring or cirrhosis, liver failure, liver enlargement, elevated transaminases, or hepatic fat accumulation in an animal.

(241) In certain embodiments, a method of treating, preventing, or ameliorating a disease associated with HSD17B13 in an individual comprises administering to the individual a compound

comprising a HSD17B13 specific inhibitor, thereby treating, preventing, or ameliorating the disease. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13. In certain embodiments, the disease is a liver disease. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an animal.

(242) In certain embodiments, a method of treating, preventing, or ameliorating liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an animal comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby treating, preventing, or ameliorating liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13.

(243) In certain embodiments, a method of inhibiting expression of HSD17B13 in an individual having, or at risk of having, a disease associated with HSD17B13 comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby inhibiting expression of HSD17B13 in the individual. In certain embodiments, administering the compound inhibits expression of HSD17B13 in the liver. In certain embodiments, the disease is a liver disease. In certain embodiments, the individual has, or is at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the individual has, or is at risk of having, liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In

certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, administering the compound improves, preserves, or prevents liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation.

(244) In certain embodiments, a method of inhibiting expression of HSD17B13 in a cell comprises contacting the cell with a compound comprising a HSD17B13 specific inhibitor, thereby inhibiting expression of HSD17B13 in the cell. In certain embodiments, the cell is a hepatocyte. In certain embodiments, the cell is in the liver. In certain embodiments, the cell is in the liver of an individual who has, or is at risk of having, liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

(245) In certain embodiments, a method of reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, a disease associated with HSD17B13 comprises administering to the individual a compound comprising a HSD17B13 specific inhibitor, thereby reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in the individual. In certain embodiments, the individual has, or is at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally. In certain embodiments, the individual is identified as having, or at risk of having, a disease associated with HSD17B13.

(246) Certain embodiments are drawn to a compound comprising a HSD17B13 specific inhibitor for use in treating a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound. In certain embodiments, the compound is administered to the individual parenterally.

(247) Certain embodiments are drawn to a compound comprising a HSD17B13 specific inhibitor for use in reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

(248) Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the manufacture or preparation of a medicament for treating a disease associated with HSD17B13. Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the preparation of a medicament for treating a disease associated with HSD17B13. In certain embodiments, the disease is a liver disease. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence

of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

(249) Certain embodiments are drawn to the use of a compound comprising a HSD17B13 specific inhibitor for the manufacture or preparation of a medicament for reducing or inhibiting liver damage, steatosis, liver fibrosis, cirrhosis, elevated transaminases, or hepatic fat accumulation in an individual having, or at risk of having, a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. Certain embodiments are drawn to use of a compound comprising a HSD17B13 specific inhibitor for the preparation of a medicament for treating a disease associated with HSD17B13. In certain embodiments, the disease is liver disease, NAFLD, NASH, alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis. In certain embodiments, the compound comprises an antisense compound targeted to HSD17B13. In certain embodiments, the compound comprises an oligonucleotide targeted to HSD17B13. In certain embodiments, a compound comprises consisting of 8 to 80 linked nucleosides and having a nucleobase sequence comprising at least 8 contiguous nucleobases of any of the nucleobase sequences of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of 16 to 80 linked nucleosides and having a nucleobase sequence comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In certain embodiments, a compound comprises a modified oligonucleotide consisting of the nucleobase sequence of any one of SEQ ID NOs: 8-2896. In any of the foregoing embodiments, the compound can be single-stranded or double-stranded. In any of the foregoing embodiments, the compound can be an antisense compound or oligomeric compound.

(250) In any of the foregoing methods or uses, the compound can be targeted to HSD17B13. In certain embodiments, the compound comprises or consists of a modified oligonucleotide, for example, a modified oligonucleotide consisting of 8 to 80 linked nucleosides, 10 to 30 linked nucleosides, 12 to 30 linked nucleosides, or 20 linked nucleosides. In certain embodiments, the modified oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4. In certain embodiments, the modified oligonucleotide comprises at least one modified internucleoside linkage, at least one modified sugar and/or at least one modified nucleobase. In certain embodiments, the modified internucleoside linkage is a phosphorothioate internucleoside linkage, the modified sugar is a bicyclic sugar or a 2'-O-methoxyethyl modified sugar, and the modified nucleobase is a 5-methylcytosine. In certain embodiments, the modified oligonucleotide comprises a gap segment consisting of linked deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides, wherein the gap segment is positioned immediately adjacent to and between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar.

(251) In any of the foregoing embodiments, the modified oligonucleotide consists of 12 to 30, 15 to 30, 15 to 25, 15 to 24, 16 to 24, 17 to 24, 18 to 24, 19 to 24, 20 to 24, 19 to 22, 20 to 22, 16 to 20, or 16 or 20 linked nucleosides. In certain embodiments, the modified oligonucleotide is at least 80%, at least 85%, at least 90%, at least 95% or 100% complementary to any of the nucleobase sequences recited in SEQ ID NOs: 1-4.

(252) In any of the foregoing methods or uses, the compound comprises or consists of a modified oligonucleotide consisting of 16 to 30 linked nucleosides and having a nucleobase sequence comprising any one of SEQ ID NOs: 8-2896, wherein the modified oligonucleotide has: a gap segment consisting of linked 2'-deoxynucleosides; a 5' wing segment consisting of linked nucleosides; and a 3' wing segment consisting of linked nucleosides;

(253) wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment and wherein each nucleoside of each wing segment comprises a modified sugar. In certain embodiments, the modified oligonucleotide is 16-30 linked nucleosides in length. In certain embodiments, the modified oligonucleotide is 16 linked nucleosides in length.

(254) In any of the foregoing methods or uses, the compound can be administered parenterally. For example, in certain embodiments the compound can be administered through injection or infusion. Parenteral administration includes subcutaneous administration, intravenous administration, intramuscular administration, intraarterial administration, intraperitoneal administration, or intracranial administration, e.g. intrathecal or intracerebroventricular administration.

(255) Certain Compounds

(256) In certain embodiments, compounds described herein can be antisense compounds. In certain embodiments, the antisense compound comprises or consists of an oligomeric compound. In certain embodiments, the oligomeric compound comprises a modified oligonucleotide. In certain embodiments, the modified oligonucleotide has a nucleobase sequence complementary to that of a target nucleic acid.

(257) In certain embodiments, a compound described herein comprises or consists of a modified oligonucleotide. In certain embodiments, the modified oligonucleotide has a nucleobase sequence complementary to that of a target nucleic acid.

(258) In certain embodiments, a compound or antisense compound is single-stranded. Such a single-stranded compound or antisense compound comprises or consists of an oligomeric compound. In certain embodiments, such an oligomeric compound comprises or consists of an oligonucleotide and optionally a conjugate group. In certain embodiments, the oligonucleotide is an antisense oligonucleotide. In certain embodiments, the oligonucleotide is modified. In certain embodiments, the oligonucleotide of a single-stranded antisense compound or oligomeric compound comprises a self-complementary nucleobase sequence.

(259) In certain embodiments, compounds are double-stranded. Such double-stranded compounds comprise a first modified oligonucleotide having a region complementary to a target nucleic acid and a second modified oligonucleotide having a region complementary to the first modified oligonucleotide. In certain embodiments, the modified oligonucleotide is an RNA oligonucleotide. In such embodiments, the thymine nucleobase in the modified oligonucleotide is replaced by a uracil nucleobase. In certain embodiments, compound comprises a conjugate group. In certain embodiments, one of the modified oligonucleotides is conjugated. In certain embodiments, both the modified oligonucleotides are conjugated. In certain embodiments, the first modified oligonucleotide is conjugated. In certain embodiments, the second modified oligonucleotide is conjugated. In certain embodiments, the first modified oligonucleotide is 16-30 linked nucleosides in length and the second modified oligonucleotide is 16-30 linked nucleosides in length. In certain embodiments, one of the modified oligonucleotides has a nucleobase sequence comprising at least 8 contiguous nucleobases of any of SEQ ID NOs: 8-2035.

(260) In certain embodiments, antisense compounds are double-stranded. Such double-stranded antisense compounds comprise a first oligomeric compound having a region complementary to a target nucleic acid and a second oligomeric compound having a region complementary to the first oligomeric compound. The first oligomeric compound of such double stranded antisense compounds typically comprises or consists of a modified oligonucleotide and optionally a conjugate group. The oligonucleotide of the second oligomeric compound of such a double-stranded antisense compound may be modified or unmodified. Either or both oligomeric

compounds of a double-stranded antisense compound may comprise a conjugate group. The oligomeric compounds of double-stranded antisense compounds may include non-complementary overhanging nucleosides.

(261) Examples of single-stranded and double-stranded compounds include, but are not limited to, oligonucleotides, siRNAs, microRNA targeting oligonucleotides, and single-stranded RNAi compounds, such as small hairpin RNAs (shRNAs), single-stranded siRNAs (ssRNAs), and microRNA mimics.

(262) In certain embodiments, a compound described herein has a nucleobase sequence that, when written in the 5' to 3' direction, comprises the reverse complement of the target segment of a target nucleic acid to which it is targeted.

(263) In certain embodiments, a compound described herein comprises an oligonucleotide 12 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 12 to 22 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 14 to 30 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 14 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 15 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 15 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 18 to 30 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 18 to 20 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 20 to 30 linked subunits in length. In other words, such oligonucleotides are 12 to 30 linked subunits, 14 to 30 linked subunits, 14 to 20 subunits, 15 to 30 subunits, 15 to 20 subunits, 16 to 30 subunits, 16 to 20 subunits, 17 to 30 subunits, 17 to 20 subunits, 18 to 30 subunits, 18 to 20 subunits, or 20 to 30 subunits in length, respectively. In certain embodiments, a compound described herein comprises an oligonucleotide 14 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 16 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 17 linked subunits in length. In certain embodiments, compound described herein comprises an oligonucleotide 18 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 19 linked subunits in length. In certain embodiments, a compound described herein comprises an oligonucleotide 20 linked subunits in length. In other embodiments, a compound described herein comprises an oligonucleotide 8 to 80, 12 to 50, 13 to 30, 13 to 50, 14 to 30, 14 to 50, 15 to 30, 15 to 50, 16 to 30, 16 to 50, 17 to 30, 17 to 50, 18 to 22, 18 to 24, 18 to 30, 18 to 50, 19 to 22, 19 to 30, 19 to 50, or 20 to 30 linked subunits. In certain such embodiments, the compound described herein comprises an oligonucleotide 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 linked subunits in length, or a range defined by any two of the above values. In some embodiments the linked subunits are nucleotides, nucleosides, or nucleobases.

(264) In certain embodiments, the compound may further comprise additional features or elements, such as a conjugate group, that are attached to the oligonucleotide. In certain embodiments, such compounds are antisense compounds. In certain embodiments, such compounds are oligomeric compounds. In embodiments where a conjugate group comprises a nucleoside (i.e. a nucleoside that links the conjugate group to the oligonucleotide), the nucleoside of the conjugate group is not counted in the length of the oligonucleotide.

(265) In certain embodiments, compounds may be shortened or truncated. For example, a single

subunit may be deleted from the 5' end (5' truncation), or alternatively from the 3' end (3' truncation). A shortened or truncated compound targeted to a HSD17B13 nucleic acid may have two subunits deleted from the 5' end, or alternatively, may have two subunits deleted from the 3' end of the compound. Alternatively, the deleted nucleosides may be dispersed throughout the compound.

(266) When a single additional subunit is present in a lengthened compound, the additional subunit may be located at the 5' or 3' end of the compound. When two or more additional subunits are present, the added subunits may be adjacent to each other, for example, in a compound having two subunits added to the 5' end (5' addition), or alternatively, to the 3' end (3' addition) of the compound. Alternatively, the added subunits may be dispersed throughout the compound.

(267) It is possible to increase or decrease the length of a compound, such as an oligonucleotide, and/or introduce mismatch bases without eliminating activity (Woolf et al. *Proc. Natl. Acad. Sci. USA* 1992, 89:7305-7309; Gautschi et al. *J. Natl. Cancer Inst. March* 2001, 93:463-471; Maher and Dolnick *Nuc. Acid. Res.* 1998, 16:3341-3358). However, seemingly small changes in oligonucleotide sequence, chemistry and motif can make large differences in one or more of the many properties required for clinical development (Seth et al. *J. Med. Chem.* 2009, 52, 10; Egli et al. *J. Am. Chem. Soc.* 2011, 133, 16642).

(268) In certain embodiments, compounds described herein are interfering RNA compounds (RNAi), which include double-stranded RNA compounds (also referred to as short-interfering RNA or siRNA) and single-stranded RNAi compounds (or ssRNA). Such compounds work at least in part through the RISC pathway to degrade and/or sequester a target nucleic acid (thus, include microRNA/microRNA-mimic compounds). As used herein, the term siRNA is meant to be equivalent to other terms used to describe nucleic acid molecules that are capable of mediating sequence-specific RNAi, for example, short interfering RNA (siRNA), double-stranded RNA (dsRNA), micro-RNA (miRNA), short hairpin RNA (shRNA), short interfering oligonucleotide, short interfering nucleic acid, short interfering modified oligonucleotide, chemically modified siRNA, post-transcriptional gene silencing RNA (ptgsRNA), and others. In addition, as used herein, the term "RNAi" is meant to be equivalent to other terms used to describe sequence-specific RNA interference, such as post transcriptional gene silencing, translational inhibition, or epigenetics.

(269) In certain embodiments, a compound described herein can comprise any of the oligonucleotide sequences targeted to HSD17B13 described herein. In certain embodiments, the compound can be double-stranded. In certain embodiments, the compound comprises a first strand comprising at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion of any one of SEQ ID NOs: 8-2035 and a second strand. In certain embodiments, the compound comprises a first strand comprising the nucleobase sequence of any one of SEQ ID NOs: 8-2035 and a second strand. In certain embodiments, the compound comprises ribonucleotides in which the first strand has uracil (U) in place of thymine (T) in any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises (i) a first strand comprising a nucleobase sequence complementary to the site on HSD17B13 to which any of SEQ ID NOs: 8-2035 is targeted, and (ii) a second strand. In certain embodiments, the compound comprises one or more modified nucleotides in which the 2' position of the sugar contains a halogen (such as fluorine group; 2'-F) or contains an alkoxy group (such as a methoxy group; 2'-OMe). In certain embodiments, the compound comprises at least one 2'-F sugar modification and at least one 2'-OMe sugar modification. In certain embodiments, the at least one 2'-F sugar modification and at least one 2'-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the dsRNA compound. In certain embodiments, the compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramidate, phosphorothioate, and phosphorodithioate linkages. The compounds may also be chemically

modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the compound contains one or two capped strands, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000.

(270) In certain embodiments, the first strand of the compound is an siRNA guide strand and the second strand of the compound is an siRNA passenger strand. In certain embodiments, the second strand of the compound is complementary to the first strand. In certain embodiments, each strand of the compound is 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides in length. In certain embodiments, the first or second strand of the compound can comprise a conjugate group.

(271) In certain embodiments, a compound described herein can comprise any of the oligonucleotide sequences targeted to HSD17B13 described herein. In certain embodiments, the compound is single stranded. In certain embodiments, such a compound is a single-stranded RNAi (ssRNAi) compound. In certain embodiments, the compound comprises at least an 8, 9, 10, 11, 12, 13, 14, 15, or 16 contiguous nucleobase portion of any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises the nucleobase sequence of any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises ribonucleotides in which uracil (U) is in place of thymine (T) in any one of SEQ ID NOs: 8-2035. In certain embodiments, the compound comprises a nucleobase sequence complementary to the site on HSD17B13 to which any of SEQ ID NOs: 8-2035 is targeted. In certain embodiments, the compound comprises one or more modified nucleotides in which the 2' position in the sugar contains a halogen (such as fluorine group; 2'-F) or contains an alkoxy group (such as a methoxy group; 2'-OMe). In certain embodiments, the compound comprises at least one 2'-F sugar modification and at least one 2'-OMe sugar modification. In certain embodiments, the at least one 2'-F sugar modification and at least one 2'-OMe sugar modification are arranged in an alternating pattern for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases along a strand of the compound. In certain embodiments, the compound comprises one or more linkages between adjacent nucleotides other than a naturally-occurring phosphodiester linkage. Examples of such linkages include phosphoramidate, phosphorothioate, and phosphorodithioate linkages. The compounds may also be chemically modified nucleic acid molecules as taught in U.S. Pat. No. 6,673,661. In other embodiments, the compound contains a capped strand, as disclosed, for example, by WO 00/63364, filed Apr. 19, 2000. In certain embodiments, the compound consists of 16, 17, 18, 19, 20, 21, 22, or 23 linked nucleosides. In certain embodiments, the compound can comprise a conjugate group.

(272) Certain Mechanisms

(273) In certain embodiments, compounds described herein comprise or consist of modified oligonucleotides. In certain embodiments, compounds described herein are antisense compounds. In certain embodiments, compounds comprise oligomeric compounds. In certain embodiments, compounds described herein are capable of hybridizing to a target nucleic acid, resulting in at least one antisense activity. In certain embodiments, compounds described herein selectively affect one or more target nucleic acid. Such compounds comprise a nucleobase sequence that hybridizes to one or more target nucleic acid, resulting in one or more desired antisense activity and does not hybridize to one or more non-target nucleic acid or does not hybridize to one or more non-target nucleic acid in such a way that results in a significant undesired antisense activity.

(274) In certain antisense activities, hybridization of a compound described herein to a target nucleic acid results in recruitment of a protein that cleaves the target nucleic acid. For example, certain compounds described herein result in RNase H mediated cleavage of the target nucleic acid. RNase H is a cellular endonuclease that cleaves the RNA strand of an RNA:DNA duplex. The DNA in such an RNA:DNA duplex need not be unmodified DNA. In certain embodiments, compounds described herein are sufficiently "DNA-like" to elicit RNase H activity. Further, in certain embodiments, one or more non-DNA-like nucleoside in the gap of a gapmer is tolerated.

(275) In certain antisense activities, compounds described herein or a portion of the compound is

loaded into an RNA-induced silencing complex (RISC), ultimately resulting in cleavage of the target nucleic acid. For example, certain compounds described herein result in cleavage of the target nucleic acid by Argonaute. Compounds that are loaded into RISC are RNAi compounds. RNAi compounds may be double-stranded (siRNA) or single-stranded (ssRNA).

(276) In certain embodiments, hybridization of compounds described herein to a target nucleic acid does not result in recruitment of a protein that cleaves that target nucleic acid. In certain such embodiments, hybridization of the compound to the target nucleic acid results in alteration of splicing of the target nucleic acid. In certain embodiments, hybridization of the compound to a target nucleic acid results in inhibition of a binding interaction between the target nucleic acid and a protein or other nucleic acid. In certain such embodiments, hybridization of the compound to a target nucleic acid results in alteration of translation of the target nucleic acid.

(277) Antisense activities may be observed directly or indirectly. In certain embodiments, observation or detection of an antisense activity involves observation or detection of a change in an amount of a target nucleic acid or protein encoded by such target nucleic acid, a change in the ratio of splice variants of a nucleic acid or protein, and/or a phenotypic change in a cell or animal.

(278) Target Nucleic Acids, Target Regions and Nucleotide Sequences

(279) In certain embodiments, compounds described herein comprise or consist of an oligonucleotide comprising a region that is complementary to a target nucleic acid. In certain embodiments, the target nucleic acid is an endogenous RNA molecule. In certain embodiments, the target nucleic acid encodes a protein. In certain such embodiments, the target nucleic acid is selected from an mRNA and a pre-mRNA, including intronic, exonic and untranslated regions. In certain embodiments, the target RNA is an mRNA. In certain embodiments, the target nucleic acid is a pre-mRNA. In certain such embodiments, the target region is entirely within an intron. In certain embodiments, the target region spans an intron/exon junction. In certain embodiments, the target region is at least 50% within an intron.

(280) Nucleotide sequences that encode HSD17B13 include, without limitation, the following: RefSeq or GENBANK Accession No. NM_178135.4 (incorporated by reference, disclosed herein as SEQ ID NO: 1), the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000 (incorporated by reference, disclosed herein as SEQ ID NO: 2), the sequence listed as SEQ ID NO: 3, or GENBANK Accession No. NM_001136230.2 (incorporated by reference, disclosed herein as SEQ ID NO: 4).

(281) Hybridization

(282) In some embodiments, hybridization occurs between a compound disclosed herein and a HSD17B13 nucleic acid. The most common mechanism of hybridization involves hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) between complementary nucleobases of the nucleic acid molecules.

(283) Hybridization can occur under varying conditions. Hybridization conditions are sequence-dependent and are determined by the nature and composition of the nucleic acid molecules to be hybridized.

(284) Methods of determining whether a sequence is specifically hybridizable to a target nucleic acid are well known in the art. In certain embodiments, the compounds provided herein are specifically hybridizable with a HSD17B13 nucleic acid.

(285) Complementarity

(286) An oligonucleotide is said to be complementary to another nucleic acid when the nucleobase sequence of such oligonucleotide or one or more regions thereof matches the nucleobase sequence of another oligonucleotide or nucleic acid or one or more regions thereof when the two nucleobase sequences are aligned in opposing directions. Nucleobase matches or complementary nucleobases, as described herein, are limited to the following pairs: adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), and 5-methyl cytosine (mC) and guanine (G), unless otherwise specified. Complementary oligonucleotides and/or nucleic acids need not have

nucleobase complementarity at each nucleoside and may include one or more nucleobase mismatches. An oligonucleotide is fully complementary or 100% complementary when such oligonucleotides have nucleobase matches at each nucleoside without any nucleobase mismatches. (287) In certain embodiments, compounds described herein comprise or consist of modified oligonucleotides. In certain embodiments, compounds described herein are antisense compounds. In certain embodiments, compounds comprise oligomeric compounds. Non-complementary nucleobases between a compound and a HSD17B13 nucleic acid may be tolerated provided that the compound remains able to specifically hybridize to a target nucleic acid. Moreover, a compound may hybridize over one or more segments of a HSD17B13 nucleic acid such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure, mismatch or hairpin structure).

(288) In certain embodiments, the compounds provided herein, or a specified portion thereof are at least, or are up to 70%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% complementary to a HSD17B13 nucleic acid, a target region, target segment, or specified portion thereof. In certain embodiments, the compounds provided herein, or a specified portion thereof, are 70% to 75%, 75% to 80%, 80% to 85%, 85% to 90%, 90% to 95%, 95% to 100%, or any number in between these ranges, complementary to a HSD17B13 nucleic acid, a target region, target segment, or specified portion thereof. Percent complementarity of a compound with a target nucleic acid can be determined using routine methods.

(289) For example, a compound in which 18 of 20 nucleobases of the compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining non-complementary nucleobases may be clustered or interspersed with complementary nucleobases and need not be contiguous to each other or to complementary nucleobases. As such, a compound which is 18 nucleobases in length having four non-complementary nucleobases which are flanked by two regions of complete complementarity with the target nucleic acid would have 77.8% overall complementarity with the target nucleic acid. Percent complementarity of a compound with a region of a target nucleic acid can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs known in the art (Altschul et al., *J. Mol. Biol.*, 1990, 215, 403 410; Zhang and Madden, *Genome Res.*, 1997, 7, 649 656). Percent homology, sequence identity or complementarity, can be determined by, for example, the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), using default settings, which uses the algorithm of Smith and Waterman (*Adv. Appl. Math.*, 1981, 2, 482 489).

(290) In certain embodiments, compounds described herein, or specified portions thereof, are fully complementary (i.e. 100% complementary) to a target nucleic acid, or specified portion thereof. For example, a compound may be fully complementary to a HSD17B13 nucleic acid, or a target region, or a target segment or target sequence thereof. As used herein, “fully complementary” means each nucleobase of a compound is complementary to the corresponding nucleobase of a target nucleic acid. For example, a 20 nucleobase compound is fully complementary to a target sequence that is 400 nucleobases long, so long as there is a corresponding 20 nucleobase portion of the target nucleic acid that is fully complementary to the compound. “Fully complementary” can also be used in reference to a specified portion of the first and/or the second nucleic acid. For example, a 20 nucleobase portion of a 30 nucleobase compound can be “fully complementary” to a target sequence that is 400 nucleobases long. The 20 nucleobase portion of the 30 nucleobase compound is fully complementary to the target sequence if the target sequence has a corresponding 20 nucleobase portion wherein each nucleobase is complementary to the 20 nucleobase portion of the compound. At the same time, the entire 30 nucleobase compound may or may not be fully complementary to the target sequence, depending on whether the remaining 10 nucleobases of the compound are also complementary to the target sequence.

(291) In certain embodiments, compounds described herein comprise one or more mismatched nucleobases relative to the target nucleic acid. In certain such embodiments, antisense activity against the target is reduced by such mismatch, but activity against a non-target is reduced by a greater amount. Thus, in certain such embodiments, selectivity of the compound is improved. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, or 8 from the 5'-end of the gap region. In certain such embodiments, the mismatch is at position 9, 8, 7, 6, 5, 4, 3, 2, 1 from the 3'-end of the gap region. In certain such embodiments, the mismatch is at position 1, 2, 3, or 4 from the 5'-end of the wing region. In certain such embodiments, the mismatch is at position 4, 3, 2, or 1 from the 3'-end of the wing region. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide not having a gapmer motif. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 5'-end of the oligonucleotide. In certain such embodiments, the mismatch is at position, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 3'-end of the oligonucleotide.

(292) The location of a non-complementary nucleobase may be at the 5' end or 3' end of the compound. Alternatively, the non-complementary nucleobase or nucleobases may be at an internal position of the compound. When two or more non-complementary nucleobases are present, they may be contiguous (i.e. linked) or non-contiguous. In one embodiment, a non-complementary nucleobase is located in the wing segment of a gapmer oligonucleotide.

(293) In certain embodiments, compounds described herein that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleobases in length comprise no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a HSD17B13 nucleic acid, or specified portion thereof.

(294) In certain embodiments, compounds described herein that are, or are up to 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleobases in length comprise no more than 6, no more than 5, no more than 4, no more than 3, no more than 2, or no more than 1 non-complementary nucleobase(s) relative to a target nucleic acid, such as a HSD17B13 nucleic acid, or specified portion thereof.

(295) In certain embodiments, compounds described herein also include those which are complementary to a portion of a target nucleic acid. As used herein, "portion" refers to a defined number of contiguous (i.e. linked) nucleobases within a region or segment of a target nucleic acid. A "portion" can also refer to a defined number of contiguous nucleobases of a compound. In certain embodiments, the compounds, are complementary to at least an 8 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 9 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 10 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least an 11 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 12 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 13 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 14 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 15 nucleobase portion of a target segment. In certain embodiments, the compounds are complementary to at least a 16 nucleobase portion of a target segment. Also contemplated are compounds that are complementary to at least a 9, 10, 17, 18, 19, 20, or more nucleobase portion of a target segment, or a range defined by any two of these values.

(296) Identity

(297) The compounds provided herein may also have a defined percent identity to a particular nucleotide sequence, SEQ ID NOs, or compound represented by a specific ION number, or portion thereof. In certain embodiments, compounds described herein are antisense compounds or oligomeric compounds. In certain embodiments, compounds described herein are modified

oligonucleotides. As used herein, a compound is identical to the sequence disclosed herein if it has the same nucleobase pairing ability. For example, a RNA which contains uracil in place of thymidine in a disclosed DNA sequence would be considered identical to the DNA sequence since both uracil and thymidine pair with adenine. Shortened and lengthened versions of the compounds described herein as well as compounds having non-identical bases relative to the compounds provided herein also are contemplated. The non-identical bases may be adjacent to each other or dispersed throughout the compound. Percent identity of an compound is calculated according to the number of bases that have identical base pairing relative to the sequence to which it is being compared.

(298) In certain embodiments, compounds described herein, or portions thereof, are, or are at least, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to one or more of the compounds or SEQ ID NOs, or a portion thereof, disclosed herein. In certain embodiments, compounds described herein are about 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical, or any percentage between such values, to a particular nucleotide sequence, SEQ ID NO, or compound represented by a specific ION number, or portion thereof, in which the compounds comprise an oligonucleotide having one or more mismatched nucleobases. In certain such embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 5'-end of the oligonucleotide. In certain such embodiments, the mismatch is at position, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 from the 3'-end of the oligonucleotide.

(299) In certain embodiments, compounds described herein comprise or consist of antisense compounds. In certain embodiments, a portion of the antisense compound is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

(300) In certain embodiments, compounds described herein comprise or consist of oligonucleotides. In certain embodiments, a portion of the oligonucleotide is compared to an equal length portion of the target nucleic acid. In certain embodiments, an 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleobase portion is compared to an equal length portion of the target nucleic acid.

(301) Certain Modified Compounds

(302) In certain embodiments, compounds described herein comprise or consist of oligonucleotides consisting of linked nucleosides. Oligonucleotides may be unmodified oligonucleotides (RNA or DNA) or may be modified oligonucleotides. Modified oligonucleotides comprise at least one modification relative to unmodified RNA or DNA (i.e., comprise at least one modified nucleoside (comprising a modified sugar moiety and/or a modified nucleobase) and/or at least one modified internucleoside linkage).

(303) A. Modified Nucleosides

(304) Modified nucleosides comprise a modified sugar moiety or a modified nucleobase or both a modified sugar moiety and a modified nucleobase.

(305) 1. Modified Sugar Moieties

(306) In certain embodiments, sugar moieties are non-bicyclic modified sugar moieties. In certain embodiments, modified sugar moieties are bicyclic or tricyclic sugar moieties. In certain embodiments, modified sugar moieties are sugar surrogates. Such sugar surrogates may comprise one or more substitutions corresponding to those of other types of modified sugar moieties.

(307) In certain embodiments, modified sugar moieties are non-bicyclic modified furanosyl sugar moieties comprising one or more acyclic substituent, including, but not limited, to substituents at the 2', 4', and/or 5' positions. In certain embodiments, the furanosyl sugar moiety is a ribosyl sugar moiety. In certain embodiments, one or more acyclic substituent of non-bicyclic modified sugar moieties is branched. Examples of 2'-substituent groups suitable for non-bicyclic modified sugar moieties include but are not limited to: 2'-F, 2'-OCH₃ ("OMe" or "O-methyl"), and 2'-

O(CH.sub.2).sub.2OCH.sub.3 (“MOE”). In certain embodiments, 2'-substituent groups are selected from among: halo, allyl, amino, azido, SH, CN, OCN, CF.sub.3, OCF.sub.3, O—C.sub.1-C.sub.10 alkoxy, O—C.sub.1-C.sub.10 substituted alkoxy, O—C.sub.1-C.sub.10 alkyl, O—C.sub.1-C.sub.10 substituted alkyl, S-alkyl, N(R.sub.m)-alkyl, O-alkenyl, S-alkenyl, N(R.sub.m)-alkenyl, O-alkynyl, S-alkynyl, N(R.sub.m)-alkynyl, O-alkylenyl-O-alkyl, alkynyl, alkaryl, aralkyl, O-alkaryl, O-aralkyl, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(R.sub.m)(R.sub.m) or OCH.sub.2C(=O)—N(R.sub.m)(R.sub.n), where each R.sub.m and R.sub.n is, independently, H, an amino protecting group, or substituted or unsubstituted C.sub.1-C.sub.10 alkyl, and the 2'-substituent groups described in Cook et al., U.S. Pat. No. 6,531,584; Cook et al., U.S. Pat. No. 5,859,221; and Cook et al., U.S. Pat. No. 6,005,087. Certain embodiments of these 2'-substituent groups can be further substituted with one or more substituent groups independently selected from among: hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro (NO.sub.2), thiol, thioalkoxy, thioalkyl, halogen, alkyl, aryl, alkenyl and alkynyl. Examples of 4'-substituent groups suitable for linearly non-bicyclic modified sugar moieties include, but are not limited to, alkoxy (e.g., methoxy), alkyl, and those described in Manoharan et al., WO 2015/106128. Examples of 5'-substituent groups suitable for non-bicyclic modified sugar moieties include, but are not limited to: 5'-methyl (R or S), 5'-vinyl, and 5'-methoxy. In certain embodiments, non-bicyclic modified sugars comprise more than one non-bridging sugar substituent, for example, 2'-F-5'-methyl sugar moieties and the modified sugar moieties and modified nucleosides described in Migawa et al., WO 2008/101157 and Rajeev et al., US2013/0203836.

(308) In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, NH.sub.2, N.sub.3, OCF.sub.3, OCH.sub.3, O(CH.sub.2).sub.3NH.sub.2, CH.sub.2CH=CH.sub.2, OCH.sub.2CH=CH.sub.2, OCH.sub.2CH.sub.2OCH.sub.3, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(R.sub.m)(R.sub.n), O(CH.sub.2).sub.2O(CH.sub.2).sub.2N(CH.sub.3).sub.2, and N-substituted acetamide (OCH.sub.2C(=O)—N(R.sub.m)(R.sub.n)), where each R.sub.m and R.sub.n is, independently, H, an amino protecting group, or substituted or unsubstituted C.sub.1-C.sub.10 alkyl.

(309) In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, OCF.sub.3, OCH.sub.3, OCH.sub.2CH.sub.2OCH.sub.3, O(CH.sub.2).sub.2SCH.sub.3, O(CH.sub.2).sub.2ON(CH.sub.3).sub.2, O(CH.sub.2).sub.2O(CH.sub.2).sub.2N(CH.sub.3).sub.2, and OCH.sub.2C(=O)—N(H)CH.sub.3 (“NMA”).

(310) In certain embodiments, a 2'-substituted nucleoside or 2'-non-bicyclic modified nucleoside comprises a sugar moiety comprising a linear 2'-substituent group selected from: F, OCH.sub.3, and OCH.sub.2CH.sub.2OCH.sub.3.

(311) Nucleosides comprising modified sugar moieties, such as non-bicyclic modified sugar moieties, are referred to by the position(s) of the substitution(s) on the sugar moiety of the nucleoside. For example, nucleosides comprising 2'-substituted or 2'-modified sugar moieties are referred to as 2'-substituted nucleosides or 2'-modified nucleosides.

(312) Certain modified sugar moieties comprise a bridging sugar substituent that forms a second ring resulting in a bicyclic sugar moiety. In certain such embodiments, the bicyclic sugar moiety comprises a bridge between the 4' and the 2' furanose ring atoms. In certain such embodiments, the furanose ring is a ribose ring. Examples of such 4' to 2' bridging sugar substituents include, but are not limited to: 4'-CH.sub.2-2', 4'-(CH.sub.2).sub.2-2', 4'-(CH.sub.2).sub.3-2', 4'-CH.sub.2—O-2' (“LNA”), 4'-CH.sub.2—S-2', 4'-(CH.sub.2).sub.2—O-2' (“ENA”), 4'-CH(CH.sub.3)—O-2' (referred to as “constrained ethyl” or “cEt” when in the S configuration), 4'-CH.sub.2—O—CH.sub.2-2', 4'-CH.sub.2—N(R)-2', 4'-CH(CH.sub.2OCH.sub.3)—O-2' (“constrained MOE” or “cMOE”) and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 7,399,845, Bhat et al., U.S. Pat. No. 7,569,686, Swayze et al., U.S. Pat. No. 7,741,457, and Swayze et al., U.S. Pat. No. 8,022,193),

4'-C(CH₃)(CH₃)—O-2' and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 8,278,283), 4'-CH₂—N(OCH₃)-2' and analogs thereof (see, e.g., Prakash et al., U.S. Pat. No. 8,278,425), 4'-CH₂—O—N(CH₃)-2' (see, e.g., Allerson et al., U.S. Pat. No. 7,696,345 and Allerson et al., U.S. Pat. No. 8,124,745), 4'-CH₂—C(H)(CH₃)-2' (see, e.g., Zhou, et al., *J. Org. Chem.*, 2009, 74, 118-134), 4'-CH₂—C(=CH₂)-2' and analogs thereof (see e.g., Seth et al., U.S. Pat. No. 8,278,426), 4'-C(R_aR_b)—N(R)—O-2', 4'-C(R_aR_b)—O—N(R)-2', 4'-CH₂—O—N(R)-2', and 4'-CH₂—N(R)—O-2', wherein each R, R_a, and R_b is, independently, H, a protecting group, or C₁-C₁₂ alkyl (see, e.g. Imanishi et al., U.S. Pat. No. 7,427,672).

(313) In certain embodiments, such 4' to 2' bridges independently comprise from 1 to 4 linked groups independently selected from: —[C(R_a)(R_b)]_n—, —[C(R_a)(R_b)]_n—O—, —C(R_a)=C(R_b)—, —C(R_a)=N—, —C(=NR_a)—, —C(=O)—, —C(=S)—, —O—, —Si(R_a)₂—, —S(=O)_x—, and —N(R_a)—; wherein: x is 0, 1, or 2; n is 1, 2, 3, or 4; each R_a and R_b is, independently, H, a protecting group, hydroxyl, C₁-C₁₂ alkyl, substituted C₁-C₁₂ alkyl, C₂-C₁₂ alkenyl, substituted C₂-C₁₂ alkenyl, C₂-C₁₂ alkynyl, substituted C₂-C₁₂ alkynyl, C₅-C₂₀ aryl, substituted C₅-C₂₀ aryl, heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C₅-C₇ alicyclic radical, substituted C₅-C₇ alicyclic radical, halogen, OJ₁, NJ₁J₂, SJ₁, N₃, COOJ₁, acyl (C(=O)—H), substituted acyl, CN, sulfonyl (S(=O)₂-J₁), or sulfoxyl (S(=O)-J₁); and each J₁ and J₂ is, independently, H, C₁-C₁₂ alkyl, substituted C₁-C₁₂ alkyl, C₂-C₁₂ alkenyl, substituted C₂-C₁₂ alkenyl, C₂-C₁₂ alkynyl, substituted C₂-C₁₂ alkynyl, C₅-C₂₀ aryl, substituted C₅-C₂₀ aryl, acyl (C(=O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C₁-C₁₂ aminoalkyl, substituted C₁-C₁₂ aminoalkyl, or a protecting group.

(314) Additional bicyclic sugar moieties are known in the art, see, for example: Freier et al., *Nucleic Acids Research*, 1997, 25(22), 4429-4443, Albaek et al., *J Org. Chem.*, 2006, 71, 7731-7740, Singh et al., *Chem. Commun.*, 1998, 4, 455-456; Koshkin et al., *Tetrahedron*, 1998, 54, 3607-3630; Wahlestedt et al., *Proc. Natl. Acad. Sci. U.S.A.*, 2000, 97, 5633-5638; Kumar et al., *Bioorg. Med. Chem. Lett.*, 1998, 8, 2219-2222; Singh et al., *J. Org. Chem.*, 1998, 63, 10035-10039; Srivastava et al., *J. Am. Chem. Soc.*, 2007, 129, 8362-8379; Elayadi et al., *Curr. Opinion Invens. Drugs*. 2001, 2, 558-561; Braasch et al., *Chem. Biol.*, 2001, 8, 1-7; Orum et al., *Curr. Opinion Mol. Ther.* 2001, 3, 239-243; Wengel et al., U.S. Pat. No. 7,053,207, Imanishi et al., U.S. Pat. No. 6,268,490, Imanishi et al. U.S. Pat. No. 6,770,748, Imanishi et al., U.S. RE44,779; Wengel et al., U.S. Pat. No. 6,794,499, Wengel et al., U.S. Pat. No. 6,670,461; Wengel et al., U.S. Pat. No. 7,034,133, Wengel et al., U.S. Pat. No. 8,080,644; Wengel et al., U.S. Pat. No. 8,034,909; Wengel et al., U.S. Pat. No. 8,153,365; Wengel et al., U.S. Pat. No. 7,572,582; and Ramasamy et al., U.S. Pat. No. 6,525,191, Torsten et al., WO 2004/106356, Wengel et al., WO 1999/014226; Seth et al., WO 2007/134181; Seth et al., U.S. Pat. No. 7,547,684; Seth et al., U.S. Pat. No. 7,666,854; Seth et al., U.S. Pat. No. 8,088,746; Seth et al., U.S. Pat. No. 7,750,131; Seth et al., U.S. Pat. No. 8,030,467; Seth et al., U.S. Pat. No. 8,268,980; Seth et al., U.S. Pat. No. 8,546,556; Seth et al., U.S. Pat. No. 8,530,640; Migawa et al., U.S. Pat. No. 9,012,421; Seth et al., U.S. Pat. No. 8,501,805; Allerson et al., US2008/0039618; and Migawa et al., US2015/0191727.

(315) In certain embodiments, bicyclic sugar moieties and nucleosides incorporating such bicyclic sugar moieties are further defined by isomeric configuration. For example, an LNA nucleoside (described herein) may be in the α-L configuration or in the β-D configuration.

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α-L-methyleneoxy (4'-CH₂—O—2') or α-L-LNA bicyclic nucleosides have been incorporated into oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research*. 2003,

21, 6365-6372). Herein, general descriptions of bicyclic nucleosides include both isomeric configurations. When the positions of specific bicyclic nucleosides (e.g., LNA or cEt) are identified in exemplified embodiments herein, they are in the β -D configuration, unless otherwise specified. (317) In certain embodiments, modified sugar moieties comprise one or more non-bridging sugar substituent and one or more bridging sugar substituent (e.g., 5'-substituted and 4'-2' bridged sugars).

(318) In certain embodiments, modified sugar moieties are sugar surrogates. In certain such embodiments, the oxygen atom of the sugar moiety is replaced, e.g., with a sulfur, carbon or nitrogen atom. In certain such embodiments, such modified sugar moieties also comprise bridging and/or non-bridging substituents as described herein. For example, certain sugar surrogates comprise a 4'-sulfur atom and a substitution at the 2'-position (see, e.g., Bhat et al., U.S. Pat. No. 7,875,733 and Bhat et al., U.S. Pat. No. 7,939,677) and/or the 5' position.

(319) In certain embodiments, sugar surrogates comprise rings having other than 5 atoms. For example, in certain embodiments, a sugar surrogate comprises a six-membered tetrahydropyran ("THP"). Such tetrahydropyrans may be further modified or substituted. Nucleosides comprising such modified tetrahydropyrans include, but are not limited to, hexitol nucleic acid ("HNA"), anitol nucleic acid ("ANA"), manitol nucleic acid ("MNA") (see e.g., Leumann, *CJ. Bioorg. & Med. Chem.* 2002, 10, 841-854), fluoro HNA:

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("F-HNA", see e.g., Swayze et al., U.S. Pat. No. 8,088,904; Swayze et al., U.S. Pat. No. 8,440,803; and Swayze et al., U.S. Pat. No. 9,005,906) F-HNA can also be referred to as a F-THP or 3'-fluoro tetrahydropyran, and nucleosides comprising additional modified THP compounds having the formula:

(321) ##STR00057##

wherein, independently, for each of said modified THP nucleoside: Bx is a nucleobase moiety; T.sub.3 and T.sub.4 are each, independently, an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide or one of T.sub.3 and T.sub.4 is an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide and the other of T.sub.3 and T.sub.4 is H, a hydroxyl protecting group, a linked conjugate group, or a 5' or 3'-terminal group; q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each, independently, H, C.sub.1-C.sub.6 alkyl, substituted C.sub.1-C.sub.6 alkyl, C.sub.2-C.sub.6 alkenyl, substituted C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, or substituted C.sub.2-C.sub.6 alkynyl; and each of R.sub.1 and R.sub.2 is independently selected from among: hydrogen, halogen, substituted or unsubstituted alkoxy, NJ.sub.1J.sub.2, SJ.sub.1, N.sub.3, OC(=X)J.sub.1, OC(=X)NJ.sub.1J.sub.2, NJ.sub.3C(=X)NJ.sub.1J.sub.2, and CN, wherein X is O, S or NJ.sub.1, and each J.sub.1, J.sub.2, and J.sub.3 is, independently, H or C.sub.1-C.sub.6 alkyl.

(322) In certain embodiments, modified THP nucleosides are provided wherein q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is other than H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is methyl. In certain embodiments, modified THP nucleosides are provided wherein one of R, and R.sub.2 is F. In certain embodiments, R.sub.1 is F and R.sub.2 is H, in certain embodiments, R.sub.1 is methoxy and R.sub.2 is H, and in certain embodiments, R.sub.1 is methoxyethoxy and R.sub.2 is H.

(323) In certain embodiments, sugar surrogates comprise rings having more than 5 atoms and more than one heteroatom. For example, nucleosides comprising morpholino sugar moieties and their use in oligonucleotides have been reported (see, e.g., Braasch et al., *Biochemistry*, 2002, 41, 4503-4510 and Summerton et al., U.S. Pat. No. 5,698,685; Summerton et al., U.S. Pat. No. 5,166,315; Summerton et al., U.S. Pat. No. 5,185,444; and Summerton et al., U.S. Pat. No. 5,034,506. As used

here, the term “morpholino” means a sugar surrogate having the following structure:

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In certain embodiments, morpholinos may be modified, for example, by adding or altering various substituent groups from the above morpholino structure. Such sugar surrogates are referred to herein as “modified morpholinos.”

(325) In certain embodiments, sugar surrogates comprise acyclic moieties. Examples of nucleosides and oligonucleotides comprising such acyclic sugar surrogates include, but are not limited to: peptide nucleic acid (“PNA”), acyclic butyl nucleic acid (see, e.g., Kumar et al., *Org. Biomol. Chem.*, 2013, 11, 5853-5865), and nucleosides and oligonucleotides described in Manoharan et al., US2013/130378.

(326) Many other bicyclic and tricyclic sugar and sugar surrogate ring systems are known in the art that can be used in modified nucleosides.

(327) In certain embodiments, modified oligonucleotides comprise a 2'-OMe modified nucleoside in the gap of a cEt gapmer. In certain embodiments, the 2'-OMe modified nucleoside is at position 2 or 3 of the gap. In certain embodiments, the 2'-OMe modified nucleoside is at position 2 of the gap. In certain embodiments, modified oligonucleotides have a sugar motif kkkdmddddd kkk, wherein each k represents a cEt nucleoside, each d represents a 2'-β-D-deoxyribosyl nucleoside, and each m represents a nucleoside comprising a 2'-OMe sugar moiety (“2'-OMe nucleoside”).

(328) In certain embodiments, modified oligonucleotides have a formula A-B—C, wherein A is a 5'-region, B is a central region, and C is a 3'-region. In certain embodiments, A and C each consist of 1-5 linked nucleosides and B consists of 7-11 linked nucleosides. In certain embodiments, A and C consist of 3 linked cEt nucleosides. In certain embodiments, the second nucleoside of the central region comprises a 2'-OMe modified sugar moiety, and each of the remaining central region nucleosides comprises a 2'-p-D-deoxyribosyl sugar moiety.

(329) In certain embodiments, modified oligonucleotides comprise a neutral phosphonate linkage at one or more positions. In certain embodiments, modified oligonucleotides comprise a methoxypropyl phosphonate linkage at one or more positions. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer, and the internucleoside linkage motif comprises at least one methoxypropyl phosphonate linkage in the gap. In certain embodiments, the methoxypropyl phosphonate linkage is between the second and third nucleosides in the gap. In certain embodiments, the methoxypropyl phosphonate linkage is between the third and fourth nucleosides in the gap.

(330) 2. Modified Nucleobases

(331) Nucleobase (or base) modifications or substitutions are structurally distinguishable from, yet functionally interchangeable with, naturally occurring or synthetic unmodified nucleobases. Both natural and modified nucleobases are capable of participating in hydrogen bonding. Such nucleobase modifications can impart nuclease stability, binding affinity or some other beneficial biological property to antisense compounds.

(332) In certain embodiments, compounds described herein comprise modified oligonucleotides. In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising an unmodified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleoside comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleosides that does not comprise a nucleobase, referred to as an abasic nucleoside.

(333) In certain embodiments, modified nucleobases are selected from: 5-substituted pyrimidines, 6-azapyrimidines, alkyl or alkynyl substituted pyrimidines, alkyl substituted purines, and N-2, N-6 and O-6 substituted purines. In certain embodiments, modified nucleobases are selected from: 2-aminopropyladenine, 5-hydroxymethyl cytosine, 5-methylcytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-N-methylguanine, 6-N-methyladenine, 2-propyladenine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-propynyl (C≡C—CH₂) uracil, 5-propynylcytosine, 6-

azouracil, 6-azocytosine, 6-azothymine, 5-ribosyluracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl, 8-aza and other 8-substituted purines, 5-halo, particularly, 5-bromo, 5-trifluoromethyl, 5-halouracil, and 5-halocytosine, 7-methylguanine, 7-methyladenine, 2-F-adenine, 2-aminoadenine, 7-deazaguanine, 7-deazaadenine, 3-deazaguanine, 3-deazaadenine, 6-N-benzoyladenine, 2-N-isobutyrylguanine, 4-N-benzoylcytosine, 4-N-benzoyluracil, 5-methyl 4-N-benzoylcytosine, 5-methyl 4-N-benzoyluracil, universal bases, hydrophobic bases, promiscuous bases, size-expanded bases, and fluorinated bases. Further modified nucleobases include tricyclic pyrimidines, such as 1,3-diazaphenoxazine-2-one, 1,3-diazaphenothiazine-2-one, and 9-(2-aminoethoxy)-1,3-diazaphenoxazine-2-one (G-clamp). Modified nucleobases may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example, 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in Merigan et al., U.S. Pat. No. 3,687,808, those disclosed in The Concise Encyclopedia Of Polymer Science And Engineering, Kroschwitz, J. I., Ed., John Wiley & Sons, 1990, 858-859; Englisch et al., Angewandte Chemie, International Edition, 1991, 30, 613; Sanghvi, Y. S., Chapter 15, Antisense Research and Applications, Crooke, S. T. and Lebleu, B., Eds., CRC Press, 1993, 273-288; and those disclosed in Chapters 6 and 15, Antisense Drug Technology, Crooke S. T., Ed., CRC Press, 2008, 163-166 and 442-443.

(334) Publications that teach the preparation of certain of the above noted modified nucleobases, as well as other modified nucleobases include without limitation, Manoharan et al., US2003/0158403, Manoharan et al., US2003/0175906; Dinh et al., U.S. Pat. No. 4,845,205; Spielvogel et al., U.S. Pat. No. 5,130,302; Rogers et al., U.S. Pat. No. 5,134,066; Bischofberger et al., U.S. Pat. No. 5,175,273; Urdea et al., U.S. Pat. No. 5,367,066; Benner et al., U.S. Pat. No. 5,432,272; Matteucci et al., U.S. Pat. No. 5,434,257; Gmeiner et al., U.S. Pat. No. 5,457,187; Cook et al., U.S. Pat. No. 5,459,255; Froehler et al., U.S. Pat. No. 5,484,908; Matteucci et al., U.S. Pat. No. 5,502,177; Hawkins et al., U.S. Pat. No. 5,525,711; Haralambidis et al., U.S. Pat. No. 5,552,540; Cook et al., U.S. Pat. No. 5,587,469; Froehler et al., U.S. Pat. No. 5,594,121; Switzer et al., U.S. Pat. No. 5,596,091; Cook et al., U.S. Pat. No. 5,614,617; Froehler et al., U.S. Pat. No. 5,645,985; Cook et al., U.S. Pat. No. 5,681,941; Cook et al., U.S. Pat. No. 5,811,534; Cook et al., U.S. Pat. No. 5,750,692; Cook et al., U.S. Pat. No. 5,948,903; Cook et al., U.S. Pat. No. 5,587,470; Cook et al., U.S. Pat. No. 5,457,191; Matteucci et al., U.S. Pat. No. 5,763,588; Froehler et al., U.S. Pat. No. 5,830,653; Cook et al., U.S. Pat. No. 5,808,027; Cook et al., U.S. Pat. No. 6,166,199; and Matteucci et al., U.S. Pat. No. 6,005,096.

(335) In certain embodiments, compounds targeted to a HSD17B13 nucleic acid comprise one or more modified nucleobases. In certain embodiments, the modified nucleobase is 5-methylcytosine. In certain embodiments, each cytosine is a 5-methylcytosine.

(336) Modified Internucleoside Linkages

(337) The naturally occurring internucleoside linkage of RNA and DNA is a 3' to 5' phosphodiester linkage. In certain embodiments, compounds described herein having one or more modified, i.e. non-naturally occurring, internucleoside linkages are often selected over compounds having naturally occurring internucleoside linkages because of desirable properties such as, for example, enhanced cellular uptake, enhanced affinity for target nucleic acids, and increased stability in the presence of nucleases.

(338) Representative internucleoside linkages having a chiral center include but are not limited to alkylphosphonates and phosphorothioates. Modified oligonucleotides comprising internucleoside linkages having a chiral center can be prepared as populations of modified oligonucleotides comprising stereorandom internucleoside linkages, or as populations of modified oligonucleotides comprising phosphorothioate linkages in particular stereochemical configurations. In certain embodiments, populations of modified oligonucleotides comprise phosphorothioate internucleoside linkages wherein all of the phosphorothioate internucleoside linkages are stereorandom. Such modified oligonucleotides can be generated using synthetic methods that result in random selection

of the stereochemical configuration of each phosphorothioate linkage. Nonetheless, as is well understood by those of skill in the art, each individual phosphorothioate of each individual oligonucleotide molecule has a defined stereoconfiguration. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising one or more particular phosphorothioate internucleoside linkages in a particular, independently selected stereochemical configuration. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 65% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 70% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 80% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 90% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 99% of the molecules in the population. Such chirally enriched populations of modified oligonucleotides can be generated using synthetic methods known in the art, e.g., methods described in Oka et al., *JACS* 125, 8307 (2003), Wan et al. *Nuc. Acid. Res.* 42, 13456 (2014), and WO 2017/015555. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one indicated phosphorothioate in the (Sp) configuration. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one phosphorothioate in the (Rp) configuration. In certain embodiments, modified oligonucleotides comprising (Rp) and/or (Sp) phosphorothioates comprise one or more of the following formulas, respectively, wherein “B” indicates a nucleobase:

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Unless otherwise indicated, chiral internucleoside linkages of modified oligonucleotides described herein can be stereorandom or in a particular stereochemical configuration.

(340) In certain embodiments, compounds targeted to a HSD17B13 nucleic acid comprise one or more modified internucleoside linkages. In certain embodiments, the modified internucleoside linkages are phosphorothioate linkages. In certain embodiments, each internucleoside linkage of an antisense compound is a phosphorothioate internucleoside linkage.

(341) In certain embodiments, compounds described herein comprise oligonucleotides.

Oligonucleotides having modified internucleoside linkages include internucleoside linkages that retain a phosphorus atom as well as internucleoside linkages that do not have a phosphorus atom. Representative phosphorus containing internucleoside linkages include, but are not limited to, phosphodiester, phosphotriesters, methylphosphonates, phosphoramidate, and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing linkages are well known.

(342) In certain embodiments, nucleosides of modified oligonucleotides may be linked together using any internucleoside linkage. The two main classes of internucleoside linking groups are defined by the presence or absence of a phosphorus atom. Representative phosphorus-containing internucleoside linkages include, but are not limited to, phosphates, which contain a phosphodiester bond (“P=O”) (also referred to as unmodified or naturally occurring linkages), phosphotriesters, methylphosphonates, phosphoramidates, and phosphorothioates (“P=S”), and phosphorodithioates (“HS—P=S”). Representative non-phosphorus containing internucleoside linking groups include, but are not limited to, methylenemethylimino (—CH₂—N(CH₂)—O—CH₂), thiodiester, thionocarbamate (—O—C(=O)(NH)—S—); siloxane (—O—SiH₂—O—); and N,N'-dimethylhydrazine (—CH₂—N(CH₂)—N(CH₂)—). Modified internucleoside linkages, compared to naturally occurring phosphate linkages, can be used to alter, typically increase, nuclease resistance of the oligonucleotide. In certain embodiments, internucleoside linkages having a chiral atom can be prepared as a racemic mixture, or as separate enantiomers. Representative chiral internucleoside linkages include, but are not limited to, alkylphosphonates

and phosphorothioates. Methods of preparation of phosphorous-containing and non-phosphorous-containing internucleoside linkages are well known to those skilled in the art.

(343) Neutral internucleoside linkages include, without limitation, phosphotriesters, methylphosphonates, MMI (3'-CH₂-N(CH₃)-O-5'), amide-3 (3'-CH₂-C(=O)-N(H)-5'), amide-4 (3'-CH₂-N(H)-C(=O)-5'), formacetal (3'-O-CH₂-O-5'), methoxypropyl, and thioformacetal (3'-S-CH₂-O-5'). Further neutral internucleoside linkages include nonionic linkages comprising siloxane (dialkylsiloxane), carboxylate ester, carboxamide, sulfide, sulfonate ester and amides (See, for example: Carbohydrate Modifications in Antisense Research; Y. S. Sanghvi and P. D. Cook, Eds., ACS Symposium Series 580; Chapters 3 and 4, 40-65). Further neutral internucleoside linkages include nonionic linkages comprising mixed N, O, S and CH₂ component parts.

(344) In certain embodiments, oligonucleotides comprise modified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or modified internucleoside linkage motif. In certain embodiments, internucleoside linkages are arranged in a gapped motif. In such embodiments, the internucleoside linkages in each of two wing regions are different from the internucleoside linkages in the gap region. In certain embodiments, the internucleoside linkages in the wings are phosphodiester and the internucleoside linkages in the gap are phosphorothioate. The nucleoside motif is independently selected, so such oligonucleotides having a gapped internucleoside linkage motif may or may not have a gapped nucleoside motif and, if it does have a gapped nucleoside motif, the wing and gap lengths may or may not be the same.

(345) In certain embodiments, oligonucleotides comprise a region having an alternating internucleoside linkage motif. In certain embodiments, oligonucleotides comprise a region of uniformly modified internucleoside linkages. In certain such embodiments, the oligonucleotide comprises a region that is uniformly linked by phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide is uniformly linked by phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate. In certain embodiments, each internucleoside linkage of the oligonucleotide is selected from phosphodiester and phosphorothioate and at least one internucleoside linkage is phosphorothioate.

(346) In certain embodiments, the oligonucleotide comprises at least 6 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 8 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least 10 phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 6 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 8 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 10 consecutive phosphorothioate internucleoside linkages. In certain embodiments, the oligonucleotide comprises at least one block of at least 12 consecutive phosphorothioate internucleoside linkages. In certain such embodiments, at least one such block is located at the 3' end of the oligonucleotide. In certain such embodiments, at least one such block is located within 3 nucleosides of the 3' end of the oligonucleotide.

(347) In certain embodiments, oligonucleotides comprise one or more methylphosphonate linkages. In certain embodiments, oligonucleotides having a gapmer nucleoside motif comprise a linkage motif comprising all phosphorothioate linkages except for one or two methylphosphonate linkages. In certain embodiments, one methylphosphonate linkage is in the central gap of an oligonucleotide having a gapmer nucleoside motif.

(348) In certain embodiments, it is desirable to arrange the number of phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages to maintain nuclease resistance. In certain embodiments, it is desirable to arrange the number and position of phosphorothioate internucleoside linkages and the number and position of phosphodiester

internucleoside linkages to maintain nuclease resistance. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased. In certain embodiments, the number of phosphorothioate internucleoside linkages may be decreased and the number of phosphodiester internucleoside linkages may be increased while still maintaining nuclease resistance. In certain embodiments, it is desirable to decrease the number of phosphorothioate internucleoside linkages while retaining nuclease resistance. In certain embodiments, it is desirable to increase the number of phosphodiester internucleoside linkages while retaining nuclease resistance.

(349) 3. Certain Motifs

(350) In certain embodiments, compounds described herein comprise oligonucleotides.

Oligonucleotides can have a motif, e.g. a pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified sugar. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more modified internucleoside linkage. In such embodiments, the modified, unmodified, and differently modified sugar moieties, nucleobases, and/or internucleoside linkages of a modified oligonucleotide define a pattern or motif. In certain embodiments, the patterns of sugar moieties, nucleobases, and internucleoside linkages are each independent of one another. Thus, a modified oligonucleotide may be described by its sugar motif, nucleobase motif and/or internucleoside linkage motif (as used herein, nucleobase motif describes the modifications to the nucleobases independent of the sequence of nucleobases).

(351) a. Certain Sugar Motifs

(352) In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise one or more type of modified sugar and/or unmodified sugar moiety arranged along the oligonucleotide or region thereof in a defined pattern or sugar motif. In certain instances, such sugar motifs include, but are not limited to, any of the sugar modifications discussed herein.

(353) In certain embodiments, modified oligonucleotides comprise or consist of a region having a gapmer motif, which comprises two external regions or “wings” and a central or internal region or “gap”. The three regions of a gapmer motif (the 5'-wing, the gap, and the 3'-wing) form a contiguous sequence of nucleosides, wherein at least some of the sugar moieties of the nucleosides of each of the wings differ from at least some of the sugar moieties of the nucleosides of the gap. Specifically, at least the sugar moieties of the nucleosides of each wing that are closest to the gap (the 3'-most nucleoside of the 5'-wing and the 5'-most nucleoside of the 3'-wing) differ from the sugar moiety of the neighboring gap nucleosides, thus defining the boundary between the wings and the gap (i.e., the wing/gap junction). In certain embodiments, the sugar moieties within the gap are the same as one another. In certain embodiments, the gap includes one or more nucleosides having a sugar moiety that differs from the sugar moiety of one or more other nucleosides of the gap. In certain embodiments, the sugar motifs of the two wings are the same as one another (symmetric gapmer). In certain embodiments, the sugar motif of the 5-wing differs from the sugar motif of the 3-wing (asymmetric gapmer).

(354) In certain embodiments, the wings of a gapmer comprise 1-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 2-5 nucleosides. In certain embodiments, the wings of a gapmer comprise 3-5 nucleosides. In certain embodiments, the nucleosides of a gapmer are all modified nucleosides.

(355) In certain embodiments, the gap of a gapmer comprises 7-12 nucleosides. In certain embodiments, the gap of a gapmer comprises 7-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 8-10 nucleosides. In certain embodiments, the gap of a gapmer comprises 10 nucleosides. In certain embodiments, each nucleoside of the gap of a gapmer is an unmodified 2'-

deoxy nucleoside.

(356) In certain embodiments, the gapmer is a deoxy gapmer. In such embodiments, the nucleosides on the gap side of each wing/gap junction are unmodified 2'-deoxy nucleosides and the nucleosides on the wing sides of each wing/gap junction are modified nucleosides. In certain such embodiments, each nucleoside of the gap is an unmodified 2'-deoxy nucleoside. In certain such embodiments, each nucleoside of each wing is a modified nucleoside.

(357) In certain embodiments, a modified oligonucleotide has a fully modified sugar motif wherein each nucleoside of the modified oligonucleotide comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif wherein each nucleoside of the region comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a region having a fully modified sugar motif, wherein each nucleoside within the fully modified region comprises the same modified sugar moiety, referred to herein as a uniformly modified sugar motif. In certain embodiments, a fully modified oligonucleotide is a uniformly modified oligonucleotide. In certain embodiments, each nucleoside of a uniformly modified comprises the same 2'-modification.

(358) b. Certain Nucleobase Motifs

(359) In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise modified and/or unmodified nucleobases arranged along the oligonucleotide or region thereof in a defined pattern or motif. In certain embodiments, each nucleobase is modified. In certain embodiments, none of the nucleobases are modified. In certain embodiments, each purine or each pyrimidine is modified. In certain embodiments, each adenine is modified. In certain embodiments, each guanine is modified. In certain embodiments, each thymine is modified. In certain embodiments, each uracil is modified. In certain embodiments, each cytosine is modified. In certain embodiments, some or all of the cytosine nucleobases in a modified oligonucleotide are 5-methylcytosines.

(360) In certain embodiments, modified oligonucleotides comprise a block of modified nucleobases. In certain such embodiments, the block is at the 3'-end of the oligonucleotide. In certain embodiments, the block is within 3 nucleosides of the 3'-end of the oligonucleotide. In certain embodiments, the block is at the 5'-end of the oligonucleotide. In certain embodiments, the block is within 3 nucleosides of the 5'-end of the oligonucleotide.

(361) In certain embodiments, oligonucleotides having a gapmer motif comprise a nucleoside comprising a modified nucleobase. In certain such embodiments, one nucleoside comprising a modified nucleobase is in the central gap of an oligonucleotide having a gapmer motif. In certain such embodiments, the sugar moiety of said nucleoside is a 2'-deoxyribose moiety. In certain embodiments, the modified nucleobase is selected from: a 2-thiopyrimidine and a 5-propynepyrimidine.

(362) c. Certain Internucleoside Linkage Motifs

(363) In certain embodiments, compounds described herein comprise oligonucleotides. In certain embodiments, oligonucleotides comprise modified and/or unmodified internucleoside linkages arranged along the oligonucleotide or region thereof in a defined pattern or motif. In certain embodiments, essentially each internucleoside linking group is a phosphate internucleoside linkage ($P=O$). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is a phosphorothioate ($P=S$). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is independently selected from a phosphorothioate and phosphate internucleoside linkage. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer and the internucleoside linkages within the gap are all modified. In certain such embodiments, some or all of the internucleoside linkages in the wings are unmodified phosphate linkages. In certain embodiments, the terminal internucleoside linkages are modified. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer, and the internucleoside linkage motif comprises at least one phosphodiester internucleoside linkage in at least one wing,

wherein the at least one phosphodiester linkage is not a terminal internucleoside linkage, and the remaining internucleoside linkages are phosphorothioate internucleoside linkages. In certain such embodiments, all of the phosphorothioate linkages are stereorandom. In certain embodiments, all of the phosphorothioate linkages in the wings are (Sp) phosphorothioates, and the gap comprises at least one Sp, Sp, Rp motif. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising such internucleoside linkage motifs.

(364) 4. Certain Modified Oligonucleotides

(365) In certain embodiments, compounds described herein comprise modified oligonucleotides. In certain embodiments, the above modifications (sugar, nucleobase, internucleoside linkage) are incorporated into a modified oligonucleotide. In certain embodiments, modified oligonucleotides are characterized by their modification, motifs, and overall lengths. In certain embodiments, such parameters are each independent of one another. Thus, unless otherwise indicated, each internucleoside linkage of an oligonucleotide having a gapmer sugar motif may be modified or unmodified and may or may not follow the gapmer modification pattern of the sugar modifications. For example, the internucleoside linkages within the wing regions of a sugar gapmer may be the same or different from one another and may be the same or different from the internucleoside linkages of the gap region of the sugar motif. Likewise, such gapmer oligonucleotides may comprise one or more modified nucleobases independent of the gapmer pattern of the sugar modifications. Furthermore, in certain instances, an oligonucleotide is described by an overall length or range and by lengths or length ranges of two or more regions (e.g., a regions of nucleosides having specified sugar modifications). In such circumstances, it may be possible to select numbers for each range that result in an oligonucleotide having an overall length falling outside the specified range. In such circumstances, both elements must be satisfied. For example, in certain embodiments, a modified oligonucleotide consists of 15-20 linked nucleosides and has a sugar motif consisting of three regions, A, B, and C, wherein region A consists of 2-6 linked nucleosides having a specified sugar motif, region B consists of 6-10 linked nucleosides having a specified sugar motif, and region C consists of 2-6 linked nucleosides having a specified sugar motif. Such embodiments do not include modified oligonucleotides where A and C each consist of 6 linked nucleosides and B consists of 10 linked nucleosides (even though those numbers of nucleosides are permitted within the requirements for A, B, and C) because the overall length of such oligonucleotide will be 22, which exceeds the upper limit of the overall length of the modified oligonucleotide (20). Herein, if a description of an oligonucleotide is silent with respect to one or more parameters, such parameter is not limited. Thus, a modified oligonucleotide described only as having a gapmer sugar motif without further description may have any length, internucleoside linkage motif, and nucleobase motif. Unless otherwise indicated, all modifications are independent of nucleobase sequence.

(366) Certain Conjugated Compounds

(367) In certain embodiments, the compounds described herein comprise or consist of an oligonucleotide (modified or unmodified) and, optionally, one or more conjugate groups and/or terminal groups. Conjugate groups consist of one or more conjugate moiety and a conjugate linker which links the conjugate moiety to the oligonucleotide. Conjugate groups may be attached to either or both ends of an oligonucleotide and/or at any internal position. In certain embodiments, conjugate groups are attached to the 2-position of a nucleoside of a modified oligonucleotide. In certain embodiments, conjugate groups that are attached to either or both ends of an oligonucleotide are terminal groups. In certain such embodiments, conjugate groups or terminal groups are attached at the 3' and/or 5'-end of oligonucleotides. In certain such embodiments, conjugate groups (or terminal groups) are attached at the 3'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 3'-end of oligonucleotides. In certain embodiments, conjugate groups (or terminal groups) are attached at the 5'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 5'-end of oligonucleotides.

(368) In certain embodiments, the oligonucleotide is modified. In certain embodiments, the oligonucleotide of a compound has a nucleobase sequence that is complementary to a target nucleic acid. In certain embodiments, oligonucleotides are complementary to a messenger RNA (mRNA). In certain embodiments, oligonucleotides are complementary to a pre-mRNA. In certain embodiments, oligonucleotides are complementary to a sense transcript.

(369) Examples of terminal groups include but are not limited to conjugate groups, capping groups, phosphate moieties, protecting groups, modified or unmodified nucleosides, and two or more nucleosides that are independently modified or unmodified.

(370) In certain embodiments, oligonucleotides are covalently attached to one or more conjugate groups. In certain embodiments, conjugate groups modify one or more properties of the attached oligonucleotide, including but not limited to pharmacodynamics, pharmacokinetics, stability, binding, absorption, tissue distribution, cellular distribution, cellular uptake, charge and clearance. In certain embodiments, conjugate groups impart a new property on the attached oligonucleotide, e.g., fluorophores or reporter groups that enable detection of the oligonucleotide. Certain conjugate groups and conjugate moieties have been described previously, for example: cholesterol moiety (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86, 6553-6556), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4, 1053-1060), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660, 306-309; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3, 2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20, 533-538), an aliphatic chain, e.g., do-decan-diol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10, 1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259, 327-330; Svinarchuk et al., *Biochimie*, 1993, 75, 49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36, 3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18, 3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14, 969-973), or adamantane acetic acid a palmitoyl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264, 229-237), an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277, 923-937), a tocopherol group (Nishina et al., *Molecular Therapy Nucleic Acids*, 2015, 4, e220; and Nishina et al., *Molecular Therapy*, 2008, 16, 734-740), or a GalNAc cluster (e.g., WO2014/179620).

(371) 1. Conjugate Moieties

(372) Conjugate moieties include, without limitation, intercalators, reporter molecules, polyamines, polyamides, peptides, carbohydrates (e.g., GalNAc), vitamin moieties, polyethylene glycols, thioethers, polyethers, cholesterol, thiocholesterols, cholic acid moieties, folate, lipids, phospholipids, biotin, phenazine, phenanthridine, anthraquinone, adamantane, acridine, fluoresceins, rhodamines, coumarins, fluorophores, and dyes.

(373) In certain embodiments, a conjugate moiety comprises an active drug substance, for example, aspirin, warfarin, phenylbutazone, ibuprofen, suprofen, fen-bufen, ketoprofen, (S)-(+)-pranoprofen, carprofen, dansylsarcosine, 2,3,5-triiodobenzoic acid, fingolimod, flufenamic acid, folinic acid, a benzothiadiazide, chlorothiazide, a diazepine, indo-methicin, a barbiturate, a cephalosporin, a sulfa drug, an antidiabetic, an antibacterial or an antibiotic.

(374) 2. Conjugate Linkers

(375) Conjugate moieties are attached to oligonucleotides through conjugate linkers. In certain compounds, the conjugate linker is a single chemical bond (i.e., the conjugate moiety is attached directly to an oligonucleotide through a single bond). In certain compounds, a conjugate moiety is attached to an oligonucleotide via a more complex conjugate linker comprising one or more conjugate linker moieties, which are sub-units making up a conjugate linker. In certain embodiments, the conjugate linker comprises a chain structure, such as a hydrocarbyl chain, or an oligomer of repeating units such as ethylene glycol, nucleosides, or amino acid units.

(376) In certain embodiments, a conjugate linker comprises one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether, and hydroxylamino. In

certain such embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and amide groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and ether groups. In certain embodiments, the conjugate linker comprises at least one phosphorus moiety. In certain embodiments, the conjugate linker comprises at least one phosphate group. In certain embodiments, the conjugate linker includes at least one neutral linking group.

(377) In certain embodiments, conjugate linkers, including the conjugate linkers described above, are bifunctional linking moieties, e.g., those known in the art to be useful for attaching conjugate groups to parent compounds, such as the oligonucleotides provided herein. In general, a bifunctional linking moiety comprises at least two functional groups. One of the functional groups is selected to bind to a particular site on a parent compound and the other is selected to bind to a conjugate group. Examples of functional groups used in a bifunctional linking moiety include but are not limited to electrophiles for reacting with nucleophilic groups and nucleophiles for reacting with electrophilic groups. In certain embodiments, bifunctional linking moieties comprise one or more groups selected from amino, hydroxyl, carboxylic acid, thiol, alkyl, alkenyl, and alkynyl.

(378) Examples of conjugate linkers include but are not limited to pyrrolidine, 8-amino-3,6-dioxaoctanoic acid (ADO), succinimidyl 4-(N-maleimidomethyl) cyclohexane-1-carboxylate (SMCC) and 6-aminohexanoic acid (AHEX or AHA). Other conjugate linkers include but are not limited to substituted or unsubstituted C.sub.1-C.sub.10 alkyl, substituted or unsubstituted C.sub.2-C.sub.10 alkenyl or substituted or unsubstituted C.sub.2-C.sub.10 alkynyl, wherein a nonlimiting list of preferred substituent groups includes hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro, thiol, thioalkoxy, halogen, alkyl, aryl, alkenyl and alkynyl.

(379) In certain embodiments, conjugate linkers comprise 1-10 linker-nucleosides. In certain embodiments, such linker-nucleosides are modified nucleosides. In certain embodiments such linker-nucleosides comprise a modified sugar moiety. In certain embodiments, linker-nucleosides are unmodified. In certain embodiments, linker-nucleosides comprise an optionally protected heterocyclic base selected from a purine, substituted purine, pyrimidine or substituted pyrimidine. In certain embodiments, a cleavable moiety is a nucleoside selected from uracil, thymine, cytosine, 4-N-benzoylcytosine, 5-methylcytosine, 4-N-benzoyl-5-methyl-cytosine, adenine, 6-N-benzoyladenine, guanine and 2-N-isobutyrylguanine. It is typically desirable for linker-nucleosides to be cleaved from the compound after it reaches a target tissue. Accordingly, linker-nucleosides are typically linked to one another and to the remainder of the compound through cleavable bonds. In certain embodiments, such cleavable bonds are phosphodiester bonds.

(380) Herein, linker-nucleosides are not considered to be part of the oligonucleotide. Accordingly, in embodiments in which an compound comprises an oligonucleotide consisting of a specified number or range of linked nucleosides and/or a specified percent complementarity to a reference nucleic acid and the compound also comprises a conjugate group comprising a conjugate linker comprising linker-nucleosides, those linker-nucleosides are not counted toward the length of the oligonucleotide and are not used in determining the percent complementarity of the oligonucleotide for the reference nucleic acid. For example, a compound may comprise (1) a modified oligonucleotide consisting of 8-30 nucleosides and (2) a conjugate group comprising 1-10 linker-nucleosides that are contiguous with the nucleosides of the modified oligonucleotide. The total number of contiguous linked nucleosides in such a compound is more than 30. Alternatively, a compound may comprise a modified oligonucleotide consisting of 8-30 nucleosides and no conjugate group. The total number of contiguous linked nucleosides in such a compound is no more than 30. Unless otherwise indicated conjugate linkers comprise no more than 10 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 5 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 3 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 2 linker-

nucleosides. In certain embodiments, conjugate linkers comprise no more than 1 linker-nucleoside. (381) In certain embodiments, it is desirable for a conjugate group to be cleaved from the oligonucleotide. For example, in certain circumstances compounds comprising a particular conjugate moiety are better taken up by a particular cell type, but once the compound has been taken up, it is desirable that the conjugate group be cleaved to release the unconjugated or parent oligonucleotide. Thus, certain conjugate linkers may comprise one or more cleavable moieties. In certain embodiments, a cleavable moiety is a cleavable bond. In certain embodiments, a cleavable moiety is a group of atoms comprising at least one cleavable bond. In certain embodiments, a cleavable moiety comprises a group of atoms having one, two, three, four, or more than four cleavable bonds. In certain embodiments, a cleavable moiety is selectively cleaved inside a cell or subcellular compartment, such as a lysosome. In certain embodiments, a cleavable moiety is selectively cleaved by endogenous enzymes, such as nucleases.

(382) In certain embodiments, a cleavable bond is selected from among: an amide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, or a disulfide. In certain embodiments, a cleavable bond is one or both of the esters of a phosphodiester. In certain embodiments, a cleavable moiety comprises a phosphate or phosphodiester. In certain embodiments, the cleavable moiety is a phosphate linkage between an oligonucleotide and a conjugate moiety or conjugate group.

(383) In certain embodiments, a cleavable moiety comprises or consists of one or more linker-nucleosides. In certain such embodiments, the one or more linker-nucleosides are linked to one another and/or to the remainder of the compound through cleavable bonds. In certain embodiments, such cleavable bonds are unmodified phosphodiester bonds. In certain embodiments, a cleavable moiety is 2'-deoxy nucleoside that is attached to either the 3' or 5'-terminal nucleoside of an oligonucleotide by a phosphate internucleoside linkage and covalently attached to the remainder of the conjugate linker or conjugate moiety by a phosphate or phosphorothioate linkage. In certain such embodiments, the cleavable moiety is 2'-deoxyadenosine.

(384) 3. Certain Cell-Targeting Conjugate Moieties

(385) In certain embodiments, a conjugate group comprises a cell-targeting conjugate moiety. In certain embodiments, a conjugate group has the general formula:

(386) ##STR00060##

(387) wherein n is from 1 to about 3, m is 0 when n is 1, m is 1 when n is 2 or greater, j is 1 or 0, and k is 1 or 0.

(388) In certain embodiments, n is 1, j is 1 and k is 0. In certain embodiments, n is 1, j is 0 and k is 1. In certain embodiments, n is 1, j is 1 and k is 1. In certain embodiments, n is 2, j is 1 and k is 0. In certain embodiments, n is 2, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 3, j is 1 and k is 0. In certain embodiments, n is 3, j is 0 and k is 1. In certain embodiments, n is 3, j is 1 and k is 1.

(389) In certain embodiments, conjugate groups comprise cell-targeting moieties that have at least one tethered ligand. In certain embodiments, cell-targeting moieties comprise two tethered ligands covalently attached to a branching group. In certain embodiments, cell-targeting moieties comprise three tethered ligands covalently attached to a branching group.

(390) In certain embodiments, the cell-targeting moiety comprises a branching group comprising one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain embodiments, the branching group comprises a branched aliphatic group comprising groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether and hydroxylamino groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain such embodiments, the branched aliphatic group comprises groups selected from alkyl and ether groups. In certain embodiments, the branching

group comprises a mono or polycyclic ring system.

(391) In certain embodiments, each tether of a cell-targeting moiety comprises one or more groups selected from alkyl, substituted alkyl, ether, thioether, disulfide, amino, oxo, amide, phosphodiester, and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, thioether, disulfide, amino, oxo, amide, and polyethylene glycol, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, phosphodiester, ether, amino, oxo, and amide, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, ether, amino, oxo, and amid, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl, amino, and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and oxo, in any combination. In certain embodiments, each tether is a linear aliphatic group comprising one or more groups selected from alkyl and phosphodiester, in any combination. In certain embodiments, each tether comprises at least one phosphorus linking group or neutral linking group. In certain embodiments, each tether comprises a chain from about 6 to about 20 atoms in length. In certain embodiments, each tether comprises a chain from about 10 to about 18 atoms in length. In certain embodiments, each tether comprises about 10 atoms in chain length.

(392) In certain embodiments, each ligand of a cell-targeting moiety has an affinity for at least one type of receptor on a target cell. In certain embodiments, each ligand has an affinity for at least one type of receptor on the surface of a mammalian liver cell. In certain embodiments, each ligand has an affinity for the hepatic asialoglycoprotein receptor (ASGP-R). In certain embodiments, each ligand is a carbohydrate. In certain embodiments, each ligand is, independently selected from galactose, N-acetyl galactosamine (GalNAc), mannose, glucose, glucoseamine and fucose. In certain embodiments, each ligand is N-acetyl galactosamine (GalNAc). In certain embodiments, the cell-targeting moiety comprises 3 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 2 GalNAc ligands. In certain embodiments, the cell-targeting moiety comprises 1 GalNAc ligand.

(393) In certain embodiments, each ligand of a cell-targeting moiety is a carbohydrate, carbohydrate derivative, modified carbohydrate, polysaccharide, modified polysaccharide, or polysaccharide derivative. In certain such embodiments, the conjugate group comprises a carbohydrate cluster (see, e.g., Maier et al., "Synthesis of Antisense Oligonucleotides Conjugated to a Multivalent Carbohydrate Cluster for Cellular Targeting," *Bioconjugate Chemistry*, 2003, 14, 18-29 or Rensen et al., "Design and Synthesis of Novel N-Acetylgalactosamine-Terminated Glycolipids for Targeting of Lipoproteins to the Hepatic Asialoglycoprotein Receptor," *J. Med. Chem.* 2004, 47, 5798-5808). In certain such embodiments, each ligand is an amino sugar or a thio sugar. For example, amino sugars may be selected from any number of compounds known in the art, such as sialic acid, α -D-galactosamine, β -muramic acid, 2-deoxy-2-methylamino-L-glucopyranose, 4,6-dideoxy-4-formamido-2,3-di-O-methyl-D-mannopyranose, 2-deoxy-2-sulfoamino-D-glucopyranose and N-sulfo-D-glucosamine, and N-glycoloyl- α -neuraminic acid. For example, thio sugars may be selected from 5-Thio- β -D-glucopyranose, methyl 2,3,4-tri-O-acetyl-1-thio-6-O-trityl- α -D-glucopyranoside, 4-thio- β -D-galactopyranose, and ethyl 3,4,6,7-tetra-O-acetyl-2-deoxy-1,5-dithio- α -D-glucopyranoside.

(394) In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

(395) ##STR00061##

(396) In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

(397) ##STR00062##

(398) In certain embodiments, conjugate groups comprise a cell-targeting moiety having the

formula:

(399) ##STR00063##

(400) In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

(401) ##STR00064##

(402) In certain embodiments, conjugate groups comprise a cell-targeting moiety having the formula:

(403) ##STR00065##

(404) In certain embodiments, compounds comprise a conjugate group described herein as "LICA-1". LICA-1 has the formula:

(405) ##STR00066##

In certain embodiments, compounds described herein comprise LICA-1 and a cleavable moiety within the conjugate linker have the formula:

(406) ##STR00067##

(407) wherein oligo is an oligonucleotide.

(408) Representative United States patents, United States patent application publications, international patent application publications, and other publications that teach the preparation of certain of the above noted conjugate groups, compounds comprising conjugate groups, tethers, conjugate linkers, branching groups, ligands, cleavable moieties as well as other modifications include without limitation, U.S. Pat. Nos. 5,994,517, 6,300,319, 6,660,720, 6,906,182, 7,262,177, 7,491,805, 8,106,022, 7,723,509, US 2006/0148740, US 2011/0123520, WO 2013/033230 and WO 2012/037254, Biessen et al., *J. Med. Chem.* 1995, 38, 1846-1852, Lee et al., *Bioorganic & Medicinal Chemistry* 2011, 19, 2494-2500, Rensen et al., *J. Biol. Chem.* 2001, 276, 37577-37584, Rensen et al., *J. Med. Chem.* 2004, 47, 5798-5808, Slidregt et al., *J. Med. Chem.* 1999, 42, 609-618, and Valentijn et al., *Tetrahedron*, 1997, 53, 759-770.

(409) In certain embodiments, modified oligonucleotides comprise a gapmer or fully modified sugar motif and a conjugate group comprising at least one, two, or three GalNAc ligands. In certain embodiments, compounds comprise a conjugate group found in any of the following references: Lee, *Carbohydr Res*, 1978, 67, 509-514; Connolly et al., *J Biol Chem*, 1982, 257, 939-945; Pavia et al., *Int J Pep Protein Res*, 1983, 22, 539-548; Lee et al., *Biochem*, 1984, 23, 4255-4261; Lee et al., *Glycoconjugate J*, 1987, 4, 317-328; Toyokuni et al., *Tetrahedron Lett*, 1990, 31, 2673-2676; Biessen et al., *J Med Chem*, 1995, 38, 1538-1546; Valentijn et al., *Tetrahedron*, 1997, 53, 759-770; Kim et al., *Tetrahedron Lett*, 1997, 38, 3487-3490; Lee et al., *Bioconjug Chem*, 1997, 8, 762-765; Kato et al., *Glycobiol*, 2001, 11, 821-829; Rensen et al., *J Biol Chem*, 2001, 276, 37577-37584; Lee et al., *Methods Enzymol*, 2003, 362, 38-43; Westerlind et al., *Glycoconj J*, 2004, 21, 227-241; Lee et al., *Bioorg Med Chem Lett*, 2006, 16(19), 5132-5135; Maierhofer et al., *Bioorg Med Chem*, 2007, 15, 7661-7676; Khorev et al., *Bioorg Med Chem*, 2008, 16, 5216-5231; Lee et al., *Bioorg Med Chem*, 2011, 19, 2494-2500; Kornilova et al., *Analyt Biochem*, 2012, 425, 43-46; Pujol et al., *Angew Chemie Int Ed Engl*, 2012, 51, 7445-7448; Biessen et al., *J Med Chem*, 1995, 38, 1846-1852; Slidregt et al., *J Med Chem*, 1999, 42, 609-618; Rensen et al., *J Med Chem*, 2004, 47, 5798-5808; Rensen et al., *Arterioscler Thromb Vasc Biol*, 2006, 26, 169-175; van Rossenberg et al., *Gene Ther*, 2004, 11, 457-464; Sato et al., *J Am Chem Soc*, 2004, 126, 14013-14022; Lee et al., *J Org Chem*, 2012, 77, 7564-7571; Biessen et al., *FASEB J*, 2000, 14, 1784-1792; Rajur et al., *Bioconjug Chem*, 1997, 8, 935-940; Duff et al., *Methods Enzymol*, 2000, 313, 297-321; Maier et al., *Bioconjug Chem*, 2003, 14, 18-29; Jayaprakash et al., *Org Lett*, 2010, 12, 5410-5413; Manoharan, *Antisense Nucleic Acid Drug Dev*, 2002, 12, 103-128; Merwin et al., *Bioconjug Chem*, 1994, 5, 612-620; Tomiya et al., *Bioorg Med Chem*, 2013, 21, 5275-5281; International applications WO1998/013381; WO2011/038356; WO1997/046098; WO2008/098788; WO2004/101619; WO2012/037254; WO2011/120053; WO2011/100131; WO2011/163121; WO2012/177947; WO2013/033230; WO2013/075035; WO2012/083185; WO2012/083046; WO2009/082607;

WO2009/134487; WO2010/144740; WO2010/148013; WO1997/020563; WO2010/088537; WO2002/043771; WO2010/129709; WO2012/068187; WO2009/126933; WO2004/024757; WO2010/054406; WO2012/089352; WO2012/089602; WO2013/166121; WO2013/165816; U.S. Pat. Nos. 4,751,219; 8,552,163; 6,908,903; 7,262,177; 5,994,517; 6,300,319; 8,106,022; 7,491,805; 7,491,805; 7,582,744; 8,137,695; 6,383,812; 6,525,031; 6,660,720; 7,723,509; 8,541,548; 8,344,125; 8,313,772; 8,349,308; 8,450,467; 8,501,930; 8,158,601; 7,262,177; 6,906,182; 6,620,916; 8,435,491; 8,404,862; 7,851,615; Published U.S. Patent Application Publications US2011/0097264; US2011/0097265; US2013/0004427; US2005/0164235; US2006/0148740; US2008/0281044; US2010/0240730; US2003/0119724; US2006/0183886; US2008/0206869; US2011/0269814; US2009/0286973; US2011/0207799; US2012/0136042; US2012/0165393; US2008/0281041; US2009/0203135; US2012/0035115; US2012/0095075; US2012/0101148; US2012/0128760; US2012/0157509; US2012/0230938; US2013/0109817; US2013/0121954; US2013/0178512; US2013/0236968; US2011/0123520; US2003/0077829; US2008/0108801; and US2009/0203132.

(410) In certain embodiments, compounds are single-stranded. In certain embodiments, compounds are double-stranded.

(411) Compositions and Methods for Formulating Pharmaceutical Compositions

(412) Compounds described herein may be admixed with pharmaceutically acceptable active or inert substances for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

(413) Certain embodiments provide pharmaceutical compositions comprising one or more compounds or a salt thereof. In certain embodiments, the compounds are antisense compounds or oligomeric compounds. In certain embodiments, the compounds comprise or consist of a modified oligonucleotide. In certain such embodiments, the pharmaceutical composition comprises a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutical composition comprises a sterile saline solution and one or more compound. In certain embodiments, such pharmaceutical composition consists of a sterile saline solution and one or more compound. In certain embodiments, the sterile saline is pharmaceutical grade saline. In certain embodiments, a pharmaceutical composition comprises one or more compound and sterile water. In certain embodiments, a pharmaceutical composition consists of one compound and sterile water. In certain embodiments, the sterile water is pharmaceutical grade water. In certain embodiments, a pharmaceutical composition comprises one or more compounds and phosphate-buffered saline (PBS). In certain embodiments, a pharmaceutical composition consists of one or more compound and sterile PBS. In certain embodiments, the sterile PBS is pharmaceutical grade PBS. Compositions and methods for the formulation of pharmaceutical compositions are dependent upon a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

(414) A compound described herein targeted to HSD17B13 nucleic acid can be utilized in pharmaceutical compositions by combining the compound with a suitable pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutically acceptable diluent is water, such as sterile water suitable for injection. Accordingly, in one embodiment, employed in the methods described herein is a pharmaceutical composition comprising a compound targeted to HSD17B13 nucleic acid and a pharmaceutically acceptable diluent. In certain embodiments, the pharmaceutically acceptable diluent is water. In certain embodiments, the compound comprises or consists of a modified oligonucleotide provided herein.

(415) Pharmaceutical compositions comprising compounds provided herein encompass any pharmaceutically acceptable salts, esters, or salts of such esters, or any other oligonucleotide which, upon administration to an animal, including a human, is capable of providing (directly or

indirectly) the biologically active metabolite or residue thereof. In certain embodiments, the compounds are antisense compounds or oligomeric compounds. In certain embodiments, the compound comprises or consists of a modified oligonucleotide. Accordingly, for example, the disclosure is also drawn to pharmaceutically acceptable salts of compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts.

(416) A prodrug can include the incorporation of additional nucleosides at one or both ends of a compound which are cleaved by endogenous nucleases within the body, to form the active compound.

(417) In certain embodiments, the compounds or compositions further comprise a pharmaceutically acceptable carrier or diluent.

EXAMPLES

(418) The Examples below describe the screening process to identify compounds targeted to HSD17B13.

(419) Non-Limiting Disclosure and Incorporation by Reference

(420) Although the sequence listing accompanying this filing identifies each sequence as either “RNA” or “DNA” as required, in reality, those sequences may be modified with any combination of chemical modifications. One of skill in the art will readily appreciate that such designation as “RNA” or “DNA” to describe modified oligonucleotides is, in certain instances, arbitrary. For example, an oligonucleotide comprising a nucleoside comprising a 2'-OH sugar moiety and a thymine base could be described as a DNA having a modified sugar (2'-OH for the natural 2'-H of DNA) or as an RNA having a modified base (thymine (methylated uracil) for natural uracil of RNA).

(421) Accordingly, nucleic acid sequences provided herein, including, but not limited to, those in the sequence listing, are intended to encompass nucleic acids containing any combination of natural or modified RNA and/or DNA, including, but not limited to, such nucleic acids having modified nucleobases. By way of further example and without limitation, an oligonucleotide having the nucleobase sequence “ATCGATCG” encompasses any oligonucleotides having such nucleobase sequence, whether modified or unmodified, including, but not limited to, such compounds comprising RNA bases, such as those having sequence “AUCGAUCG” and those having some DNA bases and some RNA bases such as “AUCGATCG” and compounds having other modified nucleobases, such as “AT.sup.mCGAUCG,” wherein .sup.mC indicates a cytosine base comprising a methyl group at the 5-position.

(422) Certain compounds described herein (e.g. modified oligonucleotides) have one or more asymmetric centers and thus give rise to enantiomers, diastereomers, and other stereoisomeric configurations that may be defined, in terms of absolute stereochemistry, as (R) or (S), as α or β , such as for sugar anomers, or as (D) or (L), such as for amino acids, etc. Compounds provided herein that are drawn or described as having certain stereoisomeric configurations include only the indicated compounds. Compounds provided herein that are drawn or described with undefined stereochemistry include all such possible isomers, including their stereorandom and optically pure forms. Likewise, all tautomeric forms of the compounds provided herein are included unless otherwise indicated. Unless otherwise indicated, oligomeric compounds and modified oligonucleotides described herein are intended to include corresponding salt forms.

(423) Compounds described herein include variations in which one or more atoms are replaced with a non-radioactive isotope or radioactive isotope of the indicated element. For example, compounds herein that comprise hydrogen atoms encompass all possible deuterium substitutions for each of the .sup.1H hydrogen atoms. Isotopic substitutions encompassed by the compounds herein include, but are not limited to: .sup.2H or .sup.3H in place of .sup.1H, .sup.13C or .sup.14C in place of .sup.12C, .sup.15N in place of .sup.14N, .sup.17O or .sup.18O in place of .sup.16O, and .sup.33S, .sup.34S, .sup.35S, or .sup.36S in place of .sup.32S.

GCTGGAACAATATGGG 36 20 1245406 670 685 12088 12103 CCCAAGGCCTGAAGTT 61 21
1245432 759 774 15640 15655 ATACAGGCCATAATCT 0 22 1245458 790 805 15671 15686
ATCAGACTTCTTACGA 45 23 1245484 842 857 15723 15738 ATTGATATACGATGGA 31 24
1245510* 907 922 20722 20737 TTCTGCATACGATTTA 83 25 1245536* 973 988 20788 20803
TCTGGCTGGAGCTTAT 85 26 1245562 1014 1029 20829 20844 CATTGATTTCGAAACTA 40
27 1245588 1088 1103 20903 20918 TTGACTGCTGCTAGTG 30 28 1245614 1260 1275 21075
21090 GGTAGCTTTTGTCCAC 66 29 1245640 1314 1329 21129 21144
CAGTCTTAAACCTTCC 70 30 1245666 1341 1356 21156 21171 TGGCTACAGATTGGAA 55
31 1245692 1385 1400 21200 21215 TTAGCTGTGCACTCAT 61 32 1245718 1423 1438 21238
21253 CCAGGTTGAGATAAAG 49 33 1245744 1499 1514 21314 21329
AGAGTTGCACCGTTTT 76 34 1245770 1558 1573 21373 21388 CACTTTTGGTGGACTT 21
35 1245796 1617 1632 21432 21447 CGGTCACCTTTCATAA 69 36 1245822 1710 1725 21525
21540 ATCTCTGGGACCAAGG 72 37 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 55 38 1245848 1772 1787 21587 21602 ATAGCCAGTACAGTTC 65
39 1245874 2177 2192 21992 22007 GGAGTCGGATTATTTT 41 40 1245900 2244 2259 22059
22074 GTCCATGCAAAAGCAT 62 41 1245926 N/A N/A 3405 3420 AATACATATCTTACTC 37
42 1245952 N/A N/A 3552 3567 GTCATAAAAATCGCTG 72 43 1245978 N/A N/A 3948 3963
GAAAAGCCTGACTCAC 48 44 1246004 N/A N/A 4517 4532 CATCAAGCCCTTTTCA 23 45
1246030 N/A N/A 5078 5093 CTAAAGGAGATCTGAG 47 46 1246056 N/A N/A 5222 5237
TAGCTTAAACTCCAAT 60 47 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 87 48
1246082 N/A N/A 5817 5832 CTCAAAGTCAGTATCC 70 49 1246108 N/A N/A 5989 6004
GGCAAGGATACCTGAA 67 50 1246134 N/A N/A 6233 6248 CTCATTTATGTACCAA 87 51
1246160 N/A N/A 6474 6489 TATGTAATAGGCAGTA 62 52 1246186 N/A N/A 7482 7497
GACAATTAACATTTCGG 69 53 1246212 N/A N/A 7951 7966 CTAGGAAGTGATCCAT 66 54
1246238 N/A N/A 8243 8258 CTAATAACTACATGAC 18 55 1246264 N/A N/A 8694 8709
TCCAAAGATGAGAGTC 17 56 1246290 N/A N/A 9002 9017 GCAAAGATCTGGCCAG 22 57
1246316 N/A N/A 9348 9363 TCCAAAAGTGTCCTCG 31 58 1246342 N/A N/A 9518 9533
CTACACTAATATTGAG 20 59 1246368 N/A N/A 9715 9730 AGTGAACATACATTGT 48 60
1246394 N/A N/A 9981 9996 ATTAAGACAGTTGAGT 34 61 1246420 N/A N/A 10211 10226
ATTTATTGGTATGGTA 46 62 1246446 N/A N/A 10730 10745 CTGTAATTGGCTCTTG 28 63
1246472 N/A N/A 11261 11276 TCTTATCTTGGGCACC 44 64 1246498 N/A N/A 11742 11757
TCCATTGAATCTTCAA 57 65 1246524 N/A N/A 11985 12000 GAAACCAATCCTCAGC 3 66
1246550 N/A N/A 12421 12436 TGGTAACGGTGATCAA 74 67 1246576 N/A N/A 12624 12639
TTAAGCAGATGGCTTA 0 68 1246602 N/A N/A 12761 12776 ACAGAGAATTGTTTAG 39 69
1246628 N/A N/A 13605 13620 AGACAATACAGGATAG 15 70 1246654 N/A N/A 13862 13877
CCTTAGGAAAGCTCAT 29 71 1246680 N/A N/A 14041 14056 ATCAATGCCTTAGCCC 38 72
1246706 N/A N/A 14199 14214 TATTATGTGATTGAGT 86 73 1246732 N/A N/A 14399 14414
ATTCATAAACATAGGC 36 74 1246758 N/A N/A 14730 14745 ACTCTAAATACCCTTG 16 75
1246784 N/A N/A 15316 15331 GGATTAATCATGGGAC 44 76 1246810 N/A N/A 15405 15420
TCATAGCTCACTTAGT 0 77 1246836 N/A N/A 15775 15790 TTTAGTATTTGGGTGT 21 78
1246862 N/A N/A 16407 16422 CAATTGCTCTATAGAT 15 79 1246888 N/A N/A 17767 17782
AGCATATTCATTTGGC 61 80 1246914 N/A N/A 18033 18048 AGTTTATATGGATTTG 67 81
1246940 N/A N/A 18816 18831 CTAGTAATTGCATCTG 57 82 1246966 N/A N/A 19445 19460
CTGGGATAGTGGAGGA 21 83 1246992 N/A N/A 19686 19701 GCTAAAAGCTCACCAA 26
84 1247018 N/A N/A 19902 19917 GATACCCAGGTTGCTT 53 85 1247044 N/A N/A 20186
20201 ATTAGAAGTCAGCCCA 42 86 1247070 N/A N/A 20401 20416 GCTATAGTAATTGCTA
33 87

(430) TABLE-US-00002 TABLE 2 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID NO:
NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop Sequence (% ID

Number Site Site Site Site (5' to 3') Inhibition No 1245069 2 17 3096 3111
GGGAGGAGGTACTGTC 35 88 1245095 61 76 3155 3170 ATGATGTTTCATGGCTT 72 89
1245121 112 127 3206 3221 GACTCCAAGTAGGAGT 0 90 1245147 182 197 3276 3291
AGCTCCAGTAATGAGA 26 91 1245173 263 278 N/A N/A GCGCTTATTAATATCC 3 92
1245199 300 315 7589 7604 CTAGTTTTTCGGCACTC 33 93 1245225 360 375 7649 7664
GAGAGCGATAGATCTC 1 94 1245251 403 418 8797 8812 ACCACGATTGTTACAT 34 95
1245277 433 448 8827 8842 TCGGCTGGATATACTG 23 96 1245303 476 491 8870 8885
CTCAAATGTCTTGGA 29 97 1245329 516 531 10402 10417 GAAGTGCTTTTGTGAT 34 98
1245355 562 577 10448 10463 GCCACTGTGACGATGT 9 99 1245381 615 630 N/A N/A
TGCTGGAACAATATGG 36 100 1245407 671 686 12089 12104 TCCCAAGGCCTGAAGT 43
101 1245433 760 775 15641 15656 AATACAGGCCATAATC 16 102 1245459 791 806 15672
15687 TATCAGACTTCTTACG 59 103 1245485 843 858 15724 15739 TATTGATATACGATGG
16 104 1245511* 908 923 20723 20738 ATTCTGCATACGATTT 89 105 1245537* 974 989 20789
20804 CTCTGGCTGGAGCTTA 86 106 1245563 1015 1030 20830 20845
GCATTGATTCGAAACT 66 107 1245589 1089 1104 20904 20919 TTTGACTGCTGCTAGT 21
108 1245615 1261 1276 21076 21091 AGGTAGCTTTTGTCCA 56 109 1245641 1315 1330
21130 21145 ACAGTCTTAAACCTTC 76 110 1245667 1342 1357 21157 21172
ATGGCTACAGATTGGA 72 111 1245693 1386 1401 21201 21216 CTTAGCTGTGCACTCA 60
112 1245719 1425 1440 21240 21255 GTCCAGGTTGAGATAA 60 113 1245745 1500 1515
21315 21330 TAGAGTTGCACCGTTT 58 114 1245771 1559 1574 21374 21389
CCACTTTTGGTGGACT 17 115 1245797 1618 1633 21433 21448 TCGGTCACCTTTCATA 64
116 1245823 1711 1726 21526 21541 CATCTCTGGGACCAAG 51 117 1245841 1765 1780
21580 21595 GTACAGTTCCTTTTCC 46 38 1245849 1773 1788 21588 21603
AATAGCCAGTACAGTT 46 118 1245875 2178 2193 21993 22008 GGGAGTCGGATTATTT 33
119 1245901 2245 2260 22060 22075 AGTCCATGCAAAAGCA 77 120 1245927 N/A N/A 3433
3448 AGTAGATGGTAAGTCA 90 121 1245953 N/A N/A 3553 3568 TGTCATAAAAATCGCT
57 122 1245979 N/A N/A 3962 3977 TACTAATGTCCAAGGA 77 123 1246005 N/A N/A 4529
4544 AAGGTTAGATTTCATC 80 124 1246031 N/A N/A 5079 5094 CCTAAAGGAGATCTGA
48 125 1246057 N/A N/A 5225 5240 CTTTAGCTTAAACTCC 69 126 1246065 N/A N/A 5659
5674 ATAAGTTACCAGAGCA 84 48 1246083 N/A N/A 5833 5848 GATGTTAGGACCCAGT 64
127 1246109 N/A N/A 5990 6005 AGGCAAGGATACCTGA 42 128 1246135 N/A N/A 6298 6313
GGAGATATAACATTAC 70 129 1246161 N/A N/A 6475 6490 CTATGTAATAGGCAGT 73 130
1246187 N/A N/A 7483 7498 AGACAATTAACATTCG 62 131 1246213 N/A N/A 7952 7967
ACTAGGAAGTGATCCA 58 132 1246239 N/A N/A 8244 8259 CCTAATAACTACATGA 3 133
1246265 N/A N/A 8695 8710 ATCCAAAGATGAGAGT 5 134 1246291 N/A N/A 9003 9018
GGCAAAGATCTGGCCA 10 135 1246317 N/A N/A 9349 9364 GTCCAAAAGTGTCCTC 35
136 1246343 N/A N/A 9526 9541 CCTAGATTCTACACTA 14 137 1246369 N/A N/A 9718 9733
GATAGTGAACATACAT 54 138 1246395 N/A N/A 9983 9998 GTATTAAGACAGTTGA 45 139
1246421 N/A N/A 10212 10227 AATTTATTGGTATGGT 30 140 1246447 N/A N/A 10831 10846
GTAAACGACTCTGTAA 8 141 1246473 N/A N/A 11308 11323 GCACAAGCACACTGTA 7 142
1246499 N/A N/A 11758 11773 CTACATGTAAGGTTTT 11 143 1246525 N/A N/A 12000 12015
TAGCTAAGGGAGAGTG 9 144 1246551 N/A N/A 12429 12444 AGACAGGGTGGTAACG 51
145 1246577 N/A N/A 12625 12640 GTTAAGCAGATGGCTT 41 146 1246603 N/A N/A 12765
12780 CTCAACAGAGAATTGT 29 147 1246629 N/A N/A 13612 13627
GCTTTGAAGACAATAC 42 148 1246655 N/A N/A 13863 13878 CCCTTAGGAAAGCTCA 37
149 1246681 N/A N/A 14042 14057 TATCAATGCCTTAGCC 15 150 1246707 N/A N/A 14200
14215 TTATTATGTGATTGAG 23 151 1246733 N/A N/A 14432 14447
ACTGAAGGCTGTGTAC 14 152 1246759 N/A N/A 14731 14746 AACTCTAAATACCCTT 28
153 1246785 N/A N/A 15317 15332 AGGATTAATCATGGGA 37 154 1246811 N/A N/A 15407
15422 TTTCATAGCTCACTTA 16 155 1246837 N/A N/A 15787 15802 CTCTATTGGTGTTTTA

32 156 1246863 N/A N/A 16431 AATACTTCCCAATTGC 0 157 1246889 N/A N/A 17780
17795 AGAAGATATTATCAGC 64 158 1246915 N/A N/A 18087 18102
CACCAATGCAGTTTGT 42 159 1246941 N/A N/A 18817 18832 GCTAGTAATTGCATCT 47
160 1246967 N/A N/A 19474 19489 GCTATAACTGGAAGGA 58 161 1246993 N/A N/A 19687
19702 TGCTAAAAGCTCACCA 49 162 1247019 N/A N/A 19906 19921
TTAAGATACCCAGGTT 37 163 1247045 N/A N/A 20187 20202 TATTAGAAGTCAGCCC 27
164 1247071 N/A N/A 20402 20417 GGCTATAGTAATTGCT 13 165
(431) TABLE-US-00003 TABLE 3 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1 and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID NO:
NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop Sequence (% ID
Number Site Site Site Site (5' to 3') Inhibition) NO 1245070 5 20 3099 3114
CTAGGGAGGAGGTACT 29 166 1245096 62 77 3156 3171 GATGATGTTTCATGGCT 86 167
1245122 113 128 3207 3222 CGACTCCAAGTAGGAG 10 168 1245148 200 215 3294 3309
CCTGCCTATTCCATGC 24 169 1245174 264 279 N/A N/A CGCGCTTATTAATATC 6 170
1245200 301 316 7590 7605 CCTAGTTTTTCGGCACT 62 171 1245226 361 376 7650 7665
AGAGAGCGATAGATCT 15 172 1245252 404 419 8798 8813 CACCACGATTGTTACA 36 173
1245278 434 449 8828 8843 ATCGGCTGGATATACT 10 174 1245304 477 492 8871 8886
CCTCAAATGTCTTGGT 6 175 1245330 524 539 10410 10425 CGATGGAAGAAGTGCT 3 176
1245356 563 578 10449 10464 AGCCACTGTGACGATG 10 177 1245382 616 631 N/A N/A
TTGCTGGAACAATATG 20 178 1245408 688 703 12106 12121 GTTTTGATAACCAGTTT 67 179
1245434 761 776 15642 15657 CAATACAGGCCATAAT 0 180 1245460 792 807 15673 15688
CTATCAGACTTCTTAC 39 181 1245486 844 859 15725 15740 ATATTGATATACGATG 0 182
1245512* 909 924 20724 20739 TATTCTGCATACGATT 81 183 1245538* 975 990 20790 20805
TCTCTGGCTGGAGCTT 90 184 1245564 1016 1031 20831 20846 AGCATTGATTCGAAAC 70
185 1245590 1090 1105 20905 20920 GTTTGACTGCTGCTAG 58 186 1245616 1262 1277
21077 21092 GAGGTAGCTTTTGTCC 59 187 1245642 1316 1331 21131 21146
AACAGTCTTAAACCTT 82 188 1245668 1343 1358 21158 21173 CATGGCTACAGATTGG 54
189 1245694 1387 1402 21202 21217 TCTTAGCTGTGCACTC 69 190 1245720 1426 1441
21241 21256 TGTCCAGGTTGAGATA 48 191 1245746 1501 1516 21316 21331
ATAGAGTTGCACCGTT 57 192 1245772 1560 1575 21375 21390 TCCACTTTTGGTGGAC 16
193 1245798 1619 1634 21434 21449 GTCGGTCACCTTTCAT 85 194 1245824 1712 1727
21527 21542 ACATCTCTGGGACCAA 78 195 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 64 38 1245850 1774 1789 21589 21604 TAATAGCCAGTACAGT 64
196 1245876 2179 2194 21994 22009 TGGGAGTCGGATTATT 44 197 1245902 2246 2261
22061 22076 TAGTCCATGCAAAAGC 66 198 1245928 N/A N/A 3434 3449
AAGTAGATGGTAAGTC 85 199 1245954 N/A N/A 3554 3569 ATGTCATAAAAATCGC 53 200
1245980 N/A N/A 3963 3978 ATACTAATGTCCAAGG 89 201 1246006 N/A N/A 4535 4550
GATTTGAAGGTTAGAT 47 202 1246032 N/A N/A 5080 5095 GCCTAAAGGAGATCTG 67 203
1246058 N/A N/A 5239 5254 TAGCAAAACACTTGCT 21 204 1246065 N/A N/A 5659 5674
ATAAGTTACCAGAGCA 87 48 1246084 N/A N/A 5834 5849 GGATGTTAGGACCCAG 81 205
1246110 N/A N/A 6035 6050 ATACTTTTCCGTCTCA 90 206 1246136 N/A N/A 6303 6318
CTCTAGGAGATATAAC 49 207 1246162 N/A N/A 6476 6491 ACTATGTAATAGGCAG 85 208
1246188 N/A N/A 7681 7696 AAATTTGTGAACCTGC 26 209 1246214 N/A N/A 7953 7968
AACTAGGAAGTGATCC 58 210 1246240 N/A N/A 8245 8260 ACCTAATAACTACATG 0 211
1246266 N/A N/A 8703 8718 CCAATATAATCCAAAG 28 212 1246292 N/A N/A 9006 9021
TAAGGCAAAGATCTGG 46 213 1246318 N/A N/A 9350 9365 CGTCCAAAAGTGTCCT 53
214 1246344 N/A N/A 9527 9542 CCCTAGATTCTACACT 22 215 1246370 N/A N/A 9719 9734
GGATAGTGAACATACA 62 216 1246396 N/A N/A 9984 9999 TGTATTAAGACAGTTG 34 217
1246422 N/A N/A 10335 10350 CCAAATCAGGTAGTTT 13 218 1246448 N/A N/A 10832 10847
AGTAAACGACTCTGTA 33 219 1246474 N/A N/A 11385 11400 TTTTAGTCAGGTAGAG 3

220 1246500 N/A N/A 11776 11776 CCTACATGTAAGGT 24 221 1246526 N/A N/A 12002
12017 GTTAGCTAAGGGAGAG 18 222 1246552 N/A N/A 12433 12448
TTAAAGACAGGGTGGT 21 223 1246578 N/A N/A 12626 12641 TGTTAAGCAGATGGCT 39
224 1246604 N/A N/A 12777 12792 AATGAGGCGGCACTCA 20 225 1246630 N/A N/A 13639
13654 TGACAATGTGCAGCTC 64 226 1246656 N/A N/A 13864 13879
ACCCTTAGGAAAGCTC 52 227 1246682 N/A N/A 14043 14058 ATATCAATGCCTTAGC 32
228 1246708 N/A N/A 14202 14217 GATTATTATGTGATTG 39 229 1246734 N/A N/A 14435
14450 TATACTGAAGGCTGTG 44 230 1246760 N/A N/A 14800 14815
GAATTGCTCACCTTT 24 231 1246786 N/A N/A 15318 15333 TAGGATTAATCATGGG 47
232 1246812 N/A N/A 15418 15433 CAGTAGGTGTGTTTCA 47 233 1246838 N/A N/A 15793
15808 AAAAAGCTCTATTGGT 0 234 1246864 N/A N/A 16424 16439 GATATGTCAATACTCC
59 235 1246890 N/A N/A 17857 17872 CATTTGAAGTCTATAC 21 236 1246916 N/A N/A 18091
18106 GATACACCAATGCAGT 73 237 1246942 N/A N/A 18818 18833
TGCTAGTAATTGCATC 17 238 1246968 N/A N/A 19475 19490 GGCTATAACTGGAAGG 52
239 1246994 N/A N/A 19688 19703 ATGCTAAAAGCTCACC 50 240 1247020 N/A N/A 19907
19922 CTTAAGATACCCAGGT 36 241 1247046 N/A N/A 20188 20203
TTATTAGAAGTCAGCC 36 242 1247072 N/A N/A 20403 20418 TGGCTATAGTAATTGC 34
243

(432) TABLE-US-00004 TABLE 4 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop Sequence (% ID
Number Site Site Site Site (5' to 3') Inhibition) NO 1245071 6 21 3100 3115
CCTAGGGAGGAGGTAC 34 244 1245097 63 78 3157 3172 GGATGATGTTTCATGGC 82 245
1245123 114 129 3208 3223 ACGACTCCAAGTAGGA 24 246 1245149 202 217 3296 3311
TGCCTGCCTATTCCAT 40 247 1245175 265 280 N/A N/A CCGCGCTTATTAATAT 21 248
1245201 302 317 7591 7606 GCCTAGTTTTTCGGCAC 12 249 1245227 362 377 7651 7666
TAGAGAGCGATAGATC 22 250 1245253 405 420 8799 8814 TCACCACGATTGTTAC 34 251
1245279 435 450 8829 8844 GATCGGCTGGATATAC 7 252 1245305 478 493 8872 8887
ACCTCAAATGTCTTGG 17 253 1245331 525 540 10411 10426 TCGATGGAAGAAGTGC 34
254 1245357 566 581 10452 10467 TGAAGCCACTGTGACG 49 255 1245383 623 638 12041
12056 GGCAAATTTGCTGGAA 54 256 1245409 689 704 12107 12122
GGTTTTGATACCAGTT 58 257 1245435 762 777 15643 15658 CCAATACAGGCCATAA 38
258 1245461 793 808 15674 15689 TCTATCAGACTTCTTA 49 259 1245487 845 860 15726
15741 GATATTGATATACGAT 2 260 1245513* 910 925 20725 20740 ATATTCTGCATACGAT
36 261 1245539* 977 992 20792 20807 CATCTCTGGCTGGAGC 79 262 1245565 1017 1032
20832 20847 CAGCATTGATTCGAAA 69 263 1245591 1091 1106 20906 20921
CGTTTGACTGCTGCTA 58 264 1245617 1263 1278 21078 21093 GGAGGTAGCTTTTGTC 43
265 1245643 1317 1332 21132 21147 GAACAGTCTTAAACCT 64 266 1245669 1344 1359
21159 21174 GCATGGCTACAGATTG 56 267 1245695 1388 1403 21203 21218
CTCTTAGCTGTGCACT 53 268 1245721 1427 1442 21242 21257 ATGTCCAGGTTGAGAT 55
269 1245747 1502 1517 21317 21332 AATAGAGTTGCACCGT 48 270 1245773 1561 1576
21376 21391 GTCCACTTTTGGTGGA 14 271 1245799 1620 1635 21435 21450
AGTCGGTCACCTTTCA 66 272 1245825 1713 1728 21528 21543 AACATCTCTGGGACCA 74
273 1245851 1775 1790 21590 21605 GTAATAGCCAGTACAG 77 274 1245877 2180 2195
21995 22010 GTGGGAGTCGGATTAT 37 275 1245903 2251 2266 22066 22081
GAGGATAGTCCATGCA 59 276 1245929 N/A N/A 3437 3452 GATAAGTAGATGGTAA 59 277
1245955 N/A N/A 3573 3588 TTCTATCAACCTGCAC 62 278 1245981 N/A N/A 3964 3979
AATACTAATGTCCAAG 65 279 1246007 N/A N/A 4572 4587 ATATAGCCCTTTCCCC 26 280
1246033 N/A N/A 5081 5096 CGCCTAAAGGAGATCT 59 281 1246059 N/A N/A 5241 5256
GTTAGCAAAACACTTG 71 282 1246085 N/A N/A 5840 5855 AAACATGGATGTTAGG 56 283

TACTCTAGGAGATATA 33 285 1246163 N/A N/A 6478 6493 CAACTATGTAATAGGC 79 286
1246189 N/A N/A 7684 7699 AAGAAATTTGTGAACC 21 287 1246215 N/A N/A 7973 7988
AGTTAATAGGACTAAA 13 288 1246241 N/A N/A 8246 8261 AACCTAATAACTACAT 0 289
1246267 N/A N/A 8705 8720 GACCAATATAATCCAA 34 290 1246293 N/A N/A 9009 9024
GATTAAGGCCAAAGATC 25 291 1246319 N/A N/A 9373 9388 CTAGAACACTTGCCTC 25 292
1246345 N/A N/A 9540 9555 GAATCCAGATCTGCCC 29 293 1246371 N/A N/A 9721 9736
GAGGATAGTGAACATA 50 294 1246397 N/A N/A 9985 10000 CTGTATTAAGACAGTT 41
295 1246423 N/A N/A 10336 10351 TCCAAATCAGGTAGTT 22 296 1246449 N/A N/A 10833
10848 GAGTAAACGACTCTGT 43 297 1246475 N/A N/A 11386 11401
CTTTTAGTCAGGTAGA 33 298 1246501 N/A N/A 11780 11795 GCTTAATGTTTCAGTTT 50
299 1246527 N/A N/A 12009 12024 AAGCATTGTTAGCTAA 11 300 1246553 N/A N/A 12465
12480 ATTTATTTGCTGGTCC 47 301 1246579 N/A N/A 12647 12662 TATACTAGGATTAGAA
2 302 1246605 N/A N/A 12778 12793 AAATGAGGCGGCACTC 20 303 1246631 N/A N/A 13704
13719 AATGTAAGAAGCCACG 26 304 1246657 N/A N/A 13881 13896
CCCAAGAGTGGCAGGA 29 305 1246683 N/A N/A 14044 14059 CATATCAATGCCTTAG 29
306 1246709 N/A N/A 14205 14220 GCAGATTATTATGTGA 59 307 1246735 N/A N/A 14436
14451 TTATACTGAAGGCTGT 23 308 1246761 N/A N/A 14916 14931 ATACATTAGCAAGCTA
69 309 1246787 N/A N/A 15319 15334 CTAGGATTAATCATGG 65 310 1246813 N/A N/A 15420
15435 TCCAGTAGGTGTGTTT 47 311 1246839 N/A N/A 16116 16131
CTCTATTGGGCCAGGC 46 312 1246865 N/A N/A 16461 16476 GATATTATGTTCTTGG 59
313 1246891 N/A N/A 17909 17924 GGAAATTGTTGCTGTT 46 314 1246917 N/A N/A 18097
18112 GAAATTGATACACCAA 67 315 1246943 N/A N/A 18819 18834
ATGCTAGTAATTGCAT 16 316 1246969 N/A N/A 19484 19499 AGAATTAAGGGCTATA 45
317 1246995 N/A N/A 19689 19704 GATGCTAAAAGCTCAC 46 318 1247021 N/A N/A 19908
19923 TCTTAAGATACCCAGG 58 319 1247047 N/A N/A 20189 20204
GTTATTAGAAGTCAGC 56 320 1247073 N/A N/A 20415 20430 AATTATGCCTTGTGGC 19
321

(433) TABLE-US-00005 TABLE 5 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop Sequence (% ID
Number Site Site Site Site (5' to 3') Inhibition) NO 1245072 7 22 3101 3116
TCCTAGGGAGGAGGTA 33 322 1245098 64 79 3158 3173 AGGATGATGTTTCATGG 88 323
1245124 115 130 3209 3224 AACGACTCCAAGTAGG 27 324 1245150 205 220 3299 3314
GTCTGCCTGCCTATTC 53 325 1245176 266 281 N/A N/A ACCGCGCTTATTAATA 4 326
1245202 303 318 7592 7607 CGCCTAGTTTTTCGGCA 23 327 1245228 363 378 7652 7667
TTAGAGAGCGATAGAT 23 328 1245254 406 421 8800 8815 TTCACCACGATTGTTA 40 329
1245280 436 451 8830 8845 AGATCGGCTGGATATA 27 330 1245306 479 494 8873 8888
GACCTCAAATGTCTTG 12 331 1245332 526 541 10412 10427 ATCGATGGAAGAAGTG 39
332 1245358 573 588 10459 10474 CGCACACTGAAGCCAC 56 333 1245384 624 639 12042
12057 CGGCAAATTTGCTGGA 29 334 1245410 690 705 12108 12123
AGGTTTTGATACCAGT 60 335 1245436 763 778 15644 15659 TCCAATACAGGCCATA 62
336 1245462 795 810 15676 15691 CATCTATCAGACTTCT 38 337 1245488 846 861 15727
15742 AGATATTGATATACGA 20 338 1245514* 911 926 20726 20741 AATATTCTGCATACGA
70 339 1245540* 978 993 20793 20808 ACATCTCTGGCTGGAG 85 340 1245566 1018 1033
20833 20848 GCAGCATTGATTCGAA 63 341 1245592 1105 1120 20920 20935
TAATTAATCTTGTTTCG 63 342 1245618 1264 1279 21079 21094 GGGAGGTAGCTTTTGT 10
343 1245644 1318 1333 21133 21148 TGAACAGTCTTAAACC 56 344 1245670 1345 1360
21160 21175 GGCATGGCTACAGATT 69 345 1245696 1389 1404 21204 21219
TCTCTTAGCTGTGCAC 79 346 1245722 1428 1443 21243 21258 TATGTCCAGGTTGAGA 64

347 1245748 151548 21333 GAATAGAGTTGCACCG 55 348 1245774 1562 1577
21377 21392 GGTCCACTTTTGGTGG 13 349 1245800 1621 1636 21436 21451
GAGTCGGTCACCTTTC 82 350 1245826 1715 1730 21530 21545 TAAACATCTCTGGGAC 51
351 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 58 38 1245852 1776 1791 21591
21606 TGTAATAGCCAGTACA 46 352 1245878 2181 2196 21996 22011
AGTGGGAGTCGGATTA 50 353 1245904 2252 2267 22067 22082 AGAGGATAGTCCATGC 74
354 1245930 N/A N/A 3439 3454 AAGATAAGTAGATGGT 79 355 1245956 N/A N/A 3602 3617
AGCTTGGAAGGAGACT 69 356 1245982 N/A N/A 3999 4014 GTTAATGTAGTGTTTA 77 357
1246008 N/A N/A 4573 4588 AATATAGCCCTTTCCC 44 358 1246034 N/A N/A 5082 5097
TCGCCTAAAGGAGATC 70 359 1246060 N/A N/A 5267 5282 TGCAAAATGTGATGCC 69 360
1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 90 48 1246086 N/A N/A 5841 5856
CAAACATGGATGTTAG 49 361 1246112 N/A N/A 6037 6052 TTATACTTTTCCGTCT 61 362
1246138 N/A N/A 6323 6338 ATAGGGAGACCATGCT 64 363 1246164 N/A N/A 6567 6582
AAAACATTGCTCTTCG 59 364 1246190 N/A N/A 7751 7766 TAAATCAGGCAGTCAT 35 365
1246216 N/A N/A 7974 7989 AAGTTAATAGGACTAA 17 366 1246242 N/A N/A 8248 8263
ATAACCTAATAACTAC 26 367 1246268 N/A N/A 8714 8729 AAGTTTTGGGACCAAT 11 368
1246294 N/A N/A 9012 9027 CTTGATTAAGGCAAAG 38 369 1246320 N/A N/A 9375 9390
ATCTAGAACACTTGCC 27 370 1246346 N/A N/A 9588 9603 CTGGTAAAGGTAAGGG 58
371 1246372 N/A N/A 9722 9737 AGAGGATAGTGAACAT 59 372 1246398 N/A N/A 9986
10001 ACTGTATTAAGACAGT 20 373 1246424 N/A N/A 10337 10352
TTCCAAATCAGGTAGT 12 374 1246450 N/A N/A 10834 10849 TGAGTAAACGACTCTG 23
375 1246476 N/A N/A 11387 11402 ACTTTTAGTCAGGTAG 26 376 1246502 N/A N/A 11781
11796 GGCTTAATGTTTCAGTT 54 377 1246528 N/A N/A 12011 12026 TTAAGCATTGTTAGCT
16 378 1246554 N/A N/A 12466 12481 AATTTATTTGCTGGTC 27 379 1246580 N/A N/A 12648
12663 ATATACTAGGATTAGA 33 380 1246606 N/A N/A 12779 12794
CAAATGAGGCGGCACT 32 381 1246632 N/A N/A 13707 13722 GCAAATGTAAGAAGCC 4
382 1246658 N/A N/A 13894 13909 CTATCATGCCTTCCCC 1 383 1246684 N/A N/A 14046
14061 TACATATCAATGCCTT 38 384 1246710 N/A N/A 14206 14221 TGCAGATTATTATGTG
38 385 1246736 N/A N/A 14437 14452 CTTATACTGAAGGCTG 48 386 1246762 N/A N/A
14934 14949 CATTATAAGCTAACTA 39 387 1246788 N/A N/A 15321 15336
GGCTAGGATTAATCAT 44 388 1246814 N/A N/A 15421 15436 ATCCAGTAGGTGTGTT 41
389 1246840 N/A N/A 16123 16138 ATAATAGCTCTATTGG 46 390 1246866 N/A N/A 16533
16548 CAAGACTTAAACACCA 55 391 1246892 N/A N/A 17910 17925
GGGAAATTGTTGCTGT 42 392 1246918 N/A N/A 18099 18114 CTGAAATTGATACACC 67
393 1246944 N/A N/A 18827 18842 AAACCTTCATGCTAGT 27 394 1246970 N/A N/A 19485
19500 TAGAATTAAGGGCTAT 31 395 1246996 N/A N/A 19699 19714 GATTAATCTTGATGCT
26 396 1247022 N/A N/A 19909 19924 ATCTTAAGATACCCAG 70 397 1247048 N/A N/A 20192
20207 GCAGTTATTAGAAGTC 76 398 1247074 N/A N/A 20419 20434
GTAAAATTATGCCTTG 50 399

(434) TABLE-US-00006 TABLE 6 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop Sequence (% ID
Number Site Site Site Site (5' to 3') Inhibition) NO 1245073 8 23 3102 3117
GTCCTAGGGAGGAGGT 36 400 1245099 65 80 3159 3174 TAGGATGATGTTTCATG 72 401
1245125 116 131 3210 3225 CAACGACTCCAAGTAG 0 402 1245151 206 221 3300 3315
AGTCTGCCTGCCTATT 47 403 1245177 267 282 N/A N/A CACCGCGCTTATTAAT 7 404
1245203 304 319 7593 7608 ACGCCTAGTTTTTCGGC 7 405 1245229 364 379 7653 7668
TTTAGAGAGCGATAGA 13 406 1245255 407 422 8801 8816 ATTCACCACGATTGTT 0 407
1245281 437 452 8831 8846 AAGATCGGCTGGATAT 1 408 1245307 481 496 8875 8890
TTGACCTCAAATGTCT 0 409 1245333 527 542 10413 10428 CATCGATGGAAGAAGT 16 410

1245359 574 589 10459 10460 10461 CCGCACTGAAGCCA 34 411 1245385 625 640 12043 12058
GCGGCAAATTTGCTGG 0 412 1245411 691 706 12109 12124 GAGGTTTTGATACCAG 59
413 1245437 764 779 15645 15660 CTCCAATACAGGCCAT 20 414 1245463 796 811 15677
15692 CCATCTATCAGACTTC 78 415 1245489 847 862 15728 15743 AAGATATTGATATACG
45 416 1245515* 912 927 20727 20742 GAATATTCTGCATACG 68 417 1245541* 979 994
20794 20809 TACATCTCTGGCTGGA 86 418 1245567 1019 1034 20834 20849
TGCAGCATTGATTCTGA 47 419 1245593 1106 1121 20921 20936 GTAATTAATCTTGTTTC 52
420 1245619 1265 1280 21080 21095 AGGGAGGTAGCTTTTG 46 421 1245645 1319 1334
21134 21149 TTGAACAGTCTTAAAC 33 422 1245671 1346 1361 21161 21176
TGGCATGGCTACAGAT 57 423 1245697 1390 1405 21205 21220 ATCTCTTAGCTGTGCA 63
424 1245723 1429 1444 21244 21259 ATATGTCCAGGTTGAG 61 425 1245749 1504 1519
21319 21334 AGAATAGAGTTGCACC 70 426 1245775 1563 1578 21378 21393
GGGTCCACTTTTGGTG 13 427 1245801 1622 1637 21437 21452 AGAGTCGGTCACCTTT 71
428 1245827 1718 1733 21533 21548 GTCTAAACATCTCTGG 58 429 1245841 1765 1780
21580 21595 GTACAGTTCCTTTTCC 57 38 1245853 1777 1792 21592 21607
ATGTAATAGCCAGTAC 52 430 1245879 2182 2197 21997 22012 TAGTGGGAGTCGGATT 54
431 1245905 2253 2268 22068 22083 AAGAGGATAGTCCATG 65 432 1245931 N/A N/A 3474
3489 AGGGAATTTATCAAAC 47 433 1245957 N/A N/A 3610 3625 CATTTAGCAGCTTGGA
88 434 1245983 N/A N/A 4002 4017 GTTGTTAATGTAGTGT 77 435 1246009 N/A N/A 4641
4656 ATACGACTTCCTTCTA 44 436 1246035 N/A N/A 5104 5119 GAAAACTCAGCCAGCA
78 437 1246061 N/A N/A 5278 5293 AGCTAGACAATTGCAA 30 438 1246065 N/A N/A 5659
5674 ATAAGTTACCAGAGCA 87 48 1246087 N/A N/A 5842 5857 CCAAACATGGATGTTA 43
439 1246113 N/A N/A 6038 6053 ATTATACTTTTCCGTC 75 440 1246139 N/A N/A 6327 6342
TTTCATAGGGAGACCA 71 441 1246165 N/A N/A 6607 6622 CTAAAGCAGGCTACTT 32 442
1246191 N/A N/A 7752 7767 TTAAATCAGGCAGTCA 45 443 1246217 N/A N/A 7975 7990
AAAGTTAATAGGACTA 12 444 1246243 N/A N/A 8249 8264 CATAACCTAATAACTA 10 445
1246269 N/A N/A 8717 8732 TCAAAGTTTTGGGACC 42 446 1246295 N/A N/A 9080 9095
TCCACTTGAATTCTGT 32 447 1246321 N/A N/A 9376 9391 GATCTAGAACACTTGC 27 448
1246347 N/A N/A 9589 9604 GCTGGTAAAGGTAAGG 65 449 1246373 N/A N/A 9751 9766
CTCCAAATTCCCAACC 31 450 1246399 N/A N/A 9998 10013 AAATCTTGTAAGTACTG 41
451 1246425 N/A N/A 10359 10374 GACAATGACATAGACG 0 452 1246451 N/A N/A 10835
10850 CTGAGTAAACGACTCT 33 453 1246477 N/A N/A 11389 11404
CAACTTTTAGTCAGGT 66 454 1246503 N/A N/A 11782 11797 TGGCTTAATGTTTCAGT 50
455 1246529 N/A N/A 12012 12027 ATTAAGCATTGTTAGC 0 456 1246555 N/A N/A 12467
12482 CAATTTATTTGCTGGT 31 457 1246581 N/A N/A 12649 12664 TATATACTAGGATTAG
10 458 1246607 N/A N/A 12781 12796 CTCAAATGAGGCGGCA 56 459 1246633 N/A N/A
13711 13726 CGTAGCAAATGTAAGA 0 460 1246659 N/A N/A 13896 13911
ATCTATCATGCCTTCC 6 461 1246685 N/A N/A 14047 14062 TTACATATCAATGCCT 12 462
1246711 N/A N/A 14243 14258 GTTAAACTTCATTCC 5 463 1246737 N/A N/A 14438 14453
CCTTATACTGAAGGCT 41 464 1246763 N/A N/A 14935 14950 TCATTATAAGCTAACT 0 465
1246789 N/A N/A 15328 15343 GAGGAATGGCTAGGAT 47 466 1246815 N/A N/A 15424
15439 CATATCCAGTAGGTGT 37 467 1246841 N/A N/A 16128 16143 CCATAATAATAGCTCT
48 468 1246867 N/A N/A 16755 16770 AAAGTTAGTTGGGCGA 26 469 1246893 N/A N/A
17911 17926 TGGGAAATTGTTGCTG 38 470 1246919 N/A N/A 18100 18115
ACTGAAATTGATACAC 49 471 1246945 N/A N/A 18828 18843 GAAACCTTCATGCTAG 31
472 1246971 N/A N/A 19486 19501 CTAGAATTAAGGGCTA 42 473 1246997 N/A N/A 19700
19715 TGATTAATCTTGATGC 15 474 1247023 N/A N/A 19910 19925 CATCTTAAGATACCCA
76 475 1247049 N/A N/A 20248 20263 CCTAAACAAACACTAT 8 476 1247075 N/A N/A 20422
20437 ACAGTAAAATTATGCC 73 477

(435) TABLE-US-00007 TABLE 7 Inhibition of HSD17B13 RNA by 3-10-3 cEt

gapmers targeting SEQ ID NO: 1, and 2 SEQ ID NO: 1 ID NO:
NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number Site
Site Site Site Sequence (5' to 3') Inhibition) NO 1245074 10 25 3104 3119
TAGTCCTAGGGAGGAG 68 478 1245100 66 81 3160 3175 CTAGGATGATGTTTCAT 47 479
1245126 117 132 3211 3226 CCAACGACTCCAAGTA 20 480 1245152 207 222 3301 3316
TAGTCTGCCTGCCTAT 44 481 1245178 268 283 N/A N/A ACACCGCGCTTATTAA 7 482
1245204 316 331 7605 7620 GCATGCGCAGTGACGC 36 483 1245230 365 380 7654 7669
ATTTAGAGAGCGATAG 14 484 1245256 408 423 8802 8817 TATTCACCACGATTGT 2 485
1245282 438 453 8832 8847 GAAGATCGGCTGGATA 16 486 1245308 482 497 8876 8891
GTTGACCTCAAATGTC 0 487 1245334 528 543 10414 10429 TCATCGATGGAAGAAG 24
488 1245360 575 590 10461 10476 GCCGCACACTGAAGCC 25 489 1245386 626 641 12044
12059 AGCGGCAAATTTGCTG 5 490 1245412 692 707 12110 12125 TGAGGTTTTGATACCA
63 491 1245438 765 780 15646 15661 TCTCCAATACAGGCCA 21 492 1245464 797 812 15678
15693 TCCATCTATCAGACTT 42 493 1245490* 858 873 15739 15754
GTAGTCTCAGAAAGAT 45 494 1245516* 926 941 20741 20756 CACTGCTTCAAATTGA 64
495 1245542* 980 995 20795 20810 ATACATCTCTGGCTGG 88 496 1245568 1020 1035 20835
20850 TTGCAGCATTGATTCG 59 497 1245594 1107 1122 20922 20937
GGTAATTAATCTTGTT 58 498 1245620 1266 1281 21081 21096 TAGGGAGGTAGCTTTT 46
499 1245646 1320 1335 21135 21150 CTTGAACAGTCTTAAA 39 500 1245672 1347 1362
21162 21177 GTGGCATGGCTACAGA 56 501 1245698 1391 1406 21206 21221
GATCTCTTAGCTGTGC 64 502 1245724 1430 1445 21245 21260 AATATGTCCAGGTTGA 64
503 1245750 1505 1520 21320 21335 CAGAATAGAGTTGCAC 50 504 1245776 1564 1579
21379 21394 AGGGTCCACTTTTGGT 11 505 1245802 1623 1638 21438 21453
TAGAGTCGGTCACCTT 65 506 1245828 1719 1734 21534 21549 TGTCTAAACATCTCTG 58
507 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 51 38 1245854 1778 1793 21593
21608 TATGTAATAGCCAGTA 50 508 1245880 2183 2198 21998 22013
GTAGTGGGAGTCGGAT 48 509 1245906 2254 2269 22069 22084 CAAGAGGATAGTCCAT 58
510 1245932 N/A N/A 3487 3502 AGACTAAGGGACCAGG 80 511 1245958 N/A N/A 3611
3626 TCATTTAGCAGCTTGG 79 512 1245984 N/A N/A 4123 4138 ACAATTTTTCCAATCC 86
513 1246010 N/A N/A 4642 4657 AATACGACTTCCTTCT 26 514 1246036 N/A N/A 5105 5120
TGAAAACTCAGCCAGC 77 515 1246062 N/A N/A 5287 5302 TTATAACTGAGCTAGA 8 516
1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 89 48 1246088 N/A N/A 5843 5858
CCCAAACATGGATGTT 72 517 1246114 N/A N/A 6039 6054 AATTATACTTTTCCGT 66 518
1246140 N/A N/A 6328 6343 CTTTCATAGGGAGACC 73 519 1246166 N/A N/A 6608 6623
TCTAAAGCAGGCTACT 32 520 1246192 N/A N/A 7757 7772 CCAATTTAAATCAGGC 73 521
1246218 N/A N/A 7976 7991 CAAAGTTAATAGGACT 50 522 1246244 N/A N/A 8280 8295
GTCTAAAGTGCTGGTT 39 523 1246270 N/A N/A 8747 8762 GAACACTTTCCTGTTG 0 524
1246296 N/A N/A 9110 9125 ACCAAGTAGCTTACAT 2 525 1246322 N/A N/A 9377 9392
GGATCTAGAACAATTG 32 526 1246348 N/A N/A 9598 9613 ATGTGAAGAGCTGGTA 42 527
1246374 N/A N/A 9761 9776 CCCAAGCTACCTCCAA 15 528 1246400 N/A N/A 10001 10016
GTGAAATCTTGTGTAA 44 529 1246426 N/A N/A 10360 10375 AGACAATGACATAGAC 14
530 1246452 N/A N/A 10836 10851 ACTGAGTAAACGACTC 45 531 1246478 N/A N/A 11428
11443 CCCTTTATGTCTTTGG 35 532 1246504 N/A N/A 11790 11805 CTATTGATTGGCTTAA
31 533 1246530 N/A N/A 12038 12053 AAATTTGCTGGAAGT 15 534 1246556 N/A N/A
12478 12493 AGCTAGATACTCAATT 2 535 1246582 N/A N/A 12652 12667
GAATATATACTAGGAT 5 536 1246608 N/A N/A 12782 12797 CCTCAAATGAGGCGGC 47
537 1246634 N/A N/A 13730 13745 ATTAAGTACTGTGAGA 1 538 1246660 N/A N/A 13897
13912 CATCTATCATGCCTTC 18 539 1246686 N/A N/A 14048 14063 ATTACATATCAATGCC
36 540 1246712 N/A N/A 14250 14265 AGCGAATGTTAAACT 16 541 1246738 N/A N/A
14439 14454 TCCTTATACTGAAGGC 22 542 1246764 N/A N/A 14936 14951

CTCATTATAAGCTAAC 13 543 1246790 N/A N/A 15350 15365 AGTAGTGGAGCCAGAC 39
544 1246816 N/A N/A 15425 15440 TCATATCCAGTAGGTG 32 545 1246842 N/A N/A 16129
16144 TCCATAATAATAGCTC 69 546 1246868 N/A N/A 16905 16920
CGTCAAAAACCGTCAA 20 547 1246894 N/A N/A 17923 17938 GAGTATTTCTCTCTGGG 23
548 1246920 N/A N/A 18506 18521 ACTAATTTAGTCAACT 18 549 1246946 N/A N/A 18846
18861 TTAACTACCCTCACA 0 550 1246972 N/A N/A 19487 19502 TCTAGAATTAAGGGCT
45 551 1246998 N/A N/A 19701 19716 CTGATTAATCTTGATG 32 552 1247024 N/A N/A 19911
19926 GCATCTTAAGATACCC 82 553 1247050 N/A N/A 20249 20264
GCCTAAACAAACACTA 37 554 1247076 N/A N/A 20481 20496 CATAGTGGACTTCATT 34
555
(436) TABLE-US-00008 TABLE 8 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245075 11 26 3105 3120
GTAGTCCTAGGGAGGA 77 556 1245101 67 82 3161 3176 TCTAGGATGATGTTCA 64 557
1245127 118 133 3212 3227 ACCAACGACTCCAAGT 16 558 1245153 208 223 3302 3317
GTAGTCTGCCTGCCTA 68 559 1245179 269 284 N/A N/A CACACCGCGCTTATTA 11 560
1245205 317 332 7606 7621 CGCATGCGCAGTGACG 46 561 1245231 366 381 7655 7670
GATTTAGAGAGCGATA 44 562 1245257 409 424 8803 8818 TTATTCACCACGATTG 40 563
1245283 439 454 8833 8848 AGAAGATCGGCTGGAT 29 564 1245309 483 498 8877 8892
TGTTGACCTCAAATGT 14 565 1245335 529 544 10415 10430 ATCATCGATGGAAGAA 30
566 1245361 576 591 10462 10477 GGCCGCACACTGAAGC 9 567 1245387 632 647 12050
12065 GCCAACAGCGGCAAAT 30 568 1245413 693 708 12111 12126
ATGAGGTTTTGATACC 47 569 1245439 766 781 15647 15662 GTCTCCAATACAGGCC 47
570 1245465 798 813 15679 15694 TTCCATCTATCAGACT 33 571 1245491* 866 881 N/A N/A
AACTTCTGTAGTCTC 67 572 1245517* 927 942 20742 20757 CCACTGCTTCAAATTG 77
573 1245543* 981 996 20796 20811 CATACTCTCTGGCTG 69 574 1245569 1021 1036 20836
20851 TTTGCAGCATTGATTC 53 575 1245595 1109 1124 20924 20939
CAGGTAATTAATCTTG 54 576 1245621 1267 1282 21082 21097 TTAGGGAGGTAGCTTT 54
577 1245647 1321 1336 21136 21151 ACTTGAACAGTCTTAA 63 578 1245673 1348 1363
21163 21178 TGTGGCATGGCTACAG 24 579 1245699 1392 1407 21207 21222
TGATCTCTTAGCTGTG 59 580 1245725 1431 1446 21246 21261 AAATATGTCCAGGTTG 68
581 1245751 1506 1521 21321 21336 CCAGAATAGAGTTGCA 62 582 1245777 1565 1580
21380 21395 GAGGGTCCACTTTTGG 23 583 1245803 1624 1639 21439 21454
ATAGAGTCGGTCACCT 60 584 1245829 1721 1736 21536 21551 ATTGTCTAAACATCTC 84
585 1245855 1779 1794 21594 21609 TTATGTAATAGCCAGT 77 586 1245881 2184 2199 21999
22014 TGTAGTGGGAGTCGGA 62 587 1245907 2255 2270 22070 22085
ACAAGAGGATAGTCCA 64 588 1245933 N/A N/A 3488 3503 CAGACTAAGGGACCAG 75
589 1245959 N/A N/A 3612 3627 CTCATTTAGCAGCTTG 70 590 1245985 N/A N/A 4136 4151
CAAGGAGTACTTCACA 66 591 1246011 N/A N/A 4643 4658 TAATACGACTTCCTTC 32 592
1246037 N/A N/A 5125 5140 GCAAAAGGAGCCAGTT 57 593 1246063 N/A N/A 5290 5305
GTTTTATAACTGAGCT 84 594 1246089 N/A N/A 5871 5886 CTAAACCCTGGAGCAG 42 595
1246115 N/A N/A 6040 6055 AAATTATACTTTTCCG 85 596 1246141 N/A N/A 6337 6352
GAAGAAGGACTTTCAT 40 597 1246167 N/A N/A 6609 6624 CTCTAAAGCAGGCTAC 51
598 1246193 N/A N/A 7758 7773 TCCAATTTAAATCAGG 61 599 1246219 N/A N/A 7977 7992
GCAAAGTTAATAGGAC 52 600 1246245 N/A N/A 8282 8297 TGGTCTAAAGTGCTGG 61
601 1246271 N/A N/A 8895 8910 CCCAAAAATGTCCTAG 18 602 1246297 N/A N/A 9111 9126
GACCAAGTAGCTTACA 28 603 1246323 N/A N/A 9379 9394 ATGGATCTAGAACACT 43 604
1246349 N/A N/A 9600 9615 ATATGTGAAGAGCTGG 51 605 1246375 N/A N/A 9762 9777
ACCCAAGCTACCTCCA 9 606 1246401 N/A N/A 10002 10017 AGTGAAATCTTGTGTA 41

607 1246427 N/A N/A 10380 10395 GATCATTTGCAGAAAGA 23 608 1246453 N/A N/A 10849
10864 ACATATGGTGTACACT 19 609 1246479 N/A N/A 11468 11483
GCAAATACAGCAGTAC 56 610 1246505 N/A N/A 11791 11806 TCTATTGATTGGCTTA 39
611 1246531 N/A N/A 12039 12054 CAAATTTGCTGGAAGT 39 612 1246557 N/A N/A 12482
12497 AAGGAGCTAGATACTC 19 613 1246583 N/A N/A 12654 12669
CAGAATATATACTAGG 55 614 1246609 N/A N/A 12795 12810 TAAGAGTCAGTATCCT 52
615 1246635 N/A N/A 13731 13746 TATTAAGTACTGTGAG 0 616 1246661 N/A N/A 13915
13930 CATTATCCTTACTCCC 16 617 1246687 N/A N/A 14049 14064 CATTACATATCAATGC
29 618 1246713 N/A N/A 14262 14277 AGTTAGGGAGCCAGCG 38 619 1246739 N/A N/A
14465 14480 CTGGAGGTATGTCATA 23 620 1246765 N/A N/A 15022 15037
TGATAAAATAGGGTGC 44 621 1246791 N/A N/A 15351 15366 GAGTAGTGGAGCCAGA 44
622 1246817 N/A N/A 15427 15442 TTTCATATCCAGTAGG 36 623 1246843 N/A N/A 16189
16204 AATAACTGTTCTCCCC 20 624 1246869 N/A N/A 16956 16971
GACAACAAACAATGGG 39 625 1246895 N/A N/A 17924 17939 AGAGTATTTCTCTCTGG 32
626 1246921 N/A N/A 18507 18522 CACTAATTTAGTCAAC 7 627 1246947 N/A N/A 18847
18862 TTTTAACTACCCTCAC 2 628 1246973 N/A N/A 19488 19503 ATCTAGAATTAAGGGC
41 629 1246999 N/A N/A 19702 19717 GCTGATTAATCTTGAT 48 630 1247025 N/A N/A 19921
19936 TCCAAGGAGTGCATCT 46 631 1247051 N/A N/A 20356 20371
TTATTGCCTGAACACA 58 632 1247077 N/A N/A 20482 20497 TCATAGTGGACTTCAT 36
633

(437) TABLE-US-00009 TABLE 9 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245076 12 27 3106 3121
TGTAGTCCTAGGGAGG 86 634 1245102 68 83 3162 3177 TTCTAGGATGATGTTT 72 635
1245128 119 134 3213 3228 CACCAACGACTCCAAG 16 636 1245154 211 226 3305 3320
TAAGTAGTCTGCCTGC 19 637 1245180 271 286 N/A N/A TCCACACCGCGCTTAT 32 638
1245206 318 333 7607 7622 ACGCATGCGCAGTGAC 52 639 1245232 367 382 7656 7671
TGATTTAGAGAGCGAT 35 640 1245258 410 425 8804 8819 ATTATTCACCACGATT 29 641
1245284 440 455 8834 8849 GAGAAGATCGGCTGGA 6 642 1245310 484 499 8878 8893
ATGTTGACCTCAAATG 0 643 1245336 530 545 10416 10431 CATCATCGATGGAAGA 14 644
1245362 577 592 10463 10478 TGGCCGCACACTGAAG 0 645 1245388 638 653 12056 12071
GTGAAAGCCAACAGCG 62 646 1245414 696 711 12114 12129 GACATGAGGTTTTGAT 52
647 1245440 767 782 15648 15663 TGTCTCCAATACAGGC 39 648 1245466 799 814 15680
15695 ATTCCATCTATCAGAC 7 649 1245492* 867 882 N/A N/A GAAACTTCTGTAGTCT 86
650 1245518* 931 946 20746 20761 CCAACCACTGCTTCAA 98 651 1245544* 982 997 20797
20812 GCATACATCTCTGGCT 27 652 1245570 1023 1038 20838 20853
GCTTTGCAGCATTGAT 61 653 1245596 1110 1125 20925 20940 ACAGGTAATTAATCTT 39
654 1245622 1268 1283 21083 21098 TTTAGGGAGGTAGCTT 24 655 1245648 1322 1337
21137 21152 TACTTGAACAGTCTTA 48 656 1245674 1349 1364 21164 21179
CTGTGGCATGGCTACA 66 657 1245700 1393 1408 21208 21223 TTGATCTCTTAGCTGT 74
658 1245726 1461 1476 21276 21291 CTAGGGAAATCTTTCA 48 659 1245752 1507 1522
21322 21337 TCCAGAATAGAGTTGC 46 660 1245778 1566 1581 21381 21396
AGAGGGTCCACTTTTG 34 661 1245804 1625 1640 21440 21455 AATAGAGTCGGTCACC 58
662 1245830 1728 1743 21543 21558 GCCTAAAATTGTCTAA 32 663 1245841 1765 1780
21580 21595 GTACAGTTCCTTTTCC 49 38 1245856 1780 1795 21595 21610
CTTATGTAATAGCCAG 74 664 1245882 2185 2200 22000 22015 ATGTAGTGGGAGTCGG 72
665 1245908 2256 2271 22071 22086 AACAAGAGGATAGTCC 54 666 1245934 N/A N/A 3489
3504 ACAGACTAAGGGACCA 68 667 1245960 N/A N/A 3613 3628 CCTCATTTAGCAGCTT
81 668 1245986 N/A N/A 4138 4153 CTCAAGGAGTACTTCA 82 669 1246012 N/A N/A 4644

4659 TTAATCACTTCTT 35 670 1246038 N/A N/A 5141 GGCAAAGGAGCCAGT
43 671 1246064 N/A N/A 5655 5670 GTTACCAGAGCATTCA 73 672 1246065 N/A N/A 5659
5674 ATAAGTTACCAGAGCA 89 48 1246090 N/A N/A 5872 5887 GCTAAACCCTGGAGCA 5
673 1246116 N/A N/A 6089 6104 CAAGTTTTTCACCTCAG 93 674 1246142 N/A N/A 6357 6372
GGGAAAGAAGTCTCCT 13 675 1246168 N/A N/A 6610 6625 TCTCTAAAGCAGGCTA 60
676 1246194 N/A N/A 7770 7785 TAGCAGCTAAAATCCA 66 677 1246220 N/A N/A 7978 7993
TGCAAAGTTAATAGGA 43 678 1246246 N/A N/A 8487 8502 CTATATAAATTCTGAC 0 679
1246272 N/A N/A 8939 8954 TAAGAGAAGGTGCACA 12 680 1246298 N/A N/A 9130 9145
GATACTGGTGGACAAG 13 681 1246324 N/A N/A 9410 9425 GTATAGCTGCATTTCT 35 682
1246350 N/A N/A 9601 9616 CATATGTGAAGAGCTG 25 683 1246376 N/A N/A 9776 9791
AACTAGAACTCCCAAC 0 684 1246402 N/A N/A 10025 10040 GGCTTTATCATTCTAA 20 685
1246428 N/A N/A 10384 10399 TAAGGATCATTGCAGA 0 686 1246454 N/A N/A 10855 10870
CCATTTACATATGGTG 28 687 1246480 N/A N/A 11469 11484 GGCAAATACAGCAGTA 64
688 1246506 N/A N/A 11792 11807 CTCTATTGATTGGCTT 65 689 1246532 N/A N/A 12168
12183 CTTACCTTGTGCTTGG 34 690 1246558 N/A N/A 12486 12501
GACTAAGGAGCTAGAT 25 691 1246584 N/A N/A 12655 12670 GCAGAATATATACTAG 55
692 1246610 N/A N/A 12799 12814 GCTGTAAGAGTCAGTA 89 693 1246636 N/A N/A 13733
13748 CTTATTAAGTACTGTG 0 694 1246662 N/A N/A 13916 13931 CCATTATCCTTACTCC
27 695 1246688 N/A N/A 14089 14104 CTACATGGATCCAACA 5 696 1246714 N/A N/A 14266
14281 GAGGAGTTAGGGAGCC 36 697 1246740 N/A N/A 14469 14484
ACAACTGGAGGTATGT 0 698 1246766 N/A N/A 15023 15038 TTGATAAAATAGGGTG 46
699 1246792 N/A N/A 15353 15368 AGGAGTAGTGGAGCCA 16 700 1246818 N/A N/A 15443
15458 TATCAGAACTTATAC 0 701 1246844 N/A N/A 16191 16206 GAAATAACTGTTCTCC
32 702 1246870 N/A N/A 16969 16984 TCCAATATATACTGAC 72 703 1246896 N/A N/A 17930
17945 GCTGAGAGAGTATTC 51 704 1246922 N/A N/A 18508 18523
ACACTAATTTAGTCAA 16 705 1246948 N/A N/A 18848 18863 CTTTTAACTACCCTCA 16
706 1246974 N/A N/A 19490 19505 CCATCTAGAATTAAGG 0 707 1247000 N/A N/A 19731
19746 ACTGAAGGTCTGAGCT 20 708 1247026 N/A N/A 19922 19937
CTCCAAGGAGTGCATC 53 709 1247052 N/A N/A 20359 20374 AAATTATTGCCTGAAC 32
710 1247078 N/A N/A 20484 20499 CTTCATAGTGGACTTC 79 711
(438) TABLE-US-00010 TABLE 10 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245077 13 28 3107 3122
GTGTAGTCCTAGGGAG 67 712 1245103 71 86 3165 3180 GATTTCTAGGATGATG 57 713
1245129 120 135 3214 3229 TCACCAACGACTCCAA 19 714 1245155 212 227 3306 3321
ATAAGTAGTCTGCCTG 43 715 1245181 272 287 N/A N/A CTCCACACCGCGCTTA 7 716
1245207 319 334 7608 7623 TACGCATGCGCAGTGA 60 717 1245233 368 383 7657 7672
CTGATTTAGAGAGCGA 61 718 1245259 411 426 8805 8820 CATTATTCACCACGAT 33 719
1245285 441 456 8835 8850 TGAGAAGATCGGCTGG 37 720 1245311 485 500 8879 8894
GATGTTGACCTCAAAT 0 721 1245337 531 546 10417 10432 CCATCATCGATGGAAG 8 722
1245363 578 593 10464 10479 GTGGCCGCACACTGAA 41 723 1245389 642 657 12060 12075
CTCTGTGAAAGCCAAC 50 724 1245415 698 713 12116 12131 GAGACATGAGGTTTTTG 77
725 1245441 768 783 15649 15664 CTGTCTCCAATACAGG 29 726 1245467 801 816 15682
15697 GTATTCCATCTATCAG 52 727 1245493* 877 892 N/A N/A CGTTCAGGAAGAACT
70 728 1245519* 932 947 20747 20762 GCCAACCAGTCTTCA 97 729 1245545* 983 998
20798 20813 TGCATACATCTCTGGC 53 730 1245571 1024 1039 20839 20854
AGCTTTGCAGCATTGA 62 731 1245597 1111 1126 20926 20941 GACAGGTAATTAATCT 60
732 1245623 1269 1284 21084 21099 TTTTAGGGAGGTAGCT 39 733 1245649 1323 1338
21138 21153 CTACTTGAACAGTCTT 80 734 1245675 1350 1365 21165 21180

TCTGTGGCATGGGTAC 67 735 1245701 1394 1409 21224 CTTGATCTCTTAGCTG 51
736 1245727 1462 1477 21277 21292 GCTAGGGAAATCTTTC 66 737 1245753 1508 1523
21323 21338 GTCCAGAATAGAGTTG 60 738 1245779 1567 1582 21382 21397
TAGAGGGTCCACTTTT 36 739 1245805 1626 1641 21441 21456 AAATAGAGTCGGTCAC 59
740 1245831 1729 1744 21544 21559 AGCCTAAAATTGTCTA 53 741 1245857 1781 1796
21596 21611 TCTTATGTAATAGCCA 77 742 1245883 2186 2201 22001 22016
GATGTAGTGGGAGTCG 67 743 1245909 2257 2272 22072 22087 AAACAAGAGGATAGTC
48 744 1245935 N/A N/A 3492 3507 TATACAGACTAAGGGA 24 745 1245961 N/A N/A 3629
3644 GATAATATGTGACTTG 68 746 1245987 N/A N/A 4139 4154 TCTCAAGGAGTACTTC 80
747 1246013 N/A N/A 4645 4660 TTTAATACGACTTCCT 72 748 1246039 N/A N/A 5156 5171
CAAAGTGTACTGTGT 57 749 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 91 48
1246091 N/A N/A 5875 5890 TAGGCTAAACCCTGGA 64 750 1246117 N/A N/A 6134 6149
CTCTAATGTTTCTTGA 53 751 1246143 N/A N/A 6422 6437 TAAACTGTATGCCTCT 49 752
1246169 N/A N/A 6622 6637 CAGTAGGCAGTTTCTC 72 753 1246195 N/A N/A 7776 7791
TTCAGTTAGCAGCTAA 72 754 1246221 N/A N/A 7979 7994 CTGCAAAGTTAATAGG 57 755
1246247 N/A N/A 8529 8544 GTTTATCAGTAAAGTG 64 756 1246273 N/A N/A 8940 8955
TTAAGAGAAGGTGCAC 38 757 1246299 N/A N/A 9146 9161 GCCAAGCTTTGTGTCA 64
758 1246325 N/A N/A 9413 9428 TCTGTATAGCTGCATT 47 759 1246351 N/A N/A 9603 9618
CACATATGTGAAGAGC 71 760 1246377 N/A N/A 9780 9795 ATAGAACTAGAACTCC 22 761
1246403 N/A N/A 10058 10073 GATAATTTGGTGTGCAT 59 762 1246429 N/A N/A 10385 10400
TTAAGGATCATTGCAG 14 763 1246455 N/A N/A 10931 10946 CATAAGGCCGATTCAT 32
764 1246481 N/A N/A 11482 11497 GGAAAGGATATTTGGC 58 765 1246507 N/A N/A 11793
11808 ACTCTATTGATTGGCT 57 766 1246533 N/A N/A 12178 12193 TGATTTTGACCTTACC
11 767 1246559 N/A N/A 12487 12502 TGAATAAGGAGCTAGA 31 768 1246585 N/A N/A
12685 12700 ATTAAGCTTAGTATAT 10 769 1246611 N/A N/A 12810 12825
AGCTAACTCAGGCTGT 14 770 1246637 N/A N/A 13738 13753 GCATACTTATTAAGTA 32
771 1246663 N/A N/A 13917 13932 TCCATTATCCTTACTC 15 772 1246689 N/A N/A 14090
14105 TCTACATGGATCCAAC 26 773 1246715 N/A N/A 14280 14295
TGTAAGGCTGGGTGA 26 774 1246741 N/A N/A 14473 14488 TATGACAACTGGAGGT 14
775 1246767 N/A N/A 15024 15039 GTTGATAAAATAGGGT 42 776 1246793 N/A N/A 15354
15369 AAGGAGTAGTGGAGCC 38 777 1246819 N/A N/A 15445 15460
GTTATCAGAACTTAT 31 778 1246845 N/A N/A 16219 16234 TTATAGACTGGGTAGG 51
779 1246871 N/A N/A 17017 17032 GTTTAGCAATTTAGCA 50 780 1246897 N/A N/A 17956
17971 ATTTAGAACTGCTCC 21 781 1246923 N/A N/A 18521 18536
GCTTAGTACCAAGACA 62 782 1246949 N/A N/A 18849 18864 CCTTTTAACTACCCTC 22
783 1246975 N/A N/A 19533 19548 AGGCTAAAATGGTCAT 36 784 1247001 N/A N/A 19736
19751 CTAAAACTGAAGGTCT 40 785 1247027 N/A N/A 19936 19951
TAGGAACATTCCCTCT 24 786 1247053 N/A N/A 20362 20377 TCTAAATTATTGCCTG 43
787 1247079 N/A N/A 20485 20500 ACTTCATAGTGGACTT 61 788
(439) TABLE-US-00011 TABLE 11 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245078 14 29 3108 3123
TGTGTAGTCCTAGGGA 50 789 1245104 72 87 3166 3181 GGATTTCTAGGATGAT 58 790
1245130 121 136 3215 3230 TTCACCAACGACTCCA 62 791 1245156 213 228 3307 3322
CATAAGTAGTCTGCCT 31 792 1245182 273 288 N/A N/A CCTCCACACCGCGCTT 16 793
1245208 320 335 7609 7624 ATACGCATGCGCAGTG 38 794 1245234 369 384 7658 7673
CCTGATTTAGAGAGCG 42 795 1245260 412 427 8806 8821 GCATTATTCACCACGA 59 796
1245286 442 457 8836 8851 CTGAGAAGATCGGCTG 37 797 1245312 486 501 8880 8895
GGATGTTGACCTCAAA 5 798 1245338 532 547 10418 10433 TCCATCATCGATGGAA 31 799

1245364 579 594 10465 10480 CGTGGCCGCACACTGA 42 800 1245390 645 660 12063 12078
GACCTCTGTGAAAGCC 46 801 1245416 715 730 12133 12148 TTCACAAAAACTGGGC 37
802 1245442 769 784 15650 15665 TCTGTCTCCAATACAG 13 803 1245468 802 817 15683
15698 AGTATTCCATCTATCA 62 804 1245494* 891 906 20706 20721
AAATCGCTGAGGCGCG 89 805 1245520* 933 948 20748 20763 GGCCAACCACTGCTTC 93
806 1245546* 984 999 20799 20814 ATGCATACATCTCTGG 53 807 1245572 1025 1040 20840
20855 AAGCTTTGCAGCATTG 59 808 1245598 1112 1127 20927 20942
AGACAGGTAATTAATC 29 809 1245624 1270 1285 21085 21100 CTTTTAGGGAGGTAGC 62
810 1245650 1324 1339 21139 21154 GCTACTTGAACAGTCT 66 811 1245676 1351 1366
21166 21181 TTCTGTGGCATGGCTA 78 812 1245702 1395 1410 21210 21225
ACTTGATCTCTTAGCT 59 813 1245728 1463 1478 21278 21293 GGCTAGGGAAATCTTT 57
814 1245754 1509 1524 21324 21339 AGTCCAGAATAGAGTT 58 815 1245780 1568 1583
21383 21398 ATAGAGGGTCCACTTT 21 816 1245806 1627 1642 21442 21457
AAAATAGAGTCGGTCA 71 817 1245832 1730 1745 21545 21560 GAGCCTAAAATTGTCT 49
818 1245858 1782 1797 21597 21612 TTCTTATGTAATAGCC 71 819 1245884 2187 2202 22002
22017 TGATGTAGTGGGAGTC 65 820 1245910 2349 2364 22164 22179
CTACAAGAGGTTATTT 31 821 1245936 N/A N/A 3493 3508 ATATACAGACTAAGGG 24 822
1245962 N/A N/A 3671 3686 AAGTTATTATCCCCAC 60 823 1245988 N/A N/A 4220 4235
AACTAAACATGACAGC 72 824 1246014 N/A N/A 4646 4661 TTTTAATACGACTTCC 57 825
1246040 N/A N/A 5164 5179 ATAAGAGCCAAACTGC 49 826 1246066 N/A N/A 5660 5675
CATAAGTTACCAGAGC 80 827 1246092 N/A N/A 5878 5893 CTGTAGGCTAAACCCT 47 828
1246118 N/A N/A 6135 6150 GCTCTAATGTTTCTTG 62 829 1246144 N/A N/A 6423 6438
GTAAACTGTATGCCTC 74 830 1246170 N/A N/A 6624 6639 TACAGTAGGCAGTTTC 60 831
1246196 N/A N/A 7777 7792 CTTCAGTTAGCAGCTA 50 832 1246222 N/A N/A 8039 8054
ACTACATTAACACCAA 46 833 1246248 N/A N/A 8534 8549 TGCAAGTTTATCAGTA 70 834
1246274 N/A N/A 8941 8956 CTTAAGAGAAGGTGCA 43 835 1246300 N/A N/A 9188 9203
CTTAACAAACCTCCAC 3 836 1246326 N/A N/A 9424 9439 ATTATATGAGGTCTGT 59 837
1246352 N/A N/A 9604 9619 TCACATATGTGAAGAG 27 838 1246378 N/A N/A 9781 9796
CATAGAACTAGAACTC 19 839 1246404 N/A N/A 10059 10074 TGATAATTTGGTGTCA 3 840
1246430 N/A N/A 10386 10401 CTTAAGGATCATTGCA 29 841 1246456 N/A N/A 10971 10986
GTAATCTTGAGGCAGG 51 842 1246482 N/A N/A 11511 11526 TATTATGAGGATCTGG 44
843 1246508 N/A N/A 11853 11868 AGCAGAATTGTGAACG 67 844 1246534 N/A N/A 12194
12209 CATACCCATTCTAACT 6 845 1246560 N/A N/A 12488 12503 TTGACTAAGGAGCTAG
20 846 1246586 N/A N/A 12686 12701 CATTAAAGCTTAGTATA 19 847 1246612 N/A N/A 12813
12828 CATAGCTAACTCAGGC 29 848 1246638 N/A N/A 13748 13763 CTTACATATTGCATAC
35 849 1246664 N/A N/A 13951 13966 CTAAGTTAGCCCCCAG 15 850 1246690 N/A N/A
14091 14106 ATCTACATGGATCCAA 12 851 1246716 N/A N/A 14283 14298
GAATGTAAAGGCTGGG 42 852 1246742 N/A N/A 14477 14492 GGAGTATGACAACTGG 28
853 1246768 N/A N/A 15170 15185 CGTTTAAATCAGAGCA 54 854 1246794 N/A N/A 15355
15370 TAAGGAGTAGTGGAGC 18 855 1246820 N/A N/A 15492 15507
ATTAATGCCACCCTAC 6 856 1246846 N/A N/A 16223 16238 CTTTTTATAGACTGGG 67 857
1246872 N/A N/A 17019 17034 AGGTTTAGCAATTTAG 54 858 1246898 N/A N/A 17985 18000
ATAGTTATCTTCTCAC 48 859 1246924 N/A N/A 18585 18600 TATTAATATGACTTGC 25 860
1246950 N/A N/A 18850 18865 GCCTTTTAACTACCCT 0 861 1246976 N/A N/A 19534 19549
TAGGCTAAAATGGTCA 59 862 1247002 N/A N/A 19739 19754 GATCTAAAACCTGAAGG 44
863 1247028 N/A N/A 19970 19985 AATTTCATACAGTTTCG 66 864 1247054 N/A N/A 20363
20378 GTCTAAATTATTGCCT 45 865 1247080 N/A N/A 20489 20504 CATAACTTCATAGTGG
31 866

(440) TABLE-US-00012 TABLE 12 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID

NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245079 15 30 3109 3124
TTGTGTAGTCCTAGGG 64 867 1245105 75 90 3169 3184 GAAGGATTTCTAGGAT 71 868
1245131 122 137 3216 3231 CTTACCAACGACTCC 8 869 1245157 214 229 3308 3323
TCATAAGTAGTCTGCC 62 870 1245183 274 289 7563 7578 TCCTCCACACCGCGCT 20 871
1245209 321 336 7610 7625 CATACGCATGCGCAGT 60 872 1245235 370 385 7659 7674
ACCTGATTTAGAGAGC 65 873 1245261 413 428 8807 8822 AGCATTATTCACCACG 53 874
1245287 443 458 8837 8852 GCTGAGAAGATCGGCT 10 875 1245313 487 502 8881 8896
AGGATGTTGACCTCAA 4 876 1245339 533 548 10419 10434 CTCCATCATCGATGGA 0 877
1245365 593 608 10479 10494 GTAAGGAATCCCTTCG 22 878 1245391 646 661 12064 12079
AGACCTCTGTGAAAGC 50 879 1245417 723 738 12141 12156 ACCCAGTATTCACAAA 10
880 1245443 770 785 15651 15666 ATCTGTCTCCAATACA 55 881 1245469 803 818 15684
15699 AAGTATTCCATCTATC 52 882 1245495* 892 907 20707 20722
AAAATCGCTGAGGCGC 97 883 1245521* 934 949 20749 20764 TGGCCAACCACTGCTT 98
884 1245547* 985 1000 20800 20815 CATGCATACATCTCTG 55 885 1245573 1029 1044 20844
20859 AATAAAGCTTTGCAGC 26 886 1245599 1117 1132 20932 20947
CAGGAAGACAGGTAAT 60 887 1245625 1271 1286 21086 21101 ACTTTTAGGGAGGTAG
20 888 1245651 1325 1340 21140 21155 TGCTACTTGAACAGTC 81 889 1245677 1353 1368
21168 21183 TATTCTGTGGCATGGC 71 890 1245703 1396 1411 21211 21226
AACTTGATCTCTTAGC 64 891 1245729 1464 1479 21279 21294 AGGCTAGGGAAATCTT 62
892 1245755 1510 1525 21325 21340 AAGTCCAGAATAGAGT 60 893 1245781 1569 1584
21384 21399 TATAGAGGGTCCACTT 17 894 1245807 1628 1643 21443 21458
TAAAATAGAGTCGGTC 81 895 1245833 1731 1746 21546 21561 TGAGCCTAAAATTGTC 30
896 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 57 38 1245859 1784 1799 21599
21614 GTTCTTATGTAATAG 46 897 1245885 2188 2203 22003 22018
TTGATGTAGTGGGAGT 68 898 1245911 2350 2365 22165 22180 ACTACAAGAGGTTATT 46
899 1245937 N/A N/A 3494 3509 CATATACAGACTAAGG 57 900 1245963 N/A N/A 3672 3687
TAAGTTATTATCCCCA 70 901 1245989 N/A N/A 4249 4264 CTCTAAGTGTAAGAGA 28 902
1246015 N/A N/A 4647 4662 TTTTAAATACGACTTC 52 903 1246041 N/A N/A 5165 5180
CATAAGAGCCAAACTG 27 904 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 90 48
1246067 N/A N/A 5661 5676 GCATAAGTTACCAGAG 79 905 1246093 N/A N/A 5889 5904
GTATTTTCGCTACTGTA 76 906 1246119 N/A N/A 6149 6164 CCTAAGATAAAATTGC 18 907
1246145 N/A N/A 6425 6440 TGGTAAACTGTATGCC 76 908 1246171 N/A N/A 6625 6640
GTACAGTAGGCAGTTT 50 909 1246197 N/A N/A 7782 7797 AACTACTTCAGTTAGC 53 910
1246223 N/A N/A 8043 8058 GAAGACTACATTAACA 36 911 1246249 N/A N/A 8562 8577
AAATTTTCAGTACCCAG 46 912 1246275 N/A N/A 8942 8957 TCTTAAGAGAAGGTGC 39 913
1246301 N/A N/A 9189 9204 CCTTAACAAACCTCCA 27 914 1246327 N/A N/A 9426 9441
CAATTATATGAGGTCT 36 915 1246353 N/A N/A 9610 9625 GCCAATTCACATATGT 34 916
1246379 N/A N/A 9887 9902 TATATGTTGATGTTAC 34 917 1246405 N/A N/A 10060 10075
ATGATAATTTGGTGTC 55 918 1246431 N/A N/A 10387 10402 TCTTAAGGATCATTGC 0 919
1246457 N/A N/A 10973 10988 TTGTAATCTTGAGGCA 27 920 1246483 N/A N/A 11512 11527
TTATTATGAGGATCTG 50 921 1246509 N/A N/A 11854 11869 AAGCAGAATTGTGAAC 18
922 1246535 N/A N/A 12201 12216 CATACCACATACCCAT 0 923 1246561 N/A N/A 12489
12504 CTTGACTAAGGAGCTA 24 924 1246587 N/A N/A 12687 12702
TCATTAAGCTTAGTAT 31 925 1246613 N/A N/A 12814 12829 ACATAGCTAACTCAGG 38
926 1246639 N/A N/A 13763 13778 GTACTATAGTATTTAC 27 927 1246665 N/A N/A 13952
13967 TCTAAGTTAGCCCCCA 36 928 1246691 N/A N/A 14092 14107
CATCTACATGGATCCA 63 929 1246717 N/A N/A 14316 14331 TTAAGATGTCCAGGCA 35
930 1246743 N/A N/A 14479 14494 AAGGAGTATGACAAC 28 931 1246769 N/A N/A 15211
15226 CATAAAGCTGGATTGT 0 932 1246795 N/A N/A 15356 15371

GTAAGTAGTGAG 37 933 1246821 N/A N/A 15493 15508 CATTAAATGCCACCCTA 8
 934 1246847 N/A N/A 16254 16269 AGTAGATTTGGTAGAG 67 935 1246873 N/A N/A 17041
 17056 GCTTATCGAGTTTAGC 17 936 1246899 N/A N/A 17987 18002 GTATAGTTATCTTCTC
 59 937 1246925 N/A N/A 18662 18677 ATACTATTCTCCAAC 25 938 1246951 N/A N/A 18871
 18886 ATAAGAAGAGAGCTCA 14 939 1246977 N/A N/A 19544 19559
 GTTTAGCTGATAGGCT 47 940 1247003 N/A N/A 19749 19764 TATGAGTAAAGATCTA 10
 941 1247029 N/A N/A 20004 20019 GTAATCAGTTTTTCCTC 79 942 1247055 N/A N/A 20373
 20388 GTAAGGTAAAGTCTAA 37 943 1247081 N/A N/A 20490 20505
 ACATAACTTCATAGTG 27 944
 (441) TABLE-US-00013 TABLE 13 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
 NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
 Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245080 16 31 3110 3125
 CTTGTGTAGTCCTAGG 71 945 1245106 91 106 3185 3200 ATGGTGATCAGAAGCA 85 946
 1245132 123 138 3217 3232 ACTTCACCAACGACTC 15 947 1245158 215 230 3309 3324
 TTCATAAGTAGTCTGC 64 948 1245184 275 290 7564 7579 TTCCTCCACACCGCGC 47 949
 1245210 322 337 7611 7626 ACATACGCATGCGCAG 20 950 1245236 371 386 7660 7675
 CACCTGATTTAGAGAG 11 951 1245262 416 431 8810 8825 CCCAGCATTATTCACC 19 952
 1245288 444 459 8838 8853 TGCTGAGAAGATCGGC 7 953 1245314 488 503 8882 8897
 TAGGATGTTGACCTCA 19 954 1245340 534 549 10420 10435 TCTCCATCATCGATGG 24
 955 1245366 594 609 10480 10495 GGTAAGGAATCCCTTC 43 956 1245392 649 664 12067
 12082 GTCAGACCTCTGTGAA 34 957 1245418 724 739 12142 12157
 AACCCAGTATTCACAA 0 958 1245444 771 786 15652 15667 CATCTGTCTCCAATAC 32 959
 1245470 804 819 15685 15700 TAAGTATTCCATCTAT 31 960 1245496* 893 908 20708 20723
 TAAAATCGCTGAGGCG 87 961 1245522* 935 950 20750 20765 GTGGCCAACCACTGCT 99
 962 1245548* 986 1001 20801 20816 TCATGCATACATCTCT 70 963 1245574 1055 1070 20870
 20885 ATATTATCAGGACTGA 47 964 1245600 1136 1151 20951 20966
 CGTAAATATTCTTGAG 61 965 1245626 1272 1287 21087 21102 TACTTTTAGGGAGGTA 28
 966 1245652 1326 1341 21141 21156 ATGCTACTTGAACAGT 47 967 1245678 1354 1369
 21169 21184 ATATTCTGTGGCATGG 69 968 1245704 1403 1418 21218 21233
 CTGCTGAAACTTGATC 50 969 1245730 1465 1480 21280 21295 GAGGCTAGGGAAATCT 56
 970 1245756 1516 1531 21331 21346 GTAATAAAGTCCAGAA 66 971 1245782 1570 1585
 21385 21400 ATATAGAGGGTCCACT 18 972 1245808 1629 1644 21444 21459
 TTAAAATAGAGTCGGT 65 973 1245834 1732 1747 21547 21562 TTGAGCCTAAAATTGT 36
 974 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 58 38 1245860 1792 1807 21607
 21622 GGTCCATTGTTTCTTA 36 975 1245886 2189 2204 22004 22019
 CTTGATGTAGTGGGAG 83 976 1245912 2351 2366 22166 22181 AACTACAAGAGGTTAT 57
 977 1245938 N/A N/A 3495 3510 ACATATACAGACTAAG 46 978 1245964 N/A N/A 3673 3688
 CTAAGTTATTATCCCC 70 979 1245990 N/A N/A 4296 4311 GTATAGTATTCTTCAC 81 980
 1246016 N/A N/A 4649 4664 CATTTTTAATACGACT 57 981 1246042 N/A N/A 5166 5181
 GCATAAGAGCCAAACT 67 982 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 87 48
 1246068 N/A N/A 5662 5677 AGCATAAGTTACCAGA 87 983 1246094 N/A N/A 5891 5906
 TTGTATTTTCGCTACTG 58 984 1246120 N/A N/A 6170 6185 CCTAAATTCAAAGACC 64 985
 1246146 N/A N/A 6429 6444 GTAATGGTAAACTGTA 70 986 1246172 N/A N/A 6737 6752
 TGAGATTACCCCTGGC 53 987 1246198 N/A N/A 7785 7800 CATAACTACTTCAGTT 3 988
 1246224 N/A N/A 8121 8136 CTTAGTTCATTGTGAT 69 989 1246250 N/A N/A 8564 8579
 CTAAATTTTCAGTACCC 33 990 1246276 N/A N/A 8943 8958 ATCTTAAGAGAAGGTG 43 991
 1246302 N/A N/A 9190 9205 CCCTTAACAAACCTCC 43 992 1246328 N/A N/A 9429 9444
 AACCAATTATATGAGG 38 993 1246354 N/A N/A 9632 9647 GTGTAGTTTGGTGGGC 33 994
 1246380 N/A N/A 9908 9923 CCTATTTGTAATATTG 0 995 1246406 N/A N/A 10062 10077

TTATTGATTGTTGGTGG 0 996 1246432 N/A N/A 10388 10403 ATCTTAAGGATCATTG 7 997
1246458 N/A N/A 10976 10991 GCATTGTAATCTTGAG 35 998 1246484 N/A N/A 11513 11528
TTTATTATGAGGATCT 43 999 1246510 N/A N/A 11856 11871 TTAAGCAGAATTGTGA 7
1000 1246536 N/A N/A 12203 12218 ATCATACCACATACCC 45 1001 1246562 N/A N/A 12501
12516 GCAGAAATTCACCTTG 74 1002 1246588 N/A N/A 12690 12705
GAATCATTAAGCTTAG 63 1003 1246614 N/A N/A 12816 12831 CCACATAGCTAACTCA 53
1004 1246640 N/A N/A 13768 13783 CAATAGTACTATAGTA 14 1005 1246666 N/A N/A 13953
13968 CTCTAAGTTAGCCCCC 38 1006 1246692 N/A N/A 14126 14141
AATTACTCCGTTCTGC 6 1007 1246718 N/A N/A 14317 14332 ATTAAGATGTCCAGGC 31
1008 1246744 N/A N/A 14480 14495 CAAGGAGTATGACAAC 6 1009 1246770 N/A N/A 15213
15228 TCCATAAAGCTGGATT 13 1010 1246796 N/A N/A 15358 15373
TGGTAAGGAGTAGTGG 34 1011 1246822 N/A N/A 15494 15509 GCATTAATGCCACCCT 36
1012 1246848 N/A N/A 16255 16270 CAGTAGATTTGGTAGA 49 1013 1246874 N/A N/A 17042
17057 AGCTTATCGAGTTTAG 54 1014 1246900 N/A N/A 17988 18003
AGTATAGTTATCTTCT 67 1015 1246926 N/A N/A 18663 18678 GATACTATTCTCCAAC 44
1016 1246952 N/A N/A 18872 18887 GATAAGAAGAGAGCTC 36 1017 1246978 N/A N/A
19549 19564 ATATTGTTTAGCTGAT 14 1018 1247004 N/A N/A 19751 19766
GATATGAGTAAAGATC 23 1019 1247030 N/A N/A 20006 20021 GAGTAATCAGTTTTCC 65
1020 1247056 N/A N/A 20374 20389 AGTAAGGTAAAGTCTA 55 1021 1247082 N/A N/A 20492
20507 TGACATAACTTCATAG 51 1022

(442) TABLE-US-00014 TABLE 14 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245081 17 32 3111 3126
CCTTGTGTAGTCCTAG 67 1023 1245107 92 107 3186 3201 GATGGTGATCAGAAGC 52 1024
1245133 124 139 3218 3233 AACTTCACCAACGACT 0 1025 1245159 216 231 3310 3325
ATTCATAAGTAGTCTG 35 1026 1245185 276 291 7565 7580 TTTCTCCACACCGCG 18
1027 1245211 323 338 7612 7627 CACATACGCATGCGCA 59 1028 1245237 372 387 7661 7676
TCACCTGATTTAGAGA 13 1029 1245263 418 433 8812 8827 GTCCCAGCATTATTCA 21
1030 1245289 445 460 8839 8854 GTGCTGAGAAGATCGG 0 1031 1245315 489 504 8883 8898
CTAGGATGTTGACCTC 0 1032 1245341 535 550 10421 10436 CTCTCCATCATCGATG 0 1033
1245367 595 610 10481 10496 AGGTAAGGAATCCCTT 0 1034 1245393 650 665 12068 12083
TGTCAGACCTCTGTGA 0 1035 1245419 725 740 12143 12158 GAACCCAGTATTCACA 0
1036 1245445 777 792 15658 15673 CGACTTCATCTGTCTC 19 1037 1245471 805 820 15686
15701 GTAAGTATTCCATCTA 45 1038 1245497* 894 909 20709 20724
TTAAAATCGCTGAGGC 67 1039 1245523* 936 951 20751 20766 TGTGGCCAACCACTGC 75
1040 1245549* 987 1002 20802 20817 ATCATGCATACATCTC 62 1041 1245575 1075 1090
20890 20905 GTGCCAAACCAATGTT 43 1042 1245601 1150 1165 20965 20980
CCTATGAAAACTACG 36 1043 1245627 1277 1292 21092 21107 GTATTTACTTTTAGGG 61
1044 1245653 1327 1342 21142 21157 AATGCTACTTGAACAG 54 1045 1245679 1355 1370
21170 21185 GATATTCTGTGGCATG 46 1046 1245705 1404 1419 21219 21234
CCTGCTGAAACTTGAT 19 1047 1245731 1466 1481 21281 21296 AGAGGCTAGGGAAATC
63 1048 1245757 1517 1532 21332 21347 AGTAATAAAGTCCAGA 43 1049 1245783 1571 1586
21386 21401 AATATAGAGGGTCCAC 7 1050 1245809 1630 1645 21445 21460
TTTAAAATAGAGTCGG 69 1051 1245835 1733 1748 21548 21563 TTTGAGCCTAAAATTG
26 1052 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 47 38 1245861 1793 1808
21608 21623 GGGTCCATTGTTTCTT 21 1053 1245887 2190 2205 22005 22020
TCTTGATGTAGTGGA 44 1054 1245913 2352 2367 22167 22182 TAACTACAAGAGGTTA
22 1055 1245939 N/A N/A 3507 3522 AACTAACCTGACACAT 20 1056 1245965 N/A N/A 3674
3689 CCTAAGTTATTATCCC 18 1057 1245991 N/A N/A 4360 4375 ATACATCAACTCCTCT 64

124617 N/A N/A 4650 4665 TCATTTTAACTACGAC 48 1059 1246043 N/A N/A 5167
5182 TGCATAAGAGCCAAAC 56 1060 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA
84 48 1246069 N/A N/A 5663 5678 GAGCATAAGTTACCAG 61 1061 1246095 N/A N/A 5916
5931 CTTGAATGAGTGGTCT 64 1062 1246121 N/A N/A 6184 6199 CAATTTACTTTCAGCC
71 1063 1246147 N/A N/A 6437 6452 AACTAGTGGTAATGGT 32 1064 1246173 N/A N/A 6741
6756 GCAATGAGATTACCCC 83 1065 1246199 N/A N/A 7786 7801 ACATAACTACTTCAGT
28 1066 1246225 N/A N/A 8122 8137 GCTTAGTTCATTGTGA 45 1067 1246251 N/A N/A 8587
8602 GGGATAAACTGTTCTT 47 1068 1246277 N/A N/A 8944 8959 TATCTTAAGAGAAGGT
29 1069 1246303 N/A N/A 9191 9206 ACCCTTAACAAACCTC 36 1070 1246329 N/A N/A 9430
9445 TAACCAATTATATGAG 0 1071 1246355 N/A N/A 9636 9651 CTCCGTGTAGTTTGGT 28
1072 1246381 N/A N/A 9922 9937 GAATTCACATCTCTCC 38 1073 1246407 N/A N/A 10063
10078 CTTATGATAATTTGGT 2 1074 1246433 N/A N/A 10389 10404 GATCTTAAGGATCATT
0 1075 1246459 N/A N/A 11000 11015 TAAATCCAAATCACTC 39 1076 1246485 N/A N/A
11514 11529 TTTTATTATGAGGATC 18 1077 1246511 N/A N/A 11857 11872
ATTAAGCAGAATTGTG 8 1078 1246537 N/A N/A 12205 12220 TTATCATACCACATAC 19
1079 1246563 N/A N/A 12512 12527 CAAGAGTTCTTGCAGA 17 1080 1246589 N/A N/A
12692 12707 TCGAATCATTAAGCTT 34 1081 1246615 N/A N/A 12817 12832
ACCACATAGCTAACTC 50 1082 1246641 N/A N/A 13769 13784 GCAATAGTACTATAGT 39
1083 1246667 N/A N/A 13954 13969 CCTCTAAGTTAGCCCC 97 1084 1246693 N/A N/A 14127
14142 AAATTACTCCGTTCTG 28 1085 1246719 N/A N/A 14321 14336
TCAAATTAAGATGTCC 6 1086 1246745 N/A N/A 14499 14514 TGAGAATTTAAGAGGG 25
1087 1246771 N/A N/A 15217 15232 GTATTCCATAAAGCTG 10 1088 1246797 N/A N/A 15359
15374 ATGGTAAGGAGTAGTG 17 1089 1246823 N/A N/A 15495 15510
TGCATTAATGCCACCC 22 1090 1246849 N/A N/A 16257 16272 GACAGTAGATTTGGTA 34
1091 1246875 N/A N/A 17043 17058 AAGCTTATCGAGTTTA 26 1092 1246901 N/A N/A 17989
18004 TAGTATAGTTATCTTC 28 1093 1246927 N/A N/A 18666 18681 TGGGATACTATTCTCC
18 1094 1246953 N/A N/A 18873 18888 GGATAAGAAGAGAGCT 46 1095 1246979 N/A N/A
19550 19565 GATATTGTTTAGCTGA 59 1096 1247005 N/A N/A 19753 19768
CTGATATGAGTAAAGA 21 1097 1247031 N/A N/A 20008 20023 GAGAGTAATCAGTTTTT 27
1098 1247057 N/A N/A 20375 20390 AAGTAAGGTAAAGTCT 20 1099 1247083 N/A N/A
20571 20586 TTTAACTTTCCTCCGT 16 1100
(443) TABLE-US-00015 TABLE 15 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ ID SEQ ID SEQ NO: 1 ID
NO: NO: 2 ID NO: HSD17B13 SEQ Compound Start 1 Stop Start 2 Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245082 18 33 3112 3127
TCCTTGTGTAGTCCTA 61 1101 1245108 93 108 3187 3202 TGATGGTGATCAGAAG 0 1102
1245134 125 140 3219 3234 AAACCTTCACCAACGAC 27 1103 1245160 217 232 3311 3326
AATTCATAAGTAGTCT 28 1104 1245186 277 292 7566 7581 GTTTCCTCCACACCGC 45
1105 1245212 324 339 7613 7628 CCACATACGCATGCGC 68 1106 1245238 373 388 N/A N/A
TTCACCTGATTTAGAG 9 1107 1245264 419 434 8813 8828 TGTCCCAGCATTATTC 31 1108
1245290 446 461 8840 8855 GGTGCTGAGAAGATCG 33 1109 1245316 490 505 8884 8899
CCTAGGATGTTGACCT 3 1110 1245342 536 551 10422 10437 TCTCTCCATCATCGAT 24
1111 1245368 596 611 10482 10497 GAGGTAAGGAATCCCT 27 1112 1245394 651 666 12069
12084 ATGTCAGACCTCTGTG 19 1113 1245420 726 741 12144 12159
TGAACCCAGTATTCAC 10 1114 1245446 778 793 15659 15674 ACGACTTCATCTGTCT 8
1115 1245472 806 821 15687 15702 GGTAAGTATTCCATCT 12 1116 1245498* 895 910 20710
20725 TTTAAAATCGCTGAGG 24 1117 1245524* 937 952 20752 20767
TTGTGGCCAACCACTG 83 1118 1245550* 988 1003 20803 20818 TATCATGCATACATCT 45
1119 1245576 1076 1091 20891 20906 AGTGCCAAACCAATGT 62 1120 1245602 1154 1169
20969 20984 CAGACCTATGAAAAAC 26 1121 1245628 1296 1311 21111 21126

GTGTAATAAGTCTCTC 61 1122 1245654 1328 1343 21158 21158 GAATGCTACTTGAACA 44
1123 1245680 1356 1371 21171 21186 TGATATTCTGTGGCAT 69 1124 1245706 1405 1420
21220 21235 GCCTGCTGAAACTTGA 18 1125 1245732 1483 1498 21298 21313
GGGCTAATGAAAAAGG 14 1126 1245758 1520 1535 21335 21350 TCAAGTAATAAAGTCC
49 1127 1245784 1572 1587 21387 21402 AAATATAGAGGGTCCA 0 1128 1245810 1631 1646
21446 21461 ATTTAAAATAGAGTCG 24 1129 1245836 1738 1753 21553 21568
TAATTTTTTGAGCCTAA 41 1130 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 55
38 1245862 1794 1809 21609 21624 TGGGTCCATTGTTTCT 35 1131 1245888 2191 2206 22006
22021 GTCTTGATGTAGTGGG 67 1132 1245914 2353 2368 22168 22183
ATAACTACAAGAGGTT 64 1133 1245940 N/A N/A 3508 3523 TAACTAACCTGACACA 57
1134 1245966 N/A N/A 3675 3690 CCCTAAGTTATTATCC 25 1135 1245992 N/A N/A 4361
4376 TATACATCAACTCCTC 39 1136 1246018 N/A N/A 4652 4667 AGTCATTTTTTAATACG 13
1137 1246044 N/A N/A 5168 5183 TTGCATAAGAGCCAAA 46 1138 1246065 N/A N/A 5659
5674 ATAAGTTACCAGAGCA 75 48 1246070 N/A N/A 5736 5751 TAGTTGGTAGCTTGCT 66
1139 1246096 N/A N/A 5918 5933 ACCTTGAATGAGTGGT 26 1140 1246122 N/A N/A 6185
6200 GCAATTTACTTTCAGC 66 1141 1246148 N/A N/A 6441 6456 TCTAAACTAGTGGTAA
24 1142 1246174 N/A N/A 6743 6758 TTGCAATGAGATTACC 29 1143 1246200 N/A N/A 7787
7802 GACATAACTACTTCAG 66 1144 1246226 N/A N/A 8133 8148 TTTGTTAGAGTGCTTA
80 1145 1246252 N/A N/A 8596 8611 CATAAAGGTGGGATAA 23 1146 1246278 N/A N/A 8945
8960 GTATCTTAAGAGAAGG 30 1147 1246304 N/A N/A 9208 9223 AAAGATGTACACTGAC
57 1148 1246330 N/A N/A 9431 9446 CTAACCAATTATATGA 21 1149 1246356 N/A N/A 9648
9663 AATGTATGAGGTCTCC 65 1150 1246382 N/A N/A 9926 9941 GATTGAATTCACTATC 0
1151 1246408 N/A N/A 10064 10079 CCTTATGATAATTTGG 29 1152 1246434 N/A N/A 10390
10405 TGATCTTAAGGATCAT 0 1153 1246460 N/A N/A 11030 11045 CATTCTAAACTACTAT
11 1154 1246486 N/A N/A 11542 11557 GACAATGGTTGCCACT 63 1155 1246512 N/A N/A
11858 11873 GATTAAGCAGAATTGT 37 1156 1246538 N/A N/A 12209 12224
CAATTTATCATACCAC 37 1157 1246564 N/A N/A 12513 12528 ACAAGAGTTCTTGCAG 29
1158 1246590 N/A N/A 12693 12708 TTCGAATCATTAAAGCT 63 1159 1246616 N/A N/A 12862
12877 CACCATGAAGGACTCC 48 1160 1246642 N/A N/A 13770 13785
TGCAATAGTACTATAG 24 1161 1246668 N/A N/A 13975 13990 ATTTAGCATTGCCTGT 10
1162 1246694 N/A N/A 14128 14143 TAAATTACTCCGTTCT 36 1163 1246720 N/A N/A 14362
14377 GTGCAAGATGATGACA 19 1164 1246746 N/A N/A 14541 14556
GTATTTGAGATTCACC 41 1165 1246772 N/A N/A 15244 15259 GCTAAGTATACTTTCT 37
1166 1246798 N/A N/A 15360 15375 GATGGTAAGGAGTAGT 28 1167 1246824 N/A N/A 15496
15511 CTGCATTAATGCCACC 67 1168 1246850 N/A N/A 16272 16287
CAAAGTTTGGGCAGAG 44 1169 1246876 N/A N/A 17046 17061 AAAAAGCTTATCGAGT 8
1170 1246902 N/A N/A 17991 18006 GTTAGTATAGTTATCT 66 1171 1246928 N/A N/A 18671
18686 TTTACTGGGATACTAT 25 1172 1246954 N/A N/A 18901 18916
AGGGAAGATCTGGCTG 4 1173 1246980 N/A N/A 19563 19578 AGCTATTGTCTTTGAT 39
1174 1247006 N/A N/A 19760 19775 CTCTTATCTGATATGA 7 1175 1247032 N/A N/A 20037
20052 GCATTAAACTCTCAT 35 1176 1247058 N/A N/A 20376 20391
CAAGTAAGGTAAAGTC 33 1177 1247084 N/A N/A 20572 20587 GTTTAACTTTCCTCCG 30
1178

(444) TABLE-US-00016 TABLE 16 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245083 19 34 3113 3128
GTCCTTGTGTAGTCCT 62 1179 1245109 98 113 3192 3207
GTAGATGATGGTGATC 36 1180 1245135 126 141 3220 3235
AAAACCTTCACCAACGA 29 1181 1245161 240 255 3334 3349

CCAATGTCCTGTGTCG 65 1182 1245187 278 293 7567 7582
AGTTTCCTCCACACCG 28 1183 1245213 325 340 7614 7629
ACCACATACGCATGCG 45 1184 1245239 374 389 N/A N/A CTTCACCTGATTAGA 7
1185 1245265 421 436 8815 8830 ACTGTCCCAGCATTAT 40 1186 1245291 452
467 8846 8861 ATCCTTGGTGCTGAGA 13 1187 1245317 491 506 8885 8900
TCCTAGGATGTTGACC 0 1188 1245343 537 552 10423 10438 TTCTCTCCATCATCGA
35 1189 1245369 597 612 10483 10498 TGAGGTAAGGAATCCC 2 1190 1245395 652
667 12070 12085 GATGTCAGACCTCTGT 6 1191 1245421 727 742 12145 12160
GTGAACCCAGTATTCA 1 1192 1245447 779 794 15660 15675
TACGACTTCATCTGTC 29 1193 1245473 807 822 15688 15703 TGGTAAGTATTCCATC
26 1194 1245499* 896 911 20711 20726 ATTTAAAATCGCTGAG 30 1195 1245525* 938
953 20753 20768 TTTGTGGCCAACCACT 88 1196 1245551* 989 1004 20804 20819
TTATCATGCATACATC 52 1197 1245577 1077 1092 20892 20907 TAGTGCCAAACCAATG 46
1198 1245603 1180 1195 20995 21010 GTTTTAAAGAGGCATG 63 1199 1245629 1298 1313
21113 21128 CTGTGTAAATAAGTTC 38 1200 1245655 1329 1344 21144 21159
GGAATGCTACTTGAAC 63 1201 1245681 1357 1372 21172 21187 TTGATATTCTGTGGCA 87
1202 1245707 1406 1421 21221 21236 TGCCTGCTGAAACTTG 31 1203 1245733 1488 1503
21303 21318 GTTTTGGGCTAATGAA 54 1204 1245759 1538 1553 21353 21368
GTTATACAGAAGACAG 54 1205 1245785 1573 1588 21388 21403 GAAATATAGAGGGTCC
22 1206 1245811 1654 1669 21469 21484 GGGCTAGAACTTAAAA 35 1207 1245837 1742 1757
21557 21572 GCTTTAATTTTTGAGC 26 1208 1245863 1795 1810 21610 21625
TTGGGTCCATTGTTTC 44 1209 1245889 2192 2207 22007 22022 AGTCTTGATGTAGTGG 63
1210 1245915 2354 2369 22169 22184 TATAACTACAAGAGGT 46 1211 1245941 N/A N/A
3509 3524 CTAATAACCTGACAC 23 1212 1245967 N/A N/A 3703 3718
TAATTTTCAGATCCCG 75 1213 1245993 N/A N/A 4362 4377 ATATACATCAACTCCT 43
1214 1246019 N/A N/A 4991 5006 GAATTTTTTGAGGTTGG 77 1215 1246045 N/A N/A
5173 5188 TTAGATTGCATAAGAG 61 1216 1246071 N/A N/A 5738 5753
TTTAGTTGGTAGCTTG 63 1217 1246097 N/A N/A 5919 5934 AACCTTGAATGAGTGG
50 1218 1246123 N/A N/A 6204 6219 CATAGGACATGGAGAC 69 1219 1246149 N/A N/A
6443 6458 ATTCTAAACTAGTGGT 39 1220 1246175 N/A N/A 6784 6799
CTCCTTATTTGTTAGA 26 1221 1246201 N/A N/A 7788 7803 TGACATAACTACTTCA 22
1222 1246227 N/A N/A 8137 8152 AAAGTTTGTTAGAGTG 75 1223 1246253 N/A N/A
8597 8612 TCATAAAGGTGGGATA 40 1224 1246279 N/A N/A 8959 8974
GTTATAAGTTTCATGT 21 1225 1246305 N/A N/A 9215 9230 GCATTGAAAAGATGTA
0 1226 1246331 N/A N/A 9476 9491 TCTATACAAAGGCAGT 53 1227 1246357 N/A N/A
9649 9664 TAATGTATGAGGTCTC 51 1228 1246383 N/A N/A 9928 9943
ATGATTGAATTCACCTA 33 1229 1246409 N/A N/A 10112 10127 TAAACAATATTAAGGG 0
1230 1246435 N/A N/A 10494 10509 CCAATATGGGATGAGG 60 1231 1246461 N/A N/A 11074
11089 TTACAACCTGGTTTCA 25 1232 1246487 N/A N/A 11543 11558
TGACAATGGTTGCCAC 40 1233 1246513 N/A N/A 11859 11874 TGATTAAGCAGAATTG 57
1234 1246539 N/A N/A 12210 12225 TCAATTATCATACCA 26 1235 1246565 N/A N/A 12514
12529 GACAAGAGTTCTTGCA 22 1236 1246591 N/A N/A 12694 12709
TTTCGAATCATTAAGC 30 1237 1246617 N/A N/A 12875 12890 ACCTATGGTCTAACAC 14
1238 1246643 N/A N/A 13771 13786 TTGCAATAGTACTATA 3 1239 1246669 N/A N/A 13976
13991 TATTTAGCATTGCCTG 12 1240 1246695 N/A N/A 14129 14144
CTAAATTACTCCGTTC 20 1241 1246721 N/A N/A 14377 14392 TAGAATTGTCTGTTAG 9
1242 1246747 N/A N/A 14542 14557 CGTATTTGAGATTAC 59 1243 1246773 N/A N/A 15245
15260 AGCTAAGTATACTTTC 18 1244 1246799 N/A N/A 15361 15376
AGATGGTAAGGAGTAG 26 1245 1246825 N/A N/A 15511 15526 ATATTCCAGGATTTGC 26
1246 1246851 N/A N/A 16273 16288 GCAAAGTTTGGGCAGA 50 1247 1246877 N/A N/A

17047 17062 17063 17064 AATGCTTATCTGAG 7 1248 1246903 N/A N/A 17992 18007
AGTTAGTATAGTTATC 74 1249 1246929 N/A N/A 18695 18710 AAAGACTTTGAGACTC 68
1250 1246955 N/A N/A 18925 18940 GAAAGATGGAATGAGC 54 1251 1246981 N/A N/A
19588 19603 AGGTAAACTAAAGTAC 6 1252 1247007 N/A N/A 19778 19793
CATACCACTCTTCTCA 24 1253 1247033 N/A N/A 20039 20054 ATGCATTAAAACCTCTC 12
1254 1247059 N/A N/A 20377 20392 ACAAGTAAGGTAAAGT 38 1255 1247085 N/A N/A
20589 20604 AGAGTTATCTGGTTTG 56 1256
(445) TABLE-US-00017 TABLE 17 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245084 20 35 3114 3129
AGTCCTTGTGTAGTCC 64 1257 1245110 100 115 3194 3209
GAGTAGATGATGGTGA 28 1258 1245136 127 142 3221 3236
AAAAACTTCACCAACG 8 1259 1245162 241 256 3335 3350
ACCAATATGCTCTGTC 67 1260 1245188 289 304 7578 7593
CACTCAGCTGCAGTTT 13 1261 1245214 326 341 7615 7630
TACCACATACGCATGC 33 1262 1245240 375 390 N/A N/A TCTTCACCTGATTAG 24
1263 1245266 422 437 8816 8831 TACTGTCCCAGCATTA 21 1264 1245292 453
468 8847 8862 CATCCTTGGTGCTGAG 12 1265 1245318 492 507 8886 8901
GTCCTAGGATGTTGAC 1 1266 1245344 538 553 10424 10439 TTTCTCTCCATCATCG
27 1267 1245370 598 613 10484 10499 ATGAGGTAAGGAATCC 0 1268 1245396 654
669 12072 12087 CTGATGTCAGACCTCT 35 1269 1245422 728 743 12146 12161
GGTGAACCCAGTATTC 7 1270 1245448 780 795 15661 15676
TTACGACTTCATCTGT 36 1271 1245474 810 825 15691 15706 TATTGGTAAGTATTCC
67 1272 1245500* 897 912 20712 20727 GATTTAAATCGCTGA 49 1273 1245526* 939
954 20754 20769 TTTTGTGGCCAACCAC 77 1274 1245552* 992 1007 20807 20822
TCATTATCATGCATAC 67 1275 1245578 1078 1093 20893 20908 CTAGTGCCAAACCAAT 49
1276 1245604 1183 1198 20998 21013 GAAGTTTTTAAGAGGC 64 1277 1245630 1299 1314
21114 21129 CCTGTGTAAATAAGTT 18 1278 1245656 1331 1346 21146 21161
TTGGAATGCTACTTGA 72 1279 1245682 1358 1373 21173 21188 GTTGATATTCTGTGGC 80
1280 1245708 1408 1423 21223 21238 GCTGCCTGCTGAAACT 38 1281 1245734 1489 1504
21304 21319 CGTTTTGGGCTAATGA 48 1282 1245760 1540 1555 21355 21370
GAGTTATACAGAAGAC 61 1283 1245786 1574 1589 21389 21404 GGAAATATAGAGGGTC
50 1284 1245812 1668 1683 21483 21498 AAAAGGTTATCATGGG 49 1285 1245838 1750 1765
21565 21580 CTGTGTTAGCTTTAAT 67 1286 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 61 38 1245864 1796 1811 21611 21626 CTTGGGTCCATTGTTT
44 1287 1245890 2193 2208 22008 22023 TAGTCTTGATGTAGTG 55 1288 1245916 2355 2370
22170 22185 TTATACTACAAGAGG 68 1289 1245942 N/A N/A 3511 3526
ATCTAACTAACCTGAC 0 1290 1245968 N/A N/A 3704 3719 GTAATTTTTCAGATCCC
82 1291 1245994 N/A N/A 4363 4378 TATATACATCAACTCC 61 1292 1246020 N/A N/A
5029 5044 GCTTAAATGTTTAACC 67 1293 1246046 N/A N/A 5175 5190
GTTTAGATTGCATAAG 46 1294 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA
88 48 1246072 N/A N/A 5739 5754 ATTTAGTTGGTAGCTT 35 1295 1246098 N/A N/A
5920 5935 AAACCTTGAATGAGTG 37 1296 1246124 N/A N/A 6205 6220
CCATAGGACATGGAGA 68 1297 1246150 N/A N/A 6444 6459 AATTCTAAACTAGTGG
31 1298 1246176 N/A N/A 6795 6810 TAAAAATTGGCCTCCT 29 1299 7104 7119
1246202 N/A N/A 7804 7819 ATATATCCTTCCTTGG 20 1300 1246228 N/A N/A 8139
8154 GTAAAGTTTGTTAGAG 54 1301 1246254 N/A N/A 8598 8613
GTCATAAAGGTGGGAT 28 1302 1246280 N/A N/A 8960 8975 CGTTATAAGTTTCATG
33 1303 1246306 N/A N/A 9256 9271 AATCATGGTGTGGAGG 11 1304 1246332 N/A N/A

9497 9512 GGAATGAATCAATA 32 1305 1246358 N/A N/A 9650 9665
CTAATGTATGAGGTCT 46 1306 1246384 N/A N/A 9937 9952 TAATGAATCATGATTG
2 1307 1246410 N/A N/A 10145 10160 GAATCAAATAGATCCT 46 1308 1246436 N/A N/A
10495 10510 ACCAATATGGGATGAG 24 1309 1246462 N/A N/A 11078 11093
ATAATTACAACCTGGT 44 1310 1246488 N/A N/A 11547 11562 CTCTTGACAATGGTTG 50
1311 11572 11587 1246514 N/A N/A 11861 11876 CATGATTAAGCAGAAT 42 1312 1246540
N/A N/A 12211 12226 ATCAATTTATCATACC 11 1313 1246566 N/A N/A 12515 12530
AGACAAGAGTTCTTGC 30 1314 1246592 N/A N/A 12701 12716 AAATTGGTTTCGAATC 32
1315 1246618 N/A N/A 12889 12904 CTTTAGTCAACAGTAC 12 1316 1246644 N/A N/A 13783
13798 AACAATCAGGAGTTGC 31 1317 1246670 N/A N/A 13997 14012
CTCTATAAAATCAATC 0 1318 1246696 N/A N/A 14130 14145 TCTAAATTACTCCGTT 15
1319 1246722 N/A N/A 14378 14393 GTAGAATTGTCTGTTA 51 1320 1246748 N/A N/A 14556
14571 TCCGGTAAGAATTTTCG 63 1321 1246774 N/A N/A 15246 15261
TAGCTAAGTATACTTT 10 1322 1246800 N/A N/A 15364 15379 ATTAGATGGTAAGGAG 4
1323 1246826 N/A N/A 15570 15585 GTAAATGATGGCAGGA 34 1324 1246852 N/A N/A
16274 16289 AGCAAAGTTTGGGCAG 44 1325 1246878 N/A N/A 17077 17092
AAATTAGAGGCCCAGG 43 1326 1246904 N/A N/A 17994 18009 GAAGTTAGTATAGTTA 59
1327 1246930 N/A N/A 18732 18747 CTAGTTACAAACCACC 43 1328 1246956 N/A N/A 18958
18973 AGATTTACTTCCTTGT 41 1329 1246982 N/A N/A 19590 19605
TAAGGTAAACTAAAGT 9 1330 1247008 N/A N/A 19781 19796 ATACATAACCACTCTTC 18
1331 1247034 N/A N/A 20087 20102 AGTAACTTTATTAACG 12 1332 1247060 N/A N/A 20381
20396 AATCACAAGTAAGGTA 52 1333 1247086 N/A N/A 20599 20614
GAATACGCTGAGAGTT 11 1334

(446) TABLE-US-00018 TABLE 18 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245085 21 36 3115 3130
CAGTCCTTGTGTAGTC 70 1335 1245111 101 116 3195 3210
GGAGTAGATGATGGTG 31 1336 1245137 154 169 3248 3263
GCCACAGATTTTCTCC 49 1337 1245163 242 257 3336 3351
AACCAATATGCTCTGT 51 1338 1245189 290 305 7579 7594
GCACTCAGCTGCAGTT 29 1339 1245215 327 342 7616 7631
CTACCACATACGCATG 25 1340 1245241 392 407 8786 8801
TACATCACCCACTTCT 37 1341 1245267 423 438 8817 8832
ATACTGTCCCAGCATT 32 1342 1245293 454 469 8848 8863
TCATCCTTGGTGCTGA 12 1343 1245319 493 508 8887 8902
TGTCCTAGGATGTTGA 41 1344 1245345 548 563 10434 10449 GTGGCCATGATTTCTC
40 1345 1245371 599 614 10485 10500 GATGAGGTAAGGAATC 0 1346 1245397 655
670 12073 12088 TCTGATGTCAGACCTC 42 1347 1245423 729 744 12147 12162
TGGTGAACCCAGTATT 0 1348 1245449 781 796 15662 15677
CTTACGACTTCATCTG 38 1349 1245475 812 827 15693 15708 CTTATTGGTAAGTATT
22 1350 1245501* 898 913 20713 20728 CGATTTAAAATCGCTG 40 1351 1245527* 940
955 20755 20770 ATTTTGTGGCCAACCA 59 1352 1245553 996 1011 20811 20826
CATATCATTATCATGC 66 1353 1245579 1079 1094 20894 20909 GCTAGTGCCAAACCAA 70
1354 1245605 1192 1207 21007 21022 GTAAGCACAGAAGTTT 55 1355 1245631 1300 1315
21115 21130 CCCTGTGTAAATAAGT 41 1356 1245657 1332 1347 21147 21162
ATTGGAATGCTACTTG 53 1357 1245683 1359 1374 21174 21189 TGTTGATATTCTGTGG 68
1358 1245709 1412 1427 21227 21242 TAAAGCTGCCTGCTGA 44 1359 1245735 1490 1505
21305 21320 CCGTTTTGGGCTAATG 58 1360 1245761 1545 1560 21360 21375
CTTCAGAGTTATACAG 35 1361 1245787 1575 1590 21390 21405 AGGAAATATAGAGGGT

54 1362 1245813 1675 21685 21680 GAAAGGTTATCATG 31 1363 1245839 1751 1766
21566 21581 CCTGTGTTAGCTTTAA 67 1364 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 58 38 1245865 1797 1812 21612 21627 TCTTGGGTCCATTGTT
60 1365 1245891 2194 2209 22009 22024 TTAGTCTTGATGTAGT 61 1366 1245917 2378 2393
22193 22208 GTTTAGAAGTCAAACG 0 1367 1245943 N/A N/A 3512 3527
CATCTAACTAACCTGA 0 1368 1245969 N/A N/A 3750 3765 TAAAGGTTTCTGTTGC
66 1369 1245995 N/A N/A 4405 4420 ATTCGAATTTCTTCAA 65 1370 1246021 N/A N/A
5042 5057 GAATCATTAGTAGCT 65 1371 1246047 N/A N/A 5177 5192
CAGTTTAGATTGCATA 60 1372 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA
85 48 1246073 N/A N/A 5740 5755 GATTTAGTTGGTAGCT 63 1373 1246099 N/A
N/A 5921 5936 CAAACCTTGAATGAGT 53 1374 1246125 N/A N/A 6206 6221
ACCATAGGACATGGAG 62 1375 1246151 N/A N/A 6445 6460 TAATTCTAACTAGTG
11 1376 1246177 N/A N/A 6796 6811 ATAAAAATTGGCCTCC 41 1377 1246203 N/A N/A
7828 7843 TGGATTACATACAAGA 68 1378 1246229 N/A N/A 8140 8155
TGTAAGTTTGTTAGA 35 1379 1246255 N/A N/A 8599 8614 TGTCATAAAGGTGGGA
48 1380 1246281 N/A N/A 8961 8976 CCGTTATAAGTTTCAT 54 1381 1246307 N/A N/A
9257 9272 AAATCATGGTGTGGAG 1 1382 1246333 N/A N/A 9499 9514
CTGGAAGATTAATCAT 31 1383 1246359 N/A N/A 9652 9667 AGCTAATGTATGAGGT
56 1384 1246385 N/A N/A 9938 9953 CTAATGAATCATGATT 20 1385 1246411 N/A N/A
10180 10195 CTAATCTCAGTACTCC 53 1386 1246437 N/A N/A 10496 10511
TACCAATATGGGATGA 1 1387 1246463 N/A N/A 11079 11094 TATAATTACAACCTGG 46
1388 1246489 N/A N/A 11548 11563 TCTCTTGACAATGGTT 70 1389 11573 11588 1246515
N/A N/A 11865 11880 CTTACATGATTAAGCA 44 1390 1246541 N/A N/A 12238 12253
GCCTAAACATTTCTCA 2 1391 1246567 N/A N/A 12516 12531 GAGACAAGAGTTCTTG
19 1392 1246593 N/A N/A 12703 12718 AAAAATTGGTTTCGAA 13 1393 1246619 N/A N/A
13551 13566 CTCTCTAAAGGGCTGC 55 1394 1246645 N/A N/A 13813 13828
TTTTATTGGGCCCAAT 18 1395 1246671 N/A N/A 14007 14022 ATCTAGCAACCTCTAT 12
1396 1246697 N/A N/A 14131 14146 TTCTAAATTACTCCGT 53 1397 1246723 N/A N/A 14380
14395 GTGTAGAATTGTCTGT 42 1398 1246749 N/A N/A 14557 14572
TTCCGGTAAGAATTTTC 33 1399 1246775 N/A N/A 15248 15263 TATAGCTAAGTATACT 23
1400 1246801 N/A N/A 15365 15380 CATTAGATGGTAAGGA 32 1401 1246827 N/A N/A 15571
15586 AGTAAATGATGGCAGG 40 1402 1246853 N/A N/A 16287 16302
TAGAATAGTCTTCAGC 70 1403 1246879 N/A N/A 17135 17150 AATAGCCATGTAGCTA 28
1404 1246905 N/A N/A 17995 18010 TGAAGTTAGTATAGTT 43 1405 1246931 N/A N/A 18733
18748 CCTAGTTACAAACCAC 51 1406 1246957 N/A N/A 18962 18977
TGCAAGATTTACTTCC 45 1407 1246983 N/A N/A 19598 19613 CTCTATAATAAGGTAA 0
1408 1247009 N/A N/A 19782 19797 GATACATACCACTCTT 52 1409 1247035 N/A N/A 20110
20125 CAAGACATTCTAGCCT 58 1410 1247061 N/A N/A 20383 20398
GTAATCACAAGTAAGG 70 1411 1247087 N/A N/A 20600 20615 AGAATACGCTGAGAGT 56
1412

(447) TABLE-US-00019 TABLE 19 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245086 22 37 3116 3131
TCAGTCCTTGTGTAGT 72 1413 1245112 103 118 3197 3212
TAGGAGTAGATGATGG 15 1414 1245138 166 181 3260 3275
ACAATCTCCCCAGCCA 67 1415 1245164 243 258 3337 3352
GAACCAATATGCTCTG 61 1416 1245190 291 306 7580 7595
GGCACTCAGCTGCAGT 49 1417 1245216 328 343 7617 7632
TCTACCACATACGCAT 55 1418 1245242 393 408 8787 8802

TTCATCACCCACTTC 37 1419 1245268 424 439 8818 8833
TATACTGTCCCAGCAT 40 1420 1245294 456 471 8850 8865
CTTCATCCTTGGTGCT 30 1421 1245320 494 509 8888 8903
ATGTCCTAGGATGTTG 50 1422 1245346 551 566 10437 10452 GATGTGGCCATGATTT
9 1423 1245372 606 621 10492 10507 AATATGGGATGAGGTA 53 1424 1245398 656
671 12074 12089 TTCTGATGTCAGACCT 42 1425 1245424 730 745 12148 12163
TTGGTGAACCCAGTAT 31 1426 1245450 782 797 15663 15678 TCTTACGACTTCATCT
34 1427 1245476 813 828 15694 15709 TCTTATTGGTAAGTAT 25 1428 1245502* 899
914 20714 20729 ACGATTAAATCGCT 44 1429 1245528* 941 956 20756 20771
GATTTTGTGGCCAACC 91 1430 1245554 1006 1021 20821 20836 CGAAACTATTCATATC 66
1431 1245580 1080 1095 20895 20910 TGCTAGTGCCAAACCA 78 1432 1245606 1198 1213
21013 21028 GTTTATGTAAGCACAG 67 1433 1245632 1301 1316 21116 21131
TCCCTGTGTAAATAAG 35 1434 1245658 1333 1348 21148 21163 GATTGGAATGCTACTT 58
1435 1245684 1364 1379 21179 21194 GTTCTTGTTGATATTC 84 1436 1245710 1414 1429
21229 21244 GATAAAGCTGCCTGCT 42 1437 1245736 1491 1506 21306 21321
ACCGTTTTGGGCTAAT 78 1438 1245762 1547 1562 21362 21377 GACTTCAGAGTTATAC 16
1439 1245788 1589 1604 21404 21419 GACTATAAAAAGGGAG 40 1440 1245814 1702 1717
21517 21532 GACCAAGGATATATGA 48 1441 1245840 1752 1767 21567 21582
TCCTGTGTTAGCTTTA 56 1442 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC 57
38 1245866 1798 1813 21613 21628 CTCTTGGGTCCATTGT 40 1443 1245892 2195 2210
22010 22025 ATTAGTCTTGATGTAG 67 1444 1245918 2379 2394 22194 22209
AGTTTAGAAGTCAAAC 23 1445 1245944 N/A N/A 3530 3545 TCTTAATTACATTCCC
64 1446 1245970 N/A N/A 3751 3766 CTAAAGGTTTCTGTGTG 34 1447 1245996 N/A N/A
4406 4421 CATTCGAATTTCTTCA 77 1448 1246022 N/A N/A 5045 5060
AGTGAATCATTAGTA 86 1449 1246048 N/A N/A 5178 5193 ACAGTTTAGATTGCAT
84 1450 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA 82 48 1246074 N/A
N/A 5741 5756 TGATTTAGTTGGTAGC 79 1451 1246100 N/A N/A 5923 5938
GGCAAACCTTGAATGA 61 1452 1246126 N/A N/A 6207 6222 AACCATAGGACATGGA
71 1453 1246152 N/A N/A 6446 6461 GTAATTCTAAACTAGT 39 1454 1246178 N/A N/A
6914 6929 CTCGGGAAGTTTAGAC 47 1455 1246204 N/A N/A 7829 7844
GTGGATTACATACAAG 57 1456 1246230 N/A N/A 8141 8156 ATGTAAAGTTTGTTAG
12 1457 1246256 N/A N/A 8610 8625 CAAGGGAACACTGTCA 58 1458 1246282 N/A N/A
8962 8977 TCCGTTATAAGTTTCA 65 1459 1246308 N/A N/A 9260 9275
ACAAAATCATGGTGTG 26 1460 1246334 N/A N/A 9510 9525 ATATTGAGGCACTGGA
53 1461 1246360 N/A N/A 9653 9668 TAGCTAATGTATGAGG 48 1462 1246386 N/A N/A
9939 9954 ACTAATGAATCATGAT 39 1463 1246412 N/A N/A 10181 10196
ACTAATCTCAGTACTC 45 1464 1246438 N/A N/A 10498 10513 CTTACCAATATGGGAT 11
1465 1246464 N/A N/A 11155 11170 TGGAAACATGCTTGGC 62 1466 1246490 N/A N/A 11549
11564 TTCTCTTGACAATGGT 67 1467 11574 11589 1246516 N/A N/A 11892 11907
GTATAAATGGCAATTC 56 1468 1246542 N/A N/A 12251 12266 TATTAGTTGGCCTGCC 14
1469 1246568 N/A N/A 12598 12613 TCTTATCCACACCTTC 31 1470 1246594 N/A N/A 12704
12719 AAAAAATTGTTTTCGA 20 1471 1246620 N/A N/A 13574 13589
CTATCAGAACAGTTCA 48 1472 1246646 N/A N/A 13814 13829 TTTTATTGGGCCCAA 51
1473 1246672 N/A N/A 14008 14023 AATCTAGCAACCTCTA 22 1474 1246698 N/A N/A 14132
14147 TTTCTAAATTACTCCG 54 1475 1246724 N/A N/A 14382 14397
TAGTGTAGAATTGTCT 46 1476 1246750 N/A N/A 14600 14615 AGAAATCTAAGAACCC 25
1477 1246776 N/A N/A 15252 15267 ATTCTATAGCTAAGTA 23 1478 1246802 N/A N/A 15366
15381 GCATTAGATGGTAAGG 64 1479 1246828 N/A N/A 15633 15648
CCATAATCTGTGATTA 9 1480 1246854 N/A N/A 16289 16304 GTTAGAATAGTCTTCA 46
1481 1246880 N/A N/A 17136 17151 GAATAGCCATGTAGCT 21 1482 1246906 N/A N/A 17996

1801AATGATTAGTATAGT 55 1483 1246932 N/A N/A 18734 18749
GCCTAGTTACAAACCA 56 1484 1246958 N/A N/A 18968 18983 GACTAATGCAAGATTT 55
1485 1246984 N/A N/A 19599 19614 ACTCTATAATAAGGTA 42 1486 1247010 N/A N/A 19785
19800 ACTGATACATAACCACT 52 1487 1247036 N/A N/A 20112 20127
ATCAAGACATTCTAGC 61 1488 1247062 N/A N/A 20385 20400 TAGTAATCACAAGTAA 30
1489 1247088 N/A N/A 20601 20616 TAGAATACGCTGAGAG 47 1490
(448) TABLE-US-00020 TABLE 20 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245087 23 38 3117 3132
TTCAGTCCTTGTGTAG 45 1491 1245113 104 119 3198 3213
GTAGGAGTAGATGATG 7 1492 1245139 167 182 3261 3276
AACAATCTCCCCAGCC 59 1493 1245165 244 259 3338 3353
AGAACCAATATGCTCT 38 1494 1245191 292 307 7581 7596
CGGCACTCAGCTGCAG 55 1495 1245217 329 344 7618 7633
GTCTACCACATACGCA 66 1496 1245243 394 409 8788 8803
GTTACATCACCCACTT 41 1497 1245269 425 440 8819 8834
ATATACTGTCCCAGCA 43 1498 1245295 457 472 8851 8866
TCTTCATCCTTGGTGC 48 1499 1245321 495 510 8889 8904
AATGTCCTAGGATGTT 70 1500 1245347 552 567 10438 10453 CGATGTGGCCATGATT
44 1501 1245373 607 622 10493 10508 CAATATGGGATGAGGT 62 1502 1245399 657
672 12075 12090 GTTCTGATGTCAGACC 55 1503 1245425 731 746 12149 12164
TTTGGTGAACCCAGTA 34 1504 1245451 783 798 15664 15679 TTCTTACGACTTCATC
47 1505 1245477 835 850 15716 15731 TACGATGGAACAAAAA 20 1506 1245503* 900
915 20715 20730 TACGATTTAAAATCGC 0 1507 1245529* 942 957 20757 20772
TGATTTTGTGGCCAAC 72 1508 1245555 1007 1022 20822 20837 TCGAAACTATTCATAT 64
1509 1245581 1081 1096 20896 20911 CTGCTAGTGCCAAACC 74 1510 1245607 1202 1217
21017 21032 GTATGTTTATGTAAGC 80 1511 1245633 1305 1320 21120 21135
ACCTTCCCTGTGTAAA 60 1512 1245659 1334 1349 21149 21164 AGATTGGAATGCTACT 63
1513 1245685 1366 1381 21181 21196 GTGTTCTTGTTGATAT 81 1514 1245711 1415 1430
21230 21245 AGATAAAGCTGCCTGC 48 1515 1245737 1492 1507 21307 21322
CACCGTTTTGGGCTAA 85 1516 1245763 1548 1563 21363 21378 GGACTTCAGAGTTATA
36 1517 1245789 1599 1614 21414 21429 TATCTTATAAGACTAT 43 1518 1245815 1703 1718
21518 21533 GGACCAAGGATATATG 56 1519 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 67 38 1245867 1799 1814 21614 21629 TCTCTTGGGTCCATTG
63 1520 1245893 2196 2211 22011 22026 GATTAGTCTTGATGTA 65 1521 1245919 N/A N/A
3356 3371 ACCTTATTAATATCCC 71 1522 1245945 N/A N/A 3532 3547
GTTCTTAATTACATTC 64 1523 1245971 N/A N/A 3753 3768 CACTAAAGGTTTCTGT 56
1524 1245997 N/A N/A 4407 4422 TCATTCGAATTTCTTC 76 1525 1246023 N/A N/A
5046 5061 GAGTGAATCATTCAGT 87 1526 1246049 N/A N/A 5179 5194
AACAGTTTAGATTGCA 71 1527 1246075 N/A N/A 5742 5757 CTGATTTAGTTGGTAG
78 1528 1246101 N/A N/A 5928 5943 TTAGGGCAAACCTTG 67 1529 1246127 N/A N/A
6208 6223 TAACCATAGGACATGG 70 1530 1246153 N/A N/A 6447 6462
AGTAATTCTAAACTAG 31 1531 1246179 N/A N/A 7101 7116 AAATTGGCCTCCTGGG
44 1532 1246205 N/A N/A 7846 7861 ATAATGGGCAGAGCTG 53 1533 1246231 N/A N/A
8165 8180 TTATTGTAGGGATTGA 63 1534 1246257 N/A N/A 8612 8627
GCCAAGGGAACACTGT 61 1535 1246283 N/A N/A 8975 8990 GAAGTATGTGAACTCC
56 1536 1246309 N/A N/A 9312 9327 ATTACTTCTGATGTCC 65 1537 1246335 N/A N/A
9511 9526 AATATTGAGGCACTGG 54 1538 1246361 N/A N/A 9656 9671
AGGTAGCTAATGTATG 53 1539 1246387 N/A N/A 9940 9955 CACTAATGAATCATGA

25 1540 1246413 N/A N/A 10197 CACTAATCTCAGTACT 48 1541 1246439 N/A N/A
10501 10516 ATACTTACCAATATGG 0 1542 1246465 N/A N/A 11156 11171
ATGGAACATGCTTGG 48 1543 1246491 N/A N/A 11565 11580 CAATGGTTGATAGTTT 47
1544 1246517 N/A N/A 11894 11909 GAGTATAAATGGCAAT 53 1545 1246543 N/A N/A 12252
12267 CTATTAGTTGGCCTGC 32 1546 1246569 N/A N/A 12600 12615
GTTCTTATCCACACCT 57 1547 1246595 N/A N/A 12705 12720 TAAAAAATTGGTTTTCG 22
1548 1246621 N/A N/A 13575 13590 ACTATCAGAACAGTTC 62 1549 1246647 N/A N/A 13815
13830 GTTTTTTATTGGGCCCA 67 1550 1246673 N/A N/A 14009 14024
CAATCTAGCAACCTCT 51 1551 1246699 N/A N/A 14166 14181 AGCAAGCCAACAGAGA 40
1552 1246725 N/A N/A 14384 14399 CTTAGTGTAGAATTGT 56 1553 1246751 N/A N/A 14614
14629 AATTATTTCTTGCGAG 16 1554 1246777 N/A N/A 15253 15268
AATTCTATAGCTAAGT 9 1555 1246803 N/A N/A 15367 15382 GGCATTAGATGGTAAG 62
1556 1246829 N/A N/A 15634 15649 GCCATAATCTGTGATT 0 1557 1246855 N/A N/A
16290 16305 AGTTAGAATAGTCTTC 67 1558 1246881 N/A N/A 17139 17154
ATTGAATAGCCATGTA 39 1559 1246907 N/A N/A 17997 18012 TATGAAGTTAGTATAG 30
1560 1246933 N/A N/A 18746 18761 TATATACCTGTTGCCT 55 1561 1246959 N/A N/A 19388
19403 ATGAGTAGGCAACTGA 61 1562 1246985 N/A N/A 19608 19623
CTATTATGCACTCTAT 32 1563 1247011 N/A N/A 19786 19801 CACTGATACATAACCAC 64
1564 1247037 N/A N/A 20114 20129 GCATCAAGACATTCTA 72 1565 1247063 N/A N/A 20387
20402 TATAGTAATCACAAGT 53 1566 1247089 N/A N/A 20602 20617
TTAGAATACGCTGAGA 48 1567
(449) TABLE-US-00021 TABLE 21 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245088 24 39 3118 3133
GTTTCAGTCCTTGTGTA 38 1568 1245114 105 120 3199 3214
AGTAGGAGTAGATGAT 0 1569 1245140 168 183 3262 3277
GAACAATCTCCCCAGC 63 1570 1245166 245 260 3339 3354
CAGAACCAATATGCTC 56 1571 1245192 293 308 7582 7597
TCGGCACTCAGCTGCA 52 1572 1245218 330 345 7619 7634
AGTCTACCACATACGC 65 1573 1245244 395 410 8789 8804
TGTTACATCACCCACT 35 1574 1245270 426 441 8820 8835
GATATACTGTCCCAGC 34 1575 1245296 458 473 8852 8867
CTCTTCATCCTTGGTG 25 1576 1245322 496 511 8890 8905
AAATGTCCTAGGATGT 36 1577 1245348 553 568 10439 10454 ACGATGTGGCCATGAT
52 1578 1245374 608 623 N/A N/A ACAATATGGGATGAGG 71 1579 1245400 658
673 12076 12091 AGTTCTGATGTCAGAC 40 1580 1245426 749 764 N/A N/A
TAATCTTGCTTGGGA 54 1581 1245452 784 799 15665 15680 CTTCTTACGACTTCAT
45 1582 1245478 836 851 15717 15732 ATACGATGGAACAAAA 27 1583 1245504* 901
916 20716 20731 ATACGATTTAAAATCG 17 1584 1245530* 966 981 20781 20796
GGAGCTTATTTATTCA 38 1585 1245556 1008 1023 20823 20838 TTCGAAACTATTCATA 44
1586 1245582 1082 1097 20897 20912 GCTGCTAGTGCCAAAC 67 1587 1245608 1250 1265
21065 21080 GTCCACCTTTAAATGG 33 1588 1245634 1306 1321 21121 21136
AACCTTCCCTGTGTAA 38 1589 1245660 1335 1350 21150 21165 CAGATTGGAATGCTAC 74
1590 1245686 1367 1382 21182 21197 TGTGTTCTTGTTGATA 69 1591 1245712 1416 1431
21231 21246 GAGATAAAGCTGCCTG 67 1592 1245738 1493 1508 21308 21323
GCACCGTTTTGGGCTA 74 1593 1245764 1551 1566 21366 21381 GGTGGACTTCAGAGTT
26 1594 1245790 1600 1615 21415 21430 GTATCTTATAAGACTA 81 1595 1245816 1704 1719
21519 21534 GGGACCAAGGATATAT 17 1596 1245841 1765 1780 21580 21595
GTACAGTTCCTTTTCC 54 38 1245842 1766 1781 21581 21596 AGTACAGTTCCTTTTC

AAATGTCCTAGTG 0 1655 1245349 554 569 10440 10455
GACGATGTGGCCATGA 42 1656 1245375 609 624 N/A N/A AACAAATATGGGATGAG 39
1657 1245401 659 674 12077 12092 AAGTTCTGATGTCAGA 7 1658 1245427 750
765 N/A N/A ATAATCTTGTGCTTGG 36 1659 1245453 785 800 15666 15681
ACTTCTTACGACTTCA 43 1660 1245479 837 852 15718 15733 TATACGATGGAACAAA
4 1661 1245505* 902 917 20717 20732 CATACGATTTAAAATC 13 1662 1245531*
967 982 20782 20797 TGGAGCTTATTTATTC 39 1663 1245557 1009 1024 20824 20839
ATTCGAAACTATTCAT 24 1664 1245583 1083 1098 20898 20913 TGCTGCTAGTGCCAAA 58
1665 1245609 1251 1266 21066 21081 TGTCACCTTTAAATG 29 1666 1245635 1307 1322
21122 21137 AACCTTCCCTGTGTA 34 1667 1245661 1336 1351 21151 21166
ACAGATTGGAATGCTA 50 1668 1245687 1379 1394 21194 21209 GTGCACTCATTCTGTG
6 1669 1245713 1418 1433 21233 21248 TTGAGATAAAGCTGCC 49 1670 1245739 1494 1509
21309 21324 TGCACCGTTTTGGGCT 36 1671 1245765 1552 1567 21367 21382
TGGTGGACTTCAGAGT 17 1672 1245791 1601 1616 21416 21431 TGTATCTTATAAGACT 45
1673 1245817 1705 1720 21520 21535 TGGGACCAAGGATATA 20 1674 1245841 1765 1780
21580 21595 GTACAGTTCCTTTTCC 48 38 1245843 1767 1782 21582 21597
CAGTACAGTTCCTTTT 43 1675 1245869 1801 1816 21616 21631 CTTCTCTTGGGTCCAT 51
1676 1245895 2198 2213 22013 22028 AAGATTAGTCTTGATG 25 1677 1245921 N/A N/A N/A
N/A TCACCTTATTAATATC 0 1678 1245947 N/A N/A 3546 3561
AAAATCGCTGCTTAGT 35 1679 1245973 N/A N/A 3755 3770 GGCCTAAAGGTTTCT
62 1680 1245999 N/A N/A 4409 4424 GTTCATTCGAATTTCT 76 1681 1246025 N/A N/A
5053 5068 AGCCTTAGAGTGAATC 74 1682 1246051 N/A N/A 5187 5202
CCCAACGCAACAGTTT 66 1683 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA
84 48 1246077 N/A N/A 5759 5774 AATAGGAGAGTCTTTC 47 1684 1246103 N/A
N/A 5930 5945 TATTTAGGGCAAACCT 40 1685 1246129 N/A N/A 6210 6225
CATAACCATAGGACAT 41 1686 1246155 N/A N/A 6454 6469 GCTAAGTAGTAATTCT
57 1687 1246181 N/A N/A 7119 7134 GATTAGTTAACCTTTT 58 1688 1246207 N/A N/A
7848 7863 TAATAATGGGCAGAGC 30 1689 1246233 N/A N/A 8171 8186
ACAGAGTTATTGTAGG 58 1690 1246259 N/A N/A 8670 8685 ATTGTATGAGGTCTCT
49 1691 1246285 N/A N/A 8977 8992 CAGAAGTATGTGAACT 22 1692 1246311 N/A N/A
9329 9344 CTATATACATCCAAGT 6 1693 1246337 N/A N/A 9513 9528
CTAATATTGAGGCACT 51 1694 1246363 N/A N/A 9661 9676 GCTATAGGTAGCTAAT 20
1695 1246389 N/A N/A 9942 9957 CACACTAATGAATCAT 30 1696 1246415 N/A N/A
10185 10200 CAACACTAATCTCAGT 4 1697 1246441 N/A N/A 10520 10535
ATAACATGGCTGGCAT 14 1698 1246467 N/A N/A 11235 11250 CATAGACATGTGTTCA 52
1699 1246493 N/A N/A 11568 11583 TGACAATGGTTGATAG 16 1700 1246519 N/A N/A 11953
11968 GTATCATTTGGCTTAA 29 1701 1246545 N/A N/A 12254 12269
TTCTATTAGTTGGCCT 28 1702 1246571 N/A N/A 12610 12625 TAATTTTCGGGTTCTT 40
1703 1246597 N/A N/A 12723 12738 GGATTAATTCCTTCC 10 1704 1246623 N/A N/A 13594
13609 GATAGAGCACATGGCC 20 1705 1246649 N/A N/A 13842 13857
CACGAAGCCTGAATGC 24 1706 1246675 N/A N/A 14011 14026 CACAATCTAGCAACCT 40
1707 1246701 N/A N/A 14170 14185 GAATAGCAAGCCAACA 64 1708 1246727 N/A N/A
14391 14406 ACATAGGCTTAGTGTA 0 1709 1246753 N/A N/A 14616 14631
CGAATTATTTCTTGCG 58 1710 1246779 N/A N/A 15280 15295 CTCAAAGCCTGTTGGA 10
1711 1246805 N/A N/A 15380 15395 AAAATGGTGGCGAGGC 0 1712 1246831* N/A N/A
15751 15766 GTACTTACTTCTGTAG 0 1713 1246857 N/A N/A 16292 16307
TTAGTTAGAATAGTCT 8 1714 1246883 N/A N/A 17144 17159 AGTAAATTGAATAGCC 39
1715 1246909 N/A N/A 18000 18015 GCTTATGAAGTTAGTA 67 1716 1246935 N/A N/A 18798
18813 CGTAAAATTGTGTCTC 43 1717 1246961 N/A N/A 19397 19412
GCATAAGAGATGAGTA 33 1718 1246987 N/A N/A 19610 19625 ATCTATTATGCACTCT 24

1719 1247013 N/A N/A 19819 19834 CATAGTTCAAATTGGA 39 1720 1247039 N/A N/A 20138
20153 GGGAAAAAGTGTGTTC 20 1721 1247065 N/A N/A 20395 20410
GTAATTGCTATAGTAA 33 1722 1247091 N/A N/A 20604 20619 ATTTAGAATACGCTGA 19
1723
(451) TABLE-US-00023 TABLE 23 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245090 26 41 3120 3135
TGGTTCAGTCCTTGTG 50 1724 1245116 107 122 3201 3216
CAAGTAGGAGTAGATG 5 1725 1245142 172 187 3266 3281
ATGAGAACAATCTCCC 65 1726 1245168 247 262 3341 3356
CACAGAACCAATATGC 54 1727 1245194 295 310 7584 7599
TTTCGGCACTCAGCTG 29 1728 1245220 355 370 7644 7659
CGATAGATCTCTTCTC 15 1729 1245246 398 413 8792 8807
GATTGTTACATCACCC 65 1730 1245272 428 443 8822 8837
TGGATATACTGTCCCA 24 1731 1245298 465 480 8859 8874
TGGTAATCTCTTCATC 23 1732 1245324 500 515 8894 8909
CCAAAAATGTCCTAGG 38 1733 1245350 557 572 10443 10458
TGTGACGATGTGCCCA 42 1734 1245376 610 625 N/A N/A GAACAATATGGGATGA 26
1735 1245402 660 675 12078 12093 GAAGTTCTGATGTCAG 0 1736 1245428 751
766 N/A N/A CATAATCTTGTGCTTG 37 1737 1245454 786 801 15667 15682
GACTTCTTACGACTTC 38 1738 1245480 838 853 15719 15734 ATATACGATGGAACAA
0 1739 1245506* 903 918 20718 20733 GCATACGATTTAAAT 0 1740 1245532*
969 984 20784 20799 GCTGGAGCTTATTTAT 16 1741 1245558 1010 1025 20825 20840
GATTTCGAACTATTCA 57 1742 1245584 1084 1099 20899 20914 CTGCTGCTAGTGCCAA 57
1743 1245610 1256 1271 21071 21086 GCTTTTGTCCACCTTT 68 1744 1245636 1308 1323
21123 21138 TAAACCTTCCCTGTGT 17 1745 1245662 1337 1352 21152 21167
TACAGATTGGAATGCT 46 1746 1245688 1380 1395 21195 21210 TGTGCACTCATTCTGT 18
1747 1245714 1419 1434 21234 21249 GTTGAGATAAAGCTGC 54 1748 1245740 1495 1510
21310 21325 TTGCACCGTTTTGGGC 45 1749 1245766 1553 1568 21368 21383
TTGGTGGACTTCAGAG 8 1750 1245792 1602 1617 21417 21432 ATGTATCTTATAAGAC
0 1751 1245818 1706 1721 21521 21536 CTGGGACCAAGGATAT 32 1752 1245841 1765
1780 21580 21595 GTACAGTTCCTTTTCC 50 38 1245844 1768 1783 21583 21598
CCAGTACAGTTCCTTT 66 1753 1245870 1802 1817 21617 21632 TCTTCTCTTGGGTCCA 59
1754 1245896 2199 2214 22014 22029 CAAGATTAGTCTTGAT 27 1755 1245922 N/A N/A
3357 3372 TACCTTATTAATATCC 54 1756 1245948 N/A N/A 3547 3562
AAAAATCGCTGCTTAG 25 1757 1245974 N/A N/A 3834 3849 GACTACGGACTGGCAA
60 1758 1246000 N/A N/A 4414 4429 AGAAAGTTCATTCGAA 61 1759 1246026 N/A N/A
5057 5072 TTACAGCCTTAGAGTG 21 1760 1246052 N/A N/A 5200 5215
TTTGAACCGTATTCCC 69 1761 1246065 N/A N/A 5659 5674 ATAAGTTACCAGAGCA
77 48 1246078 N/A N/A 5760 5775 GAATAGGAGAGTCTTT 58 1762 1246104 N/A
N/A 5955 5970 ATTGAGATGACAGTGG 70 1763 1246130 N/A N/A 6211 6226
CCATAACCATAGGACA 63 1764 1246156 N/A N/A 6455 6470 TGCTAAGTAGTAATTC
32 1765 1246182 N/A N/A 7121 7136 AAGATTAGTTAACCTT 24 1766 1246208 N/A N/A
7851 7866 CAATAATAATGGGCAG 43 1767 1246234 N/A N/A 8205 8220
GAGTCATAATTTGCTT 57 1768 1246260 N/A N/A 8672 8687 CTATTGTATGAGGTCT 42
1769 1246286 N/A N/A 8978 8993 CCAGAAGTATGTGAAC 8 1770 1246312 N/A N/A
9330 9345 CCTATATACATCCAAG 53 1771 1246338 N/A N/A 9514 9529
ACTAATATTGAGGCAC 19 1772 1246364 N/A N/A 9662 9677 AGCTATAGGTAGCTAA
0 1773 1246390 N/A N/A 9943 9958 CCACACTAATGAATCA 43 1774 1246416 N/A N/A

10202 10217 TATGGTAAGCCCATG 30 1775 1246442 N/A N/A 105237 10537
 ATATAACATGGCTGGC 29 1776 1246468 N/A N/A 11236 11251 ACATAGACATGTGTTC 35
 1777 1246494 N/A N/A 11590 11605 CTCAATAACTGAGTTA 4 1778 1246520 N/A N/A
 11962 11977 GAGAAGGCGGTATCAT 48 1779 1246546 N/A N/A 12255 12270
 CTTCTATTAGTTGGCC 11 1780 1246572 N/A N/A 12611 12626 TTAATTTTCGGGTTCT 31
 1781 1246598 N/A N/A 12724 12739 AGGATTAATTCCCTTC 19 1782 1246624 N/A N/A 13595
 13610 GGATAGAGCACATGGC 33 1783 1246650 N/A N/A 13845 13860
 GTACACGAAGCCTGAA 42 1784 1246676 N/A N/A 14012 14027 TCACAATCTAGCAACC
 4 1785 1246702 N/A N/A 14171 14186 TGAATAGCAAGCCAAC 4 1786 1246728 N/A N/A
 14393 14408 AAACATAGGCTTAGTG 16 1787 1246754 N/A N/A 14724 14739
 AATACCCTTGGTGGAG 16 1788 1246780 N/A N/A 15312 15327 TAATCATGGGACAGGA 15
 1789 1246806 N/A N/A 15381 15396 CAAAATGGTGGCGAGG 10 1790 1246832 N/A N/A
 15771 15786 GTATTTGGGTGTTCTG 36 1791 1246858 N/A N/A 16293 16308
 TTTAGTTAGAATAGTC 15 1792 1246884 N/A N/A 17227 17242 GTACAAATTTATGCCA 45
 1793 1246910 N/A N/A 18001 18016 GGCTTATGAAGTTAGT 59 1794 1246936 N/A N/A 18812
 18827 TAATTGCATCTGCTCG 46 1795 1246962 N/A N/A 19398 19413
 TGCATAAGAGATGAGT 19 1796 1246988 N/A N/A 19611 19626 TATCTATTATGCACTC 48
 1797 1247014 N/A N/A 19820 19835 TCATAGTTCAAATTGG 47 1798 1247040 N/A N/A 20172
 20187 CAATTATGAATCTGCA 25 1799 1247066 N/A N/A 20396 20411
 AGTAATTGCTATAGTA 48 1800 1247092 N/A N/A 20605 20620 CATTAGAATACGCTG 25
 1801

(452) TABLE-US-00024 TABLE 24 Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO: ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245091 27 42 3121 3136

CTGGTTCAGTCCTTGT	63	1802	1245117	108	123	3202	3217				
CCAAGTAGGAGTAGAT	19	1803	1245143	175	190	3269	3284				
GTAATGAGAACAATCT	48	1804	1245169	248	263	3342	3357				
CCACAGAACCAATATG	49	1805	1245195	296	311	7585	7600				
TTTTCGGCACTCAGCT	13	1806	1245221	356	371	7645	7660				
GCGATAGATCTCTTCT	24	1807	1245247	399	414	8793	8808				
CGATTGTTACATCACC	47	1808	1245273	429	444	8823	8838				
CTGGATATACTGTCCC	20	1809	1245299	469	484	8863	8878				
GTCTTGGTAATCTCTT	37	1810	1245325	511	526	N/A	N/A	GCTTTTGTGATCCAAA	43		
1811 1245351	558	573	10444	10459	CTGTGACGATGTGGCC	28	1812	1245377	611		
626 N/A N/A	GGAACAATATGGGATG	36	1813	1245403	665	680	12083	12098			
GGCCTGAAGTTCTGAT	0	1814	1245429	752	767	N/A	N/A	CCATAATCTTGTGCTT	32		
1815 1245455	787	802	15668	15683	AGACTTCTTACGACTT	41	1816	1245481	839		
854 15720 15735	GATATACGATGGAACA	46	1817	1245507*	904	919	20719	20734			
TGCATACGATTTAAAA	18	1818	1245533*	970	985	20785	20800	GGCTGGAGCTTATTTA			
57 1819 1245559	1011	1026	20826	20841	TGATTCGAAACTATTC	29	1820	1245585	1085	1100	
20900 20915	ACTGCTGCTAGTGCCA	47	1821	1245611	1257	1272	21072	21087			
AGCTTTTGTCCACCTT	57	1822	1245637	1310	1325	21125	21140	CTTAAACCTTCCCTGT	22		
1823 1245663	1338	1353	21153	21168	CTACAGATTGGAATGC	52	1824	1245689	1381	1396	
21196 21211	CTGTGCACTCATTCTG	51	1825	1245715	1420	1435	21235	21250			
GGTTGAGATAAAGCTG	76	1826	1245741	1496	1511	21311	21326	GTTGCACCGTTTTGGG			
56 1827 1245767	1554	1569	21369	21384	TTTGGTGGACTTCAGA	16	1828	1245793	1613	1628	
21428 21443	CACCTTTCATAATGTA	59	1829	1245819	1707	1722	21522	21537			
TCTGGGACCAAGGATA	30	1830	1245841	1765	1780	21580	21595	GTACAGTTCCTTTTCC			
50 38 1245845	1769	1784	21584	21599	GCCAGTACAGTTCCTT	61	1831	1245871	2174		

2189 21989 22004 GTCGATTATTTTTC 61 1832 1245897 2200 2215 22015 22030
 ACAAGATTAGTCTTGA 45 1833 1245923 N/A N/A 3368 3383 GATGTATACATTACCT 67
 1834 1245949 N/A N/A 3549 3564 ATAAAAATCGCTGCTT 11 1835 1245975 N/A N/A
 3877 3892 ATCATATGCTAAGTGC 73 1836 1246001 N/A N/A 4415 4430
 CAGAAAGTTCATTCTGA 67 1837 1246027 N/A N/A 5061 5076 TACATTACAGCCTTAG
 75 1838 1246053 N/A N/A 5204 5219 GTTTTTTTGAACCGTAT 71 1839 1246065 N/A N/A
 5659 5674 ATAAGTTACCAGAGCA 74 48 1246079 N/A N/A 5761 5776
 AGAATAGGAGAGTCTT 64 1840 1246105 N/A N/A 5957 5972 GTATTGAGATGACAGT
 68 1841 1246131 N/A N/A 6212 6227 CCCATAACCATAGGAC 47 1842 1246157 N/A N/A
 6464 6479 GCAGTAAAGTGCTAAG 83 1843 1246183 N/A N/A 7122 7137
 GAAGATTAGTTAACCT 13 1844 1246209 N/A N/A 7938 7953 CATATTCCCTGATGAT 14
 1845 1246235 N/A N/A 8238 8253 AACTACATGACCTGGG 27 1846 1246261 N/A N/A
 8674 8689 TTCTATTGTATGAGGT 0 1847 1246287 N/A N/A 8980 8995
 GTCCAGAAGTATGTGA 9 1848 1246313 N/A N/A 9331 9346 TCCTATATACATCCAA
 45 1849 1246339 N/A N/A 9515 9530 CACTAATATTGAGGCA 31 1850 1246365 N/A N/A
 9663 9678 CAGCTATAGGTAGCTA 0 1851 1246391 N/A N/A 9966 9981
 TAGTGTAAGCTGAGAG 75 1852 1246417 N/A N/A 10203 10218 GTATGGTAAGCCCCAT 36
 1853 1246443 N/A N/A 10523 10538 TATATAACATGGCTGG 8 1854 1246469 N/A N/A 11238
 11253 TGACATAGACATGTGT 42 1855 1246495 N/A N/A 11621 11636
 CTGTAGGACTCTGCTC 48 1856 1246521 N/A N/A 11963 11978 TGAGAAGGCGGTATCA 20
 1857 1246547 N/A N/A 12337 12352 CCCTAATGTGTTTTCC 25 1858 1246573 N/A N/A 12612
 12627 CTTAATTTTCGGGTTC 37 1859 1246599 N/A N/A 12725 12740
 TAGGATTAATTCCCTT 0 1860 1246625 N/A N/A 13601 13616 AATACAGGATAGAGCA
 0 1861 1246651 N/A N/A 13859 13874 TAGGAAAGCTCATGGT 27 1862 1246677 N/A N/A
 14018 14033 GGAAATTCACAATCTA 22 1863 1246703 N/A N/A 14172 14187
 TTGAATAGCAAGCCAA 31 1864 1246729 N/A N/A 14394 14409 TAAACATAGGCTTAGT 34
 1865 1246755 N/A N/A 14726 14741 TAAATACCCTTGGTGG 7 1866 1246781 N/A N/A
 15313 15328 TTAATCATGGGACAGG 27 1867 1246807 N/A N/A 15382 15397
 TCAAAATGGTGGCGAG 31 1868 1246833 N/A N/A 15772 15787 AGTATTTGGGTGTTCT 17
 1869 1246859 N/A N/A 16327 16342 TCACTATATTGCATTC 35 1870 1246885 N/A N/A 17239
 17254 AGGGAATGTTATGTAC 49 1871 1246911 N/A N/A 18012 18027
 AAAAGGCCCAAGGCTT 30 1872 1246937 N/A N/A 18813 18828 GTAATTGCATCTGCTC 51
 1873 1246963 N/A N/A 19399 19414 ATGCATAAGAGATGAG 30 1874 1246989 N/A N/A 19649
 19664 CTATGCAAGCCTTCAC 19 1875 1247015 N/A N/A 19827 19842
 GTAATATTCATAGTTC 49 1876 1247041 N/A N/A 20173 20188 CCAATTATGAATCTGC 61
 1877 1247067 N/A N/A 20397 20412 TAGTAATTGCTATAGT 7 1878 1247093 N/A N/A 20606
 20621 TCATTTAGAATACGCT 20 1879

(453) TABLE-US-00025 TABLE 25 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
 ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number
 Site Site Site Site Sequence (5' to 3') Inhibition) NO 1245092 31 46 3125 3140
 CCTTCTGGTTCAGTCC 72 1880 1245118 109 124 3203 3218
 TCCAAGTAGGAGTAGA 24 1881 1245144 176 191 3270 3285
 AGTAATGAGAACAATC 45 1882 1245170 249 264 3343 3358
 CCCACAGAACCAATAT 16 1883 1245196 297 312 7586 7601
 GTTTTCGGCACTCAGC 51 1884 1245222 357 372 7646 7661
 AGCGATAGATCTCTTC 31 1885 1245248 400 415 8794 8809
 ACGATTGTTACATCAC 53 1886 1245274 430 445 8824 8839
 GCTGGATATACTGTCC 36 1887 1245300 471 486 8865 8880
 ATGTCTTGGTAATCTC 23 1888 1245326 513 528 N/A N/A GTGCTTTTGTGATCCA 62

1849 1245352 559 574 10445 10460 TGTGTACGATGTGGC 40 1890 1245378 612
627 N/A N/A TGGAACAATATGGGAT 46 1891 1245404 667 682 12085 12100
AAGGCCTGAAGTTCTG 5 1892 1245430 755 770 N/A N/A AGGCCATAATCTTGTG
33 1893 1245456 788 803 15669 15684 CAGACTTCTTACGACT 33 1894 1245482 840
855 15721 15736 TGATATACGATGGAAC 13 1895 1245508* 905 920 20720 20735
CTGCATACGATTTAAA 32 1896 1245534* 971 986 20786 20801 TGGCTGGAGCTTATTT
87 1897 1245560 1012 1027 20827 20842 TTGATTCGAAACTATT 55 1898 1245586 1086 1101
20901 20916 GACTGCTGCTAGTGCC 67 1899 1245612 1258 1273 21073 21088
TAGCTTTTGTCCACCT 60 1900 1245638 1312 1327 21127 21142 GTCTTAAACCTTCCCT 63
1901 1245664 1339 1354 21154 21169 GCTACAGATTGGAATG 47 1902 1245690 1383 1398
21198 21213 AGCTGTGCACTCATTC 57 1903 1245716 1421 1436 21236 21251
AGGTTGAGATAAAGCT 74 1904 1245742 1497 1512 21312 21327 AGTTGCACCGTTTTGG
55 1905 1245768 1556 1571 21371 21386 CTTTTGGTGGACTTCA 10 1906 1245794 1615 1630
21430 21445 GTCACCTTTCATAATG 45 1907 1245820 1708 1723 21523 21538
CTCTGGGACCAAGGAT 51 1908 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC
55 38 1245846 1770 1785 21585 21600 AGCCAGTACAGTTCCT 58 1909 1245872 2175
2190 21990 22005 AGTCGGATTATTTTTT 42 1910 1245898 2201 2216 22016 22031
AACAAGATTAGTCTTG 40 1911 1245924 N/A N/A 3401 3416 CATATCTTACTCTGTG 47
1912 1245950 N/A N/A 3550 3565 CATAAAAATCGCTGCT 45 1913 1245976 N/A N/A
3882 3897 CGTTTATCATATGCTA 88 1914 1246002 N/A N/A 4416 4431
CCAGAAAGTTCATTTCG 81 1915 1246028 N/A N/A 5062 5077 TTACATTACAGCCTTA
67 1916 1246054 N/A N/A 5218 5233 TTAAACTCCAATGTGT 36 1917 1246065 N/A N/A
5659 5674 ATAAGTTACCAGAGCA 82 48 1246080 N/A N/A 5800 5815
CTCAAATAGCCTTTTG 64 1918 1246106 N/A N/A 5979 5994 CCTGAAAAGCCCTATT
0 1919 1246132 N/A N/A 6222 6237 ACCAAACTTGCCCAT 46 1920 1246158 N/A N/A
6465 6480 GGCAGTAAAGTGCTAA 62 1921 1246184 N/A N/A 7123 7138
TGAAGATTAGTTAACC 21 1922 1246210 N/A N/A 7941 7956 ATCCATATTCCCTGAT 26
1923 1246236 N/A N/A 8239 8254 TAACTACATGACCTGG 37 1924 1246262 N/A N/A
8688 8703 GATGAGAGTCTTTCTT 26 1925 1246288 N/A N/A 9000 9015
AAAGATCTGGCCAGTT 3 1926 1246314 N/A N/A 9332 9347 ATCCTATATACATCCA
76 1927 1246340 N/A N/A 9516 9531 AACTAATATTGAGGC 68 1928 1246366 N/A N/A
9712 9727 GAACATACATTGTTGC 58 1929 1246392 N/A N/A 9972 9987
GTTGAGTAGTGTAAGC 53 1930 1246418 N/A N/A 10209 10224 TTATTGGTATGGTAAG 14
1931 1246444 N/A N/A 10728 10743 GTAATTGGCTCTTGAG 44 1932 1246470 N/A N/A 11239
11254 ATGACATAGACATGTG 23 1933 1246496 N/A N/A 11652 11667
CTACAAAACCTGGTATC 0 1934 1246522 N/A N/A 11982 11997 ACCAATCCTCAGCTTT 11
1935 1246548 N/A N/A 12353 12368 GATCATTCCTGCTAGT 35 1936 1246574 N/A N/A 12613
12628 GCTTAATTTTCGGGTT 36 1937 1246600 N/A N/A 12726 12741
TTAGGATTAATTCCCT 14 1938 1246626 N/A N/A 13603 13618 ACAATACAGGATAGAG 23
1939 1246652 N/A N/A 13860 13875 TTAGGAAAGCTCATGG 26 1940 1246678 N/A N/A
14026 14041 CTTACTAAGGAAATTC 19 1941 1246704 N/A N/A 14174 14189
CCTTGAATAGCAAGCC 61 1942 1246730 N/A N/A 14395 14410 ATAAACATAGGCTTAG 41
1943 1246756 N/A N/A 14728 14743 TCTAAATACCCTTGGT 26 1944 1246782 N/A N/A 15314
15329 ATTAATCATGGGACAG 20 1945 1246808 N/A N/A 15383 15398
ATCAAAATGGTGGCGA 20 1946 1246834 N/A N/A 15773 15788 TAGTATTTGGGTGTTT 27
1947 1246860 N/A N/A 16377 16392 ACCAAACTTCCAGCAG 31 1948 1246886 N/A N/A
17675 17690 CAAAACCTTTGTTGCC 14 1949 1246912 N/A N/A 18015 18030
TCAAAAAGGCCCAAGG 46 1950 1246938 N/A N/A 18814 18829 AGTAATTGCATCTGCT 42
1951 1246964 N/A N/A 19404 19419 CAGGAATGCATAAGAG 33 1952 1246990 N/A N/A
19662 19677 ACCAGAATCCATTCTA 60 1953 1247016 N/A N/A 19860 19875

AATCAATGTCTGTGAA 34 1954 20174 20189 CCAATTATGAATCTG 62
 1955 1247068 N/A N/A 20398 20413 ATAGTAATTGCTATAG 36 1956 1247094 N/A N/A 20607
 20622 GTCATTTAGAATACGC 46 1957
 (454) TABLE-US-00026 TABLE 26 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
 ID NO: ID NO: 1 1 2 2 HSD17B13 SEQ Compound Start Stop Start Stop (% ID Number Site
 Site Site Site Sequence (5' to 3') Inhibition) NO 1245093 58 73 3152 3167
 ATGTTTCATGGCTTTGCT 73 1958 1245119 110 125 3204 3219
 CTCCAAGTAGGAGTAG 5 1959 1245145 180 195 3274 3289
 CTCCAGTAATGAGAAC 40 1960 1245171 250 265 3344 3359
 TCCCACAGAACCAATA 24 1961 1245197 298 313 7587 7602
 AGTTTTTCGGCACTCAG 16 1962 1245223 358 373 7647 7662
 GAGCGATAGATCTCTT 0 1963 1245249 401 416 8795 8810
 CACGATTGTTACATCA 37 1964 1245275 431 446 8825 8840
 GGCTGGATATACTGTC 18 1965 1245301 472 487 8866 8881
 AATGTCTTGATAATCT 20 1966 1245327 514 529 N/A N/A AGTGCTTTTGTGATCC 25
 1967 1245353 560 575 10446 10461 CACTGTGACGATGTGG 10 1968 1245379 613
 628 N/A N/A CTGGAACAATATGGGA 38 1969 1245405 669 684 12087 12102
 CCAAGGCCTGAAGTTC 20 1970 1245431 758 773 N/A N/A TACAGGCCATAATCTT 16
 1971 1245457 789 804 15670 15685 TCAGACTTCTTACGAC 36 1972 1245483 841
 856 15722 15737 TTGATATACGATGGAA 38 1973 1245509* 906 921 20721 20736
 TCTGCATACGATTTAA 44 1974 1245535* 972 987 20787 20802 CTGGCTGGAGCTTATT
 82 1975 1245561 1013 1028 20828 20843 ATTGATTGAACTAT 22 1976 1245587 1087 1102
 20902 20917 TGA CTGCTGCTAGTGC 50 1977 1245613 1259 1274 21074 21089
 GTAGCTTTTGTCCACC 63 1978 1245639 1313 1328 21128 21143 AGTCTTAAACCTTCCC 77
 1979 1245665 1340 1355 21155 21170 GGCTACAGATTGGAAT 23 1980 1245691 1384 1399
 21199 21214 TAGCTGTGCACTCATT 40 1981 1245717 1422 1437 21237 21252
 CAGGTTGAGATAAAGC 49 1982 1245743 1498 1513 21313 21328 GAGTTGCACCGTTTTG
 48 1983 1245769 1557 1572 21372 21387 ACTTTTGGTGGACTTC 0 1984 1245795 1616
 1631 21431 21446 GGTCACCTTTCATAAT 57 1985 1245821 1709 1724 21524 21539
 TCTCTGGGACCAAGGA 56 1986 1245841 1765 1780 21580 21595 GTACAGTTCCTTTTCC
 44 38 1245847 1771 1786 21586 21601 TAGCCAGTACAGTTCC 52 1987 1245873 2176
 2191 21991 22006 GAGTCGGATTATTTTT 44 1988 1245899 2204 2219 22019 22034
 CAAAACAAGATTAGTC 18 1989 1245925 N/A N/A 3403 3418 TACATATCTTACTCTG
 40 1990 1245951 N/A N/A 3551 3566 TCATAAAAATCGCTGC 26 1991 1245977 N/A N/A
 3902 3917 GTTTATCATGTGCCAC 83 1992 1246003 N/A N/A 4434 4449
 GTAAATTCTTGCTATC 40 1993 1246029 N/A N/A 5063 5078 GTTACATTACAGCCTT 71
 1994 1246055 N/A N/A 5221 5236 AGCTTAAACTCCAATG 40 1995 1246065 N/A N/A
 5659 5674 ATAAGTTACCAGAGCA 80 48 1246081 N/A N/A 5816 5831
 TCAAAGTCAGTATCCC 79 1996 1246107 N/A N/A 5983 5998 GATACCTGAAAAGCCC
 49 1997 1246133 N/A N/A 6223 6238 TACCAAACCTTGCCCAT 33 1998 1246159 N/A N/A
 6473 6488 ATGTAATAGGCAGTAA 33 1999 1246185 N/A N/A 7481 7496
 ACAATTAACATTTCGGC 77 2000 1246211 N/A N/A 7950 7965 TAGGAAGTGATCCATA
 30 2001 1246237 N/A N/A 8240 8255 ATA ACTACATGACCTG 29 2002 1246263 N/A N/A
 8693 8708 CCAAAGATGAGAGTCT 19 2003 1246289 N/A N/A 9001 9016
 CAAAGATCTGGCCAGT 1 2004 1246315 N/A N/A 9346 9361
 CAAAAGTGTCCTCGAT 5 2005 1246341 N/A N/A 9517 9532 TACACTAATATTGAGG
 30 2006 1246367 N/A N/A 9713 9728 TGAACATACATTGTTG 29 2007 1246393 N/A N/A
 9976 9991 GACAGTTGAGTAGTGT 46 2008 1246419 N/A N/A 10210 10225
 TTTATTGGTATGGTAA 25 2009 1246445 N/A N/A 10729 10744 TGTAATTGGCTCTTGA 0

2010 1246471 N/A N/A 11248 11263 ACCAAGGCCATGACAT 17 2011 1246497 N/A N/A 11653
11668 TCTACAAAACCTGGTAT 9 2012 1246523 N/A N/A 11983 11998
AACCAATCCTCAGCTT 0 2013 1246549 N/A N/A 12411 12426 GATCAAATGTATGTGC 25
2014 1246575 N/A N/A 12614 12629 GGCTTAATTTTCGGGT 17 2015 1246601 N/A N/A 12727
12742 TTTAGGATTAATTCCC 16 2016 1246627 N/A N/A 13604 13619
GACAATACAGGATAGA 43 2017 1246653 N/A N/A 13861 13876 CTTAGGAAAGCTCATG
0 2018 1246679 N/A N/A 14027 14042 CCTTACTAAGGAAATT 28 2019 1246705 N/A N/A
14194 14209 TGTGATTGAGTTCTCC 62 2020 1246731 N/A N/A 14397 14412
TCATAAACATAGGCTT 58 2021 1246757 N/A N/A 14729 14744 CTCTAAATACCCTTGG 7
2022 1246783 N/A N/A 15315 15330 GATTAATCATGGGACA 9 2023 1246809 N/A N/A
15384 15399 TATCAAATGGTGGCG 18 2024 1246835 N/A N/A 15774 15789
TTAGTATTTGGGTGTT 29 2025 1246861 N/A N/A 16382 16397 TCCCAACCAAACCTTCC 82
2026 1246887 N/A N/A 17766 17781 GCATATTCATTTGGCC 6 2027 1246913 N/A N/A 18018
18033 GTTTCAAAAAGGCCCA 39 2028 1246939 N/A N/A 18815 18830
TAGTAATTGCATCTGC 51 2029 1246965 N/A N/A 19436 19451 TGGAGGAAAGCTTCAA 10
2030 1246991 N/A N/A 19675 19690 ACCAAGAGACACCACC 41 2031 1247017 N/A N/A
19889 19904 CTTACGACAGGTCATC 39 2032 1247043 N/A N/A 20185 20200
TTAGAAGTCAGCCCAA 37 2033 1247069 N/A N/A 20400 20415 CTATAGTAATTGCTAT 0
2034 1247095 N/A N/A 20608 20623 GGTCATTTAGAATACG 39 2035

Example 2: Dose-Dependent Inhibition of Human HSD17B13 in HepaRG Cells by cEt Gapmers
(455) Certain modified oligonucleotides described in the studies above exhibiting significant in vitro inhibition of HSD17B13 RNA were selected and tested at various doses in HepaRG cells.
(456) The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with modified oligonucleotides diluted to different concentrations as specified in the Tables below. After a treatment period of approximately 24 hours, HSD17B13 RNA levels were measured as previously described using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total RNA, as measured by RIBOGREEN®. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells.
(457) The half maximal inhibitory concentration (IC_{sub}.50) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

(458) TABLE-US-00027 TABLE 27 Multi-dose assay of modified oligonucleotides in HepaRG cells ION % Inhibition IC₅₀ No. 6 nM 32 nM 160 nM 800 nM 4000 nM (μM) 1245651 0 6 24 57 82 0.6 1245927 0 19 55 80 90 0.2 1245930 0 0 30 77 86 0.4 1245957 8 8 49 77 87 0.2 1245984 12 14 62 78 90 0.1 1246065 0 10 46 65 89 0.3 1246066 0 13 35 58 76 0.5 1246068 0 5 30 61 83 0.5 1246110 3 12 44 82 87 0.2 1246115 8 14 45 75 89 0.2 1246116 6 17 48 72 83 0.2 1246134 3 21 55 79 91 0.2 1246163 2 15 43 75 92 0.2

(459) TABLE-US-00028 TABLE 28 Multi-dose assay of modified oligonucleotides in HepaRG cells ION % Inhibition IC₅₀ No. 6 nM 32 nM 160 nM 800 nM 4000 nM (μM) 1245681 0 12 38 72 90 0.3 1245790 1 8 37 67 93 0.3 1245968 15 23 49 75 93 0.2 1245976 0 16 49 73 88 0.2 1245977 0 10 41 75 77 0.3 1245999 1 14 48 82 90 0.2 1246022 13 24 56 83 93 0.1 1246023 13 24 66 90 93 0.1 1246104 0 0 35 67 87 0.4 1246157 0 22 40 75 84 0.3 1246226 11 7 40 71 88 0.3 1246667 89 95 97 98 100 <0.006 1246853 0 2 11 57 82 0.7 1246861 4 0 12 31 67 2.6

Example 3: Antisense Inhibition of Human HSD17B13 in HepaRG Cells by cEt Gapmers
(460) Additional modified oligonucleotides complementary to an HSD17B13 nucleic acid were synthesized and tested for their effect on HSD17B13 RNA levels in vitro. The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below.

(461) Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with 1,000 nM of modified oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and HSD17B13 RNA levels were measured by quantitative real-time RTPCR. Human primer probe set RTS43553 was used to measure RNA levels. HSD17B13 RNA levels were adjusted according to total RNA content, as measured by RIBOGREEN® S. Results are presented as percent inhibition of HSD17B13 relative to untreated control cells. As used herein, a value of '0' indicates that treatment with the modified oligonucleotide did not inhibit HSD17B13 mRNA levels. An asterisk (*) indicates that the modified oligonucleotide is complementary to the target transcript within the amplicon region of the primer probe set and so, the associated data is not reliable. In such instances, additional using alternate primer probe sets must be performed to accurately assess the potency and efficacy of such modified oligonucleotides.

(462) The modified oligonucleotides described in the tables below were 3-10-3 cEt gapmers. The internucleoside linkages throughout each modified oligonucleotide were phosphorothioate (P=S) linkages. All cytosine residues throughout each modified oligonucleotide were 5-methylcytosines.

(463) "Start site" indicates the 5'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. "Stop site" indicates the 3'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. As shown in the Tables below, the modified oligonucleotide are complementary to either the human HSD17B13 mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession No. NM_178135.4) or to the human HSD17B13 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000) or to both. 'N/A' indicates that the modified oligonucleotide is not complementary to that particular target sequence with 100% complementarity.

(464) TABLE-US-00029 TABLE 29 Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2

SEQ ID NO:	Compound	Start	Stop	% Inhibition	Seq ID NO:	Start	Stop	% Inhibition	Seq ID NO:	Start	Stop	% Inhibition	
1	1245927	N/A	N/A	3433	1	91	121	1246023	N/A	N/A	5046	5061	GAGTGAATCATTTCAGT
2	1340071	N/A	N/A	20191	2	89	1526	1340071	N/A	N/A	20191	20206	CAGTTATTAGAAGTCA
1	20588	20603	GAGTTATCTGGTTTGC	84	2037	1340086	N/A	N/A	9335	9350			
2	2039	1340099	1763	1778	21578	21593	ACAGTTCCTTTTCCTG	58	2040	1340104	N/A	N/A	
1	10500	10515	TACTTACCAATATGGG	13	2041	1340120	N/A	N/A	11585	11600			
2	23	2043	1340170	N/A	N/A	9657	9672	TAGGTAGCTAATGTAT	34	2044	1340194	N/A	N/A
1	11077	11092	TAATTACAACCTGGTT	52	2045	1340201	N/A	N/A	14279	14294			
2	2047	1340209	N/A	N/A	18520	18535	CTTAGTACCAAGACAC	60	2048	1340231	N/A	N/A	
1	16134	16149	GATGCTCCATAATAAT	42	2049	1340239	N/A	N/A	19912	19927			
2	2051	1340256	N/A	N/A	19532	19547	GGCTAAAATGGTCATC	75	2052	1340276	N/A	N/A	19394
1	19409	TAAGAGATGAGTAGGC	46	2053	1340278	N/A	N/A	15250	15265				
2	2055	1340290	809	824	15690	15705	ATTGGTAAGTATTCCA	50	2056	1340305	1177	1192	
1	20992	21007	TTTAAGAGGCATGAAA	40	2057	1340308	N/A	N/A	20111	20126			
2	2059	1340360	N/A	N/A	12800	12815	GGCTGTAAGAGTCAGT	88	2060	1340368	N/A	N/A	
1	13827	13842	CCTACCAAGAGGGTTT	34	2061	1340380	N/A	N/A	12005	12020			
2	ATTGTTAGCTAAGGGA	22	2062	1340390	N/A	N/A	14175	14190	GCCTTGAATAGCAAGC	32			

2063 1340405 N/A N/A 13598 N/A 13613 ACAGGATAGAGCAT 44 2064 1340412 N/A N/A
 14914 14929 ACATTAGCAAGCTAAG 52 2065 1340423 N/A N/A 9538 9553
 ATCCAGATCTGCCCTA 39 2066 1340427 N/A N/A 8535 8550 TTGCAAGTTTATCAGT
 66 2067 1340432 N/A N/A 4225 4240 GTTTCAACTAAACATG 65 2068 1340438 N/A N/A
 3479 3494 GGACCAGGGAATTTAT 55 2069 1340439 1255 1270 21070 21085
 CTTTTGTCCACCTTTA 76 2070 1340448 N/A N/A 5672 5687 GTCAGTAGAGAGCATA
 85 2071 1340453 N/A N/A 6084 6099 TTTCACCTCAGGTGAC 52 2072 1340459 N/A N/A
 5083 5098 CTCGCCTAAAGGAGAT 56 2073 1340465 N/A N/A 5895 5910
 TCCTTTGTATTTCGCT 68 2074 1340479 N/A N/A 6785 6800 CCTCCTTATTTGTTAG 41
 2075 1340491 N/A N/A 3699 3714 TTTCAGATCCCGTTCT 45 2076 1340525 N/A N/A
 9965 9980 AGTGTAAGCTGAGAGT 71 2077 1340542 N/A N/A 5186 5201
 CCAACGCAACAGTTTA 77 2078 1340546 N/A N/A 5647 5662 AGCATTTCATCAGATGT
 83 2079 1340556 N/A N/A 5268 5283 TTGCAAAATGTGATGC 75 2080 1340563 N/A N/A
 3990 4005 GTGTTTACAAGTAAGA 79 2081 1340567 N/A N/A 6260 6275
 AGATGGGCAAGGCCAC 59 2082 1340576 N/A N/A 11499 11514 CTGGAGAAGAGTTTAC
 38 2083 1340580 N/A N/A 8641 8656 AGACAGACTGTTGAGC 74 2084 1340597 N/A N/A
 18036 18051 CATAGTTTATATGGAT 56 2085 1340601 N/A N/A 18138 18153
 TGTTATCTCAAGTCAG 81 2086 1340613 N/A N/A 5832 5847 ATGTTAGGACCCAGTC
 60 2087 1340661 N/A N/A 9201 9216 TACTGACAACCCTT 56 2088 1340665 N/A N/A
 14601 14616 GAGAAATCTAAGAACC 42 2089 1340667 N/A N/A 9936 9951
 AATGAATCATGATTGA 33 2090 1340669 N/A N/A 10000 10015 TGAAATCTTGTGTAAC 43
 2091 1340684 N/A N/A 8125 8140 AGTGCTTAGTTCATTG 84 2092 1340685 N/A N/A
 16385 16400 CAATCCCAACCAAACT 49 2093 1340689 N/A N/A 3600 3615
 CTTGGAAGGAGACTGG 60 2094 1340697 N/A N/A 4661 4676 CCGCCCTTAAGTCATT
 53 2095 1340704 N/A N/A 6329 6344 ACTTTCATAGGGAGAC 80 2096 1340705 74
 89 3168 3183 AAGGATTCTAGGATG 71 2097 1340722 N/A N/A 5737 5752
 TTAGTTGGTAGCTTGC 72 2098 1340767 N/A N/A 9441 9456 TCCCACAAAACCTAACC
 24 2099 1340773 N/A N/A 9773 9788 TAGAACTCCCAACCCA 44 2100 1340776 N/A N/A
 6732 6747 TTACCCCTGGCTTTTC 26 2101 1340785 N/A N/A 7781 7796
 ACTACTTCAGTTAGCA 79 2102 1340792 N/A N/A 9115 9130 GACAGACCAAGTAGCT
 39 2103 1340812 N/A N/A 10264 10279 CTTCCAAGCATTCCAT 57 2104 1340813 N/A N/A
 7526 7541 CCGAAAAAAGTGAGG 31 2105 1340836 N/A N/A 18694 18709
 AAGACTTTGAGACTCT 62 2106 1340850 N/A N/A 8744 8759 CACTTTCAGTGGGTGT
 7 2107 1340852 1424 1439 21239 21254 TCCAGGTTGAGATAAA 70 2108 1340872 1614
 1629 21429 21444 TCACCTTTCATAATGT 63 2109 1340878 N/A N/A 12627 12642
 TTGTTAAGCAGATGGC 84 2110 1340889 N/A N/A 6597 6612 CTACTTTCAAACCTTG
 57 2111 1340903 N/A N/A 20390 20405 TGCTATAGTAATCACA 65 2112
 (465) TABLE-US-00030 TABLE 30 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
 ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
 Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448
 AGTAGATGTAAGTCA 84 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTAGT
 89 1526 1340078 N/A N/A 16295 16310 CCTTTAGTTAGAATAG 37 2113 1340094 N/A N/A
 19547 19562 ATTGTTTAGCTGATAG 61 2114 1340113 N/A N/A 12667 12682
 TGTTAACCTGCAGCAG 39 2115 1340122 N/A N/A 14192 14207 TGATTGAGTTCTCCAC 48
 2116 1340126 N/A N/A 19779 19794 ACATACCACTCTTCTC 35 2117 1340130 N/A N/A 17954
 17969 TTAGAAACTGCTCCTC 48 2118 1340146 N/A N/A 8132 8147
 TTGTTAGAGTGCTTAG 76 2119 1340149 N/A N/A 11545 11560 CTTGACAATGGTTGCC 58
 2120 1340152 N/A N/A 10138 10153 ATAGATCCTTTTTTGG 6 2121 1340154 N/A N/A
 6284 6299 ACATTCTTCAGGTGTT 88 2122 1340156 N/A N/A 20194 20209

TTGCAGTTATTAGTAAG 13 2123 1340179 N/A N/A 14330 TTAAGATGTCCAGGCAT 17
2124 1340197 N/A N/A 10554 10569 GAAGTGTAAATGCTCCC 71 2125 1340220 N/A N/A 14611
14626 TATTTCTTGCGAGAAA 30 2126 1340241 N/A N/A 16220 16235
TTTATAGACTGGGTAG 59 2127 1340255 1783 1798 21598 21613 TTTCTTATGTAATAGC 76
2128 1340260 1714 1729 21529 21544 AAACATCTCTGGGACC 76 2129 1340277 N/A N/A
19914 19929 AGTGCATCTTAAGATA 57 2130 1340291 N/A N/A 18540 18555
GAAACTCACTTCCTAC 56 2131 1340297 N/A N/A 14917 14932 AATACATTAGCAAGCT 45
2132 1340304 697 712 12115 12130 AGACATGAGGTTTTGA 59 2133 1340310 N/A N/A 13638
13653 GACAATGTGCAGCTCT 67 2134 1340321 N/A N/A 6606 6621
TAAAGCAGGCTACTTT 9 2135 1340323 N/A N/A 19396 19411 CATAAGAGATGAGTAG
44 2136 1340342 N/A N/A 5279 5294 GAGCTAGACAATTGCA 73 2137 1340350 N/A N/A
9716 9731 TAGTGAACATACATTG 23 2138 1340361 N/A N/A 5935 5950
GCTGCTATTTAGGGCA 53 2139 1340364 N/A N/A 5678 5693 TTCTAGGTCAGTAGAG
58 2140 1340369 1433 1448 21248 21263 TAAAATATGTCCAGGT 75 2141 1340385 N/A N/A
4389 4404 AGACCAAAAAGATGGC 80 2142 1340395 N/A N/A 11794 11809
AACTCTATTGATTGGC 72 2143 1340416 N/A N/A 12496 12511 AATTCACCTTGACTAA 9
2144 1340419 N/A N/A 12410 12425 ATCAAATGTATGTGCT 73 2145 1340443 N/A N/A 11117
11132 CTACAGAGGAGTTTGC 27 2146 1340445 N/A N/A 5847 5862
ACTCCCCAAACATGGA 48 2147 1340458 N/A N/A 7834 7849 GCTGAGTGGATTACAT
44 2148 1340469 N/A N/A 6456 6471 GTGCTAAGTAGTAATT 58 2149 1340476 815
830 15696 15711 TTTCTTATTGGTAAGT 44 2150 1340512 N/A N/A 6736 6751
GAGATTACCCCTGGCT 72 2151 1340522 N/A N/A 6915 6930 ACTCGGGAAGTTTAGA
26 2152 1340533 N/A N/A 9456 9471 GTTGATATTTCCATCT 55 2153 1340543 N/A N/A
20115 20130 AGCATCAAGACATTCT 79 2154 1340549 N/A N/A 9968 9983
AGTAGTGTAAGCTGAG 88 2155 1340552 N/A N/A 12802 12817 CAGGCTGTAAGAGTCA 56
2156 1340555 N/A N/A 5158 5173 GCCAAACTGCTACTGT 73 2157 1340558 N/A N/A 15423
15438 ATATCCAGTAGGTGTG 55 2158 1340606 N/A N/A 18046 18061
CAAGTTTGTTTCATAGT 78 2159 1340614 1311 1326 21126 21141 TCTTAAACCTTCCCTG 21
2160 1340651 N/A N/A 9344 9359 AAAGTGTCTCTCGATCC 36 2161 1340671 N/A N/A
10293 10308 TGAGAGAACTTATACA 32 2162 1340673 N/A N/A 5191 5206
TATTCCCAACGCAACA 62 2163 1340674 N/A N/A 4663 4678 ACCCGCCCTTAAGTCA
48 2164 1340679 N/A N/A 5758 5773 ATAGGAGAGTCTTTCA 77 2165 1340681 N/A N/A
8547 8562 GCACATCATGTTTTGC 44 2166 1340683 N/A N/A 9947 9962
TACACCACACTAATGA 26 2167 1340696 N/A N/A 8656 8671 CTCAGAGTTCAGGCAA
67 2168 1340713 N/A N/A 6191 6206 GACAGAGCAATTTACT 55 2169 1340734 N/A N/A
8964 8979 ACTCCGTTATAAGTTT 42 2170 1340747 218 233 3312 3327
AAATTCATAAGTAGTC 39 2171 1340757 N/A N/A 18143 18158 CCTTATGTTATCTCAA 67
2172 1340761 N/A N/A 4068 4083 CCTTACCAGAATTTAC 63 2173 1340762 N/A N/A
5651 5666 CCAGAGCATTTCATCAG 83 2174 1340768 N/A N/A 9210 9225
GAAAAGATGTACACTG 56 2175 1340770 N/A N/A 3505 3520 CTAACCTGACACATAT
18 2176 1340771 1179 1194 20994 21009 TTTTAAAGAGGCATGA 60 2177 1340775 N/A N/A
9593 9608 AAGAGCTGGTAAAGGT 47 2178 1340798 N/A N/A 3619 3634
GACTTGCCTCATTTAG 69 2179 1340802 N/A N/A 9793 9808 CCTATGCAAATTCATA 34
2180 1340803 N/A N/A 6333 6348 AAGGACTTTCATAGGG 71 2181 1340824 N/A N/A
20421 20436 CAGTAAAATTATGCCT 38 2182 1340831 N/A N/A 3702 3717
AATTTTCAGATCCCGT 74 2183 1340833 N/A N/A 9117 9132 AAGACAGACCAAGTAG
16 2184 1340837 N/A N/A 13955 13970 ACCTCTAAGTTAGCCC 28 2185 1340845 N/A N/A
18697 18712 CAAAAGACTTTGAGAC 22 2186 1340863 N/A N/A 20598 20613
AATACGCTGAGAGTTA 48 2187 1340874 N/A N/A 7528 7543 TTCCGAAAAAAGTGGA
13 2188 1340891 N/A N/A 16895 16910 CGTCAAATAGGGCTGG 53 2189

(466) TABLE-US-00031 TABLE 31 Inhibition of HSD17B13 RNA by 3-10-3 cEt gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO: ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448

AGTAGATGGTAAGTCA 91 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTTCAGT 83 1526 1340082 N/A N/A 16329 16344 CTTCACATATATTGCAT 50 2190 1340083 N/A N/A 20116 20131 CAGCATCAAGACATTC 72 2191 1340092 N/A N/A 16221 16236

TTTTATAGACTGGGTA 53 2192 1340098 N/A N/A 19818 19833 ATAGTTCAAATTGGAT 48 2193 1340125 N/A N/A 15778 15793 TGTTTTAGTATTTGGG 48 2194 1340131 N/A N/A 18720 18735 CACCATGACAGCTTCA 65 2195 1340141 N/A N/A 11213 11228

CTTCTGTAGGTTTTGC 60 2196 1340142 N/A N/A 17984 17999 TAGTTATCTTCTCACT 52 2197 1340148 2240 2255 22055 22070 ATGCAAAAGCATTCTA 50 2198 1340164 N/A N/A 11546 11561 TCTTGACAATGGTTGC 77 2199 1340183 N/A N/A 11795 11810

TAACCTCTATTGATTGG 51 2200 1340187 N/A N/A 20628 20643 GACAACAGAGTTCTGT 9 2201 1340192 N/A N/A 12498 12513 GAAATTCACCTTGACT 41 2202 1340206 1330 1345 21145 21160 TGGAATGCTACTTGAA 85 2203 1340208 N/A N/A 9969 9984

GAGTAGTGTAAGCTGA 80 2204 1340215 N/A N/A 12416 12431 ACGGTGATCAAATGTA 53 2205 1340221 N/A N/A 16900 16915 AAAACCGTCAAATAGG 28 2206 1340230 N/A N/A 18658 18673 TATTCTCCAACCTCAGG 54 2207 1340235 N/A N/A 18145 18160

CTCCTTATGTTATCTC 57 2208 1340245 N/A N/A 15446 15461 TGTTATCAGAACTTA 44 2209 1340270 N/A N/A 7550 7565 GCTGTAATTAGGAAGA 24 2210 1340274 N/A N/A 10584 10599 AGAACCAGGACTCTCC 67 2211 1340275 219 234 3313 3328

CAAATTCATAAGTAGT 76 2212 1340284 1467 1482 21282 21297 AAGAGGCTAGGGAAAT 66 2213 1340292 N/A N/A 19417 19432 GACAACCTGATGTCCAG 12 2214 1340312 N/A N/A 20423 20438 GACAGTAAAATTATGC 57 2215 1340314 N/A N/A 18075 18090

TTGTACCCTGTTCTCA 64 2216 1340320 N/A N/A 14318 14333 AATTAAGATGTCCAGG 13 2217 1340324 N/A N/A 9950 9965 TTCTACACCACACTAA 12 2218 1340329 1181 1196 20996 21011 AGTTTTTAAGAGGCAT 75 2219 1340337 N/A N/A 13659 13674

TCACAAAGTCCCTGAC 11 2220 1340353 N/A N/A 7125 7140 CATGAAGATTAGTTAA 11 2221 1340355 N/A N/A 8657 8672 TCTCAGAGTTCAGGCA 54 2222 1340356 N/A N/A 5680 5695 GTTTCTAGGTCAGTAG 73 2223 1340366 N/A N/A 6457 6472

AGTGCTAAGTAGTAAT 71 2224 1340367 N/A N/A 5870 5885 TAAACCCTGGAGCAGC 66 2225 1340370 N/A N/A 5159 5174 AGCCAAACTGCTACTG 78 2226 1340426 N/A N/A 9597 9612 TGTGAAGAGCTGGTAA 41 2227 1340472 N/A N/A 6335 6350

AGAAGGACTTTCATAG 46 2228 1340475 N/A N/A 8136 8151 AAGTTTGTTAGAGTGC 74 2229 1340484 N/A N/A 12709 12724 CCAGTAAAAAATTGGT 23 2230 1340503 N/A N/A 14613 14628 ATTATTTCTTGCGAGA 60 2231 1340514 N/A N/A 9345 9360

AAAAGTGTCTCGATC 17 2232 1340515 N/A N/A 5762 5777 AAGAATAGGAGAGTCT 70 2233 1340539 N/A N/A 3721 3736 GACTTTTGTTTGTAGC 74 2234 1340560 N/A N/A 6285 6300 TACATTCTTCAGGTGT 61 2235 1340577 N/A N/A 5280 5295

TGAGCTAGACAATTGC 78 2236 1340603 699 714 12117 12132

AGAGACATGAGGTTTT 77 2237 1340604 N/A N/A 20195 20210 ATTGCAGTTATTAGAA 42 2238 1340617 N/A N/A 3516 3531 CCTTCATCTAACTAAC 30 2239 1340636 N/A N/A 12804 12819 CTCAGGCTGTAAGAGT 21 2240 1340647 N/A N/A 5195 5210

ACCGTATTCCCAACGC 90 2241 1340659 N/A N/A 8965 8980 AACTCCGTTATAAGTT 11 2242 1340664 N/A N/A 3676 3691 CCCCTAAGTTATTATC 33 2243 1340690 N/A N/A 9726 9741 TGTGAGAGGATAGTGA 50 2244 1340724 N/A N/A 4862 4877

AACTGAAACGATCCTC 52 2245 1340725 N/A N/A 9466 9481 GGCAGTTGAAGTTGAT 46 2246 1340727 N/A N/A 8589 8604 GTGGGATAAACTGTTC 52 2247 1340746 N/A N/A 19564 19579 TAGCTATTGTCTTTGA 63 2248 1340781 N/A N/A 13956 13971

TACCTCTAAGTTAGCC 28 2249 1340789 N/A N/A 14193 14208 GTGATTGAGTTCTCCA 80
2250 1340790 N/A N/A 9118 9133 CAAGACAGACCAAGTA 27 2251 1340830 N/A N/A
7835 7850 AGCTGAGTGGATTACA 29 2252 1340834 N/A N/A 6203 6218
ATAGGACATGGAGACA 82 2253 1340835 N/A N/A 5653 5668 TACCAGAGCATTTCATC
55 2254 1340838 N/A N/A 10294 10309 TTGAGAGAACTTATAC 21 2255 1340839 N/A N/A
19915 19930 GAGTGCATCTTAAGAT 43 2256 1340841 N/A N/A 4069 4084
GCCTTACCAGAATTTA 75 2257 1340864 N/A N/A 9796 9811 AATCCTATGCAAATTC 42
2258 1340865 1716 1731 21531 21546 CTAAACATCTCTGGGA 62 2259 1340870 N/A N/A
4402 4417 CGAATTTCTTCAAAGA 55 2260 1340875 N/A N/A 5974 5989
AAAGCCCTATTCTTCG 65 2261 1340876 N/A N/A 6626 6641 TGTACAGTAGGCAGTT
75 2262 1340883 N/A N/A 9224 9239 CAAGTGACAGCATTGA 48 2263 1340884 N/A N/A
6738 6753 ATGAGATTACCCCTGG 62 2264 1340885 N/A N/A 10184 10199
AACACTAATCTCAGTA 15 2265 1340892 N/A N/A 14918 14933 TAATACATTAGCAAGC 33
2266
(467) TABLE-US-00032 TABLE 32 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448
AGTAGATGGTAAGTCA 59 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTTCAGT
78 1526 1340075 N/A N/A 13957 13972 CTACCTCTAAGTTAGC 6 2267 1340077 1468 1483
21283 21298 GAAGAGGCTAGGGAAA 44 2268 1340093 N/A N/A 18744 18759
TATACCTGTTGCCTAG 11 2269 1340108 N/A N/A 10848 10863 CATATGGTGTACACTG 15
2270 1340129 N/A N/A 16904 16919 GTCAAAAACCGTCAAA 39 2271 1340144 254 269
3348 3363 AATATCCCACAGAACC 60 2272 1340151 N/A N/A 11796 11811
ATAACTCTATTGATTG 39 2273 1340157 N/A N/A 9147 9162 TGCCAAGCTTTGTGTC 38
2274 1340163 N/A N/A 14617 14632 CCGAATTATTTCTTGC 60 2275 1340166 N/A N/A 4412
4427 AAAGTTCATTCTGAATT 63 2276 1340168 N/A N/A 18086 18101
ACCAATGCAGTTTGTA 75 2277 1340169 N/A N/A 6460 6475 TAAAGTGCTAAGTAGT
50 2278 1340171 N/A N/A 11553 11568 GTTTTTCTCTTGACAA 59 2279 1340176 N/A N/A
16117 16132 GCTCTATTGGGCCAGG 47 2280 1340181 N/A N/A 18176 18191
CGCGAGATGATGTTGC 53 2281 1340211 N/A N/A 18661 18676 TACTATTCTCCAACCTC 47
2282 1340226 1182 1197 20997 21012 AAGTTTTTAAGAGGCA 77 2283 1340233 N/A N/A
12499 12514 AGAAATTCACCTTGAC 39 2284 1340268 N/A N/A 16224 16239
TCTTTTTATAGACTGG 82 2285 1340273 N/A N/A 11237 11252 GACATAGACATGTGTT 26
2286 1340283 1717 1732 21532 21547 TCTAAACATCTCTGGG 40 2287 1340286 N/A N/A
19846 19861 AACCTATGTGTGCACC 73 2288 1340327 N/A N/A 12418 12433
TAACGGTGATCAAATG 17 2289 1340332 N/A N/A 10295 10310 TTTGAGAGAACTTATA 0
2290 1340334 N/A N/A 8973 8988 AGTATGTGAAGTCCGT 81 2291 1340335 N/A N/A
5654 5669 TTACCAGAGCATTTCAT 50 2292 1340352 N/A N/A 9970 9985
TGAGTAGTGTAAGCTG 62 2293 1340375 N/A N/A 19444 19459 TGGGATAGTGGAGGAA 13
2294 1340377 N/A N/A 17986 18001 TATAGTTATCTTCTCA 46 2295 1340386 N/A N/A 14919
14934 ATAATACATTAGCAAG 7 2296 1340387 N/A N/A 5873 5888
GGCTAAACCCTGGAGC 36 2297 1340388 N/A N/A 8164 8179 TATTGTAGGGATTGAA
40 2298 1340397 N/A N/A 3741 3756 CTGTTGCAAGTTCTTC 85 2299 1340398 N/A N/A
13688 13703 TTTACTCACTTCTGGT 30 2300 1340415 N/A N/A 12815 12830
CACATAGCTAACTCAG 42 2301 1340430 N/A N/A 5197 5212 GAACCGTATTCCCAAC
72 2302 1340433 N/A N/A 9467 9482 AGGCAGTTGAAGTTGA 47 2303 1340446 N/A N/A
15491 15506 TTAATGCCACCCTACC 19 2304 1340450 N/A N/A 5712 5727
GATTGCACAAACCACA 73 2305 1340451 N/A N/A 6301 6316 CTAGGAGATATAACAT
40 2306 1340452 N/A N/A 5978 5993 CTGAAAAGCCCTATTC 11 2307 1340454 N/A N/A

7680 7695 AATTGTGACCTGCA 20 2308 1340468 N/A N/A 2012
ACATTGCAGTTATTAG 42 2309 1340471 N/A N/A 14385 14400 GCTTAGTGTAGAATTG 16
2310 1340478 N/A N/A 6740 6755 CAATGAGATTACCCCT 16 2311 1340489 N/A N/A
6388 6403 GATGATTCTGACTCAT 55 2312 1340498 N/A N/A 20470 20485
TCATTAGGAAAAGTGT 16 2313 1340521 N/A N/A 8593 8608 AAAGGTGGGATAAACT
33 2314 1340534 N/A N/A 10187 10202 GCCAACACTAATCTCA 40 2315 1340545 N/A N/A
20117 20132 ACAGCATCAAGACATT 53 2316 1340547 N/A N/A 4864 4879
TGAAGTGAACGATCC 58 2317 1340554 N/A N/A 9225 9240 ACAAGTGACAGCATTG
44 2318 1340566 700 715 12118 12133 CAGAGACATGAGGTTT 61 2319 1340574 N/A
N/A 9599 9614 TATGTGAAGAGCTGGT 63 2320 1340578 N/A N/A 3678 3693
ATCCCCTAAGTTATTA 13 2321 1340585 N/A N/A 5782 5797 GGATTGCAGAAGAGTA
76 2322 1340610 N/A N/A 20629 20644 TGACAACAGAGTTCTG 11 2323 1340628 N/A N/A
19585 19600 TAAACTAAAGTACTTG 1 2324 1340631 N/A N/A 4071 4086
TGGCCTTACCAGAATT 46 2325 1340634 N/A N/A 7127 7142 GACATGAAGATTAGTT
3 2326 1340638 N/A N/A 9727 9742 CTGTGAGAGGATAGTG 33 2327 1340645 1352
1367 21167 21182 ATTCTGTGGCATGGCT 69 2328 1340668 2242 2257 22057 22072
CCATGCAAAAGCATT 72 2329 1340701 N/A N/A 3544 3559 AATCGCTGCTTAGTTC
41 2330 1340740 N/A N/A 19999 20014 CAGTTTTCTCATGAT 65 2331 1340777 N/A N/A
14195 14210 ATGTGATTGAGTTCTC 36 2332 1340783 N/A N/A 9915 9930
CTATCTCCCTATTTGT 15 2333 1340821 N/A N/A 7942 7957 GATCCATATTCCTGA 74
2334 1340822 N/A N/A 5281 5296 CTGAGCTAGACAATTG 56 2335 1340832 N/A N/A
8668 8683 TGTATGAGGTCTCTCA 66 2336 1340873 N/A N/A 6221 6236
CCAAACTTGCCCATAA 45 2337 1340879 N/A N/A 5161 5176 AGAGCCAAACTGCTAC
55 2338 1340894 N/A N/A 9372 9387 TAGAACACTTGCCTCA 31 2339 1340895 N/A N/A
9953 9968 GAGTTCTACACCACAC 76 2340 1340901 N/A N/A 16332 16347
CTACTTCACTATATTG 33 2341 1340902 N/A N/A 6628 6643 GCTGTACAGTAGGCAG
66 2342 1340910 N/A N/A 12722 12737 GATTAATTCCCTTCCA 30 2343
(468) TABLE-US-00033 TABLE 33 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448
AGTAGATGTAAGTCA 85 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT
79 1526 1340100 N/A N/A 19467 19482 CTGGAAGGAGGATTCA 39 2344 1340117 N/A N/A
12419 12434 GTAACGGTGATCAAAT 34 2345 1340119 N/A N/A 9971 9986
TTGAGTAGTGTAAGCT 52 2346 1340123 N/A N/A 9602 9617 ACATATGTGAAGAGCT
66 2347 1340127 1185 1200 21000 21015 CAGAAGTTTTTAAGAG 30 2348 1340135 N/A N/A
19847 19862 GAACCTATGTGTGCAC 53 2349 1340145 N/A N/A 6391 6406
CTTGATGATTCTGACT 57 2350 1340189 N/A N/A 20479 20494 TAGTGGACTTCATTAG 52
2351 1340190 N/A N/A 14921 14936 CTATAATACATTAGCA 3 2352 1340193 N/A N/A 14618
14633 CCCGAATTATTTCTTG 34 2353 1340196 N/A N/A 16225 16240
CTCTTTTTATAGACTG 63 2354 1340205 N/A N/A 19586 19601 GTAAACTAAAGTACTT 0
2355 1340212 N/A N/A 10930 10945 ATAAGGCCGATTCATG 38 2356 1340218 N/A N/A 20001
20016 ATCAGTTTTCTCATG 48 2357 1340228 N/A N/A 20361 20376
CTAAATTATTGCCTGA 39 2358 1340229* N/A N/A 20685 20700 GAAGAAACCTGTACAA 85
2359 1340234 N/A N/A 16337 16352 ATTAAGTACTTCACTA 14 2360 1340238 N/A N/A 11855
11870 TAAGCAGAATTGTGAA 35 2361 1340244 N/A N/A 16968 16983
CCAATATATACTGACA 68 2362 1340246 N/A N/A 13959 13974 GTCTACCTCTAAGTTA 3
2363 1340258 N/A N/A 11570 11585 CTTGACAATGGTTGAT 54 2364 1340266 N/A N/A 18748
18763 AATATATACCTGTTGC 38 2365 1340271 N/A N/A 17990 18005 TTAGTATAGTTATCTT
37 2366 1340309 N/A N/A 14196 14211 TATGTGATTGAGTTCT 28 2367 1340311 N/A N/A

18178 18193 CTCGCGAGATGATT 40 2368 1340339 N/A N/A 6467 6482
 TAGGCAGTAAAGTGCT 48 2369 1340345 N/A N/A 18092 18107 TGATACACCAATGCAG 44
 2370 1340348 N/A N/A 5283 5298 AACTGAGCTAGACAAT 44 2371 1340349 N/A N/A
 6225 6240 TGTACCAAACCTTGCCC 53 2372 1340354 N/A N/A 9148 9163
 ATGCCAAGCTTTGTGT 50 2373 1340357 N/A N/A 3375 3390 GTTGGAAGATGTATAC
 53 2374 1340402 N/A N/A 9958 9973 GCTGAGAGTTCTACAC 45 2375 1340406 N/A N/A
 8594 8609 TAAAGGTGGGATAAAC 0 2376 1340428 N/A N/A 9921 9936
 AATTCACCTATCTCCCT 10 2377 1340440 1360 1375 21175 21190 TTGTTGATATTCTGTG 52
 2378 1340449 N/A N/A 5198 5213 TGAACCGTATTCCCAA 76 2379 1340466 N/A N/A
 6638 6653 TTGGTTGGAGGCTGTA 60 2380 1340492 N/A N/A 15497 15512
 GCTGCATTAATGCCAC 46 2381 1340501 N/A N/A 6742 6757 TGCAATGAGATTACCC
 54 2382 1340504 N/A N/A 12500 12515 CAGAAATTCACCTTGA 28 2383 1340505 N/A N/A
 6302 6317 TCTAGGAGATATAACA 48 2384 1340506 N/A N/A 12758 12773
 GAGAATTGTTTAGTTC 47 2385 1340548 N/A N/A 5715 5730 CCAGATTGCACAAACC
 72 2386 1340553 N/A N/A 10188 10203 TGCCAACACTAATCTC 30 2387 1340571 N/A N/A
 9498 9513 TGGAAGATTAATCATA 12 2388 1340582 2243 2258 22058 22073
 TCCATGCAAAAGCATT 62 2389 1340626 N/A N/A 5657 5672 AAGTTACCAGAGCATT
 70 2390 1340653 N/A N/A 5984 5999 GGATACCTGAAAAGCC 68 2391 1340655 701
 716 12119 12134 GCAGAGACATGAGGTT 55 2392 1340657 N/A N/A 4076 4091
 TTTCATGGCCTTACCA 52 2393 1340678 N/A N/A 7130 7145 TTGGACATGAAGATTA
 18 2394 1340692 N/A N/A 3756 3771 AGGCACTAAAGGTTTC 61 2395 1340693 N/A N/A
 4476 4491 GGACAAATGCCTGAGA 52 2396 1340695 N/A N/A 13689 13704
 GTTACTCACTTCTGG 29 2397 1340716 N/A N/A 8168 8183 GAGTTATTGTAGGGAT
 69 2398 1340723 N/A N/A 8669 8684 TTGTATGAGGTCTCTC 50 2399 1340750 N/A N/A
 7682 7697 GAAATTTGTGAACCTG 32 2400 1340759 N/A N/A 9755 9770
 CTACCTCCAAATTCCC 27 2401 1340763 N/A N/A 7947 7962 GAAGTGATCCATATTC
 44 2402 1340786 N/A N/A 12890 12905 TCTTTAGTCAACAGTA 68 2403 1340788 N/A N/A
 18667 18682 CTGGGATACTATTCTC 57 2404 1340800 N/A N/A 14388 14403
 TAGGCTTAGTGTAGAA 38 2405 1340805 N/A N/A 9319 9334 CCAAGTAATTACTTCT
 68 2406 1340807 1471 1486 21286 21301 AAGGAAGAGGCTAGGG 57 2407 1340808 N/A N/A
 5804 5819 TCCCCTCAAATAGCCT 60 2408 1340823 N/A N/A 20119 20134
 TTACAGCATCAAGACA 46 2409 1340842 N/A N/A 16124 16139 AATAATAGCTCTATTG 0
 2410 1340846 N/A N/A 8981 8996 TGTCCAGAAGTATGTG 0 2411 1340857 N/A N/A
 5162 5177 AAGAGCCAAACTGCTA 63 2412 1340859 N/A N/A 3568 3583
 TCAACCTGCACACCAT 72 2413 1340893 N/A N/A 5876 5891 GTAGGCTAAACCCTGG
 62 2414 1340896 N/A N/A 11290 11305 ATGTTTTGATCCAGGG 46 2415 1340898 1720 1735
 21535 21550 TTGTCTAAACATCTCT 60 2416 1340899 N/A N/A 3679 3694
 CATCCCCTAAGTTATT 24 2417 1340904 N/A N/A 9381 9396 CCATGGATCTAGAACA
 15 2418 1340905 N/A N/A 10345 10360 CGTTAGGTTTCCAAAT 0 2419 1340908 N/A N/A
 5048 5063 TAGAGTGAATCATTCA 75 2420

(469) TABLE-US-00034 TABLE 34 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
 ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
 Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448
 AGTAGATGGTAAGTCA 88 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT
 86 1526 1340080 N/A N/A 10970 10985 TAATCTTGAGGCAGGA 17 2421 1340088 N/A N/A
 12420 12435 GGTAACGGTGATCAAA 51 2422 1340111 N/A N/A 9519 9534
 TCTACACTAATATTGA 21 2423 1340121 N/A N/A 14619 14634 CCCC GAATTATTTCTT 35
 2424 1340132 N/A N/A 16379 16394 CAACCAAACCTTCCAGC 72 2425 1340133 N/A N/A
 12502 12517 TGCAGAAATTCACCTT 54 2426 1340160 N/A N/A 8675 8690

CTTCTATTGTAGG 40 2427 1340175 N/A N/A 11881 TCTTACATGATTAAGC 35
2428 1340198 N/A N/A 15019 15034 TAAAATAGGGTGCAGT 58 2429 1340199 N/A N/A 20120
20135 ATTACAGCATCAAGAC 63 2430 1340200 N/A N/A 14197 14212
TTATGTGATTGAGTTC 34 2431 1340222 734 749 12152 12167 ATTTTTGGTGAACCCA
8 2432 1340243 N/A N/A 13690 13705 CGTTTACTCACTTCTG 21 2433 1340259 N/A N/A
16126 16141 ATAATAATAGCTCTAT 1 2434 1340261 1197 1212 21012 21027
TTTATGTAAGCACAGA 70 2435 1340264 2247 2262 22062 22077 ATAGTCCATGCAAAAG
43 2436 1340269 1371 1386 21186 21201 ATTCTGTGTTCTTGTT 67 2437 1340280 N/A N/A
17993 18008 AAGTTAGTATAGTTAT 20 2438 1340303 N/A N/A 17014 17029
TAGCAATTTAGCAAGA 54 2439 1340307 N/A N/A 11571 11586 TCTTGACAATGGTTGA 75
2440 1340318 N/A N/A 19607 19622 TATTATGCACTCTATA 9 2441 1340338 N/A N/A 18672
18687 TTTTACTGGGATACTA 27 2442 1340347 N/A N/A 9924 9939
TTGAATTCACATCTC 45 2443 1340358 N/A N/A 5199 5214 TTGAACCGTATTCCCA 76
2444 1340363 N/A N/A 9975 9990 ACAGTTGAGTAGTGTA 72 2445 1340396 N/A N/A
20002 20017 AATCAGTTTTCTCAT 53 2446 1340400 N/A N/A 20483 20498
TTCATAGTGGACTTCA 51 2447 1340418 N/A N/A 20378 20393 CACAAGTAAGGTAAAG 54
2448 1340425 N/A N/A 6747 6762 CTAGTTGCAATGAGAT 65 2449 1340431 N/A N/A
10199 10214 GGTAAGCCCCATGCCA 38 2450 1340437 N/A N/A 11307 11322
CACAAGCACACTGTAA 2 2451 1340455 N/A N/A 5716 5731
GCCAGATTGCACAAAC 60 2452 1340462 N/A N/A 16262 16277 GCAGAGACAGTAGATT 63
2453 1340470 N/A N/A 6228 6243 TTATGTACCAAACTTG 53 2454 1340493 N/A N/A
5985 6000 AGGATACCTGAAAAGC 81 2455 1340508 N/A N/A 14392 14407
AACATAGGCTTAGTGT 18 2456 1340510 N/A N/A 4103 4118 ATCCTGTAAACACTTC
79 2457 1340513 N/A N/A 7949 7964 AGGAAGTGATCCATAT 60 2458 1340518 N/A N/A
9004 9019 AGGCAAAGATCTGGCC 7 2459 1340519 N/A N/A 9386 9401
GCACCCCATGGATCTA 35 2460 1340524 1544 1559 21359 21374 TTCAGAGTTATACAGA 66
2461 1340527 N/A N/A 3680 3695 CCATCCCCTAAGTTAT 14 2462 1340544 1022 1037
20837 20852 CTTTGCAGCATTGATT 47 2463 1340561 N/A N/A 3782 3797
GAACTGTGTTGCTTGT 77 2464 1340570 N/A N/A 19887 19902 TACGACAGGTCATCTT 58
2465 1340572 N/A N/A 6426 6441 ATGGTAAACTGTATGC 78 2466 1340575 N/A N/A
9606 9621 ATTCACATATGTGAAG 0 2467 1340588 N/A N/A 4481 4496
TTTTTGGACAAATGCC 50 2468 1340609 N/A N/A 7683 7698 AGAAATTTGTGAACCT
39 2469 1340620 N/A N/A 18963 18978 ATGCAAGATTACTTC 38 2470 1340640 N/A N/A
5049 5064 TTAGAGTGAATCATTC 69 2471 1340642 N/A N/A 6308 6323
TGCTACTCTAGGAGAT 60 2472 1340648 N/A N/A 5809 5824 CAGTATCCCCTCAAAT
53 2473 1340650 1722 1737 21537 21552 AATTGTCTAAACATCT 62 2474 1340670 N/A N/A
8170 8185 CAGAGTTATTGTAGGG 83 2475 1340680 N/A N/A 5169 5184
ATTGCATAAGAGCCAA 67 2476 1340687 519 534 10405 10420
GAAGAAGTGCTTTTGT 50 2477 1340691 N/A N/A 9757 9772 AGCTACCTCCAAATTC
0 2478 1340700 N/A N/A 14077 14092 AACAACCCAGCCTCGG 20 2479 1340711 N/A N/A
9959 9974 AGCTGAGAGTTCTACA 67 2480 1340717 N/A N/A 3569 3584
ATCAACCTGCACACCA 63 2481 1340742 N/A N/A 19472 19487 TATAACTGGAAGGAGG 44
2482 1340744 N/A N/A 12780 12795 TCAAATGAGGCGGCAC 46 2483 1340749 N/A N/A
6710 6725 GGAATGCTCATATTAA 19 2484 1340753 N/A N/A 15498 15513
TGCTGCATTAATGCCA 15 2485 1340756 N/A N/A 5284 5299 TAACTGAGCTAGACAA
53 2486 1340765 N/A N/A 9152 9167 GTACATGCCAAGCTTT 61 2487 1340778 N/A N/A
6468 6483 ATAGGCAGTAAAGTGC 56 2488 1340828 N/A N/A 8600 8615
CTGTCATAAAGGTGGG 59 2489 1340840 N/A N/A 9320 9335 TCCAAGTAATTACTTC
72 2490 1340843 N/A N/A 18505 18520 CTAATTTAGTCAACTT 40 2491 1340848 N/A N/A
5658 5673 TAAGTTACCAGAGCAT 77 2492 1340861 N/A N/A 3429 3444

GATGGTAAAGTCAATA 85 2493 1340871 N/A N/A 7426 7441 TTGGCCGAGGAGGGCG
 3 2494 1340909 N/A N/A 13548 13563 TCTAAAGGGCTGCTCT 33 2495 1340913 N/A N/A
 5883 5898 CGCTACTGTAGGCTAA 68 2496 1340916 N/A N/A 18094 18109
 ATTGATACACCAATGC 66 2497
 (470) TABLE-US-00035 TABLE 35 Inhibition of HSD17B13 RNA by 3-10-3 cEt
 gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
 ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
 Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA 84
 121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT 81 1526 1340081 1028 1043 20843
 20858 ATAAAGCTTTGCAGCA 54 2498 1340090 N/A N/A 9961 9976 TAAGCTGAGAGTTCTA
 64 2499 1340106 N/A N/A 19473 19488 CTATAACTGGAAGGAG 43 2500 1340110 N/A N/A
 17228 17243 TGTACAAATTTATGCC 67 2501 1340114 565 580 10451 10466
 GAAGCCACTGTGACGA 58 2502 1340118 N/A N/A 9608 9623 CAATTCACATATGTGA 11
 2503 1340128 N/A N/A 14461 14476 AGGTATGTCATACTTC 5 2504 1340143 N/A N/A 15021
 15036 GATAAAATAGGGTGCA 52 2505 1340150 N/A N/A 15501 15516
 ATTTGCTGCATTAATG 20 2506 1340165 N/A N/A 14198 14213 ATTATGTGATTGAGTT 46
 2507 1340177 N/A N/A 3682 3697 GCCCATCCCCTAAGTT 33 2508 1340185 N/A N/A 14621
 14636 TGCCCCGAATTATTTC 3 2509 1340191 N/A N/A 19890 19905
 GCTTACGACAGGTCAT 73 2510 1340195 N/A N/A 16127 16142 CATAATAATAGCTCTA 41
 2511 1340210 N/A N/A 18969 18984 TGAATAATGCAAGATT 45 2512 1340223 N/A N/A 13729
 13744 TTAAGTACTGTGAGAC 35 2513 1340254 N/A N/A 13563 13578
 GTTCAACTGAACCTCT 55 2514 1340265 N/A N/A 11897 11912 TGAGAGTATAAATGGC 67
 2515 1340272 N/A N/A 11576 11591 TATTCTCTTGACAATG 47 2516 1340294 N/A N/A 19651
 19666 TTCTATGCAAGCCTTC 51 2517 1340300 N/A N/A 17998 18013
 TTATGAAGTTAGTATA 31 2518 1340302 N/A N/A 18509 18524 GACACTAATTTAGTCA 11
 2519 1340319 2248 2263 22063 22078 GATAGTCCATGCAAAA 64 2520 1340322 735 750
 12153 12168 GATTTTTTGGTGAACCC 30 2521 1340336 N/A N/A 9925 9940
 ATTGAATTCACCTATCT 31 2522 1340372 N/A N/A 5988 6003 GCAAGGATACCTGAAA 67
 2523 1340373 N/A N/A 6309 6324 CTGCTACTCTAGGAGA 66 2524 1340379 N/A N/A 18095
 18110 AATTGATACACCAATG 55 2525 1340382 N/A N/A 3570 3585 TATCAACCTGCACACC
 62 2526 1340384 N/A N/A 18673 18688 ATTTTACTGGGATACT 49 2527 1340394 N/A N/A
 12503 12518 TTGCAGAAATTCACCT 36 2528 1340401 N/A N/A 5664 5679
 AGAGCATAAGTTACCA 88 2529 1340404 N/A N/A 9977 9992 AGACAGTTGAGTAGTG 63
 2530 1340414 N/A N/A 7954 7969 AAAGTAGGAAGTGATC 39 2531 1340422 N/A N/A 5202
 5217 TTTTGAACCGTATTC 56 2532 1340429 N/A N/A 16275 16290
 CAGCAAAGTTTGGGCA 37 2533 1340434 N/A N/A 20121 20136 GATTACAGCATCAAGA 52
 2534 1340435 N/A N/A 3784 3799 TAGAACTGTGTTGCTT 78 2535 1340482 1199 1214 21014
 21029 TGTATTATGTAAGCACA 74 2536 1340496 N/A N/A 8605 8620 GAACACTGTCATAAAG
 43 2537 1340499 N/A N/A 20380 20395 ATCACAAGTAAGGTAA 62 2538 1340500 N/A N/A
 16380 16395 CCAACCAAACCTCCAG 70 2539 1340526 N/A N/A 10974 10989
 ATTGTAATCTTGAGGC 39 2540 1340529 N/A N/A 9158 9173 CTCCTGGTACATGCCA 47
 2541 1340538 N/A N/A 6428 6443 TAATGGTAAACTGTAT 32 2542 1340584 N/A N/A 6748
 6763 CCTAGTTGCAATGAGA 77 2543 1340587 N/A N/A 12422 12437
 GTGGTAACGGTGATCA 49 2544 1340590 N/A N/A 6711 6726 GGGAATGCTCATATTA 18
 2545 1340594 N/A N/A 9057 9072 TGTCAGAGACTCTGTG 39 2546 1340607 N/A N/A 7708
 7723 CGTCCTAGGAAACAAA 45 2547 1340619 N/A N/A 8236 8251 CTACATGACCTGGGTC
 59 2548 1340630 N/A N/A 4133 4148 GGAGTACTTCACAATT 79 2549 1340646 N/A N/A 5170
 5185 GATTGCATAAGAGCCA 82 2550 1340654 N/A N/A 5810 5825 TCAGTATCCCCTCAA
 64 2551 1340677 N/A N/A 8706 8721 GGACCAATATAATCCA 20 2552 1340706 N/A N/A 9323
 9338 ACATCCAAGTAATTAC 56 2553 1340731 N/A N/A 5717 5732 TGCCAGATTGCACAAA

54 2554 1340737 N/A N/A 20486 20501 AACTTCATAGTGGACT 68 2555 1340741 N/A N/A
14165 14180 GCAAGCCAACAGAGAG 52 2556 1340748 N/A N/A 11384 11399
TTTAGTCAGGTAGAGT 30 2557 1340751 N/A N/A 9411 9426 TGTATAGCTGCATTTC 37
2558 1340772 N/A N/A 5886 5901 TTTCGCTACTGTAGGC 62 2559 1340780 N/A N/A 5054
5069 CAGCCTTAGAGTGAAT 82 2560 1340782 N/A N/A 4537 4552 GAGATTTGAAGGTTAG
82 2561 1340804 N/A N/A 20005 20020 AGTAATCAGTTTTTCCT 65 2562 1340811 1546 1561
21361 21376 ACTTCAGAGTTATACA 35 2563 1340815 N/A N/A 7473 7488
CATTCGGCTAGGCGCG 46 2564 1340816 N/A N/A 9759 9774 CAAGCTACCTCCAAAT 25
2565 1340817 1372 1387 21187 21202 CATTCTGTGTTCTTGT 60 2566 1340820 N/A N/A 3432
3447 GTAGATGGTAAGTCAA 86 2567 1340826 1723 1738 21538 21553
AAATTGTCTAAACATC 24 2568 1340847 N/A N/A 6246 6261 ACTGGATGGATTTCTC 83
2569 1340854 N/A N/A 10201 10216 ATGGTAAGCCCCATGC 22 2570 1340855 N/A N/A 6471
6486 GTAATAGGCAGTAAAG 77 2571 1340866 N/A N/A 12794 12809
AAGAGTCAGTATCCTC 63 2572 1340877 N/A N/A 9521 9536 ATTCTACACTAATATT 0 2573
1340880 N/A N/A 5285 5300 ATAAGTGAAGCTAGACA 71 2574
(471) TABLE-US-00036 TABLE 36 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA 89
121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT 91 1526 1340087 N/A N/A 11578
11593 GTTATTCTCTTGACAA 42 2575 1340089 N/A N/A 11388 11403
AACTTTTAGTCAGGTA 49 2576 1340116 N/A N/A 16381 16396 CCAACCAAACCTTCCA 62
2577 1340124 603 618 10489 10504 ATGGGATGAGGTAAGG 46 2578 1340136 N/A N/A 13782
13797 ACAATCAGGAGTTGCA 62 2579 1340153 N/A N/A 5223 5238
TTAGCTTAAACTCCAA 78 2580 1340162 N/A N/A 12504 12519 CTTGCAGAAATTCACC 40
2581 1340188 N/A N/A 19489 19504 CATCTAGAATTAAGGG 53 2582 1340203 N/A N/A 18690
18705 CTTTGAGACTCTTGTT 61 2583 1340207 N/A N/A 18002 18017
AGGCTTATGAAGTTAG 57 2584 1340216 N/A N/A 18971 18986 GTTGACTAATGCAAGA 59
2585 1340225 N/A N/A 18096 18111 AAATTGATACACCAAT 14 2586 1340227 N/A N/A 19891
19906 TGCTTACGACAGGTCA 81 2587 1340232 N/A N/A 19656 19671
ATCCATTCTATGCAAG 46 2588 1340240 1397 1412 21212 21227 AAATTGATCTCTTAG 61
2589 1340248 1726 1741 21541 21556 CTAAAATTGTCTAAAC 35 2590 1340262 N/A N/A
12423 12438 GGTGGTAACGGTGATC 31 2591 1340281 N/A N/A 20122 20137
TGATTACAGCATCAAG 69 2592 1340285 2249 2264 22064 22079 GGATAGTCCATGCAAA
64 2593 1340293 1597 1612 21412 21427 TCTTATAAGACTATAA 17 2594 1340299 N/A N/A
14905 14920 AGCTAAGAGACACTTC 41 2595 1340313 N/A N/A 13568 13583
GAACAGTTCAACTGAA 10 2596 1340326 1054 1069 20869 20884 TATTATCAGGACTGAA
63 2597 1340328 N/A N/A 20382 20397 TAATCACAAGTAAGGT 54 2598 1340331 N/A N/A
8707 8722 GGGACCAATATAATCC 31 2599 1340351 N/A N/A 9194 9209
ACAACCCTTAACAAAC 26 2600 1340371 N/A N/A 6479 6494 CCAACTATGTAATAGG 68
2601 1340378 N/A N/A 9982 9997 TATTAAGACAGTTGAG 44 2602 1340407 N/A N/A 4140
4155 TTCTCAAGGAGTACTT 76 2603 1340417 N/A N/A 9422 9437 TATATGAGGTCTGTAT
42 2604 1340436 N/A N/A 9765 9780 CCAACCCAAGCTACCT 15 2605 1340441 N/A N/A 5812
5827 AGTCAGTATCCCCTCA 82 2606 1340442 N/A N/A 5718 5733 CTGCCAGATTGCACAA
44 2607 1340485 N/A N/A 5058 5073 ATTACAGCCTTAGAGT 56 2608 1340486 N/A N/A 3571
3586 CTATCAACCTGCACAC 39 2609 1340487 N/A N/A 6749 6764 GCCTAGTTGCAATGAG
61 2610 1340509 N/A N/A 7476 7491 TAACATTTCGGCTAGGC 52 2611 1340511 N/A N/A 3933
3948 CTGCGCACATATTTTA 92 2612 1340516 N/A N/A 5667 5682 TAGAGAGCATAAGTTA
59 2613 1340523 N/A N/A 9927 9942 TGATTGAATTCACCTAT N.D. 2614 1340536 N/A N/A
11924 11939 CCAAAAATCAGCCACT 0 2615 1340550 N/A N/A 7768 7783

CGACGTAACCAAT 70 2616 1340557 N/A N/A 17672 AACCTTTGTTGCCTGG 36
2617 1340559 N/A N/A 14462 14477 GAGGTATGTCATACTT 29 2618 1340562 N/A N/A 15510
15525 TATTCCAGGATTTGCT 43 2619 1340564 N/A N/A 3435 3450 TAAGTAGATGGTAAGT
74 2620 1340581 N/A N/A 8613 8628 GGCCAAGGGAACACTG 36 2621 1340589 1200 1215
21015 21030 ATGTTTATGTAAGCAC 70 2622 1340591 N/A N/A 3693 3708
ATCCCGTTCTTGCCCA 30 2623 1340593 N/A N/A 12169 12184 CCTTACCTTGTGCTTG 20
2624 1340595 N/A N/A 8017 8032 ATCTCACAAGGGAAAT 73 2625 1340596 N/A N/A 6435
6450 CTAGTGGTAATGGTAA 62 2626 1340602 N/A N/A 10206 10221
TTGGTATGGTAAGCCC 62 2627 1340605 N/A N/A 5286 5301 TATAACTGAGCTAGAC 53
2628 1340621 N/A N/A 9327 9342 ATATACATCCAAGTAA 18 2629 1340622 N/A N/A 6247
6262 CACTGGATGGATTTCT 72 2630 1340641 N/A N/A 5176 5191 AGTTTAGATTGCATAA
50 2631 1340644 N/A N/A 16282 16297 TAGTCTTCAGCAAAGT 49 2632 1340652 N/A N/A
14168 14183 ATAGCAAGCCAACAGA 37 2633 1340675 N/A N/A 14201 14216
ATTATTATGTGATTGA 39 2634 1340682 N/A N/A 9525 9540 CTAGATTCTACACTAA 17 2635
1340688 N/A N/A 9629 9644 TAGTTTGGTGGGCATG 28 2636 1340699 N/A N/A 20487 20502
TAACCTTCATAGTGGAC 76 2637 1340707 N/A N/A 16130 16145 CTCCATAATAATAGCT 59
2638 1340710 N/A N/A 9962 9977 GTAAGCTGAGAGTTCT 49 2639 1340736 N/A N/A 4640
4655 TACGACTTCCTTCTAA 67 2640 1340774 N/A N/A 15212 15227
CCATAAAGCTGGATTG 40 2641 1340779 N/A N/A 5992 6007 AGAGGCAAGGATACCT 78
2642 1340797 N/A N/A 18517 18532 AGTACCAAGACACTAA 66 2643 1340814 N/A N/A 8237
8252 ACTACATGACCTGGGT 54 2644 1340818 N/A N/A 9077 9092 ACTTGAATTCTGTGTC
52 2645 1340849 N/A N/A 6316 6331 GACCATGCTGCTACTC 74 2646 1340853 N/A N/A
10978 10993 CAGCATTGTAATCTTG 43 2647 1340858 N/A N/A 6715 6730
GCCAGGGAATGCTCAT 38 2648 1340860 N/A N/A 12796 12811 GTAAGAGTCAGTATCC 47
2649 1340886 N/A N/A 5888 5903 TATTTGCTACTGTAG 48 2650 1340887 N/A N/A 20009
20024 AGAGAGTAATCAGTTT 39 2651
(472) TABLE-US-00037 TABLE 37 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA 83
121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT 76 1526 1340073 N/A N/A 12797
12812 TGTAAGAGTCAGTATC 43 2652 1340074 1203 1218 21018 21033
AGTATGTTTATGTAAG 48 2653 1340084 N/A N/A 18692 18707 GACTTTGAGACTCTTG 65
2654 1340091 N/A N/A 9333 9348 GATCCTATATACATCC 49 2655 1340101 N/A N/A 11391
11406 TGCAACTTTTAGTCAG 41 2656 1340103 N/A N/A 16383 16398
ATCCCAACCAAACTTC 15 2657 1340140 605 620 10491 10506 ATATGGGATGAGGTAA 37
2658 1340159 N/A N/A 6029 6044 TTCCGTCTCACAATCT 79 2659 1340174 N/A N/A 13571
13586 TCAGAACAGTTCAACT 30 2660 1340178 1598 1613 21413 21428
ATCTTATAAGACTATA 32 2661 1340202 N/A N/A 19658 19673 GAATCCATTCTATGCA 47
2662 1340217 N/A N/A 11579 11594 AGTTATTCTCTTGACA 48 2663 1340236 N/A N/A 20491
20506 GACATAACTTCATAGT 39 2664 1340237 N/A N/A 16131 16146
GCTCCATAATAATAGC 21 2665 1340250 N/A N/A 12170 12185 ACCTTACCTTGTGCTT 0
2666 1340252 N/A N/A 18518 18533 TAGTACCAAGACACTA 27 2667 1340253 N/A N/A 18003
18018 AAGGCTTATGAAGTTA 61 2668 1340317 N/A N/A 18098 18113
TGAAATTGATACACCA 58 2669 1340376 N/A N/A 5304 5319 CGGCCTATTCTTCTGT 25
2670 1340389 N/A N/A 7478 7493 ATTAACATTTCGGCTAG 17 2671 1340392 N/A N/A 19498
19513 AATATCAGCCATCTAG 24 2672 1340410 1074 1089 20889 20904
TGCCAAACCAATGTTT 32 2673 1340413 N/A N/A 9195 9210 GACAACCCTTAACAAA 24
2674 1340447 N/A N/A 3436 3451 ATAAGTAGATGGTAAG 58 2675 1340461 N/A N/A 17676
17691 TCAAAACCTTTGTTGC 6 2676 1340463 N/A N/A 6716 6731 TGCCAGGGAATGCTCA

23 2677 1340464 N/A N/A 9647 9662 ATGTATGAGGTCTCCG 63 2678 1340477 N/A N/A 13824
13839 ACCAAGAGGGTTTTTA 7 2679 1340495 N/A N/A 4648 4663 ATTTTAAATACGACTT
35 2680 1340497 N/A N/A 9963 9978 TGTAAGCTGAGAGTTC 45 2681 1340507 N/A N/A 9766
9781 CCCAACCCAAGCTACC 22 2682 1340517 N/A N/A 18983 18998
CACTTTGGTATTGTTG 53 2683 1340528 N/A N/A 8636 8651 GACTGTTGAGCTCAAA 52
2684 1340537 N/A N/A 5669 5684 AGTAGAGAGCATAAGT 52 2685 1340540 N/A N/A 8040
8055 GACTACATTAACACCA 58 2686 1340541 N/A N/A 6438 6453 AAAGTGTGGTAATGG
47 2687 1340568 N/A N/A 20123 20138 CTGATTACAGCATCAA 40 2688 1340573 N/A N/A
6762 6777 CTTGAGAGTGATTGCC 64 2689 1340579 2250 2265 22065 22080
AGGATAGTCCATGCAA 64 2690 1340592 N/A N/A 10218 10233 GATACAAATTTATTGG 33
2691 1340598 N/A N/A 5720 5735 CACTGCCAGATTGCAC 38 2692 1340608 N/A N/A 3575
3590 CTTTCTATCAACCTGC 68 2693 1340615 N/A N/A 14475 14490
AGTATGACAACCTGGAG 15 2694 1340616 N/A N/A 20046 20061 CGTAACCATGCATTAA 68
2695 1340618 N/A N/A 6248 6263 CCACTGGATGGATTTC 68 2696 1340624 N/A N/A 7772
7787 GTTAGCAGCTAAAATC 45 2697 1340635 N/A N/A 5814 5829 AAAGTCAGTATCCCCT
77 2698 1340649 N/A N/A 4222 4237 TCAACTAAACATGACA 77 2699 1340660 N/A N/A 9933
9948 GAATCATGATTGAATT 39 2700 1340672 N/A N/A 3695 3710 AGATCCCGTTCTTGCC
68 2701 1340676 N/A N/A 5182 5197 CGCAACAGTTTAGATT 57 2702 1340703 N/A N/A
14911 14926 TTAGCAAGCTAAGAGA 42 2703 1340708 N/A N/A 12424 12439
GGGTGGTAACGGTGAT 46 2704 1340712 N/A N/A 3952 3967 CAAGGAAAAGCCTGAC 59
2705 1340715 N/A N/A 8256 8271 CTGGTTTCATAACCTA 9 2706 1340718 N/A N/A 9423 9438
TTATATGAGGTCTGTA 32 2707 1340730 N/A N/A 8709 8724 TTGGGACCAATATAAT 18 2708
1340738 N/A N/A 11965 11980 TGTGAGAAGGCGGTAT 20 2709 1340743 N/A N/A 5890 5905
TGTATTTTCGCTACTGT 76 2710 1340754 N/A N/A 16284 16299 AATAGTCTTCAGCAAA 48
2711 1340758 N/A N/A 19904 19919 AAGATACCCAGGTTGC 59 2712 1340784 N/A N/A 5226
5241 GCTTTAGCTTAACTC 72 2713 1340787 N/A N/A 9081 9096 GTCCACTTGAATTCTG
18 2714 1340799 N/A N/A 5066 5081 TGAGTTACATTACAGC 90 2715 1340825 1748 1763
21563 21578 GTGTTAGCTTTAATTT 48 2716 1340844 1398 1413 21213 21228
GAAACTTGATCTCTTA 64 2717 1340851 N/A N/A 9528 9543 GCCCTAGATTCTACAC 15
2718 1340868 N/A N/A 15214 15229 TTCCATAAAGCTGGAT 5 2719 1340869 N/A N/A 11073
11088 TACAACCTGGTTTCAT 25 2720 1340881 N/A N/A 6321 6336 AGGGAGACCATGCTGC
79 2721 1340888 N/A N/A 6480 6495 ACCAACTATGTAATAG 47 2722 1340890 N/A N/A 14204
14219 CAGATTATTATGTGAT 51 2723 1340900 N/A N/A 9989 10004 GTAAGTGTATTAAAGAC
54 2724 1340906 N/A N/A 12506 12521 TTCTTGCAGAAATTCA 42 2725 1340912 N/A N/A
14169 14184 AATAGCAAGCCAACAG 9 2726 1340914 N/A N/A 20384 20399
AGTAATCACAAGTAAG 54 2727 1340915 794 809 15675 15690 ATCTATCAGACTTCTT 47
2728

(473) TABLE-US-00038 TABLE 38 Inhibition of HSD17B13 RNA by 3-10-3 cEt
gapmers targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID NO:
ID NO: ID NO: 1 1 2 2 SEQ Compound Start Stop Start Stop % ID Number Site Site Site Site
Sequence (5' to 3') Inhibition NO 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA 84
121 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT 91 1526 1340076 N/A N/A 12426
12441 CAGGGTGGTAACGGTG 55 2729 1340079 N/A N/A 12191 12206
ACCCATTCTAACTTGA 0 2730 1340105 N/A N/A 13825 13840 TACCAAGAGGGTTTTT 17
2731 1340107 N/A N/A 12798 12813 CTGTAAGAGTCAGTAT 42 2732 1340112 N/A N/A 20493
20508 TTGACATAACTTCATA 15 2733 1340115 N/A N/A 12003 12018
TGTTAGCTAAGGGAGA 0 2734 1340139 N/A N/A 17768 17783 CAGCATATTCATTG 49
2735 1340147 N/A N/A 58315846 TGTTAGGACCCAGTCT 51 2736 1340161 N/A N/A 6764
6779 TTCTTGAGAGTGATTG 52 2737 1340167 N/A N/A 4659 4674 GCCCTTAAGTCATTTT
25 2738 1340173 N/A N/A 11075 11090 ATTACAACCTGGTTTC 50 2739 1340184 N/A N/A

14913 14928 CATTAGCAAGCTAAGA 0 2740 1340186 N/A N/A 13576 13591
CACTATCAGAACAGTT 24 2741 1340214 N/A N/A 14173 14188 CTTGAATAGCAAGCCA 25
2742 1340219 N/A N/A 18519 18534 TTAGTACCAAGACACT 41 2743 1340224 N/A N/A 20109
20124 AAGACATTCTAGCCTG 63 2744 1340242 N/A N/A 12616 12631
ATGGCTTAATTTTCGG 33 2745 1340251 N/A N/A 20190 20205 AGTTATTAGAAGTCAG 54
2746 1340257 N/A N/A 20386 20401 ATAGTAATCACAAGTA 56 2747 1340267 1749 1764
21564 21579 TGTGTTAGCTTTAATT 38 2748 1340296 N/A N/A 14476 14491
GAGTATGACAACCTGGA 23 2749 1340298 808 823 15689 15704 TTGGTAAGTATTCCAT
20 2750 1340316 1175 1190 20990 21005 TAAGAGGCATGAAAGG 52 2751 1340325 N/A N/A
8713 8728 AGTTTTGGGACCAATA 31 2752 1340343 N/A N/A 11580 11595
GAGTTATTCTCTTGAC 57 2753 1340344 N/A N/A 7479 7494 AATTAACATTCGGCTA 38
2754 1340365 N/A N/A 16384 16399 AATCCCAACCAAACCTT 23 2755 1340374 N/A N/A 19673
19688 CAAGAGACACCACCAG 50 2756 1340381 N/A N/A 6439 6454
TAAACTAGTGGTAATG 14 2757 1340383 N/A N/A 5894 5909 CCTTTGTATTTCGCTA 47
2758 1340408 N/A N/A 6326 6341 TTCATAGGGAGACCAT 65 2759 1340421 N/A N/A 9964
9979 GTGTAAGCTGAGAGTT 60 2760 1340424 N/A N/A 10499 10514
ACTTACCAATATGGGA 0 2761 1340457 N/A N/A 15247 15262 ATAGCTAAGTATACTT 0
2762 1340480 N/A N/A 18102 18117 ATACTGAAATTGATAC 11 2763 1340481 N/A N/A 9651
9666 GCTAATGTATGAGGTC 78 2764 1340483 N/A N/A 9112 9127 AGACCAAGTAGCTTAC
42 2765 1340488 N/A N/A 5070 5085 GATCTGAGTTACATTA 69 2766 1340502 N/A N/A 9999
10014 GAAATCTTGTGTAAC 69 2767 1340535 N/A N/A 9198 9213 ACTGACAACCCTTAAC
10 2768 1340551 N/A N/A 16285 16300 GAATAGTCTTCAGCAA 51 2769 1340569 N/A N/A
8050 8065 TTCTACAGAAGACTAC 40 2770 1340583 N/A N/A 9432 9447
ACTAACCAATTATATG 6 2771 1340611 N/A N/A 16132 16147 TGCTCCATAATAATAG 35
2772 1340623 1603 1618 21418 21433 AATGTATCTTATAAGA 10 2773 1340632 N/A N/A
18998 19013 AGTCATATTGAAAATC 39 2774 1340637 N/A N/A 7773 7788
AGTTAGCAGCTAAAAT 33 2775 1340639 N/A N/A 5185 5200 CAACGCAACAGTTTAG 52
2776 1340656 N/A N/A 10258 10273 AGCATTCCATGATTAA 47 2777 1340658 N/A N/A 18005
18020 CCAAGGCTTATGAAGT 27 2778 1340662 N/A N/A 8279 8294
TCTAAAGTGCTGGTTG 34 2779 1340663 N/A N/A 6482 6497 GCACCAACTATGTAAT 53
2780 1340666 N/A N/A 3697 3712 TCAGATCCCGTTCTTG 59 2781 1340702 N/A N/A 3584
3599 CTCCTGAGTCTTTCTA 23 2782 1340714 N/A N/A 5227 5242 TGCTTTAGCTTAAACT
51 2783 1340720 N/A N/A 19531 19546 GCTAAAATGGTCATCT 40 2784 1340721 2356 2371
22171 22186 TTTATAACTACAAGAG 38 2785 1340732 N/A N/A 8639 8654
ACAGACTGTTGAGCTC 53 2786 1340733 1254 1269 21069 21084 TTTTGTCCACCTTTAA 68
2787 1340735 N/A N/A 9334 9349 CGATCCTATATACATC 33 2788 1340745 N/A N/A 9767
9782 TCCCAACCCAAGCTAC 8 2789 1340760 N/A N/A 19905 19920
TAAGATACCCAGGTTG 39 2790 1340764 N/A N/A 3478 3493 GACCAGGGAATTTATC 41
2791 1340766 N/A N/A 9934 9949 TGAATCATGATTGAAT 38 2792 1340793 N/A N/A 5646
5661 GCATTCATCAGATGTT 89 2793 1340794 N/A N/A 6717 6732 CTGCCAGGGAATGCTC
23 2794 1340796 N/A N/A 6033 6048 ACTTTTCCGTCTCACA 89 2795 1340806 N/A N/A 5671
5686 TCAGTAGAGAGCATAA 52 2796 1340809 1417 1432 21232 21247
TGAGATAAAGCTGCCT 63 2797 1340827 N/A N/A 5732 5747 TGGTAGCTTGCTCACT 71
2798 1340829 N/A N/A 6249 6264 GCCACTGGATGGATTT 36 2799 1340856 N/A N/A 11392
11407 CTGCAACTTTTAGTCA 21 2800 1340862 N/A N/A 9530 9545 CTGCCCTAGATTCTAC
33 2801 1340867 N/A N/A 14263 14278 GAGTTAGGGAGCCAGC 39 2802 1340882 N/A N/A
4223 4238 TTCAACTAAACATGAC 55 2803 1340897 N/A N/A 18693 18708
AGACTTTGAGACTCTT 57 2804 1340911 N/A N/A 3988 4003 GTTTACAAGTAAGAAC 42
2805
(474) Modified oligonucleotides complementary to an HSD17B13 nucleic acid with other

1341228 1333 1348 21148 21163 GATTGGAATGCTACTT
G.sub.ksA.sub.ksT.sub.dsT.sub.dsG.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.ds 53 1435
G.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.es.sup.mC.sub.ksT.sub.esT.sub.k 1341234 1356 1371 21171
21186 TGATATTCTGTGGCAT
T.sub.ksG.sub.ksA.sub.dsT.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds 46 1124
G.sub.dsT.sub.dsG.sub.ksG.sub.es.sup.mC.sub.ksA.sub.esT.sub.k 1341236 1352 1367 21167 21182
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A.sub.ksT.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.dsG.sub.ds 55 2328
.sup.mC.sub.dsA.sub.dsT.sub.ksG.sub.esG.sub.ks.sup.mC.sub.esT.sub.k 1341239 1340 1355 21155
21170 GGCTACAGATTGGAAT
G.sub.ksG.sub.ks.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds 73 1980
A.sub.dsT.sub.dsT.sub.dsG.sub.ksG.sub.esA.sub.ksA.sub.esT.sub.k 1341241 1387 1402 21202
21217 TCTTAGCTGTGCACTC
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.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ksA.sub.es.sup.mC.sub.ksT.sub.es.sup.mC.sub.k 1341247
1364 1379 21179 21194 GTTCTTGTTGATATTC
G.sub.ksT.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsT.sub.dsT.sub.ds 66 1436
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21235 21250 GGTTGAGATAAAGCTG
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21319 CGTTTTGGGCTAATGA
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21323 GCACCGTTTTGGGCTA
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21312 21327 AGTTGCACCGTTTTGG
A.sub.ksG.sub.ksT.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.m 41 1905
C.sub.dsG.sub.dsT.sub.dsT.sub.ksT.sub.esT.sub.ksG.sub.esG.sub.k 1341280 1622 1637 21437
21452 AGAGTCGGTCACCTTT
A.sub.ksG.sub.ksA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsG.sub.dsG.sub.ds 61 428
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1630 1645 21445 21460 TTAAATAGAGTCGG
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G.sub.dsA.sub.dsG.sub.ksT.sub.es.sup.mC.sub.ksG.sub.esG.sub.k 1341287 1618 1633 21433
21448 TCGGTCACCTTTCATA
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.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ks.sup.mC.sub.esA.sub.ksT.sub.esA.sub.k 1341290 1626
1641 21441 21456 AAATAGAGTCGGTCAC
A.sub.ksA.sub.ksA.sub.dsT.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ds 50 740
.sup.mC.sub.dsG.sub.dsG.sub.ksT.sub.es.sup.mC.sub.ksA.sub.es.sup.mC.sub.k 1341296 1714 1729
21529 21544 AAACATCTCTGGGACC
A.sub.ksA.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.m 36 2129
C.sub.dsT.sub.dsG.sub.dsG.sub.ksG.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341297 1710
1725 21525 21540 ATCTCTGGGACCAAGG
A.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsG.sub.dsG 65 37

.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.esA.sub.ksG.sub.esG.sub.k 1341303 1768
1783 21583 21598 CCAGTACAGTTCCTTT
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds 75 1753
G.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ks.sup.mC.sub.esT.sub.ksT.sub.esT.sub.k 1341305 1750
1765 21565 21580 CTGTGTTAGCTTTAAT
.sup.mC.sub.ksT.sub.ksG.sub.dsT.sub.dsG.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ds 47 1286
.sup.mC.sub.dsT.sub.dsT.sub.ksT.sub.esA.sub.ksA.sub.esT.sub.k 1341313 1778 1793 21593 21608
TATGTAATAGCCAGTA
T.sub.ksA.sub.ksT.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.ds 78 508
G.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.esG.sub.ksT.sub.esA.sub.k 1341315 1782 1797
21597 21612 TTCTTATGTAATAGCC
T.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsT.sub.dsG.sub.dsT.sub.ds 78 819
A.sub.dsA.sub.dsT.sub.ksA.sub.esG.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341325 2247 2262
22062 22077 ATAGTCCATGCAAAAG
A.sub.ksT.sub.ksA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 31 2436
T.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.esA.sub.ksA.sub.esG.sub.k 1341328 2193 2208
22008 22023 TAGTCTTGATGTAGTG
T.sub.ksA.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds 43 1288
T.sub.dsG.sub.dsT.sub.ksA.sub.esG.sub.ksT.sub.esG.sub.k 1341339 N/A N/A 9330 9345
CCTATATACATCCAAG
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.m 61 1771
C.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ks.sup.mC.sub.esA.sub.ksA.sub.esG.sub.k 1341343 N/A
N/A 11571 11586 TCTTGACAATGGTTGA
T.sub.ks.sup.mC.sub.ksT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds 40 2440
A.sub.dsT.sub.dsG.sub.dsG.sub.ksT.sub.esT.sub.ksG.sub.esA.sub.k 1341344 N/A N/A 11855
11870 TAAGCAGAATTGTGAA
T.sub.ksA.sub.ksA.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds 29 2361
A.sub.dsT.sub.dsT.sub.dsG.sub.ksT.sub.esG.sub.ksA.sub.esA.sub.k 1341347 N/A N/A 9964 9979
GTGTAAGCTGAGAGTT
G.sub.ksT.sub.ksG.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds 57 2760
T.sub.dsG.sub.dsA.sub.dsG.sub.ksA.sub.esG.sub.ksT.sub.esT.sub.k 1341351 N/A N/A 12423
12438 GGTGGTAACGGTGATC
G.sub.ksG.sub.ksT.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.ds.sup.m 39 2591
C.sub.dsG.sub.dsG.sub.dsT.sub.ksG.sub.esA.sub.ksT.sub.es.sup.mC.sub.k 1341354 N/A N/A
14914 14929 ACATTAGCAAGCTAAG
A.sub.ks.sup.mC.sub.ksA.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds 23 2065
A.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ksT.sub.esA.sub.ksA.sub.esG.sub.k 1341360 N/A N/A
15494 15509 GCATTAATGCCACCCT
G.sub.ks.sup.mC.sub.ksA.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.ds 24 1012
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.es.sup.mC.sub.ks.sup.mC.sub.esT.sub.k
1341361 N/A N/A 16287 16302 TAGAATAGTCTTCAGC
T.sub.ksA.sub.ksG.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ds 50 1403
.sup.mC.sub.dsT.sub.dsT.sub.ks.sup.mC.sub.esA.sub.ksG.sub.es.sup.mC.sub.k 1341375 N/A N/A
17992 18007 AGTTAGTATAGTTATC
A.sub.ksG.sub.ksT.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.ds 47 1249
A.sub.dsG.sub.dsT.sub.ksT.sub.esA.sub.ksT.sub.es.sup.mC.sub.k 1341377 N/A N/A 16380 16395
CCAACCAAACCTTCCAG
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 15 2539
A.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ks.sup.mC.sub.es.sup.mC.sub.ksA.sub.esG.sub.k
1341382 N/A N/A 19908 19923 TCTTAAGATACCCAGG

T.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsA.sub.dsT.sub.ds 35 319
A.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.esA.sub.ksG.sub.esG.sub.k 1341387 N/A N/A
18693 18708 AGACTTTGAGACTCTT
A.sub.ksG.sub.ksA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds 55 2804
G.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.es.sup.mC.sub.ksT.sub.esT.sub.k 1341389 N/A N/A 19912
19927 TGCATCTTAAGATACC
T.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 24 2050
A.sub.dsA.sub.dsG.sub.dsA.sub.ksT.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341393 N/A
N/A 20381 20396 AATCACAAGTAAGGTA
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G.sub.dsT.sub.dsA.sub.dsA.sub.ksG.sub.esG.sub.ksT.sub.esA.sub.k 1341396 N/A N/A 20116
20131 CAGCATCAAGACATTC
.sup.mC.sub.ksA.sub.ksG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds 62 2191
A.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.esT.sub.ksT.sub.es.sup.mC.sub.k 1341406 N/A
N/A 12503 12518 TTGCAGAAATTCACCT
T.sub.ksT.sub.ksG.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.dsA.sub.ds 46 2528
A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.es.sup.mC.sub.ks.sup.mC.sub.esT.sub.k 1341407
N/A N/A 20482 20497 TCATAGTGGACTTCAT
T.sub.ks.sup.mC.sub.ksA.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds 48 633
G.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.es.sup.mC.sub.ksA.sub.esT.sub.k 1341412 N/A N/A
14196 14211 TATGTGATTGAGTTCT
T.sub.ksA.sub.ksT.sub.dsG.sub.dsT.sub.dsG.sub.dsA.sub.dsT.sub.dsT.sub.ds 23 2367
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G.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.es.sup.mC.sub.ks.sup.mC.sub.esA.sub.k
1341417 N/A N/A 14170 14185 GAATAGCAAGCCAACA
G.sub.ksA.sub.ksA.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ds 27 1708
A.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.esA.sub.ks.sup.mC.sub.esA.sub.k 1341532
805 820 15686 15701 GTAAGTATTCCATCTA
G.sub.ksT.sub.ksA.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.dsT.sub.ds 53 1038
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksT.sub.es.sup.mC.sub.ksT.sub.esA.sub.k 1341534 613 628
N/A N/A CTGGAACAATATGGGA
.sup.mC.sub.ksT.sub.ksG.sub.dsG.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds 33 1969
A.sub.dsT.sub.dsA.sub.dsT.sub.ksG.sub.esG.sub.ksG.sub.esA.sub.k 1341538 1091 1106 20906
20921 CGTTTGACTGCTGCTA
.sup.mC.sub.ksG.sub.ksT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds 48 264
T.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.es.sup.mC.sub.ksT.sub.esA.sub.k 1341541 1320
1335 21135 21150 CTTGAACAGTCTTAAA
.sup.mC.sub.ksT.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds 38 500
G.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.esA.sub.ksA.sub.esA.sub.k 1341543 1308 1323
21123 21138 TAAACCTTCCCTGTGT
T.sub.ksA.sub.ksA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.m 9 1745
C.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.esT.sub.ksG.sub.esT.sub.k 1341547 1076
1091 20891 20906 AGTGCCAAACCAATGT
A.sub.ksG.sub.ksT.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ds 71 1120
A.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.esT.sub.ksG.sub.esT.sub.k 1341551 1347
1362 21162 21177 GTGGCATGGCTACAGA
G.sub.ksT.sub.ksG.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.ds 59 501
G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.esA.sub.ksG.sub.esA.sub.k 1341555 1328

1343 21143 21158 GAATGCTACTTGAACA
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C.sub.dsT.sub.dsT.sub.dsG.sub.ksA.sub.esA.sub.ks.sup.mC.sub.esA.sub.k 1341564 1425 1440
21240 21255 GTCCAGGTTGAGATAA
G.sub.ksT.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.dsT.sub.ds 36 113
T.sub.dsG.sub.dsA.sub.dsG.sub.ksA.sub.esT.sub.ksA.sub.esA.sub.k 1341565 1461 1476 21276
21291 CTAGGGAAATCTTTCA
.sup.mC.sub.ksT.sub.ksA.sub.dsG.sub.dsG.sub.dsG.sub.dsA.sub.dsA.sub.ds 46 659
A.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.esT.sub.ks.sup.mC.sub.esA.sub.k 1341567 1415
1430 21230 21245 AGATAAAGCTGCCTGC
A.sub.ksG.sub.ksA.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.dsG.sub.ds.sup.m 30 1515
C.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ks.sup.mC.sub.esT.sub.ksG.sub.es.sup.mC.sub.k 1341568
1718 1733 21533 21548 GTCTAAACATCTCTGG
G.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds 50 429
A.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ks.sup.mC.sub.esT.sub.ksG.sub.esG.sub.k 1341569 1504
1519 21319 21334 AGAATAGAGTTGCACC
A.sub.ksG.sub.ksA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.ds 59 426
T.sub.dsT.sub.dsG.sub.ks.sup.mC.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341575 1614 1629
21429 21444 TCACCTTTCATAATGT
T.sub.ks.sup.mC.sub.ksA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ds 31 2109
.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.ksA.sub.esT.sub.ksG.sub.esT.sub.k 1341578 2186 2201
22001 22016 GATGTAGTGGGAGTCG
G.sub.ksA.sub.ksT.sub.dsG.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds 38 743
G.sub.dsG.sub.dsA.sub.ksG.sub.esT.sub.ks.sup.mC.sub.esG.sub.k 1341580 2257 2272 22072
22087 AAACAAGAGGATAGTC
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G.sub.dsG.sub.dsA.sub.dsT.sub.ksA.sub.esG.sub.ksT.sub.es.sup.mC.sub.k 1341585 1774 1789
21589 21604 TAATAGCCAGTACAGT
T.sub.ksA.sub.ksA.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 44 196
A.sub.dsG.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.esA.sub.ksG.sub.esT.sub.k 1341588 N/A N/A 9511
9526 AATATTGAGGCACTGG
A.sub.ksA.sub.ksT.sub.dsA.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.ds 29 1538
G.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.esT.sub.ksG.sub.esG.sub.k 1341596 N/A N/A 9598
9613 ATGTGAAGAGCTGGTA
A.sub.ksT.sub.ksG.sub.dsT.sub.dsG.sub.dsA.sub.dsA.sub.dsG.sub.dsA.sub.ds 54 527
G.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.esG.sub.ksT.sub.esA.sub.k 1341603 N/A N/A 16134 16149
GATGCTCCATAATAAT
G.sub.ksA.sub.ksT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 17 2049
A.sub.dsT.sub.dsA.sub.dsA.sub.ksT.sub.esA.sub.ksA.sub.esT.sub.k 1341604 N/A N/A 16124
16139 AATAATAGCTCTATTG
A.sub.ksA.sub.ksT.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.m 2 2410
C.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.esT.sub.ksT.sub.esG.sub.k 1341614 N/A N/A 17987
18002 GTATAGTTATCTTCTC
G.sub.ksT.sub.ksA.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.dsA.sub.ds 49 937
T.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.es.sup.mC.sub.ksT.sub.es.sup.mC.sub.k 1341615 N/A N/A
17997 18012 TATGAAGTTAGTATAG
T.sub.ksA.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.ds 47 1560
A.sub.dsG.sub.dsT.sub.ksA.sub.esT.sub.ksA.sub.esG.sub.k

(476) TABLE-US-00040 TABLE 40 Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID

NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ds 76 121
T.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061
GAGTGAATCATTCAGT
G.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds 84 1526
A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341176 796 811 15677 15692
CCATCTATCAGACTTC
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.ds 52 415
.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ks.sup.mC.sub.esT.sub.ksT.sub.es.sup.mC.sub.k 1341182
808 823 15689 15704 TTGGTAAGTATTCCAT
T.sub.ksT.sub.ksG.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ds 13 2750
A.sub.dsT.sub.dsT.sub.ks.sup.mC.sub.es.sup.mC.sub.ksA.sub.esT.sub.k 1341183 1077 1092 20892
20907 TAGTGCCAAACCAATG
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A.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.esA.sub.ksT.sub.esG.sub.k 1341187 1081
1096 20896 20911 CTGCTAGTGCCAAACC
.sup.mC.sub.ksT.sub.ksG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds 51 1510
.sup.m.sub.kCA.sub.s.sup.m.sub.kCA.sub.sA.sub.ksA.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k
1341191 1085 1100 20900 20915 ACTGCTGCTAGTGCCA
A.sub.ks.sup.mC.sub.ksT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds 39 1821
T.sub.dsA.sub.dsG.sub.dsT.sub.ksG.sub.es.sup.mC.sub.ks.sup.mC.sub.esA.sub.k 1341202 1259
1274 21074 21089 GTAGCTTTTGTCCACC
G.sub.ksT.sub.ksA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ds 34 1978
G.sub.dsT.sub.ds.sup.mC.sub.ks.sup.mC.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341204
1313 1328 21128 21143 AGTCTTAAACCTTCCC
A.sub.ksG.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ds 43 1979
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.es.sup.mC.sub.ks.sup.mC.sub.es.sup.mC.sub.k
1341211 1325 1340 21140 21155 TGCTACTTGAACAGTC
T.sub.ksG.sub.ks.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.ds 64 889
A.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.esG.sub.ksT.sub.es.sup.mC.sub.k 1341214 1321 1336
21136 21151 ACTTGAACAGTCTTAA
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A.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.esT.sub.ksA.sub.esA.sub.k 1341218 1317 1332
21132 21147 GAACAGTCTTAAACCT
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T.sub.dsT.sub.dsA.sub.dsA.sub.ksA.sub.es.sup.mC.sub.ks.sup.mC.sub.esT.sub.k 1341222 1342
1357 21157 21172 ATGGCTACAGATTGGA
A.sub.ksT.sub.ksG.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ds 57 111
A.sub.dsG.sub.dsA.sub.dsT.sub.ksT.sub.esG.sub.ksG.sub.esA.sub.k 1341226 1329 1344 21144
21159 GGAATGCTACTTGAAC
G.sub.ksG.sub.ksA.sub.dsA.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.d 49 1201
.sub.s.sup.mC.sub.dsT.sub.dsT.sub.ksG.sub.esA.sub.ksA.sub.es.sup.mC.sub.k 1341227 1335 1350
21150 21165 CAGATTGGAATGCTAC
.sup.mC.sub.ksA.sub.ksG.sub.dsA.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsA.sub.d 57 1590
.sub.sA.sub.dsT.sub.dsG.sub.ks.sup.mC.sub.esT.sub.ksA.sub.es.sup.mC.sub.k 1341237 1353 1368
21168 21183 TATTCTGTGGCATGGC
T.sub.ksA.sub.ksT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.ds 57 890
G.sub.ds.sup.mC.sub.dsA.sub.ksT.sub.esG.sub.ksG.sub.es.sup.mC.sub.k 1341242 1366 1381

21181 21196 GTGTTCTGTGTTGATAT
G.sub.ksT.sub.ksG.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.ds 63 1514
T.sub.dsT.sub.dsCT.sub.ksA.sub.esT.sub.ksA.sub.eT.sub.k 1341248 1357 1372 21172 21187
TTGATATTCTGTGGCA
T.sub.ksT.sub.ksG.sub.dsA.sub.dsT.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.MC.sub.ds 55 1202
T.sub.dsG.sub.dsT.sub.ksG.sub.esG.sub.ks.sup.mC.sub.esA.sub.k 1341254 1421 1436 21236
21251 AGGTTGAGATAAAGCT
A.sub.ksG.sub.ksG.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.ds 60 1904
T.sub.dsA.sub.dsA.sub.ksA.sub.esG.sub.ks.sup.mC.sub.esT.sub.k 1341262 1490 1505 21305
21320 CCGTTTTGGGCTAATG
.sup.mC.sub.ks.sup.mC.sub.ksG.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.dsG.sub.dsG 72 1360
.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.esA.sub.ksT.sub.esG.sub.k 1341265 1494 1509
21309 21324 TGCACCGTTTTGGGCT
T.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsG.sub.dsT.sub.ds 29 1671
T.sub.dsT.sub.dsT.sub.dsG.sub.ksG.sub.esG.sub.ks.sup.mC.sub.esT.sub.k 1341269 1464 1479
21279 21294 AGGCTAGGGAAATCTT
A.sub.ksG.sub.ksG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsG.sub.dsG.sub.dsG.sub.d 45 892
.sub.sA.sub.dsA.sub.dsA.sub.ksT.sub.es.sup.mC.sub.ksT.sub.esT.sub.k 1341275 1619 1634 21434
21449 GTCGGTCACCTTTCAT
G.sub.ksT.sub.ks.sup.mC.sub.dsG.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds.sup.m 57 194
C.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksT.sub.es.sup.mC.sub.ksA.sub.esT.sub.k 1341277 1598
1613 21413 21428 ATCTTATAAGACTATA
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21453 TAGAGTCGGTCACCTT
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T.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.es.sup.mC.sub.ksT.sub.esT.sub.k 1341291 1711 1726
21526 21541 CATCTCTGGGACCAAG
.sup.mC.sub.ksA.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds 44 117
G.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.esA.sub.ksA.sub.esG.sub.k 1341292 1715
1730 21530 21545 TAAACATCTCTGGGAC
T.sub.ksA.sub.ksA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT 39 351
.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ksG.sub.esG.sub.ksA.sub.es.sup.mC.sub.k 1341300 1719
1734 21534 21549 TGTCTAAACATCTCTG
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.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.esT.sub.ksA.sub.esA.sub.k 1341310 1783 1798 21598
21613 TTTCTTATGTAATAGC
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21609 TTATGTAATAGCCAGT
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21590 21605 GTAATAGCCAGTACAG
G.sub.ksT.sub.ksA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.m 47 274
C.sub.dsA.sub.dsG.sub.dsT.sub.ksA.sub.es.sup.mC.sub.ksA.sub.esG.sub.k 1341329 2187 2202
22002 22017 TGATGTAGTGGGAGTC
T.sub.ksG.sub.ksA.sub.dsT.sub.dsG.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ds 46 820
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G.sub.dsA.sub.dsG.sub.ksG.sub.es.sup.mC.sub.ksA.sub.es.sup.mC.sub.k 1341334 2250 2265
22065 22080 AGGATAGTCCATGCAA
A.sub.ksG.sub.ksG.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.d 46 2690
.sub.s.sup.mC.sub.dsA.sub.dsT.sub.ksG.sub.es.sup.mC.sub.ksA.sub.esA.sub.k 1341337 N/A N/A
9332 9347 ATCCTATATACATCCA
A.sub.ksT.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.dsT 65 1927
.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksT.sub.es.sup.mC.sub.ks.sup.mC.sub.esA.sub.k 1341342
N/A N/A 9966 9981 TAGTGTAAGCTGAGAG
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11573 11588 C.sub.dsA.sub.dsA.sub.dsT.sub.ksG.sub.esG.sub.ksT.sub.esT.sub.k 100 1341350 N/A
N/A 14916 14931 ATACATTAGCAAGCTA
A.sub.ksT.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.d 22 309
.sub.s.sup.mC.sub.dsA.sub.dsA.sub.ksG.sub.es.sup.mC.sub.ksT.sub.esA.sub.k 1341358 N/A N/A
14197 14212 TTATGTGATTGAGTTC
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.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ks.sup.mC.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k
1341364 N/A N/A 16127 16142 CATAATAATAGCTCTA
.sup.mC.sub.ksA.sub.ksT.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.ds 13 2511
A.sub.dsG.sub.ds.sup.mC.sub.ksT.sub.esT.sub.ks.sup.mC.sub.esA.sub.k 1341365 N/A N/A 16289
16304 GTTAGAATAGTCTTCA
G.sub.ksT.sub.ksT.sub.dsA.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.ds 45 1481
G.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.esT.sub.ks.sup.mC.sub.esA.sub.k 1341370 N/A N/A 16382
16397 TCCCAACCAAACCTTCC
T.sub.ks.sup.mC.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.m 27
2026 C.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.esT.sub.ks.sup.mC.sub.es.sup.m
C.sub.k 1341371 N/A N/A 18095 18110 AATTGATACACCAATG
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TAGTATAGTTATCTTC

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19924 ATCTTAAGATACCCAG
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N/A 19913 19928 GTGCATCTTAAGATAC
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.sub.dsA.sub.dsA.sub.dsG.sub.ksA.sub.esT.sub.ksA.sub.es.sup.mC.sub.k 1341388 N/A N/A 18695
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.sub.sG.sub.dsA.sub.dsG.sub.ksA.sub.es.sup.mC.sub.ksT.sub.es.sup.mC.sub.k 1341391 N/A N/A
20383 20398 GTAATCACAAAGTAAGG
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13954 13969 CCTCTAAGTTAGCCCC
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ds 20 1084
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A.sub.dsA.sub.dsG.sub.ks.sup.mC.sub.es.sup.mC.sub.ksA.sub.esA.sub.k 1341529 693 708 12111
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T.sub.dsG.sub.dsA.sub.ksT.sub.esA.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341533 1022 1037 20837
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.sub.sG.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.es.sup.mC.sub.ksA.sub.esG.sub.k 1341554 1394 1409
21209 21224 CTTGATCTCTTAGCTG
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 22165 22180 ACTACAAGAGGTTATT
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 16233 TATAGACTGGGTAGGA
 T.sub.ksA.sub.ksT.sub.dsA.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.d 38 2808
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 .sub.sA.sub.dsA.sub.ds.sup.mC.sub.ksG.sub.esG.sub.ksT.sub.esG.sub.k
 (477) TABLE-US-00041 TABLE 41 Inhibition of HSD17B13 RNA by modified
 oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID
 NO: ID NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
 Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
 A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.dsT.sub.d 93 121
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 A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341165 608 623 N/A N/A
 ACAATATGGGATGAGG

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15694 TTCCATCTATCAGACT
T.sub.ksT.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.d 33
571 .sub.sT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.esA.sub.ks.sup.mC.sub.esT.sub.k 1341177 696
711 12114 12129 GACATGAGGTTTTGAT
G.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsG.sub.ds 55 647
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22011 22026 GATTAGTCTTGATGTA
G.sub.ksA.sub.ksT.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT 56 1521
.sub.dsG.sub.dsA.sub.ksT.sub.esG.sub.ksT.sub.esA.sub.k 1341584 1776 1791 21591 21606
TGTAATAGCCAGTACA
T.sub.ksG.sub.ksT.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.m 45 352
C.sub.dsA.sub.dsG.sub.ksT.sub.esA.sub.ks.sup.mC.sub.esA.sub.k 1341592 N/A N/A 11578 11593
GTTATTCTCTTGACAA
G.sub.ksT.sub.ksT.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds 27 2575
T.sub.dsT.sub.dsG.sub.ksA.sub.es.sup.mC.sub.ksA.sub.esA.sub.k 1341602 N/A N/A 16292 16307
TTAGTTAGAATAGTCT
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.sub.sT.sub.dsA.sub.ksG.sub.esT.sub.ks.sup.mC.sub.esT.sub.k 1341617 N/A N/A 17995 18010
TGAAGTTAGTATAGTT
T.sub.ksG.sub.ksA.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.d 46 1405
.sub.sA.sub.dsT.sub.ksA.sub.esG.sub.ksT.sub.esT.sub.k
(478) TABLE-US-00042 TABLE 42 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID
NO: ID NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
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GAGTGAATCATTCACT
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AGTTGCACCGTTTTGG
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CGTTTTGGGCTAATGA
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C.sub.dsT.sub.dsA.sub.dsA.sub.ksT.sub.ksG.sub.ksA.sub.e 1340990 1618 1633 21433 21448
TCGGTCACCTTTCATA
T.sub.ks.sup.mC.sub.ksG.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.m 72
116 C.sub.dsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.ksT.sub.ksA.sub.e 1340997 1622 1637
21437 21452 AGAGTCGGTCACCTTT

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21456 AAATAGAGTCGGTCAC
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C.sub.dsG.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.ks.sup.mC.sub.e 1341005 1630 1645
21445 21460 TTAAAATAGAGTCGG
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21529 21544 AAACATCTCTGGGACC
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1783 21583 21598 CCAGTACAGTTCCTTT
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21565 21580 CTGTGTTAGCTTTAAT
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TAGTCTTGATGTAGTG
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21612 TTCTTATGTAATAGCC
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CCTATATACATCCAAG
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1097 20897 20912 GCTGCTAGTGCCAAAC
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11855 11870 TAAGCAGAATTGTGAA
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TCTTGACAATGGTTGA
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 14933 TAATACATTAGCAAGC
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 1341083 1078 1093 20893 20908 CTAGTGCCAAACCAAT
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 C.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.e 1341090 N/A N/A 18697
 18712 CAAAAGACTTTGAGAC
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 19926 GCATCTTAAGATACCC
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 12801 12816 AGGCTGTAAGAGTCAG
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 1084 .sub.sT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.e
 1341136 N/A N/A 14168 14183 ATAGCAAGCCAACAGA
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21073 21088 TAGCTTTTGTCCACCT
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TGATTGAGTTCTCCAC
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10492 10507 AATATGGGATGAGGTA
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1331 21131 21146 AACAGTCTTAAACCTT
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21152 TACTTGAACAGTCTTA
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21156 ATGCTACTTGAACAGT
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21172 ATGGCTACAGATTGGA
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21204 21219 TCTCTTAGCTGTGCAC
T.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.d 69
346 .sub.sT.sub.dsG.sub.dsT.sub.dsG.sub.ks.sup.mC.sub.ksA.sub.ks.sup.mC.sub.e 1341423 1422
1437 21237 21252 CAGGTTGAGATAAAGC
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21323 GCACCGTTTTGGGCTA

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21319 21334 AGAATAGAGTTGCACC
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21276 21291 CTAGGGAAATCTTTCA
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T.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksT.sub.ks.sup.mC.sub.ksA.sub.e 1341443 1614 1629 21429
21444 TCACCTTTCATAATGT
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2109 C.sub.dsA.sub.dsT.sub.dsA.sub.dsA.sub.ksT.sub.ksG.sub.ksT.sub.e 1341444 805 820 15686
15701 GTAAGTATTCCATCTA
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21604 TAATAGCCAGTACAGT
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G.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.ksG.sub.ksT.sub.e 1341459 1074 1089 20889
20904 TGCCAAACCAATGTTT
T.sub.ksG.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.m 43
2673 C.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.ksT.sub.ksT.sub.ksT.sub.e 1341461 N/A N/A 9511
9526 AATATTGAGGCACTGG
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.sub.s.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.ksG.sub.ksG.sub.e 1341462 N/A N/A 9598
9613 ATGTGAAGAGCTGGTA
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AAACAAGAGGATAGTC
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22016 GATGTAGTGGGAGTCG
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ACGGTGATCAAATGTA
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CAGGGTGGTAACGGTG
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TATGAAGTTAGTATAG
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GAGTGCATCTTAAGAT
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GTATAGTTATCTTCTC
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20124 AAGACATTCTAGCCTG
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20134 TTACAGCATCAAGACA
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20494 TAGTGGACTTCATTAG
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.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ksT.sub.ksA.sub.ksG.sub.e
(479) TABLE-US-00043 TABLE 43 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID
NO: ID NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
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21305 21320 CCGTTTTTGGGCTAATG
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21309 21324 TGCACCGTTTTTGGGCT
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1634 21434 21449 GTCGGTCACCTTTCAT
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21442 21457 AAAATAGAGTCGGTCA
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15689 15704 TTGGTAAGTATTCCAT
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21438 21453 TAGAGTCGGTCACCTT

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1790 21590 21605 GTAATAGCCAGTACAG
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21585 21600 AGCCAGTACAGTTCCT
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22002 22017 TGATGTAGTGGGAGTC
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20894 20909 GCTAGTGCCAAACCAA
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1341046 2250 2265 22065 22080 AGGATAGTCCATGCAA
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N/A 9332 9347 ATCCTATATACATCCA
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N/A 12419 12434 GTAACGGTGATCAAAAT
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G.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.ksA.sub.ksT.sub.e 1341062 N/A N/A
11548 11563 TCTCTTGACAATGGTT
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11588 .sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.ksG.sub.ksT.sub.ksT.sub.e 100 1341080 N/A
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21147 GAACAGTCTTAAACCT
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1341159 1313 1328 21128 21143 AGTCTTAAACCTTCCC
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A.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksT.sub.ksT.sub.e 1341205 1335
1350 21150 21165 CAGATTGGAATGCTAC
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1359 21159 21174 GCATGGCTACAGATTG
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.sub.sA.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ksT.sub.ksT.sub.ksG.sub.e 1341223 696 711
12114 12129 GACATGAGGTTTTGAT
G.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.ds 55 647
G.sub.dsT.sub.dsT.sub.dsT.sub.dsT.c.sub.sG.sub.ksA.sub.ksT.sub.e 1341244 1353 1368 21168
21183 TATTCTGTGGCATGGC
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1372 21172 21187 TTGATATTCTGTGGCA
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21181 21196 GTGTTCTTGTTGATAT
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GATCTCTTAGCTGTGC
G.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 70 502
A.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.e 1341424 794 809
15675 15690 ATCTATCAGACTTCTT
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1438 21238 21253 CCAGGTTGAGATAAAG
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21279 21294 AGGCTAGGGAAATCTT
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21234 21249 GTTGAGATAAAGCTGC

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1630 21430 21445 GTCACCTTTCATAATG
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.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.ksA.sub.ksT.sub.ksG.sub.e 1341442 1498
1513 21313 21328 GAGTTGCACCGTTTTG
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.sup.mC.sub.ds.sup.mC.sub.dsG.sub.dsT.sub.dsT.sub.ksT.sub.ksT.sub.ksG.sub.e 1341448 1705
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22009 22024 TTAGTCTTGATGTAGT
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22180 ACTACAAGAGGTTATT
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.sub.sA.sub.dsG.sub.dsG.sub.dsT.sub.dsT.sub.ksA.sub.ksT.sub.ksT.sub.e 1341468 N/A N/A 14204
14219 CAGATTATTATGTGAT
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20905 GTGCCAAACCAATGTT
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N/A N/A 11858 11873 GATTAAGCAGAATTGT
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11558 TGACAATGGTTGCCAC
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15506 TTAATGCCACCCTACC
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1341489 N/A N/A 18092 18107 TGATACACCAATGCAG
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19904 19919 AAGATACCCAGGTTGC
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C.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksksT.sub.ksT.sub.e 1341502 N/A N/A 20187 20202
TATTAGAAGTCAGCCC T.sub.ksA.sub.ksT.sub.ksT.sub.dsA.sub.dsG.sub.dsA.sub.dsA.sub.dsG
32 164
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1341504 N/A N/A 20197 20212 ACATTGCAGTTATTAG
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12809 AAGAGTCAGTATCCTC
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N/A N/A 12496 12511 AATTCACCTTGACTAA
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12819 CTCAGGCTGTAAGAGT
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.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ksA.sub.ksG.sub.ksT.sub.e
(480) TABLE-US-00044 TABLE 44 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID
NO: ID NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
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T.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046
5061 GAGTGAATCATTGAGT
G.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.d 83 1526
.sub.sA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1340981 1495 1510 21310
21325 TTGCACCGTTTTGGGC
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21314 21329 AGAGTTGCACCGTTTT
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21450 AGTCGGTCACCTTTCA
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21454 ATAGAGTCGGTCACCT
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21582 TCCTGTGTTAGCTTTA
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21551 ATGTCTAAACATCTC

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20834 20849 TGCAGCATTGATTCTGA
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21599 21614 GTTCTTATGTAATAG
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9349 CGATCCTATATACATC
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T.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.e 1341043 2252
2267 22067 22082 AGAGGATAGTCCATGC
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22168 22183 ATAACACTACAAGAGGTT
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9983 AGTAGTGTAAGCTGAG
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9531 AACTAATATTGAGGC
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12436 TGGTAACGGTGATCAA
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11561 TCTTGACAATGGTTGC
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15509 GCATTAATGCCACCCT
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1341075 608 623 N/A N/A ACAATATGGGATGAGG
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.sub.sG.sub.dsA.sub.dsT.sub.dsG.sub.ksA.sub.ksG.sub.ksG.sub.e 1341076 N/A N/A 14914 14929
ACATTAGCAAGCTAAG
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A.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.ksA.sub.ksG.sub.e 1341077 1080 1095
20895 20910 TGCTAGTGCCAAACCA
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N/A N/A 16223 16238 CTTTTTATAGACTGGG
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AATCCCAACCAAACCTT
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.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.e 1341092 N/A N/A
18095 18110 AATTGATACACCAATG
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17990 18005 TTAGTATAGTTATCTT
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ATCTTAAGATACCCAG
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N/A 18693 18708 AGACTTTGAGACTCTT
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1099 20899 20914 CTGCTGCTAGTGCCAA
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N/A 12499 12514 AGAAATTACCTTGAC
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21076 21091 AGGTAGCTTTTGTCCA
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21071 21086 GCTTTTGTCCACCTTT
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21188 GTTGATATTCTGTGGC
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21223 TTGATCTCTTAGCTGT
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21259 ATATGTCCAGGTTGAG
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21321 ACCGTTTTGGGCTAAT
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21281 21296 AGAGGCTAGGGAAATC

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13949 13964 AAGTTAGCCCCCAGGA
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C.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksG.sub.ksA.sub.e
(481) TABLE-US-00045 TABLE 45 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ SEQ SEQ SEQ ID NO: ID
NO: ID NO: ID NO: % SEQ Compound 1 Start 1 Stop 2 Start 2 Stop Sequence
Chemistry Notation in- ID Number Site Site Site Site (5' to 3') (5' to 3') hibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
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21611 TCTTATGTAATAGCCA

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1792 21592 21607 ATGTAATAGCCAGTAC
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C.sub.dsA.sub.dsA.sub.dsG.sub.ksA.sub.ksG.sub.ksG.sub.e 1341051 1077 1092 20892 20907
TAGTGCCAAACCAATG
T.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.d 61 1198
.sub.sA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.ksT.sub.ks 1341059 N/A N/A 9518
9533 CTACACTAATATTGAG
.sup.mC.sub.ksT.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsA 10 59
.sub.dsT.sub.dsA.sub.dsT.sub.dsT.sub.ksG.sub.ksA.sub.ksG.sub.e 1341065 N/A N/A 11853 11868
AGCAGAATTGTGAACG
A.sub.ksG.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.dsT.sub.ds 56 844
G.sub.dsT.sub.dsG.sub.dsA.sub.ksA.sub.ks.sup.mC.sub.ksG.sub.e 1341066 N/A N/A 12423 12438
GGTGGTAACGGTGATC
G.sub.ksG.sub.ksT.sub.ksG.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds 47 2591
G.sub.dsG.sub.dsT.sub.dsG.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.e 1341072 N/A N/A 15496 15511
CTGCATTAATGCCACC
.sup.mC.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.ds 40 1168
T.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.ks.sup.mC.sub.ks.sup.mC.sub.e 1341074
N/A N/A 14197 14212 TTATGTGATTGAGTTC
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.sub.sG.sub.dsA.sub.dsG.sub.ksT.sub.ksT.sub.ks.sup.mC.sub.e 1341078 N/A N/A 14916 14931
ATACATTAGCAAGCTA
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.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.e 1341086 N/A N/A 16131
16146 GCTCCATAATAATAGC
G.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.dsA 8 2665
.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.e 1341091 N/A N/A 17991
18006 GTTAGTATAGTTATCT
G.sub.ksT.sub.ksT.sub.ksA.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.dsG 71 1171
.sub.dsT.sub.dsT.sub.dsA.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.e 1341094 N/A N/A 18097 18112

GAAATTGATACACCAA
 G.sub.ksA.sub.ksA.sub.ksA.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsT.sub.dsA 64 315
 .sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.ksA.sub.e 1341104 N/A N/A
 19910 19925 CATCTTAAGATACCCA
 .sup.mC.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ds 76 475
 A.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksA.sub.e 1341106 N/A N/A
 18695 18710 AAAGACTTTGAGACTC
 A.sub.ksA.sub.ksA.sub.ksG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ds 67 1250
 G.sub.dsA.sub.dsG.sub.dsA.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.e 1341109 1085 1100
 20900 20915 ACTGCTGCTAGTGCCA
 A.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsT 57 1821
 .sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksA.sub.e 1341111 N/A N/A
 20116 20131 CAGCATCAAGACATTC
 .sup.mC.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds 70 2191
 A.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksT.sub.ksT.sub.ks 1341112 N/A N/A 20381
 20396 AATCACAAAGTAAGGTA
 A.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.d 63 1333
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 GCAGTTATTAGAAGTC
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 A.sub.dsG.sub.dsA.sub.dsA.sub.ksG.sub.ksT.sub.ks.sup.mC.sub.e 1341124 N/A N/A 12501 12516
 GCAGAAATTCACCTTG
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 .sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ksT.sub.ksG.sub.e 1341128 N/A N/A
 20486 20501 AACTTCATAGTGGACT
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 ACAGTAAAATTATGCC
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 T.sub.dsT.sub.dsA.sub.dsT.sub.ksG.sub.ks.sup.mC.sub.ks.sup.mC.sub.e 1341135 N/A N/A 13952
 13967 TCTAAGTTAGCCCCCA
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 G.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksA.sub.e
 1341139 N/A N/A 12799 12814 GCTGTAAGAGTCAGTA
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 21010 GTTTTAAAGAGGCATG
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 .sub.sG.sub.dsG.sub.ds.sup.mC.sub.ksA.sub.ksT.sub.ksG.sub.e 1341142 1257 1272 21072 21087
 AGCTTTTGTCCACCTT
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 C.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ksT.sub.ks.sup.mC.sub.e 1341151 1315 1330 21130
 21145 ACAGTCTTAAACCTTC
 A.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.ds 73 110
 A.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ksT.sub.ks.sup.mC.sub.e 1341153 1325 1340
 21140 21155 TGCTACTTGAACAGTC
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 21126 21141 TCTTAAACCTTCCCTG
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700 715 12118 12133 CAGAGACATGAGGTTT
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ACTTGAACAGTCTTAA
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.sub.sG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.ksA.sub.ksA.sub.e 1341173 1331 1346 21146
21161 TTGGAATGCTACTTGA
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T.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.ksG.sub.ksA.sub.e 1341232 1340 1355 21155 21170
GGCTACAGATTGGAAT
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.sub.sT.sub.dsT.sub.dsG.sub.dsG.sub.ksA.sub.ksA.sub.ksT.sub.e 1341255 1351 1366 21166 21181
TTCTGTGGCATGGCTA
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812 A.sub.dsT.sub.dsG.sub.dsG.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.e 1341271 1355 1370 21170
21185 GATATTCTGTGGCATG
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21189 TGTTGATATTCTGTGG
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TTCCATCTATCAGACT
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1410 21210 21225 ACTTGATCTCTTAGCT
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21217 TCTTAGCTGTGCACTC
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190 T.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.e 1341436 1421
1436 21236 21251 AGGTTGAGATAAAGCT
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AAATATGTCCAGGTTG
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.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.ksT.sub.ksT.sub.ksG.sub.e 1341446 1488 1503 21303
21318 GTTTTGGGCTAATGAA
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GTTGCACCGTTTTGGG
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20852 CTTTGCAGCATTGATT
.sup.mC.sub.ksT.sub.ksT.c.sub.sT.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds.sup.mC 56 2463
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TCTAAACATCTCTGGG

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2287 .sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.ksG.sub.ksG.sub.e 1341460 N/A N/A
9337 9352 CCTCGATCCTATATAC
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsG.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds 11
2812 .sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.ksT.sub.ksA.sub.ks.sup.mC.sub.e 1341467
2184 2199 21999 22014 TGTAGTGGGAGTCGGA
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TGACAATGGTTGATAG
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TATAGTTATCTTCTCA
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18011 ATGAAGTTAGTATAGT
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TAGTCTTCAGCAAAGT
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ACCAAACCTCCAGCAG
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N/A N/A 19914 19929 AGTGCATCTTAAGATA
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20024 AGAGAGTAATCAGTTT
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GCAAGCCAACAGAGAG
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14175 14190 GCCTTGAATAGCAAGC
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A.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.e
(482) TABLE-US-00046 TABLE 46 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO: 1, and 2 SEQ ID SEQ ID SEQ ID
SEQ ID NO: 1 NO: 1 NO: 2 NO: 2 Compound Start Stop Start Stop Chemistry Notation
% SEQ ID Number Site Site Site Site Sequence (5' to 3') (5' to 3') inhibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ds 79 121

T.sub.dsA.sub.mC.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061
GAGTGAATCATTCACT
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21147 GAACAGTCTTAAACCT
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1328 21128 21143 AGTCUTAAACCTTCCC
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21123 21138 TAAACCTTCCCTGTGT
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.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.ksG.sub.ksT.sub.k 1341668 1335 1350 21150
21165 CAGAUTGGAATGCTAC
.sup.mC.sub.ksA.sub.ksG.sub.ksA.sub.dsU.sub.ysT.sub.dsG.sub.dsG.sub.dsA.sub.ds 70 2814
A.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.ks.sup.mC.sub.k 1341672 1329 1344 21144
21159 GGAAUGCTACTTGAAC
G.sub.ksG.sub.ksA.sub.ksA.sub.dsU.sub.ysG.sub.ds.sup.mC.sub.dsT.sub.ds 62 2815
A.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ksA.sub.ks.sup.mC.sub.k 1341677 1325
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 A.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.ksG.sub.k 1341955 N/A N/A
 12496 12511 AATTCACCTTGACTAA
 A.sub.ksA.sub.ksT.sub.ksT.sub.dsC.sub.ysA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 0 2144
 T.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.ksA.sub.k
 (483) TABLE-US-00047 TABLE 47 Inhibition of HSD17B13 RNA by modified
 oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID
 SEQ ID NO: 1 NO: 1 NO: 2 NO: 2 SEQ Compound Start Stop Start Stop Chemistry
 Notation % ID Number Site Site Site Site Sequence (5' to 3') (5' to 3') inhibition NO
 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
 A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ds 86 121
 T.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061
 GAGTGAATCATTCACT
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 A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341649 1314 1329 21129 21144
 CAGTCTTAAACCTTCC
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 1318 1333 21133 21148 TGAACAGTCTTAAACC
 T.sub.ksG.sub.ksA.sub.ksA.sub.dsC.sub.ysA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ds 57 344
 T.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ks.sup.mCk.sup.mC.sub.k 1341655 1256 1271 21071

21086 GCTTUTGTCCACCTTT
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21124 21139 TTAAACCTTCCCTGTG
T.sub.ksT.sub.ksA.sub.ksA.sub.dsA.sub.ys.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ds 23 2807
T.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ksT.sub.ksG.sub.k 1341669
1326 1341 21141 21156 ATGCUACTTGAACAGT
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T.sub.dsG.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341671 1330 1345
21145 21160 TGGAATGCTACTTGAA
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1359 21159 21174 GCATGGCTACAGATTG
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21188 GTTGATATTCTGTGGC
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21179 CTGTGGCATGGCTACA
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T.sub.dsG.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.ksA.sub.k 1341701 1389
1404 21204 21219 TCTCUTAGCTGTGCAC
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.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ksA.sub.ks.sup.mC.sub.k 1341706
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1433 21233 21248 TTGAGATAAAGCTGCC
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A.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ks.sup.mC.sub.ks.sup.mC.sub.k 1341717 1422
1437 21237 21252 CAGGUTGAGATAAAGC
.sup.mC.sub.ksA.sub.ksG.sub.ksG.sub.dsU.sub.ysT.sub.dsG.sub.dsA.sub.ds 55 2851
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21244 ATATGTCCAGGTTGAG

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21321 ACCGUTTTGGGCTAAT

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1639 ATAGAGTCGGTCACCT

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21415 GTATCTTATAAGACTA

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21446 GGTCACCTTTCATAAT

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1712 ACATCTCTGGGACCAA

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1722 TCTGGGACCAAGGATA

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.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.dsG.sub.dsA.sub.ksT.sub.ksA.sub.k 1341779 1752 1767
21567 TCCTGTGTTAGCTTTA

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 20899 20914 CTGCUGCTAGTGCCAA
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 N/A 19913 19928 GTGCATCTTAAGATAC
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 T.sub.dsA.sub.dsA.sub.dsG.sub.dsA.sub.dsT.sub.ksA.sub.ks.sup.mC.sub.k 1341906 N/A N/A
 18693 18708 AGACUTTGAAGACTCTT
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 G.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.ksT.sub.k 1341909 N/A N/A 19909
 19924 ATCTUAAGATACCCAG
 A.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.dsU.sub.ysA.sub.dsA.sub.dsG.sub.dsA.sub.ds 55 2864
 T.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.ksG.sub.k 1341911 1080
 1095 20895 20910 TGCTAGTGCCAAACCA
 T.sub.ksG.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.ysG.sub.dsT.sub.dsG.sub.ds 80 1432
 .sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.k
 1341912 N/A N/A 19905 19920 TAAGATACCCAGGTTG
 T.sub.ksA.sub.ksA.sub.ksG.sub.dsA.sub.ysT.sub.dsA.sub.ds.sup.mC.sub.ds 11 2790
 .sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.dsT.sub.ksT.sub.ksG.sub.k 1341922 N/A
 N/A 20114 20129 GCATCAAGACATTCTA
 G.sub.ks.sup.mC.sub.ksA.sub.ksT.sub.dsC.sub.ysA.sub.dsA.sub.dsG.sub.ds 51 1565
 A .sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.ksA.sub.k 1341926 N/A

N/A 19999 20014 CAGTUTTCCTCATGAT
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 .sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.ksA.sub.ksT.sub.k 1341932 1076
 1091 20891 20906 AGTGCCAAACCAATGT
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 A.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.ksG.sub.ksT.sub.k 1341933 N/A
 N/A 20388 20403 CTATAGTAATCACAAG
 .sup.mC.sub.ksT.sub.ksA.sub.ksT.sub.dsA.sub.ysG.sub.dsT.sub.dsA.sub.dsA.sub.ds 37 1644
 T.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.ksG.sub.k 1341935 N/A N/A 20190
 20205 AGTTATTAGAAGTCAG
 A.sub.ksG.sub.ksT.sub.ksT.sub.dsA.sub.ysT.sub.dsT.sub.dsA.sub.dsG.sub.ds 25 2746
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 CACAAGTAAGGTAAAG
 .sup.mC.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.dsA.sub.ysG.sub.dsT.sub.dsA.sub.ds 43 2448
 A.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.ksA.sub.ksG.sub.k 1341944 N/A N/A 20484
 20499 CTTCATAGTGGACTTC
 .sup.mC.sub.ksT.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.ysT.sub.dsA.sub.dsG.sub.ds 35 711
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 12804 12819 CTCAGGCTGTAAGAGT
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 G.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsA.sub.ksG.sub.ksT.sub.k 1341956 N/A N/A 12499
 12514 AGAAATTCACCTTGAC
 A.sub.ksG.sub.ksA.sub.ksA.sub.dsA.sub.ysT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds 9 2284
 .sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.ksA.sub.ks.sup.mC.sub.k
 (484) TABLE-US-00048 TABLE 48 Inhibition of HSD17B13 RNA by modified
 oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID
 SEQ ID NO: 1 NO: 1 NO: 2 NO: 2 Compound Start Stop Start Stop Chemistry Notation
 % SEQ Number Site Site Site Site Sequence (5' to 3') (5' to 3') inhibition ID NO
 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
 A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ds 82 121
 T.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061
 GAGTGAATCATTCACT
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 A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341651 1315 1330 21130 21145
 ACAGUCTTAAACCTTC
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 1257 1272 21072 21087 AGCTUTTGTCCACCTT
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 628 N/A N/A CTGGAACAATATGGGA
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 21157 AATGCTACTTGAACAG
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 21149 TTGAACAGTCTTAAAC
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1323 1338 21138 21153 CTACUTGAACAGTCTT
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1370 21170 21185 GATAUTCTGTGGCATG
GsA.sub.ksT.sub.ksA.sub.dsU.sub.ysT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds 60 2869
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21161 TTGGAATGCTACTTGA
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1374 21174 21189 TGTGATATTCTGTGG
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21261 AAATATGTCCAGGTTG
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GTATUCCATCTATCAG

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1507 21307 21322 CACCGTTTTGGGCTAA
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.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsG.sub.ksT.sub.ksT.sub.k 1341747 1602 1617
21417 21432 ATGTATCTTATAAGAC
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A.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ksA.sub.ks.sup.mC.sub.k 1341749 1621 1636 21436
21451 GAGTCGGTCACCTTTC
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A.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksT.sub.ks.sup.mC.sub.k 1341750 1625 1640
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1632 21432 21447 CGGTCACCTTTCATAA
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12423 12438 GGTGGTAACGGTGATC
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G.sub.dsG.sub.dsT.sub.dsG.sub.dsA.sub.ksT.sub.ks.sup.mC.sub.k 1341857 N/A N/A 14204 14219
CAGAUTATTATGTGAT
.sup.mC.sub.ksA.sub.ksG.sub.ksA.sub.dsU.sub.ysT.sub.dsA.sub.dsT.sub.dsT.sub.ds 32 2877
A.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.ksA.sub.ksT.sub.k 1341859 N/A N/A 15491 15506
TTAAUGCCACCCTACC

T.sub.ksT.sub.ksA.sub.ksA.sub.dsU.sub.ysG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 8 2878
A.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.ks.sup.mC.sub.k
1341868 N/A N/A 16282 16297 TAGTCTTCAGCAAAGT
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G.sub.ds.sup.mC.sub.dsA.sub.dsAdSA.sub.ksG.sub.ksT.sub.k 1341874 N/A N/A 16131 16146
GCTCCATAATAATAGC
G.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsC.sub.ysA.sub.dsT.sub.dsA.sub.ds 16 2665
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17991 18006 GTTAGTATAGTTATCT
G.sub.ksT.sub.ksT.sub.ksA.sub.dsG.sub.ysT.sub.dsA.sub.dsT.sub.dsA.sub.ds 52 1171
G.sub.dsT.sub.dsT.sub.dsA.sub.dsT.sub.ks.sup.mC.sub.ksT.sub.k 1341883 N/A N/A 17996 18011
ATGAAGTTAGTATAGT
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ACCAAACCTCCAGCAG
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N/A N/A 18097 18112 GAAAUTGATACACCAA
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A.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.ksA.sub.k 1341894 N/A N/A
17986 18001 TATAGTTATCTTCTCA
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18695 18710 AAAGACTTTGAGACTC
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G.sub.dsA.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.ks.sup.mC.sub.k 1341901 N/A N/A 19914
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1341915 N/A N/A 19906 19921 TTAAGATACCCAGGTT
T.sub.ksT.sub.ksA.sub.ksA.sub.dsG.sub.ysA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ds 30 163
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.ksT.sub.ksT.sub.k 1341917 N/A N/A 19910
19925 CATCUTAAGATACCCA
.sup.mC.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsU.sub.ysT.sub.dsA.sub.dsA.sub.ds 56 2881
G.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.k 1341921
N/A N/A 20116 20131 CAGCATCAAGACATTC
.sup.mC.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.ysT.sub.ds.sup.mC.sub.dsA.sub.ds 52 2191
A.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ksT.sub.ks.sup.mC.sub.k 1341925 N/A
N/A 20009 20024 AGAGAGTAATCAGTTT
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20907 TAGTGCCAAACCAATG
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A.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ksT.sub.ksG.sub.k 1341939 1085

1100 20900 20915 ACTGCTGCTAGTGCCA
A.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.dsC.sub.ysT.sub.dsG.sub.ds.sup.mC.sub.ds 64 1821
T.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.k 1341943 N/A
N/A 20486 20501 AACTUCATAGTGGACT
A.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.dsU.sub.ys.sup.mC.sub.dsA.sub.dsT.sub.ds 38 2883
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20422 20437 ACAGUAAAATTATGCC
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20381 20396 AATCACAAGTAAGGTA
A.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.ysmC.sub.dsA.sub.dsA.sub.ds 31 1333
G.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsG.sub.ksT.sub.ksA.sub.k 1341954 N/A N/A 12501
12516 GCAGAAATTCACCTTG
G.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.dsA.sub.ysA.sub.dsA.sub.dsT.sub.dsT.sub.ds 54 1002
.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.ksG.sub.k
(485) TABLE-US-00049 TABLE 49 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID
SEQ ID NO: 1 NO: 1 NO: 2 NO: 2 SEQ Compound Start Stop Start Stop Sequence
Chemistry Notation % ID Number Site Site Site Site (5' to 3') (5' to 3') inhibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
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T.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061
GAGTGAATCATTCACT
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A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1341508 N/A N/A 14189 14204
TTGAGTTCTCCACTGC
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.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.e 1341518
1251 1266 21066 21081 TGTCCACCTTTAAATG
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28 1666 T.sub.dsT.sub.dsA.sub.dsA.sub.ksA.sub.ksT.sub.kgG.sub.e 1341519 1310 1325 21125
21140 CTTAAACCTTCCCTGT
.sup.mC.sub.ksT.sub.ksT.sub.ksA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 34 1823
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1328 1343 21143 21158 GAATGCTACTTGAACA
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.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.e 1341521 1330
1345 21145 21160 TGGAATGCTACTTGAA
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21150 CTTGAACAGTCTTAAA
.sup.mC.sub.ksT.sub.ksT.sub.ksG.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds 43 500
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21124 21139 TTAAACCTTCCCTGTG
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1319 1334 21134 21149 TTGAACAGTCTTAAAC
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.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.ksA.sub.ksA.sub.ks.sup.mC.sub.e 1341525 1264 1279 21079
21094 GGGAGGTAGCTTTTGT

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TAAACCTTCCCTGTGT
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.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ksT.sub.ksG.sub.ksT.sub.e 1341527
1175 1190 20990 21005 TAAGAGGCATGAAAGG
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ATGAGGTTTTGATACC
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21166 ACAGATTGGAATGCTA
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21177 GTGGCATGGCTACAGA
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.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.ksG.sub.ksA.sub.e 1341561 1346 1361
21161 21176 TGGCATGGCTACAGAT
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.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksA.sub.ksT.sub.e 1341573 1348
1363 21163 21178 TGTGGCATGGCTACAG
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G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.e 1341576 801 816 15682
15697 GTATTCCATCTATCAG
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T.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.e 1341608 N/A N/A
19914 19929 AGTGCATCTTAAGATA
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T.sub.dsT.sub.dsA.sub.dsA.sub.ksG.sub.esA.sub.ksT.sub.esA.sub.k 1341610 N/A N/A 19915
19930 GAGTGCATCTTAAGAT
G.sub.ksA.sub.ksG.sub.dsT.sub.dSG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds 27 2256
.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.ksA.sub.esG.sub.ksA.sub.esT.sub.k 1341611 N/A N/A 18690
18705 CTTTGAGACTCTTGTT
.sup.mC.sub.ksT.sub.ksT.sub.dsT.sub.dsG.sub.dsA.sub.dSG.sub.dsA.sub.ds 42 2583
.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.esG.sub.ksT.sub.esT.sub.k 1341612 N/A N/A
19904 19919 AAGATACCCAGGTTGC
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N/A 18092 18107 TGATACACCAATGCAG
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.sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.ksG.sub.es.sup.mC.sub.ksA.sub.esG.sub.k 1341618 N/A
N/A 19906 19921 TTAAGATACCCAGGTT
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.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.esG.sub.ksT.sub.esT.sub.k 1341619 N/A N/A 20388
20403 CTATAGTAATCACAAG
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20202 TATTAGAAGTCAGCCC
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N/A 20109 20124 AAGACATTCTAGCCTG
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N/A 19916 19931 GGAGTGCATCTTAAGA
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N/A 19999 20014 CAGTTTTCTCATGAT
.sup.mC.sub.ksA.sub.ksG.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds 59 2331
.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksT.sub.esG.sub.ksA.sub.esT.sub.k 1341628 N/A N/A
20378 20393 CACAAGTAAGGTAAAG
.sup.mC.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.dSA.sub.dsG.sub.dsT.sub.dsA.sub.ds 38 2448
A.sub.dsG.sub.dsG.sub.dsT.sub.ksA.sub.esA.sub.ksA.sub.esG.sub.k 1341629 N/A N/A 12804
12819 CTCAGGCTGTAAGAGT
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G.sub.dsT.sub.dsA.sub.dsA.sub.ksGc.sub.sA.sub.ksG.sub.esT.sub.k 1341630 N/A N/A 14165
14180 GCAAGCCAACAGAGAG
G.sub.ks.sup.mC.sub.ksA.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 12 2556
A.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ksA.sub.esG.sub.ksA.sub.esG.sub.k 1341631 N/A N/A
14175 14190 GCCTTGAATAGCAAGC
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N/A 13959 13974 GTCTACCTCTAAGTTA
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13964 AAGTTAGCCCCCAGGA
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1341634 N/A N/A 12496 12511 AATTCACCTTGACTAA
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CATAACTTCATAGTGG
.sup.mC.sub.ksA.sub.ksT.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 15 866
.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.ksG.sub.esT.sub.ksG.sub.esG.sub.k 1341637 N/A N/A
12794 12809 AAGAGTCAGTATCCTC
A.sub.ksA.sub.ksG.sub.dsA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds 14 2572

T.sub.dsA.sub.dsT.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341638 1394 1409
21209 21224 CTTGATCTCTTAGCTG
.sup.mC.sub.ksT.sub.kST.sub.ksG.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds 36 736
.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.e 1341639 1416
1431 21231 21246 GAGATAAAGCTGCCTG
G.sub.ksA.sub.ksG.sub.ksA.sub.dST.sub.dsA.sub.dsA.sub.dsA.sub.dsG.sub.ds 55 1592
.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.e 1341640 1426 1441
21241 21256 TGTCCAGGTTGAGATA
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T.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.ksA.sub.ksT.sub.ksA.sub.e 1341641 1415 1430 21230
21245 AGATAAAGCTGCCTGC
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T.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.e 1341642 1425 1440
21240 21255 GTCCAGGTTGAGATAA
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T.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.ksT.sub.ksA.sub.ksA.sub.e 1341643 791 806 15672
15687 TATCAGACTTCTTACG
T.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds 28 103
T.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksA.sub.ks.sup.mC.sub.ksG.sub.e 1341644 1398
1413 21213 21228 GAAACTTGATCTCTTA
G.sub.ksA.sub.ksA.sub.ksA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds 79 2717
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21218 CTCTTAGCTGTGCACT
.sup.mC.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds 66 268
T.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.e 1341646 1384
1399 21199 21214 TAGCTGTGCACTCATT
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.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.ksT.sub.ksT.sub.e 1341647
N/A N/A 14189 14204 TTGAGTTCTCCACTGC
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.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.esT.sub.ksG.sub.es.sup.mC.sub.k 1341949 N/A
N/A 12799 12814 GCTGUAAGAGTCAGTA
G.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.dsU.sub.ysA.sub.dsA.sub.dsG.sub.dsA.sub.ds 28 2886
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20921 CGTTUGACTGCTGCTA
.sup.mC.sub.ksG.sub.ksT.sub.ksT.sub.dsU.sub.ysG.sub.dsA.sub.ds.sup.mC.sub.ds 54 2887
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14172 14187 TTGAATAGCAAGCCAA
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14190 GCCTUGAATAGCAAGC
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N/A 13949 13964 AAGTUAGCCCCCAGGA

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1341963 N/A N/A 13952 13967 TCTAAGTTAGCCCCCA
T.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.dsA.sub.ysG.sub.dsT.sub.dgT.sub.dsA.sub.ds 7 928
G.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.k
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A.sub.dsG.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ks.sup.mC.sub.ks.sup.mC.sub.k 1341965 N/A
N/A 13954 13969 CCTCUAAGTTAGCCCC
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsU.sub.ySA.sub.dsA.sub.dsG.sub.ds 8 2891
T.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.k
1341966 N/A N/A 13959 13974 GTCTACCTCTAAGTTA
G.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.ys.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ds 15 2363
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12794 12809 AAGAGTCAGTATCCTC
A.sub.ksA.sub.ksG.sub.ksA.sub.dsGv.sub.sT.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds 55 2572
T.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ks.sup.mC.sub.k 1341968 1202 1217
21017 21032 GTATGTTTATGTAAGC
G.sub.kST.sub.ksA.sub.ksT.sub.dsG.sub.ysT.sub.dST.sub.dsT.sub.dsA.sub.ds 66 1511
T.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.ksG.sub.ks.sup.mC.sub.k 1341969 1200 1215 21015
21030 ATGTUTATGTAAGCAC
A.sub.ksT.sub.ksG.sub.ksT.sub.dsU.sub.ysT.sub.dsA.sub.dsT.sub.dsG.sub.ds 45 2892
T.sub.dsA.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ksA.sub.ks.sup.mC.sub.k 1341970 N/A N/A 14168
14183 ATAGCAAGCCAACAGA
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.sup.mC.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ksA.sub.k 1341971 606 621 10492
10507 AATAUGGGATGAGGTA
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A.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.ksA.sub.ksG.sub.k 1341975 N/A N/A
14170 14185 GAATAGCAAGCCAACA
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G.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ks.sup.mC.sub.ksA.sub.k 1341976 N/A N/A
14192 14207 TGATUGAGTTCTCCAC
T.sub.ksG.sub.ksA.sub.ksT.sub.dsU.sub.ysG.sub.dsA.sub.dsG.sub.dsT.sub.ds 4 2895
T.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.ks.sup.mC.sub.k 1341977 N/A
N/A 14189 14204 TTGAGTTCTCCACTGC
T.sub.ksT.sub.ksG.sub.ksA.sub.dsG.sub.ysT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds 18 2885
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.ks.sup.mC.sub.k 1341978
1261 1276 21076 21091 AGGTAGCTTTTGTCCA
A.sub.ksG.sub.ksG.sub.ksT.sub.dsA.sub.ysG.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 59 109
T.sub.dsT.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksA.sub.k 1341979 1259 1274 21074

21089 GTAGCTTTTGTCCACC

G.sub.ksT.sub.ksA.sub.ksG.sub.dsC.sub.ysT.sub.dsT.sub.dsT.sub.dsT.sub.ds 64 1978

G.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.kS.sup.mC.sub.ks.sup.mC.sub.k 1341980

1258 1273 21073 21088 TAGCUTTTGTCCACCT

T.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.dsU.sub.ysT.sub.dsT.sub.dsT.sub.dsG.sub.ds 67 2896

T.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.ksT.sub.k

Example 4: Design of and Antisense Inhibition by Human HSD171B13 Modified Oligonucleotides in HepRG Cells

(486) Modified oligonucleotides complementary to an HSD17813 nucleic acid were synthesized and tested for their effect on HSD17B813 RNA levels in vitro, specifies the chemistry of each modified oligonucleotide; wherein subscript 'd' represents a 2'-β-D-deoxyribosyl sugar moiety, subscript 'e' represents a 2'-MOE sugar moiety, subscript 'y' represents a 2'-O-methyl sugar moiety, subscript 'k' represents a cEt modified sugar moiety, subscript 's' represents a phosphorothioate internucleoside linkage, and superscript 'in' before the cytosine residue represents a 5-methyl cytosine. "Start site" indicates the 5'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. "Stop site" indicates the 3'-most nucleoside of the target sequence to which the modified oligonucleotide is complementary. As shown in the Tables below, the modified oligonucleotide are complementary to either the human HSD17B13 mRNA, designated herein as SEQ ID NO: 1 (GENBANK Accession No.

NM_178135.4) or to the human HSD17B13 genomic sequence, designated herein as SEQ ID NO: 2 (the complement of GENBANK Accession No. NC_000004.12 truncated from nucleotides 87301001 to 87326000) or to both. 'N/A' indicates that the modified oligonucleotide is not complementary to that particular target sequence with 100% complementarity.

(487) The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000 cells per well were transfected using electroporation with 1,000 nM of modified oligonucleotide. After a treatment period of approximately 24 hours, RNA was isolated from the cells and HSD17B13 RNA levels were measured by quantitative real-time RTPCR. Human primer probe set RTS43553 was used to measure RNA levels. HSD17B3 RNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Results are presented as percent inhibition of HSD17B13 relative to untreated control cells. As used herein, a value of '0' indicates that treatment with the modified oligonucleotide did not inhibit HSD17B13 mRNA levels.

(488) TABLE-US-00050 TABLE 50 Inhibition of HSD17B13 RNA by modified oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID SEQ ID NO: 1 NO: 1 NO: 2 NO: 2 SEQ Compound Start Stop Start Stop Sequence Chemistry Notation % ID Number Site Site Site Site (5' to 3') (5' to 3') inhibition NO 1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA

A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.ds 81 121

G.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks .sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046 5061 GAGTGAATCATTCACT

G.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.ds 83 1526

T.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds A.sub.ksG.sub.ksT.sub.k 1340072 N/A N/A 19913 19928 GTGCATCTTAAGATAC

G.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.ds 47 2806

.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ds A.sub.dsT.sub.ksA.sub.ks.sup.mC.sub.k 1340095 N/A N/A 19395 19410 ATAAGAGATGAGTAGG

A.sub.ksT.sub.ksA.sub.ksA.sub.dsG.sub.dsA.sub.dsG.sub.ds 56 2897

A.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.ks G.sub.ksG.sub.k 1340097 N/A N/A 14604 14619 TGCGAGAAATCTAAGA

T.sub.ksG.sub.ksT.sub.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds 50 2898
A.sub.dsA.sub.dsT.sub.dsC.sub.dsT.sub.dsA.sub.ds A.sub.ksG.sub.ksA.sub.k 1340102 N/A N/A
10521 10536 TATAACATGGCTGGCA
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N/A N/A 9967 9982 GTAGTGTAAGCTGAGA
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A.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds A.sub.ksG.sub.ksA.sub.k 1340137 N/A
N/A 18139 18154 ATGTTATCTCAAGTCA
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.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.ds T.sub.ks.sup.mC.sub.ksA.sub.k
1340138 1632 1647 21447 21462 GATTTAAAATAGAGTC
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N/A 9442 9457 CTCCCACAAAATAAC
.sup.mC.sub.ksT.sub.ks.sup.mC.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 13 2904
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G.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds T.sub.ksG.sub.ks.sup.mC.sub.k
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1340182 N/A N/A 14915 14930 TACATTAGCAAGCTAA
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N/A 18037 18052 TCATAGTTTATATGGA
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1340282 N/A N/A 19535 19550 ATAGGCTAAAATGGTC
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A.sub.dsA.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ks T.sub.ks.sup.mC.sub.k 1340287 N/A
N/A 18696 18711 AAAAGACTTTGAGACT
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T.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.ks .sup.mC.sub.ksT.sub.k 1340288 N/A
N/A 11115 11130 ACAGAGGAGTTTGCAG
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1324 21124 21139 TTAAACCTTCCCTGTG
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A.sub.dsG.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.ks A.sub.ksA.sub.k 1340301 N/A
N/A 20113 20128 CATCAAGACATTCTAG
.sup.mC.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.dsA.sub.ds 49 2915
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1340306 N/A N/A 19732 19747 AACTGAAGGTCTGAGC
A.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.ds 55 2916
G.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsA.sub.ks G.sub.ks.sup.mC.sub.k
1340315 N/A N/A 14284 14299 TGAATGTAAAGGCTGG
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1779 21579 21594 TACAGTTCCTTTTCCT
T.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsT.sub.dsT.sub.ds 68 2918
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ds
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.k 1340340 N/A N/A 6913 6928 TCGGGAAGTTTAGACA
T.sub.ks.sup.mC.sub.ksG.sub.ksG.sub.dsG.sub.dsA.sub.dsA.sub.ds 43 2919
G.sub.dsT.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.dsA.sub.ks .sup.mC.sub.ksA.sub.k 1340341 N/A
N/A 6330 6345 GACTTTCATAGGGAGA
G.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.dsT.sub.dsT.sub.ds 72 2920
.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsG.sub.dsG.sub.ds A.sub.ksG.sub.ksA.sub.k
1340346 N/A N/A 11510 11525 ATTATGAGGATCTGGA
A.sub.ksT.sub.ksrrksA.sub.dsT.sub.dsG.sub.dsA.sub.ds 49 2921
G.sub.dsG.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ks G.sub.ksA.sub.k 1340359 N/A
N/A 13949 13964 AAGTTAGCCCCCAGGA
A.sub.ksA.sub.ksG.sub.ksT.sub.dsT.sub.dsA.sub.dsG.sub.ds 36 2811
.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds
A.sub.dsG.sub.ksG.sub.ksA.sub.k 1340362 N/A N/A 6734 6749 GATTACCCCTGGCTTT
G.sub.ksA.sub.ksT.sub.ksT.sub.dsA.sub.ds.sup.mC.sub.ds 31 2922
.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsG.sub.ds
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T.sub.ksA.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.ds 24 2923
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds
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.sup.mC.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.dsG.sub.dsA.sub.ds 66 2924
T.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.ds G.sub.ksG.sub.ksT.sub.k 1340399 N/A
N/A 15419 15434 CCAGTAGGTGTGTTTC
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.dsT.sub.dsA.sub.ds 55 2925
G.sub.dsG.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.dsT.sub.ds T.sub.ksT.sub.ks.sup.mC.sub.k
1340403 N/A N/A 5673 5688 GGTCAGTAGAGAGCAT
G.sub.ksG.sub.ksT.sub.ks.sup.mC.sub.dsA.sub.dsG.sub.dsT.sub.ds 79 2926
A.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ks A.sub.ksT.sub.k 1340409
683 698 12101 12116 GATACCAGTTTTTCCC
G.sub.ksA.sub.ksT.sub.ksA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds 75 2927
A.sub.dsG.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ds
.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.k 1340411 N/A N/A 9206 9221
AGATGTACACTGACAA A.sub.ksG.sub.ksA.sub.ksT.sub.dsG.sub.dsT.sub.dsA.sub.ds 66 2928
.sup.mC.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsA.sub.ds .sup.mC.sub.ksA.sub.ksA.sub.k
1340420 N/A N/A 20193 20208 TGCAGTTATTAGAAGT
T.sub.ksG.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsT.sub.dsT.sub.ds 46 2929

A.sub.dsT.sub.dsT.sub.dsG.sub.dsAc.sub.isA.sub.ks G.sub.ksT.sub.k 1340444 N/A N/A
8963 8978 CTCCGTTATAAGTTTC
.sup.mC.sub.ksT.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsG.sub.dsT.sub.ds 59 2930
T.sub.dsA.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ds T.sub.ksT.sub.ks.sup.mC.sub.k
1340456 N/A N/A 16218 16233 TATAGACTGGGTAGGA
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9775 9790 ACTAGAACTCCCAACC
A.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsA.sub.ds 20 2931
.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds
A.sub.ks.sup.mC.sub.ks.sup.mC.sub.k 1340467 N/A N/A 8129 8144 TTAGAGTGCTTAGTTC
T.sub.ksT.sub.ksA.sub.ksG.sub.dsA.sub.dsG.sub.dsT.sub.ds 58 2932
G.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ks T.sub.ks.sup.mC.sub.k 1340473
N/A N/A 3608 3623 TTTAGCAGCTTGGAAG
T.sub.ksT.sub.ksT.sub.ksA.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ds 43 2933
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N/A 16425 16440 TGATATGTCAATACTC
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1340490 N/A N/A 10124 10139 GGATATGTCATCTAAA
G.sub.ksG.sub.ksA.sub.ksT.sub.dsA.sub.dsT.sub.dsG.sub.ds 68 2935
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N/A N/A 5896 5911 GTCCTTTGTATTTTCGC
G.sub.ksT.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.ds 45 2936
T.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.dsT.sub.dsT.sub.ds .sup.mC.sub.ksG.sub.ks.sup.mC.sub.k
1340520 N/A N/A 3701 3716 ATTTTCAGATCCCGTT
A.sub.ksT.sub.ksT.sub.ksT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds 67 2937
G.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds G.sub.ksT.sub.ksT.sub.k
1340530 N/A N/A 3490 3505 TACAGACTAAGGGACC
T.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsA.sub.ds 68 2938
.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsG.sub.dsG.sub.ds
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.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsG.sub.dsA.sub.ds 31 2812
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AGGCTGTAAGAGTCAG A.sub.ksG.sub.ksG.sub.ks.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.ds 74
2809 A.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ks A.sub.ksG.sub.k 1340565
N/A N/A 5844 5859 CCCCAAACATGGATGT
.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsA.sub.dsA.sub.ds 62 2939
A.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.dsA.sub.ds T.sub.ksG.sub.ksT.sub.k
1340586 811 826 15692 15707 TTATTGGTAAGTATTC
T.sub.ksT.sub.ksA.sub.ksT.sub.dsT.sub.dsG.sub.dsG.sub.ds 17 2940
T.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.ks T.sub.ks.sup.mC.sub.k 1340599 N/A
N/A 6599 6614 GGCTACTTTCAAACCT
G.sub.ksG.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.ds.sup.mC.sub.ds 54 2941
T.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ds
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.k 1340600 N/A N/A 8538 8553 GTTTTGCAAGTTTATC
G.sub.ksT.sub.ksT.sub.ksT.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds 73 2942
A.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.dsT.sub.dsA.sub.ks T.sub.ks.sup.mC.sub.k 1340612 N/A
N/A 6450 6465 AGTAGTAATTCTAAAC

A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsT.sub.dsA.sub.ds 23 2943
A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.ks A.sub.ks.sup.mC.sub.k 1340625
N/A N/A 7527 7542 TCCGAAAAAAGTGGAG
T.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksG.sub.dsA.sub.dsA.sub.ds 12 2944
A.sub.dsA.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds G.sub.ksA.sub.ksG.sub.k 1340627
N/A N/A 10292 10307 GAGAGA ACTTATACAA
G.sub.ksA.sub.ksG.sub.ksA.sub.dsG.sub.dsA.sub.dsA.sub.ds 37 2945
.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.ds .sup.mC.sub.ksA.sub.ksA.sub.k 1340629
77 92 3171 3186 CAGAAGGATTCTAGG
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A.sub.dsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks G.sub.ksG.sub.k 1340633 N/A
N/A 4662 4677 CCCGCCCTTAAGTCAT
.sup.mC.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksG.sub.ds.sup.mC.sub.ds 45 2947
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.ds
G.sub.dsT.sub.ds.sup.mC.sub.ksA.sub.ksT.sub.k 1340643 N/A N/A 5277 5292
GCTAGACAATTGCAAA G.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.ds 77 2948
.sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.dsT.sub.dsG.sub.ds .sup.mC.sub.dsA.sub.ksA.sub.ksA.sub.k
1340686 N/A N/A 5189 5204 TTCCCAACGCAACAGT
T.sub.ksT.sub.ks.sup.mC.sub.ks.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 60 2949
A.sub.ds.sup.mC.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ds
.sup.mC.sub.dsA.sub.ksG.sub.ksT.sub.k 1340694 N/A N/A 5648 5663 GAGCATTCATCAGATG
G.sub.ksA.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds 64 2950
.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds A.sub.ksT.sub.ksG.sub.k 1340698
N/A N/A 6087 6102 AGTTTTCACCTCAGGT
A.sub.ksG.sub.ksT.sub.ksT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds 82 2951
A.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds G.sub.ksG.sub.ksT.sub.k
1340709 N/A N/A 9116 9131 AGACAGACCAAGTAGC
A.sub.ksG.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds 21 2952
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ds A.sub.ksG.sub.ks.sup.mC.sub.k
1340719 N/A N/A 9944 9959 ACCACACTAATGAATC
A.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.ds 54 2953
.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.dsA.sub.ds A.sub.ksT.sub.ks.sup.mC.sub.k
1340726 N/A N/A 6262 6277 TGAGATGGGCAAGGCC
T.sub.ksG.sub.ksA.sub.ksG.sub.dsA.sub.dsT.sub.dsG.sub.ds 53 2954
G.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.dsG.sub.ks .sup.mC.sub.ks.sup.mC.sub.k
1340728 N/A N/A 8645 8660 GGCAAGACAGACTGTT
G.sub.ksG.sub.ks.sup.mC.sub.ksA.sub.dsA.sub.dsG.sub.dsA.sub.ds 75 2955
.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ds G.sub.ksT.sub.ksT.sub.k 1340729
N/A N/A 14191 14206 GATTGAGTTCTCCACT
G.sub.ksA.sub.ksT.sub.ksT.sub.dsG.sub.dsA.sub.dsG.sub.ds 45 2956
T.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds
A.sub.ks.sup.mC.sub.ksT.sub.k 1340739 N/A N/A 16288 16303 TTAGAATAGTCTTCAG
T.sub.ksT.sub.ksA.sub.ksG.sub.dsA.sub.dsA.sub.dsT.sub.ds 52 2957
A.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds .sup.mC.sub.ksA.sub.ksG.sub.k 1340752
N/A N/A 9590 9605 AGCTGGTAAAGGTAAG
A.sub.ksG.sub.ks.sup.mC.sub.ksT.sub.dsG.sub.dsG.sub.dsT.sub.ds 55 2958
A.sub.dsA.sub.dsA.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.ks A.sub.ksG.sub.k 1340755 N/A N/A
3995 4010 ATGTAGTGTTTACAAG A.sub.ksT.sub.ksG.sub.ksT.sub.dsA.sub.dsG.sub.dsT.sub.ds
58 2959 G.sub.dsT.sub.dsT.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ks A.sub.ksG.sub.k
1340769 N/A N/A 12666 12681 GTTAACCTGCAGCAGA

G.sub.ksT.sub.ksA.sub.dsA.sub.ds.sup.mC.sub.ds 51 2960
.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ds
.sup.mC.sub.dsA.sub.ksG.sub.ksA.sub.k 1340791 N/A N/A 17953 17968
TAGAAACTGCTCCTCA T.sub.ksA.sub.ksG.sub.ksA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds 32
2961 T.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds
T.sub.ks.sup.mC.sub.ksA.sub.k 1340795 N/A N/A 7789 7804 GTGACATAACTACTTC
G.sub.ksT.sub.ksG.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds 68 2962
A.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ds T.sub.ksT.sub.ks.sup.mC.sub.k
1340801 N/A N/A 4246 4261 TAAGTGTAAGAGACAT
T.sub.ksA.sub.ksA.sub.ksG.sub.dsT.sub.dsG.sub.dsT.sub.ds 63 2963
A.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ks A.sub.ksT.sub.k 1340810 N/A
N/A 12480 12495 GGAGCTAGATACTCAA
G.sub.ksG.sub.ksA.sub.ksG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds 42 2964
G.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.ds .sup.mC.sub.ksA.sub.ksA.sub.k
1340819 N/A N/A 18530 18545 TCCTACAGTGCTTAGT
T.sub.ks.sup.mC.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.ds.sup.mC.sub.ds 44 2965
A.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds A.sub.ksG.sub.ksT.sub.k
1340907 N/A N/A 5121 5136 AAGGAGCCAGTTATGA
A.sub.ksA.sub.ksG.sub.rksG.sub.dsA.sub.dsG.sub.ds 57 2966
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.ds A.sub.dsT.sub.ksG.sub.ksA.sub.k
1340917 N/A N/A 9666 9681 ATGCAGCTATAGGTAG
A.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.dsG.sub.ds 41 2967
.sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsG.sub.ds T.sub.ksA.sub.ksG.sub.k
(489) TABLE-US-00051 TABLE 51 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID SEQ
ID NO: 1 NO: 1 NO: 2 NO: 2 SEQ Compound Start Stop Start Stop Sequence Chemistry
Notation % ID Number Site Site Site Site (5' to 3') (5' to 3') inhibition NO 1245927 N/A
N/A 3433 3448 AGTAGATGGTAAGTCA
A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.ds 77 121
G.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks .sup.mC.sub.ksA.sub.k 1246023 N/A
N/A 5046 5061 GAGTGAATCATTCACT
G.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.ds 81 1526
T.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds A.sub.ksG.sub.ksT.sub.k 1341164
610 625 N/A N/A GAACAATATGGGATGA
G.sub.ksA.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.ds 49 1735
A.sub.dsT.sub.dsG.sub.dsG.sub.dsG.sub.ksA.sub.esT.sub.ks G.sub.esA.sub.k 1341172 812 827
15693 15708 CTTATTGGTAAGTATT
.sup.mC.sub.ksT.sub.ksT.sub.dsA.sub.dsT.sub.dsT.sub.dsG.sub.ds 9 1350
G.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ksT.sub.esA.sub.ks T.sub.esT.sub.k 1341178 698 713
12116 12131 GAGACATGAGGTTTTG
G.sub.ksA.sub.ksG.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds 49 725
G.sub.dsA.sub.dsG.sub.dsG.sub.dsT.sub.ksT.sub.esT.sub.ks T.sub.esG.sub.k 1341186 1079 1094
20894 20909 GCTAGTGCCAAACCAA
G.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds 62 1354
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ks.sup.mC.sub.es
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.sup.mC.sub.ksA.sub.ksG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds 58 263
T.sub.dsG.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksG.sub.es A.sub.ksA.sub.esA.sub.k
1341192 1200 1215 21015 21030 ATGTTTATGTAAGCAC
A.sub.ksT.sub.ksG.sub.dsT.sub.dsT.sub.dsT.sub.dsA.sub.ds 61 2622

T.sub.dsG.sub.dsT.sub.dsA.sub.ksG.sub.es.sub.k 1341196
1088 1103 20903 20918 TTGACTGCTGCTAGTG
T.sub.ksT.sub.ksG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.ds 29 28
.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.es G.sub.ksT.sub.esG.sub.k 1341199
1083 1098 20898 20913 TGCTGCTAGTGCCAAA
T.sub.ksG.sub.ks.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds 40 1665
T.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ks .sup.mC.sub.esA.sub.ksA.sub.esA.sub.k
1341203 1311 1326 21126 21141 TCTTAAACCTTCCCTG
T.sub.ks.sup.mC.sub.ksT.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ds 24 2160
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ks.sup.mC.sub.es
.sup.mC.sub.ksT.sub.esG.sub.k 1341208 1315 1330 21130 21145 ACAGTCTTAAACCTTC
A.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.ds 46 110
T.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ks
.sup.mC.sub.esT.sub.ksT.sub.es.sup.mC.sub.k 1341209 1257 1272 21072 21087
AGCTTTTGTCCACCTT A.sub.ksG.sub.ks.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ds 62
1822 G.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.es
.sup.mC.sub.ksT.sub.esT.sub.k 1341215 1323 1338 21138 21153 CTACTTGAACAGTCTT
.sup.mC.sub.ksT.sub.ksA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 47 734
G.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ksT.sub.es .sup.mC.sub.ksT.sub.esT.sub.k
1341221 1327 1342 21142 21157 AATGCTACTTGAACAG
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.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ksA.sub.es .sup.mC.sub.ksA.sub.esG.sub.k 1341224
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T.sub.dsG.sub.dsG.sub.dsA.sub.dsA.sub.ksT.sub.esG.sub.ks .sup.mC.sub.esT.sub.k 1341229 1331
1346 21146 21161 TTGGAATGCTACTTGA
T.sub.ksT.sub.ksG.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.ds 54 1279
G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.es T.sub.ksG.sub.esA.sub.k 1341230
1351 1366 21166 21181 TTCTGTGGCATGGCTA
T.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.ds 50 812
G.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.ksG.sub.es .sup.mC.sub.ksT.sub.esA.sub.k 1341235
1355 1370 21170 21185 GATATTCTGTGGCATG
G.sub.ksA.sub.ksT.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds 52 1046
T.sub.dsG.sub.dsT.sub.dsG.sub.dsG.sub.ks.sup.mC.sub.esA.sub.ks T.sub.esG.sub.k 1341245 1391
1406 21206 21221 GATCTCTTAGCTGTGC
G.sub.ksA.sub.ksT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ds 45 502
T.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.es T.sub.ksG.sub.es.sup.mC.sub.k
1341249 1359 1374 21174 21189 TGTTGATATTCTGTGG
T.sub.ksG.sub.ksT.sub.dsT.sub.dsG.sub.dsA.sub.dsT.sub.ds 54 1358
A.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.esT.sub.ks G.sub.esG.sub.k 1341252 1431
1446 21246 21261 AAATATGTCCAGGTTG
A.sub.ksA.sub.ksA.sub.dsT.sub.dsA.sub.dsT.sub.dsG.sub.ds 61 581
T.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ksG.sub.es T.sub.ksT.sub.esG.sub.k 1341253
1419 1434 21234 21249 GTTGAGATAAAGCTGC
G.sub.ksT.sub.ksT.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.ds 40 1748
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1423 1438 21238 21253 CCAGGTTGAGATAAAG
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.dsG.sub.dsT.sub.ds 55 33
T.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.dsT.sub.ksA.sub.es A.sub.ksA.sub.esG.sub.k 1341264
1492 1507 21307 21322 CACCGTTTTGGGCTAA

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T.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsG.sub.ks.sup.mC.sub.es T.sub.ksA.sub.esA.sub.k
1341270 1602 1617 21417 21432 ATGTATCTTATAAGAC
A.sub.ksT.sub.ksG.sub.dsT.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds 18 1751
T.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.ksA.sub.esG.sub.ks A.sub.es.sup.mC.sub.k 1341272 1501
1516 21316 21331 ATAGAGTTGCACCGTT
A.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ds 45 192
T.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ks.sup.mC.sub.es G.sub.ksT.sub.esT.sub.k
1341276 1617 1632 21432 21447 CGGTCACCTTTCATAA
.sup.mC.sub.ksG.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds 70 36
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ks
A.sub.esT.sub.ksA.sub.esA.sub.k 1341278 1621 1636 21436 21451 GAGTCGGTCACCTTTC
G.sub.ksA.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsG.sub.dsG.sub.ds 62 350
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T.sub.ds.sup.mC.sub.dsG.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.es
A.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341289 1629 1644 21444 21459 TTAAAATAGAGTCGGT
T.sub.ksT.sub.ksA.sub.dsA.sub.dsA.sub.dsA.sub.dsT.sub.ds 51 973
A.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.esG.sub.ks G.sub.esT.sub.k 1341293 1713
1728 21528 21543 AACATCTCTGGGACCA
A.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds 36 273
T.sub.ds.sup.mC.sub.dsT.sub.dsG.sub.dsG.sub.dsG.sub.ksA.sub.es
.sup.mC.sub.ks.sup.mC.sub.esA.sub.k 1341294 1708 1723 21523 21538
CTCTGGGACCAAGGAT .sup.mC.sub.ksT.sub.ks.sup.mC.sub.dsT.sub.dsG.sub.dsG.sub.ds 39
1908 G.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ks
G.sub.esG.sub.ksA.sub.esT.sub.k 1341301 1777 1792 21592 21607 ATGTAATAGCCAGTAC
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A.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.es T.sub.ksA.sub.es.sup.mC.sub.k
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T.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.es T.sub.ksT.sub.es.sup.mC.sub.k
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.sup.mC.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ks A.sub.esG.sub.ksT.sub.esT.sub.k
1341319 1781 1796 21596 21611 TCTTATGTAATAGCCA
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T.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.ksG.sub.es.sup.mC.sub.ks .sup.mC.sub.esA.sub.k 1341322
2254 2269 22069 22084 CAAGAGGATAGTCCAT
.sup.mC.sub.ksA.sub.ksA.sub.dsG.sub.dsA.sub.dsG.sub.dsG.sub.ds 22 510
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T.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ksA.sub.es
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T.sub.dsG.sub.dsT.sub.dsA.sub.dsG.sub.ksT.sub.esG.sub.ks G.sub.esG.sub.k 1341333 N/A N/A
9518 9533 CTACACTAATATTGAG
.sup.mC.sub.ksT.sub.ksA.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds 5 59
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N/A 12421 12436 TGGTAACGGTGATCAA
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N/A 16131 16146 GCTCCATAATAATAGC
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1341374 N/A N/A 18099 18114 CTGAAATTGATACACC
.sup.mC.sub.ksT.sub.ksG.sub.dsA.sub.dsA.sub.dsA.sub.dsT.sub.ds 45 393
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N/A 19911 19926 GCATCTTAAGATACCC
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241 T.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksA.sub.es
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A.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ksT.sub.es .sup.mC.sub.ksT.sub.esA.sub.k
1341398 N/A N/A 20194 20209 TTGCAGTTATTAGAAG
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N/A 12501 12516 GCAGAAATTCACCTTG
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A.sub.dsA.sub.dsT.sub.dsT.sub.dsA.sub.ksT.sub.esG.sub.ks .sup.mC.sub.es.sup.mC.sub.k 1341411
N/A N/A 14194 14209 TGTGATTGAGTTCTCC
T.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsT.sub.dsT.sub.ds 11 2020
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1341414 N/A N/A 14168 14183 ATAGCAAGCCAACAGA
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1341528 1075 1090 20890 20905 GTGCCAAACCAATGTT
G.sub.ksT.sub.ksG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds 38 1042
A.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ks T.sub.esG.sub.ksT.sub.esT.sub.k
1341536 801 816 15682 15697 GTATTCCATCTATCAG
G.sub.ksT.sub.ksA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds 48 727

.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ks
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T.sub.ksA.sub.ksT.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsA.sub.ds 32 103
.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksT.sub.es A.sub.ks.sup.mC.sub.esG.sub.k
1341542 1319 1334 21134 21149 TTGAACAGTCTTAAAC
T.sub.ksT.sub.ksG.sub.dsA.sub.dsA.sub.ds.sup.MC.sub.dsA.sub.ds 37 422
G.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksA.sub.esA.sub.ks A.sub.es.sup.mC.sub.k 1341544
1264 1279 21079 21094 GGGAGGTAGCTTTTGT
G.sub.ic.sub.sG.sub.ic.sub.sG.sub.dsA.sub.dsG.sub.dsG.sub.ds 66 343
T.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ksT.sub.es T.sub.ksG.sub.esT.sub.k
1341548 1384 1399 21199 21214 TAGCTGTGCACTCATT
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G.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ks.sup.mC.sub.es T.sub.ksT.sub.esA.sub.k 1341553
1346 1361 21161 21176 TGGCATGGCTACAGAT
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G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksA.sub.es G.sub.ksA.sub.esT.sub.k 1341559
1488 1503 21303 21318 GTTTTGGGCTAATGAA
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G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.ksT.sub.esG.sub.ks A.sub.esA.sub.k 1341562 1496
1511 21311 21326 GTTGCACCGTTTTGGG
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.sup.mC.sub.ds.sup.mC.sub.dsG.sub.dsT.sub.dsT.sub.dsT.sub.ks T.sub.esG.sub.ksG.sub.esG.sub.k
1341571 1717 1732 21532 21547 TCTAAACATCTCTGGG
T.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.dsA.sub.dsA.sub.ds 21 2287
.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.ks
T.sub.esG.sub.ksG.sub.esG.sub.k 1341586 2184 2199 21999 22014 TGTAGTGGGAGTCGGA
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G.sub.dsG.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.esG.sub.ks G.sub.esA.sub.k 1341587 1726
1741 21541 21556 CTAAAATTGTCTAAAC
.sup.mC.sub.ksT.sub.ksA.sub.dsA.sub.dsA.sub.dsA.sub.dsT.sub.ds 19 2590
T.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ksA.sub.esA.sub.ks A.sub.es.sup.mC.sub.k 1341590
N/A N/A 9337 9352 CCTCGATCCTATATAC
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ds.sup.mC.sub.dsG.sub.dsA.sub.ds 3 2812
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N/A 11568 11583 TGACAATGGTTGATAG
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G.sub.dsG.sub.dsT.sub.dsT.sub.dsG.sub.ksA.sub.esT.sub.ks A.sub.esG.sub.k 1341597 N/A N/A
9971 9986 TTGAGTAGTGTAAGCT T.sub.ksT.sub.ksG.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.ds
29 2346 G.sub.dsT.sub.dsG.sub.dsT.sub.dsA.sub.ksA.sub.esG.sub.ks .sup.mC.sub.esT.sub.k
1341598 N/A N/A 16282 16297 TAGTCTTCAGCAAAGT
T.sub.ksA.sub.ksG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds 35 2632
.sup.mC.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ksA.sub.es A.sub.ksG.sub.esT.sub.k
1341599 N/A N/A 16377 16392 ACCAAACTTCCAGCAG
A.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ds 32 1948

.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ks
G.sub.es.sup.mC.sub.ksA.sub.esG.sub.k 1341605 N/A N/A 15491 15506 TTAATGCCACCCTACC
T.sub.ksT.sub.ksA.sub.dsA.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds 11 2304
.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.es
A.sub.ks.sup.mC.sub.es.sup.mC.sub.k 1341606 N/A N/A 14204 14219 CAGATTATTATGTGAT
.sup.mC.sub.ksA.sub.ksG.sub.dsA.sub.dsT.sub.dsT.sub.dsA.sub.ds 31 2723
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17986 18001 TATAGTTATCTTCTCA T.sub.ksA.sub.ksT.sub.dsA.sub.dsG.sub.dsT.sub.dsT.sub.ds
45 2295 A.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ks.sup.mC.sub.es.sup.m
T.sub.ksC.sub.esA.sub.k 1341616 N/A N/A 17996 18011 ATGAAGTTAGTATAGT
A.sub.ksT.sub.ksG.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ds 34 1483
T.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.ksT.sub.esA.sub.ks G.sub.esT.sub.k
(490) TABLE-US-00052 TABLE 52 Inhibition of HSD17B13 RNA by modified
oligonucleotides targeting SEQ ID NO.: 1, and 2 SEQ ID SEQ ID SEQ ID
SEQ ID SEQ Compound NO: 1 NO: 1 NO: 2 NO: 2 Sequence Chemistry Notation %
ID Number Start Site Stop Site Start Site Stop Site (5' to 3') (5' to 3') inhibition NO
1245927 N/A N/A 3433 3448 AGTAGATGGTAAGTCA
A.sub.ksG.sub.ksT.sub.ksA.sub.dsG.sub.dsA.sub.dsT.sub.dsG.sub.ds 79 121
G.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.ks.sup.mC.sub.ksA.sub.k 1246023 N/A N/A 5046
5061 GAGTGAATCATTGAGT
G.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.dsA.sub.dsA.sub.dsT.sub.ds 89 1526
.sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ksG.sub.ks T.sub.k 1341650 1264
1279 21079 21094 GGGAGGTAGCTTTTGT
G.sub.ksG.sub.ksG.sub.ksA.sub.dsG.sub.ysG.sub.dsT.sub.dsA.sub.ds 17 343
G.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ksG.sub.ksT.sub.k 1341653 1316 1331
21131 21146 AACAGTCTTAAACCTT
A.sub.ksA.sub.ks.sup.mC.sub.ksA.sub.dsG.sub.ysT.sub.ds.sup.mC.sub.ds 75 188
T.sub.dsT.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks T.sub.ksT.sub.k 1341660
1328 1343 21143 21158 GAATGCTACTTGAACA
G.sub.ksA.sub.ksA.sub.ksT.sub.dsG.sub.ys.sup.mC.sub.dsT.sub.ds 48 1123
A.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsA.sub.ks .sup.mC.sub.ksA.sub.k
1341664 1320 1335 21135 21150 CTTGAACAGTCTTAAA
.sup.mC.sub.ksT.sub.ksT.sub.ksG.sub.dsA.sub.ysA.sub.ds.sup.mC.sub.ds 49 500
A.sub.dsG.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.ks A.sub.ksA.sub.k 1341667 1312
1327 21127 21142 GTCTUAAACCTTCCCT
G.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.dsU.sub.ysA.sub.dsA.sub.ds 35 2968
A.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.k 1341670 1333 1348 21148 21163 GATTGGAATGCTACTT
G.sub.ksA.sub.ksT.sub.ksT.sub.dsG.sub.ysG.sub.dsA.sub.dsA.sub.ds 48 1435
T.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ksT.sub.ks T.sub.k 1341676 1324
1339 21139 21154 GCTACTTGAACAGTCT
G.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.dsC.sub.ysT.sub.dsT.sub.ds 82 811
G.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsT.sub.ks .sup.mC.sub.ksT.sub.k
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.sup.mC.sub.dsT.sub.dsG.sub.dsT.sub.dsG.sub.dsG.sub.ds.sup.mC.sub.ks A.sub.ksT.sub.k 1341680
1347 1362 21162 21177 GTGGCATGGCTACAGA
G.sub.issT.sub.ksG.sub.issG.sub.dsC.sub.ysA.sub.dsT.sub.ds 49 501
G.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ks G.sub.ksA.sub.k
1341683 1340 1355 21155 21170 GGCTACAGATTGGAAT

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G.sub.dsA.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsA.sub.ksA.sub.ks T.sub.k 1341690 1364 1379
21179 21194 GTTCUTGTTGATATTC
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T.sub.dsT.sub.dsG.sub.dsA.sub.dsT.sub.dsA.sub.dsT.sub.ksT.sub.ks .sup.mC.sub.k 1341696 1352
1367 21167 21182 ATTCUGTGGCATGGCT
A.sub.ksT.sub.ksT.sub.ks.sup.mC.sub.dsU.sub.ysG.sub.dsT.sub.ds 59 2970
G.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsG.sub.dsG.sub.ks .sup.mC.sub.ksT.sub.k
1341699 1387 1402 21202 21217 TCTTAGCTGTGCACTC
T.sub.ks.sup.mC.sub.ksT.sub.ksT.sub.dsA.sub.ysG.sub.ds.sup.mC.sub.ds 71 190
T.sub.dsG.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ks T.sub.ks.sup.mC.sub.k
1341705 1393 1408 21208 21223 TTGAUCTCTTAGCTGT
T.sub.ksT.sub.ksG.sub.ksA.sub.dsU.sub.ys.sup.mC.sub.dsT.sub.ds 50 2971
.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.ks G.sub.ksT.sub.k 1341711
1420 1435 21235 21250 GGTTGAGATAAAGCTG
G.sub.ksG.sub.ksT.sub.ksT.sub.dsG.sub.ysA.sub.dsG.sub.dsA.sub.ds 57 1826
T.sub.dsA.sub.dsA.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ksT.sub.ksG.sub.k 1341714 1425 1440
21240 21255 GTCCAGGTTGAGATAA
G.sub.ksT.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsA.sub.ysG.sub.dsG.sub.ds 68 113
T.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.dsT.sub.ksA.sub.ks A.sub.k 1341718 1461 1476
21276 21291 CTAGGGAAATCTTTCA
.sup.mC.sub.ksT.sub.ksA.sub.ksG.sub.dsG.sub.ysG.sub.dsA.sub.ds 19 659
A.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ks .sup.mC.sub.ksA.sub.k 1341721
693 708 12111 12126 ATGAGGTTTTGATACC
A.sub.ksT.sub.ksG.sub.ksA.sub.dsG.sub.ysG.sub.dsT.sub.dsT.sub.ds 49 569
T.sub.dsT.sub.dsG.sub.dsA.sub.dsT.sub.dsA.sub.ks.sup.mC.sub.ks .sup.mC.sub.k 1341725 1415
1430 21230 21245 AGATAAAGCTGCCTGC
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.sup.mC.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ksG.sub.ks .sup.mC.sub.k
1341728 1489 1504 21304 21319 CGTTUTGGGCTAATGA
.sup.mc.sub.ksG.sub.ksT.sub.ksT.sub.dsU.sub.ysT.sub.dsG.sub.ds 34 2972
G.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.ks G.sub.ksA.sub.k 1341731 1497
1512 21312 21327 AGTTGCACCGTTTTGG
A.sub.ksG.sub.ksT.sub.ksT.sub.dsG.sub.ys.sup.mC.sub.dsA.sub.ds 38 1905
.sup.mC.sub.ds.sup.mC.sub.dsG.sub.dsT.sub.dsT.sub.dsT.sub.dsT.sub.ks G.sub.ksG.sub.k 1341736
1493 1508 21308 21323 GCACCGTTTTGGGCTA
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T.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsG.sub.ds.sup.mC.sub.ks T.sub.ksA.sub.k 1341738 1622
1637 21437 21452 AGAGUCGGTCACCTTT
A.sub.ksG.sub.ksA.sub.ksG.sub.dsU.sub.ys.sup.mC.sub.dsG.sub.ds 55 2973
G.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ks T.sub.ksT.sub.k
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A.sub.ksG.sub.ksA.sub.ksA.sub.dsU.sub.ysA.sub.dsG.sub.dsA.sub.d 49 2974
.sub.sG.sub.dsT.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.dsA.sub.ks.sup.mC.sub.ks .sup.mC.sub.k
1341748 1614 1629 21429 21444 TCACCTTTCATAATGT
T.sub.ks.sup.mC.sub.ksA.sub.ks.sup.mC.sub.dsC.sub.ysT.sub.dsT.sub.dsT 47 2109
.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.ksG.sub.ks T.sub.k 1341754 1618
1633 21433 21448 TCGGUCACCTTTCATA
T.sub.ks.sup.mC.sub.ksG.sub.ksG.sub.dsU.sub.ys.sup.mC.sub.dsA.sub.ds 21 2975
.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ks T.sub.ksA.sub.k

1341761 1641 21445 21460 TTATAAATAGAGTCGG
T.sub.ksT.sub.ksT.sub.ksA.sub.dsA.sub.ysA.sub.dsA.sub.dsT.sub.ds 77 1051
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1341768 1710 1725 21525 21540 ATCTCTGGGACCAAGG
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G.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ks G.sub.ksG.sub.k
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.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ks G.sub.ksG.sub.k
1341781 1768 1783 21583 21598 CCAGUACAGTTCCTTT
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.ksG.sub.dsU.sub.ysA.sub.ds.sup.mC.sub.ds 51 2976
A.sub.dsG.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ks T.sub.ksT.sub.k 1341783
1750 1765 21565 21580 CTGTGTTAGCTTTAAT
.sup.mC.sub.ksT.sub.ksG.sub.ksT.sub.dsG.sub.ysT.sub.dsT.sub.ds 41 1286
A.sub.dsG.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsA.sub.ks A.sub.ksT.sub.k 1341787 794
809 15675 15690 ATCTATCAGACTTCTT
A.sub.ksT.sub.ks.sup.mC.sub.ksT.sub.dsA.sub.ysT.sub.ds.sup.mC.sub.ds 34 2728
A.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ks T.sub.ksT.sub.k 1341789
1778 1793 21593 21608 TATGUAATAGCCAGTA
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A.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.ksT.sub.ks A.sub.k 1341792 1774
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66 2979 G.sub.dsT.sub.dsA.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.ks .sup.mC.sub.ks.sup.mC.sub.k
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2262 22062 22077 ATAGUCCATGCAAAAG
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N/A 9330 9345 CCTAUATACATCCAAG
.sup.mC.sub.ks.sup.mC.sub.ksT.sub.ksA.sub.dsU.sub.ysA.sub.dsT.sub.ds 39 2982
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9979 GTGTAAGCTGAGAGTT G.sub.ksT.sub.ksG.sub.ksT.sub.dsA.sub.ysA.sub.dsG.sub.ds 55
2760 .sup.mC.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.dsA.sub.dsG.sub.ks T.sub.ksT.sub.k 1341840
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820 15686 15701 GTAAGTATTCCATCTA
G.sub.ksT.sub.ksA.sub.ksA.sub.dsG.sub.ysT.sub.dsA.sub.dsT.sub.ds 43 1038
T.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.ks A.sub.k 1341851
N/A N/A 14196 14211 TATGUGATTGAGTTCT
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N/A 12426 12441 CAGGGTGGTAACGGTG
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N/A 16134 16149 GATGCTCCATAATAAT
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17992 18007 AGTTAGTATAGTTATC
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16380 16395 CCAACCAAACCTTCCAG
.sup.mC.sub.ks.sup.mC.sub.ksA.sub.ksA.sub.dsC.sub.ys.sup.mC.sub.dsA.sub.ds 30 2539
A.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.ks A.sub.ksG.sub.k
1341889 N/A N/A 17987 18002 GTATAGTTATCTTCTC
G.sub.ksT.sub.ksA.sub.ksT.sub.dsA.sub.ysG.sub.dsT.sub.dsT.sub.ds 54 937
A.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ksT.sub.ks .sup.mC.sub.k 1341890

1082 1097 20897 20912 GCTGCTAGTGCCAAAC
G.sub.ks.sup.mC.sub.ksT.sub.ksG.sub.dsC.sub.ysT.sub.dsA.sub.ds 39 1587
G.sub.dsT.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.ks A.sub.ks.sup.mC.sub.k
1341895 N/A N/A 18099 18114 CTGAAATTGATACACC
.sup.mC.sub.ksT.sub.ksG.sub.ksA.sub.dsA.sub.ysA.sub.dsT.sub.ds 71 393
T.sub.dsG.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ks .sup.mC.sub.ks.sup.mC.sub.k
1341896 N/A N/A 17997 18012 TATGAAGTTAGTATAG
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19926 GCATCTTAAGATACCC
G.sub.ks.sup.mC.sub.ksA.sub.ksT.sub.dsC.sub.ysT.sub.dsT.sub.ds 65 553
A.sub.dsA.sub.dsG.sub.dsA.sub.dsT.sub.dsA.sub.ds.sup.mC.sub.ks .sup.mC.sub.ks.sup.mC.sub.k
1341907 N/A N/A 18697 18712 CAAAAGACTTTGAGAC
.sup.mC.sub.ksA.sub.ksA.sub.ksA.sub.dsA.sub.ysG.sub.dsA.sub.ds 25 2186
.sup.mC.sub.dsT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.dsG.sub.ks A.sub.ks.sup.mC.sub.k 1341908
N/A N/A 19915 19930 GAGTGCATCTTAAGAT
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T.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.dsG.sub.ks A.sub.ksT.sub.k 1341914 N/A
N/A 19907 19922 CTTAAGATACCCAGGT
.sup.mC.sub.ksT.sub.ksT.sub.ksA.sub.dsA.sub.ysG.sub.dsA.sub.ds 50 241
T.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds G.sub.ksG.sub.ksT.sub.k
1341919 N/A N/A 20109 20124 AAGACATTCTAGCCTG
A.sub.ksA.sub.ksG.sub.ksA.sub.dsC.sub.ysA.sub.dsT.sub.dsT.sub.ds 54 2744
.sup.mC.sub.dsT.sub.dsA.sub.dsG.sub.ds.sup.mC.sub.ds.sup.mC.sub.ksT.sub.ks G.sub.k 1341923
N/A N/A 20119 20134 TTACAGCATCAAGACA
T.sub.ksT.sub.ksA.sub.ks.sup.mC.sub.dsA.sub.ysG.sub.ds.sup.mC.sub.ds 46 2409
A.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.dsA.sub.ks .sup.mC.sub.ksA.sub.k
1341924 1078 1093 20893 20908 CTAGUGCCAAACCAAT
.sup.mC.sub.ksT.sub.ksA.sub.ksG.sub.dsU.sub.ysG.sub.ds.sup.mC.sub.ds 45 2988
.sup.mC.sub.dsA.sub.dsA.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds A.sub.ksA.sub.ksT.sub.k
1341928 N/A N/A 20194 20209 TTGCAGTTATTAGAAG
T.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.ysG.sub.dsT.sub.ds 17 2123
T.sub.dsA.sub.dsT.sub.dsT.sub.dsA.sub.dsG.sub.dsA.sub.ks A.sub.ksG.sub.k 1341930 N/A N/A
20383 20398 GTAAUCACAAGTAAGG
G.sub.ksT.sub.ksA.sub.ksA.sub.dsU.sub.ys.sup.mC.sub.dsA.sub.ds 20 2989
.sup.mC.sub.dsA.sub.dsA.sub.dsG.sub.dsT.sub.dsA.sub.dsA.sub.ks G.sub.ksG.sub.k 1341941 N/A
N/A 20489 20504 CATAACTTCATAGTGG
.sup.mC.sub.ksA.sub.ksT.sub.ksA.sub.dsA.sub.ys.sup.mC.sub.dsT.sub.ds 38 866
T.sub.ds.sup.mC.sub.dsA.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.ks G.sub.ksG.sub.k 1341942 1074
1089 20889 20904 TGCCAAACCAATGTTT
T.sub.ksG.sub.ks.sup.mC.sub.ks.sup.mC.sub.dsA.sub.ysA.sub.dsA.sub.ds 35 2673
.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsA.sub.dsT.sub.dsG.sub.dsT.sub.ks T.sub.ksT.sub.k 1341948
N/A N/A 12503 12518 TTGCAGAAATTCACCT
T.sub.ksT.sub.ksG.sub.ks.sup.mC.sub.dsA.sub.ysG.sub.dsA.sub.ds 29 2528
A.sub.dsA.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ks .sup.mC.sub.ksT.sub.k
1341953 1086 1101 20901 20916 GACTGCTGCTAGTGCC
G.sub.ksA.sub.ks.sup.mC.sub.ksT.sub.dsG.sub.ys.sup.mC.sub.dsT.sub.ds 60 1899
G.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsG.sub.dsT.sub.dsG.sub.ks.sup.m C.sub.ks.sup.mC.sub.k
1341957 N/A N/A 20479 20494 TAGTGGACTTCATTAG
T.sub.ksA.sub.ksG.sub.ksT.sub.dsG.sub.ysG.sub.dsA.sub.ds 34 2351

Example 5: Dose-Dependent Inhibition of Human HSD17B13 in HepaRG Cells by Modified Oligonucleotides

(491) Certain modified oligonucleotides described in the studies above exhibiting significant in vitro inhibition of HSD17B13 RNA were selected and tested at various doses in HepaRG cells.

(492) The modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Cultured HepaRG cells at a density of 30,000-35,000 cells per well were transfected using electroporation with modified oligonucleotides diluted to different concentrations as specified in the tables below. After a treatment period of approximately 24 hours, HSD17B13 RNA levels were measured as previously described using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total RNA content, as measured by human GAPDH. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells. Data indicated as “N.D.” (no data) means that no data is available for that treatment with that compound.

(493) The half maximal inhibitory concentration (IC₅₀) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

(494) TABLE-US-00053 TABLE 53 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC₅₀ No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μM)
1245094 0 5 75 68 N.D. 0.3 1245641 5 16 52 75 80 0.4 1245744 5 11 40 81 87 0.4 1245822 0 0 57 78 93 0.1 1245901 1 14 38 55 78 0.8 1245927 8 42 76 83 92 0.2 1245952 25 34 51 70 86 0.2 1245979 4 0 38 64 75 0.8 1246005 1 14 60 64 78 0.5 1246134 0 50 75 79 N.D. 0.2 1246135 10 10 42 68 88 0.5 1246161 18 20 63 75 86 0.3 1246550 14 28 58 76 88 0.3 1246706 60 69 68 76 N.D. 0.02

(495) TABLE-US-00054 TABLE 54 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC₅₀ No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μM)
1245096 3 17 70 85 N.D. 0.2 1245097 4 40 75 80 76 0.2 1245565 2 0 26 56 65 1.5 1245798 16 37 57 82 90 0.2 1245825 0 11 44 64 81 0.6 1245851 0 9 45 69 84 0.6 1245927 0 25 56 79 82 0.4 1245928 7 42 67 89 95 0.2 1245980 12 43 94 84 N.D. 0.1 1246059 1 15 44 68 81 0.5 1246084 4 18 69 89 93 0.2 1246110 10 41 71 90 N.D. 0.1 1246111 0 22 44 79 83 0.4 1246162 33 31 71 88 94 0.1 1246163 1 17 42 73 92 0.4

(496) TABLE-US-00055 TABLE 55 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC₅₀ No. 19 nM 78 nM 313 nM 1250 nM 5000 nM (μM)
1245098 25 27 71 80 71 0.2 1245463 0 0 20 63 78 0.9 1245696 15 30 52 74 89 0.3 1245800 0 30 55 82 82 0.3 1245904 12 19 51 73 97 0.3 1245927 3 11 48 68 76 0.6 1245930 7 26 58 77 89 0.3 1245957 0 0 54 75 86 0.5 1245982 7 25 45 78 92 0.3 1245983 0 18 42 67 90 0.5 1246035 0 2 51 73 75 0.6 1246113 12 0 47 75 84 0.6 1247023 0 23 46 75 90 0.4 1247048 9 22 41 74 84 0.4 1247075 0 0 4 50 68 2.0

(497) TABLE-US-00056 TABLE 56 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC₅₀ No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μM)
1245075 20 50 62 75 86 0.1 1245829 18 33 58 83 94 0.2 1245855 16 21 59 78 89 0.3 1245927 0 21 54 69 76 0.5 1245932 22 24 56 90 94 0.2 1245933 23 24 63 76 89 0.2 1245958 12 31 43 87 84 0.3 1245959 1 10 46 65 90 0.5 1245984 9 37 62 76 93 0.2 1246036 15 33 65 80 95 0.2 1246063 16 29 53 84 85 0.2 1246115 0 38 66 88 92 0.2 1246140 20 2 54 75 86 0.5 1246192 0 17 52 72 89 0.5 1247024 16 45 82 89 97 0.1

(498) TABLE-US-00057 TABLE 57 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC₅₀ No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μM)
1245076 0 0 32 66 83 0.6 1245415 0 0 22 65 88 0.9 1245649 8 23 49 79 97 0.3 1245857 0 2 35 74 91 0.6 1245927 4 25 57 82 86 0.3 1245960 14 34 54 78 88 0.2 1245986 2 2 30 78 95 0.4 1245987

26 32 58 82 91 0.2 1246013 0 0 27 68 83 0.8 1246064 19 36 63 74 92 0.2 1246116 17 41 75 92 95
0.1 1246169 0 0 30 55 86 1.0 1246195 0 0 26 63 82 0.9 1246610 5 0 2 57 82 1.3 1247078 3 4 28
66 93 0.6

(499) TABLE-US-00058 TABLE 58 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245105 0 0 24 61 88 0.8 1245651 3 22 49 78 89 0.3 1245676 0 0 34 67 91 0.7 1245806 28 27 51
69 91 0.2 1245807 1 9 47 80 90 0.4 1245858 16 39 66 86 96 0.2 1245927 0 22 46 79 89 0.4
1245988 8 26 40 65 85 0.5 1246066 26 41 63 86 94 0.1 1246067 0 0 40 51 85 0.9 1246093 0 0 24
64 86 0.7 1246144 5 30 43 78 86 0.3 1246145 0 3 40 72 85 0.6 1246248 28 32 62 86 92 0.1
1247029 0 0 11 36 73 1.9

(500) TABLE-US-00059 TABLE 59 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245080 7 13 39 70 87 0.5 1245081 4 28 44 72 85 0.4 1245106 0 28 46 80 81 0.4 1245809 0 0 1
48 77 1.6 1245886 0 4 21 62 80 0.9 1245927 16 29 64 84 90 0.2 1245964 16 22 48 72 95 0.3
1245990 15 4 29 61 86 0.7 1245991 0 9 26 63 83 0.8 1246068 14 31 49 73 92 0.3 1246095 10 16
46 69 88 0.4 1246121 0 18 37 76 86 0.5 1246173 3 26 58 83 89 0.3 1246562 5 19 26 70 88 0.5

(501) TABLE-US-00060 TABLE 60 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245212 3 1 43 63 88 0.6 1245680 0 0 22 64 82 0.9 1245681 0 17 58 80 94 0.3 1245888 17 22 41
72 87 0.4 1245927 0 16 52 78 93 0.4 1245967 0 9 41 62 89 0.6 1246019 17 26 62 92 91 0.2
1246070 7 31 46 74 94 0.3 1246123 0 16 37 58 83 0.7 1246226 12 30 56 86 89 0.2 1246227 0 6 0
55 78 1.4 1246824 0 0 17 40 81 1.5 1246902 0 17 40 74 86 0.5 1246903 35 35 60 68 84 0.2
1246929 0 18 40 65 87 0.6

(502) TABLE-US-00061 TABLE 61 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245085 18 21 49 79 95 0.3 1245162 19 5 41 78 97 0.4 1245579 0 8 39 65 87 0.6 1245656 20 33
47 80 83 0.3 1245682 23 42 53 87 93 0.2 1245838 9 0 28 77 92 0.5 1245916 1 25 26 60 82 0.7
1245927 14 36 76 77 86 0.2 1245968 24 43 69 88 88 0.1 1245969 0 9 32 67 74 0.9 1246124 7 12
31 76 90 0.5 1246203 0 0 21 58 87 1.0 1246489 2 14 44 86 86 0.4 1246853 0 0 16 52 77 1.3
1247061 0 12 32 70 93 0.5

(503) TABLE-US-00062 TABLE 62 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245580 3 9 49 58 89 0.5 1245607 0 16 33 77 87 0.5 1245684 35 51 77 92 96 0.1 1245685 13 25
59 80 93 0.2 1245736 0 0 45 81 95 0.6 1245737 13 23 53 83 94 0.3 1245919 0 0 0 72 90 1.0
1245927 0 9 37 84 77 0.4 1245996 29 53 74 91 97 0.1 1245997 16 31 54 79 90 0.2 1246022 49 54
80 93 94 0.0 1246023 0 40 89 86 95 0.2 1246048 12 0 73 73 97 0.3 1246074 7 43 73 84 94 0.2
1246075 0 0 41 70 88 0.7

(504) TABLE-US-00063 TABLE 63 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245141 12 30 57 84 90 0.2 1245374 29 7 43 65 85 0.4 1245660 7 12 36 73 94 0.4 1245738 36 38
61 80 88 0.1 1245790 50 61 68 87 95 <0.2 1245927 0 23 58 85 90 0.3 1245972 24 39 58 79 90 0.2
1245973 0 26 41 66 75 0.5 1245998 12 10 43 78 89 0.4 1245999 0 19 60 87 88 0.3 1246025 35 54
84 94 89 0.0 1246051 4 26 54 78 92 0.3 1246128 37 38 58 84 92 0.1 1246701 0 4 12 20 40 5.0
1246909 0 17 27 66 86 0.7

(505) TABLE-US-00064 TABLE 64 Multi-dose assay of modified oligonucleotides in HepaRG
cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245142 0 9 40 64 78 0.8 1245246 15 32 33 74 91 0.3 1245610 28 44 61 82 97 0.1 1245715 0 0 37
65 86 0.9 1245844 38 37 53 75 89 0.1 1245927 11 27 64 75 81 0.3 1245975 0 11 48 71 82 0.5
1246027 0 2 15 71 85 0.8 1246052 36 41 57 79 85 0.1 1246053 0 0 9 56 74 1.5 1246104 21 40 64
77 91 0.2 1246105 0 0 3 47 73 1.8 1246130 35 13 54 78 91 0.2 1246157 44 37 62 78 91 0.1

1246391 0 0 2 48 83 1.5
(506) TABLE-US-00065 TABLE 65 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1245092 9 20 39 68 89 0.4 1245639 0 0 21 65 85 0.8 1245716 20 27 40 69 86 0.4 1245927 0 0 25 74 64 1.2 1245976 9 29 57 77 86 0.3 1245977 0 21 44 76 94 0.4 1246002 38 51 64 79 91 0.1 1246028 0 0 19 56 80 1.2 1246029 0 18 50 73 86 0.4 1246081 5 19 48 75 88 0.4 1246185 13 15 43 62 79 0.6 1246314 3 8 36 72 92 0.5 1246340 28 32 37 58 85 0.4 1246705 1 14 20 42 70 1.8 1246861 0 0 7 25 63 5.0
(507) TABLE-US-00066 TABLE 66 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 9 21 57 83 87 0.3 1340071 0 0 0 56 88 1.3 1340085 0 0 27 68 91 0.8 1340154 0 20 48 80 90 0.4 1340360 0 0 0 51 42 3.8 1340385 4 35 69 82 72 0.3 1340448 0 0 27 78 74 0.9 1340543 5 15 62 78 86 0.3 1340546 0 12 40 65 56 1.2 1340549 0 4 50 78 92 0.4 1340601 0 0 31 54 81 1.1 1340684 0 0 31 58 84 0.9 1340704 0 8 25 69 83 0.8 1340762 0 13 41 71 79 0.7 1340878 1 1 30 79 85 0.6
(508) TABLE-US-00067 TABLE 67 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 0 20 55 82 88 0.4 1340206 0 22 55 71 90 0.4 1340208 12 12 51 69 92 0.4 1340226 0 0 43 73 96 0.6 1340260 0 0 33 56 78 1.0 1340268 0 0 35 55 90 0.9 1340334 0 5 30 67 86 0.8 1340397 0 0 49 73 88 0.7 1340555 1 13 41 63 76 0.7 1340585 0 13 47 77 74 0.6 1340603 18 16 47 69 85 0.4 1340647 14 35 76 83 94 0.2 1340789 10 14 32 73 83 0.5 1340834 14 18 48 74 79 0.4 1340895 0 19 32 65 87 0.6
(509) TABLE-US-00068 TABLE 68 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 0 25 47 76 85 0.4 1340168 6 0 32 80 90 0.6 1340401 4 11 50 78 79 0.4 1340449 26 8 59 81 93 0.3 1340548 0 17 51 78 92 0.4 1340626 0 5 45 80 90 0.5 1340646 6 31 57 67 58 0.6 1340653 2 0 38 69 77 0.8 1340780 6 23 36 73 83 0.5 1340820 20 35 55 84 88 0.2 1340821 0 0 21 40 60 2.7 1340847 9 20 45 60 88 0.5 1340857 0 0 11 46 66 2.3 1340859 14 32 47 71 84 0.3 1340908 16 33 60 84 92 0.2
(510) TABLE-US-00069 TABLE 69 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 0 16 42 80 87 0.5 1340153 0 18 38 75 82 0.5 1340227 0 21 57 70 90 0.4 1340435 14 27 57 78 90 0.3 1340441 6 14 32 70 74 0.7 1340493 13 23 54 70 88 0.3 1340510 11 13 27 66 87 0.6 1340511 0 0 7 52 74 1.6 1340572 19 23 53 75 89 0.3 1340630 20 25 42 75 87 0.3 1340670 16 24 51 74 93 0.3 1340782 0 26 54 77 92 0.3 1340840 0 3 23 43 67 1.9 1340848 0 0 30 71 91 0.7 1340861 22 34 47 74 90 0.2
(511) TABLE-US-00070 TABLE 70 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 3 22 55 79 84 0.3 1340159 10 13 49 83 80 0.4 1340481 0 0 34 64 82 0.8 1340502 0 4 37 71 87 0.6 1340564 5 10 34 72 85 0.5 1340635 7 21 40 79 82 0.4 1340649 0 0 42 69 89 0.6 1340672 0 20 47 72 76 0.6 1340743 12 18 46 80 90 0.3 1340779 5 7 39 68 40 0.6 1340793 3 24 66 86 91 0.2 1340796 0 19 61 88 84 0.3 1340799 16 34 72 86 89 0.2 1340827 16 31 48 72 81 0.3 1340881 10 28 52 59 79 0.4
(512) TABLE-US-00071 TABLE 71 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)
1246023 0 20 42 58 79 0.7 1341183 11 18 27 53 57 2.0 1341190 0 3 29 53 48 2.4 1341193 11 27 53 63 68 0.5 1341195 15 13 12 61 71 1.2 1341239 16 0 27 62 83 0.7 1341262 0 4 36 52 78 0.9 1341273 6 18 38 63 81 0.6 1341303 29 21 44 61 90 0.3 1341313 9 33 58 67 85 0.3 1341315 8 28 52 79 82 0.3 1341342 24 25 37 51 81 0.6 1341348 15 25 43 71 70 0.5 1341391 0 2 35 49 70 1.3 1341547 13 2 40 61 83 0.6

(513) TABLE-US-00072 TABLE 72 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)

1246023	19	1	38	77	89	0.4	1341026	10	20	43	73	83	0.4	1341032	0	13	14	74	92	0.6	1341035	5	0	36
62	74	1.0	1341036	23	28	53	78	93	0.2	1341038	17	25	57	76	85	0.3	1341052	6	13	52	69	79	0.5	
1341062	0	23	47	66	79	0.5	1341095	0	17	51	61	81	0.6	1341108	28	34	58	81	83	0.2	1341146	6	0	39
66	70	0.9	1341206	28	25	45	81	87	0.2	1341260	23	24	37	66	78	0.5	1341283	8	23	40	67	90	0.4	
1341308	18	29	54	71	86	0.3																		

(514) TABLE-US-00073 TABLE 73 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)

1246023	0	0	47	66	89	0.6	1340983	0	0	26	55	45	3.1	1340995	1	17	37	66	62	0.9	1341011	0	0	34	
81	1.0	1341043	0	3	40	50	44	2.2	1341050	0	7	39	55	76	0.9	1341057	6	9	39	57	78	0.8	1341068	0	0
11	46	79	1.5	1341075	0	0	0	39	55	3.1	1341118	0	8	13	31	74	2.2	1341145	0	17	37	69	83	0.6	
1341154	0	0	47	43	79	1.0	1341244	10	0	21	55	68	1.5	1341335	0	0	25	51	55	2.4	1341510	4	17	34	
57	66	1.0																							

(515) TABLE-US-00074 TABLE 74 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)

1246023	1	20	59	67	88	0.4	1340992	0	0	0	51	76	1.8	1341004	0	12	31	55	77	0.9	1341022	15	18	25
65	84	0.6	1341048	1	15	43	74	85	0.5	1341077	8	16	38	54	71	0.9	1341104	0	23	38	64	79	0.6	
1341140	12	17	41	64	82	0.5	1341151	13	31	52	71	80	0.3	1341271	8	0	43	65	80	0.7	1341318	23	17	
49	61	55	1.0	1341436	0	23	46	64	77	0.6	1341437	3	0	16	60	88	0.9	1341654	0	24	53	66	69	0.6
1341668	7	25	42	56	82	0.6																		

(516) TABLE-US-00075 TABLE 75 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)

1246023	2	24	40	70	92	0.4	1341651	19	35	41	67	85	0.3	1341655	0	0	13	70	89	0.9	1341671	13	27	
35	66	91	0.4	1341675	9	22	46	77	76	0.4	1341686	2	2	31	61	78	0.9	1341703	16	14	33	52	58	1.8
1341735	12	32	54	71	84	0.3	1341742	13	26	39	66	84	0.4	1341797	0	0	34	65	91	0.6	1341815	0	21	
20	65	75	0.9	1341839	25	16	39	60	87	0.5	1341899	14	28	42	66	77	0.5	1341911	7	29	46	67	90	0.4
1341913	0	2	0	48	59	3.2																		

(517) TABLE-US-00076 TABLE 76 Multi-dose assay of modified oligonucleotides in HepaRG cells % inhibition HSD17B13 ION IC.sub.50 No. 20 nM 78 nM 313 nM 1250 nM 5000 nM (μ M)

1246023	0	7	18	81	92	0.6	1341531	18	21	38	62	76	0.6	1341561	26	22	39	75	70	0.4	1341644	0	11	
60	82	90	0.4	1341645	0	20	0	39	79	1.8	1341646	3	0	8	57	76	1.3	1341968	0	0	10	72	90	0.8
1341980	8	9	26	54	78	1.0																		

Example 6: Design of Human HSD17B13 Modified Oligonucleotides

(518) Certain modified oligonucleotides described above were further modified by conjugating at the 5'-end with a LICA-1 moiety. The resulting compounds and their unconjugated counterparts are listed in the table below. The compounds have identical sequence and oligonucleotide chemistry as the "parent compounds" described above.

(519) TABLE-US-00077 TABLE 77 Design of 5'-LICA-1-conjugated modified oligonucleotides targeting human HSD17B13 GalNac Conjugated Parent SEQ Compound

Compound Sequence ID	No.	(5' to 3')	NO
1310535	1246163	CAACTATGTAATAGGC	
286	1371966	1245806	AAAATAGAGTCGGTCA
817	1371968	1246068	
AGCATAAGTTACCAGA	983	1371969	1246115
AAATTATACTTTTCCG	596	1371971	1246063
GTTTTATAACTGAGCT	594	1371973	1246019
GAATTTTGTAGGTTGG	1215	1371974	
1245987	TCTCAAGGAGTACTTC	747	1371976
1245952	GTCATAAAAATCGCTG	43	1371978
1245930	AAGATAAGTAGATGGT	355	1371979
1245972	GCACTAAAGGTTTCTG	1602	
1371981	1245980	ATACTAATGTCCAAGG	201
1373086	1245649	CTACTTGAACAGTCTT	734
1373087	1246903	AGTTAGTATAGTTATC	1249
1373089	1246134	CTCATTTATGTACCAA	51
1373090	1246162	ACTATGTAATAGGCAG	208
1373097	1245984	ACAATTTTCCAATCC	513
1373098	1246022	AGTGAATCATTTCAGTA	1449
1373099	1245996	CATTCGAATTTCTTCA	

148 1373101 1245790 GTATCTTATAAGACTA 1595 1394140 1341675
CTACUTGAACAGTCTT 2868 1394141 1340820 GTAGATGGTAAGTCAA 2567 1394142
1341283 TGATATTCTGTGGCAT 1124 1394143 1341313 TATGTAATAGCCAGTA 508 1394144
1341742 GTATCTTATAAGACTA 1595 1394145 1341038 TTCTTATGTAATAGCC 819 1394146
1341735 CACCGTTTTGGGCTAA 1516 1394147 1341032 TTATGTAATAGCCAGT 586
1394148 1341108 GCATCTTAAGATACCC 553 1394149 1340435 TAGAACTGTGTTGCTT
2535 1394150 1340908 TAGAGTGAATCATTCA 2420 1394151 1340208
GAGTAGTGTAAGCTGA 2204 1394152 1340782 GAGATTTGAAGGTTAG 2561 1371970
1246157 GCAGTAAAGTGCTAAG 1843 1371972 1246025 AGCCTTAGAGTGAATC 1682
1373094 1246036 TGAAAACCTCAGCCAGC 515 1373096 1245968 GTAATTTTCAGATCCC
1291 1373100 1245682 GTTGATATTCTGTGGC 1280

Example 7: Dose-Dependent Ex-Vivo Inhibition of Human HSD17B13 in Transgenic Mouse Hepatocytes

(520) The conjugated modified oligonucleotides described above were tested for inhibition of HSD17B13 RNA at various doses in primary mouse hepatocytes extracted from transgenic mice.

(521) A transgenic mouse model was developed in-house using a Fosmid containing the human HSD17B13 gene (NCBI Clone DB ID: ABC8-43206400A 10). The clone was digested at NotI restriction sites (2 different sites) to produce a fragment containing 19,118 bp of the human HSD17B13 gene, including 12,175 bp upstream and 7,553 bp downstream regions. That gene fragment was introduced into fertilized eggs from C57BL/6NTac strain mice by pronuclear injection to produce huHSD 17469 and 17470 founder lines. Transgenic mice derived from Founder Line 17469 was used in the experiments described herein. Human HSD17B13 RNA expression is found in the liver in this model. Mice were sacrificed and hepatocytes were collected for use in experiments testing conjugated modified oligonucleotides.

(522) The conjugated modified oligonucleotides were tested in a series of experiments using the same culture conditions. The results for each experiment are presented in separate tables shown below. Primary mouse transgenic hepatocytes described above were plated at a density of 20,000 cells per well and were treated by free uptake with conjugated modified oligonucleotides diluted to different concentrations as specified in the tables below. After an overnight incubation, HSD17B13 RNA levels were measured as previously described herein using the Human HSD17B13 primer-probe set RTS43553. HSD17B13 RNA levels were normalized to total GAPDH content. Results are presented in the tables below as percent inhibition of HSD17B13, relative to untreated control cells. The half maximal inhibitory concentration (IC_{sub.50}) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel and is also presented in the Tables below.

(523) TABLE-US-00078 TABLE 78 Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes % Inhibition ION 1.95 7.81 31.25 125 500 2000 IC_{sub.50} No. nM nM nM nM nM nM (nM) 1310535 38 53 80 87 85 90 3.2 1371966 41 52 87 88 91 89 4.1 1371968 39 56 84 94 91 90 4.1 1371969 23 60 81 88 88 92 5.2 1371971 6 45 54 70 85 76 34.1 1371973 38 61 84 93 94 96 2.2 1371974 4 31 65 74 74 82 38.6 1371976 16 62 79 87 90 93 7.3 1371978 0 39 84 92 89 88 18.4 1371979 10 38 70 82 85 87 21.0 1371981 33 68 86 94 93 95 1.9

(524) TABLE-US-00079 TABLE 79 Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes % Inhibition ION 1.95 7.81 31.25 125 500 2000 IC_{sub.50} No. nM nM nM nM nM nM (nM) 1373086 24 49 78 83 86 84 8.9 1373087 23 20 67 70 64 65 22.6 1373089 36 62 84 92 91 92 4.0 1373090 43 49 69 81 80 82 5.4 1373097 41 64 84 89 92 91 1.3 1373098 43 69 80 87 89 89 2.3 1373099 19 37 68 84 83 86 13.8 1373101 9 41 66 83 85 92 20.5 1394140 5 42 80 78 85 88 19.6 1394142 5 49 73 87 89 91 16.4 1371966 4 55 77 88 88 87 11.5

(525) TABLE-US-00080 TABLE 80 Multi-dose assay of modified oligonucleotides in primary mouse hepatocytes % Inhibition ION 1.95 7.81 31.25 125 500 2000 IC_{sub.50} No. nM nM nM nM nM nM (nM) 1394143 19 45 80 83 89 89 10.6 1394144 16 21 54 70 75 80 48.1 1394145 37 44 83

91 88 88 6.2 1394146 3 30 67 74 82 77 38.2 1394147 30 71 85 94 94 93 4.0 1394148 41 80 88 90
87 91 1.7 1394149 1 26 56 72 78 82 50.9 1394150 14 45 80 84 88 85 12.8 1394151 0 37 63 81 83
85 42.5 1394152 8 35 63 82 81 87 28.1 1371966 0 61 72 82 89 88 16.7

Example 8: Tolerability of Modified Oligonucleotides Complementary to Human HSD17B13 in CD-1 Mice

(526) CD-1 mice are a multipurpose mouse model frequently utilized for safety and efficacy testing. In this study, CD-1 mice were treated with conjugated modified oligonucleotides selected from studies described above and evaluated for changes in the levels of various plasma chemistry markers.

(527) Treatment

(528) Groups of 6- to 8-week-old male CD-1 mice were injected subcutaneously once a week for six weeks (for a total of 7 treatments) with 25 mg/kg of conjugated modified oligonucleotides. One group of male CD-1 mice was injected with PBS. Mice were euthanized 48 hours following the final administration.

(529) Study 1

(530) Plasma Chemistry Markers

(531) To evaluate the effect of conjugated modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

(532) TABLE-US-00081 TABLE 81 Plasma chemistry markers in male CD-1 mice ION Albumin ALT AST CREA BUN NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) PBS 2.5 25 52 0.2 23 1310535 2.6 32 44 0.2 24 1371966 2.7 31 40 0.2 23 1371968 2.7 26 42 0.2 23 1371969 2.6 28 59 0.2 24 1371970 2.6 277 225 0.2 24 1371971 2.8 41 74 0.2 24 1371972 2.4 810 568 0.2 24 1371973 2.3 50 78 0.2 22 1371974 2.7 60 61 0.2 25 1371976 2.7 32 61 0.2 22 1371978 2.7 51 59 0.2 21 1371979 2.7 32 70 0.2 23

Body and Organ Weights

(533) Body weights of CD-1 mice were measured at days 1 and 44, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

(534) TABLE-US-00082 TABLE 82 Body and organ weights (in grams) ION Body Weight (g) Liver Kidney Spleen No. Day 1 Day 44 (g) (g) (g) PBS 34 42 2.4 0.5 0.1 1310535 33 42 2.4 0.6 0.1 1371966 32 39 2.2 0.6 0.1 1371968 30 38 2.1 0.5 0.1 1371969 33 43 2.4 0.5 0.1 1371970 29 36 2.1 0.5 0.1 1371971 33 41 2.5 0.6 0.2 1371972 33 40 3.1 0.7 0.2 1371973 31 38 1.7 0.5 0.1 1371974 34 40 2.4 0.6 0.1 1371976 32 40 2.4 0.6 0.1 1371978 32 38 2.4 0.5 0.1 1371979 33 38 2.3 0.6 0.1

Hematology Assays

(535) Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (MIT), and platelet counts (PLT). The results are presented in the table below.

(536) TABLE-US-00083 TABLE 83 Blood cell counts in male CD-1 mice ION NO. HCT (%) PLT (10.sup.9/L) PBS 46 1155 1310535 50 1389 1371966 46 1415 1371968 47 1372 1371969 45 1059 1371970 48 1395 1371971 46 1244 1371972 44 1360 1371973 47 968 1371974 45 1362 1371976 49 1199 1371978 47 1417 1371979 48 1190

Study 2

Plasma Chemistry Markers

(537) To evaluate the effect of modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

(538) TABLE-US-00084 TABLE 84 Plasma chemistry markers in male CD-1 mice ION Albumin ALT AST CREA BUN NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) PBS 2.8 36 57 0.11 19 1371981 2.7 26 39 0.14 24 1373086 2.6 39 59 0.12 24 1373087 2.6 31 54 0.13 23 1373089 2.5 107 88 0.12 24 1373090 2.5 27 45 0.11 25 1373094 2.8 687 532 0.15 23 1373096 2.7 57 87 0.13 23 1373097 2.4 67 71 0.13 23 1373098 2.5 67 73 0.12 24 1373099 2.7 43 75 0.12 22 1373100 2.7 373 365 0.12 25 1373101 2.5 153 179 0.14 23

Body and Organ Weights

(539) Body weights of CD-1 mice were measured at days 1 and 42, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

(540) TABLE-US-00085 TABLE 85 Body and organ weights (in grams) ION Body Weight (g) Liver Kidney Spleen No. Day 1 Day 42 (g) (g) (g) PBS 27 38 1.9 0.6 0.1 1371981 27 39 2.5 0.6 0.1 1373086 27 38 2.1 0.5 0.1 1373087 27 40 2.4 0.6 0.1 1373089 26 39 2.8 0.5 0.1 1373090 27 38 2.4 0.5 0.2 1373094 28 41 2.9 0.6 0.2 1373096 27 42 2.5 0.6 0.2 1373097 27 42 2.52 0.49 0.17 1373098 27 39 2.48 0.51 0.13 1373099 27 39 2.44 0.61 0.14 1373100 26 37 2.11 0.52 0.16 1373101 29 45 2.68 0.63 0.20

Hematology Assays

(541) Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

(542) TABLE-US-00086 TABLE 86 Blood cell counts in male CD-1 mice ION HCT PLT NEU RET LYM NO. (%) (10.sup.3/ μ L) (%) (%) (%) PBS 45 1150 19 3 77 1371981 46 1398 18 3 76 1373086 44 1415 17 3 74 1373087 42 1343 21 3 73 1373089 43 1216 21 3 72 1373090 42 1332 21 3 52 1373094 44 1307 24 3 68 1373096 47 1161 19 3 69 1373097 42 1394 25 3 67 1373098 44 1432 24 4 68 1373099 44 1201 20 3 75 1373100 41 1031 16 3 77 1373101 45 1236 28 4 65

Study 3

Plasma Chemistry Markers

(543) To evaluate the effect of modified oligonucleotides on liver function, plasma levels of blood urea nitrogen (BUN), albumin, alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TBIL) and creatinine (CREA) were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The results are presented in the table below.

(544) TABLE-US-00087 TABLE 87 Plasma chemistry markers in male CD-1 mice ION Albumin ALT AST CREA BUN TBIL NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) (mg/dL) PBS 2.8 25 56 0.1 26 0.2 1394140 2.9 38 72 0.2 24 0.2 1394142 2.5 67 92 0.1 23 0.2 1394143 2.7 30 55 0.1 22 0.2 1394144 2.7 37 69 0.1 24 0.2 1394145 2.7 50 67 0.1 25 0.1 1394146 2.8 142 203 0.1 24 0.2 1394147 2.5 63 80 0.1 25 0.1 1394148 3.1 219 193 0.1 23 0.2 1394149 2.9 746 757 0.2 25 0.3 1394150 2.7 608 451 0.1 22 0.2

Body and Organ Weights

(545) Body weights of CD-1 mice were measured at days 1 and 39, and the average body weight for each group is presented in the table below. Liver, kidney and spleen weights were measured at the end of the study and are presented in the table below.

(546) TABLE-US-00088 TABLE 88 Body and organ weights (in grams) ION Body Weight (g) Liver Kidney Spleen No. Day 1 Day 39 (g) (g) (g) PBS 29 36 1.8 0.5 0.1 1394140 28 36 2.0 0.5 0.1 1394142 30 38 1.8 0.5 0.2 1394143 29 37 2.1 0.5 0.1 1394144 29 37 2.1 0.5 0.1 1394145 30 38 2.4 0.5 0.2 1394146 29 37 2.1 0.5 0.1 1394147 31 38 2.5 0.5 0.2 1394148 29 37 2.6 0.5 0.2 1394149 27 35 2.3 0.4 0.1 1394150 27 35 2.3 0.5 0.2

Hematology Assays

(547) Blood obtained from mouse groups at week 6 were sent to IDEXX BioResearch for measurement of hematocrit (HCT), platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

(548) TABLE-US-00089 TABLE 89 Blood cell counts in male CD-1 mice ION HCT PLT NEU
RET LYM NO. (%) (10.sup.3/ μ L) (%) (%) (%) PBS 53 1271 17 3 74 1394140 52 1221 20 3 72
1394142 50 1249 28 3 64 1394143 50 1165 17 3 76 1394144 49 1090 18 3 75 1394145 52 1292 21
3 73 1394146 50 1382 27 3 66 1394147 52 1343 22 3 69 1394148 52 1286 27 3 64 1394149 46
1613 26 3 66 1394150 47 1444 13 2 79

Example 9: Tolerability of Modified Oligonucleotides Targeting Human HSD17B13 in Sprague-Dawley Rats

(549) Sprague-Dawley rats are a multipurpose model used for safety and efficacy evaluations. In this study, the rats were treated with Ionis conjugated modified oligonucleotides from the studies described in the Examples above and evaluated for changes in the levels of various plasma chemistry markers.

(550) Treatment

(551) Male Sprague-Dawley rats were maintained on a 12-hour light/dark cycle and fed ad libitum with Purina normal rat chow. Groups of 4 Sprague-Dawley rats each were weekly injected subcutaneously with 25 mg/kg of conjugated modified oligonucleotide for 6 weeks (total 7 doses). 72 hours after the last dose, rats were euthanized; and organs, urine and plasma were harvested for further analysis.

(552) Study 1

(553) Plasma Chemistry Markers

(554) To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

(555) TABLE-US-00090 TABLE 90 Plasma chemistry markers in Sprague-Dawley rats ION
Albumin ALT AST CREA BUN NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) PBS 3.4 34 53 0.3 13
1310535 3.2 35 68 0.3 14 1371966 3.5 33 66 0.3 16 1371968 3.8 36 57 0.4 16 1371969 3.0 25 68
0.3 16 1371970 3.4 48 101 0.3 15 1371971 3.6 34 72 0.3 16 1371972 3.5 51 76 0.4 15 1371973 3.6
36 62 0.3 14 1371974 3.3 37 76 0.4 15 1371976 3.4 31 55 0.3 15 1371978 3.4 37 67 0.3 15
1371979 3.5 117 122 0.3 12

Hematology Assays

(556) Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT). The results are presented in the table below.

(557) TABLE-US-00091 TABLE 91 Blood Cell Count in Sprague-Dawley Rats ION NO. HCT (%)
PLT (10.sup.9/L) PBS 47 849 1310535 43 954 1371966 46 844 1371968 47 1010 1371969 43 810
1371970 48 1016 1371971 46 893 1371972 46 1019 1371973 44 857 1371974 46 930 1371976 45
843 1371978 45 900 1371979 47 1077

Kidney Function

(558) To evaluate the effect of the conjugated modified oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of total protein to creatinine (P/C ratio) are presented in the table below.

(559) TABLE-US-00092 TABLE 92 Total protein to creatinine ratio in Sprague-Dawley rats ION
NO. Urine P/C Ratio PBS 0.5 1310535 0.4 1371966 0.8 1371968 0.7 1371969 0.6 1371970 0.6
1371971 1.1 1371972 0.4 1371973 0.7 1371974 0.8 1371976 0.7 1371978 0.5 1371979 0.8

Body and Organ Weights

(560) Body weights of Sprague-Dawley rats were measured at days 1 and 40, and the average body

weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

(561) TABLE-US-00093 TABLE 93 Body and Organ weights In Sprague-Dawley rats (g) ION
Body Weight (g) Kidney Spleen No. Day 1 Day 40 Liver (g) (g) (g) PBS 140 464 18 3.2 1.1
1310535 152 476 20 3.7 1.5 1371966 150 437 17 3.1 1.0 1371968 147 419 15 2.9 1.0 1371969 151
423 16 3.5 2.0 1371970 151 440 16 2.9 1.1 1371971 138 406 17 3.1 1.4 1371972 144 424 12 2.7
1.0 1371973 141 401 14 3.0 1.3 1371974 143 443 16 3.0 1.1 1371976 126 412 16 2.8 1.0 1371978
155 452 17 3.3 1.3 1371979 148 458 19 3.1 1.3

Study 2

Plasma Chemistry Markers

(562) To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

(563) TABLE-US-00094 TABLE 94 Plasma chemistry markers in Sprague-Dawley rats ION
Albumin ALT AST CREA BUN NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) PBS 3.6 33 56 0.3 16
1371981 3.7 35 57 0.3 17 1373086 3.7 35 87 0.4 19 1373087 3.3 33 64 0.4 18 1373089 3.7 92 129
0.3 18 1373090 3.5 38 72 0.4 18 1373094 3.7 187 311 0.4 16 1373096 3.3 69 171 0.3 18 1373097
3.2 40 86 0.3 17 1373098 3.8 37 74 0.3 17 1373099 3.6 33 60 0.4 16 1373100 3.1 39 115 0.3 20
1373101 3.4 37 80 0.3 14

Hematology Assays

(564) Blood obtained from mouse groups at week 6 were sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte counts (LYM). The results are presented in the table below.

(565) TABLE-US-00095 TABLE 95 Blood Cell Count in Sprague-Dawley Rats ION HCT PLT
NEU RET LYM NO. (%) (10.sup.3/ μ L) (%) (%) (%) PBS 51 709 14 3 72 1371981 52 769 11 2 84
1373086 47 921 8 2 88 1373087 50 764 6 3 89 1373089 49 1217 7 3 88 1373090 50 766 10 4 85
1373094 53 943 14 2 81 1373096 61 376 6 3 89 1373097 49 746 5 3 90 1373098 49 929 10 3 85
1373099 47 717 8 3 85 1373100 42 823 6 4 87 1373101 48 693 7 3 87

Kidney Function

(566) To evaluate the effect of Ionis oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of tot protein to creatinine (P/C ratio) are presented in the table below.

(567) TABLE-US-00096 TABLE 96 Total protein to creatinine ratio in Sprague-Dawley rats ION
NO. Urine P/C Ratio PBS 1.0 1371981 1.7 1373086 1.5 1373087 1.7 1373089 1.5 1373090 1.3
1373094 3.3 1373096 1.5 1373097 1.3 1373098 1.2 1373099 1.9 1373100 0.9 1373101 1.7

Body and Organ Weights

(568) Body weights of Sprague-Dawley rats were measured at days 1 and 44, and the average body weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

(569) TABLE-US-00097 TABLE 97 Body and Organ weights In Sprague-Dawley rats (g) ION
Body Weight (g) Liver Kidney Spleen No. Day 1 Day 44 (g) (g) (g) PBS 146 458 19 3.1 1.0
1371981 141 424 21 3.4 1.1 1373086 151 434 19 3.0 1.8 1373087 143 433 19 3.3 1.3 1373089 149
431 21 3.5 2.1 1373090 142 453 19 3.2 1.3 1373094 142 449 19 3.2 1.4 1373096 149 460 23 3.6
3.2 1373097 136 400 17 3.3 2.2 1373098 130 389 20 2.9 1.5 1373099 152 421 16 2.9 1.6 1373100

146 453 19 4.2 2.3 1373101 140 412 17 2.9 1.5

Study 3

Plasma Chemistry Markers

(570) To evaluate the effect of the conjugated modified oligonucleotides on hepatic function, plasma levels of transaminases were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). Plasma levels of ALT (alanine transaminase) and AST (aspartate transaminase) were measured and the results are presented in the Table below expressed in IU/L. Plasma levels of creatinine (CREA), albumin, total bilirubin (TBIL) and Blood Urea Nitrogen (BUN) were also measured using the same clinical chemistry analyzer and the results are also presented in the table below.

(571) TABLE-US-00098 TABLE 98 Plasma chemistry markers in Sprague-Dawley rats ION Albumin ALT AST CREA BUN TBIL NO. (g/dL) (IU/L) (IU/L) (mg/dL) (mg/dL) (mg/dL) PBS 3.2 41 62 0.3 18 0.2 1394140 3.6 48 123 0.4 20 0.3 1394142 3.0 57 136 0.4 23 0.2 1394143 3.3 122 201 0.3 19 0.4 1394144 3.2 38 73 0.4 18 0.1 1394145 3.6 112 187 0.4 17 0.2 1394146 3.4 54 98 0.4 20 0.2 1394147 3.1 53 110 0.3 18 0.2 1394148 3.5 45 57 0.4 19 0.3 1394149 3.7 94 102 0.5 27 0.3 1394150 3.8 60 83 0.4 19 0.2

Hematology Assays

(572) Blood obtained from mouse groups at week 6 and was sent to IDEXX BioResearch for measurement of hematocrit (HCT), and platelet counts (PLT1), neutrophil counts (NEU), reticulocyte counts (RET) and lymphocyte count (LYM). The results are presented in the table below.

(573) TABLE-US-00099 TABLE 99 Blood Cell Count in Sprague-Dawley Rats ION HCT PLT NEU RET LYM NO. (%) (K/ μ L) (%) (%) (%) (%) PBS 52 795 12 3 84 1394140 45 876 8 3 88 1394142 47 768 4 3 90 1394143 59 1277 11 2 80 1394144 51 574 6 3 87 1394145 51 806 12 2 82 1394146 48 1052 11 2 81 1394147 49 922 8 3 85 1394148 49 1134 12 3 84 1394149 46 1816 14 3 80 1394150 52 1094 12 3 83

Kidney Function

(574) To evaluate the effect of Ionis oligonucleotides on kidney function, urinary levels of total protein and creatinine were measured using an automated clinical chemistry analyzer (Hitachi Olympus AU400c, Melville, NY). The ratios of total protein to creatinine (P/C ratio) are presented in the table below.

(575) TABLE-US-00100 TABLE 100 Total protein to creatinine ratio in Sprague-Dawley rats ION NO. Urine P/C Ratio PBS 0.8 1394140 1.6 1394142 1.1 1394143 1.3 1394144 1.3 1394145 1.1 1394146 1.8 1394147 1.5 1394148 1.0 1394149 1.1 1394150 1.9

Body and Organ Weights

(576) Body weights of Sprague-Dawley rats were measured at days 1 and 42, and the average body weight for each group is presented in the table below. Liver, heart, spleen and kidney weights were measured at the end of the study and are presented in the table below.

(577) TABLE-US-00101 TABLE 101 Body and Organ weights In Sprague-Dawley rats (g) ION Body Weight (g) Liver Kidney Spleen No. Day 1 Day 42 (g) (g) (g) (g) PBS 270 503 19 3.0 0.9 1394140 263 440 19 3.0 1.5 1394142 258 370 13 2.9 1.5 1394143 257 393 16 2.9 1.1 1394144 254 461 18 3.2 1.5 1394145 266 475 21 3.4 1.4 1394146 268 444 17 3.1 1.7 1394147 258 401 17 3.2 2.0 1394148 267 447 21 3.0 1.5 1394149 261 422 20 5.8 1.4 1394150 251 423 18 3.0 1.2

Example 10: Measurement of Viscosity of Modified Oligonucleotides Targeting Human HSD17B13

(578) The viscosity of select conjugated modified oligonucleotides from the studies described above was measured.

(579) Conjugated modified oligonucleotides (within a range of 32-38 mg) were placed into separate glass vials; approximately 100 μ L of water was added to each vial, and the conjugated modified oligonucleotides were dissolved into solution by heating the vials to 55° C. A 75 μ L

portion of the pre-heated sample was pipetted to a micro-viscometer (PAC Cambridge Viscosity Viscometer). The temperature of the micro-viscometer was set to 25° C. and the viscosity of the sample was measured. The 75 uL portion of each sample was then combined with the remaining portion of the original sample, which was then diluted for UV reading at 260 nM (Cary UV instrument). The data is presented in the Table below. Compounds having viscosity of less than 40 cP in this assay are generally suitable for use in therapy.

(580) TABLE-US-00102 TABLE 102 Viscosity of modified oligonucleotides Concentration
Concentration Compound by weight by UV Viscosity ID (mg/mL) (mg/mL) (cP) 1310535 300 206
6 1371966 300 223 10 1371974 300 207 8 1371976 300 254 64 1371978 300 219 7 1371979 300
244 10 1373086 300 215 7 1373087 300 219 9 1373090 300 191 8 1373097 300 222 6 1373098
300 203 5 1373099 300 242 6 1373101 300 205 11 1394140 300 197 10 1394141 300 225 9
1394142 300 271 24 1394143 300 250 11 1394144 300 257 32 1394145 300 263 10 1394146 300
289 11 1394147 300 246 15 1394148 300 205 9 1394149 300 304 14 1394150 300 202 16
1394151 300 223 8 1394152 300 225 24 1371976 200 151 16

Claims

1. An oligomeric compound, according to the following chemical structure: ##STR00068##
##STR00069## or a pharmaceutically acceptable salt thereof.
2. An oligomeric compound, according to the following chemical structure: ##STR00070##
3. An oligomeric compound comprising a modified oligonucleotide consisting of 16 linked nucleosides having a nucleobase sequence SEQ ID NO: 3003, wherein the modified oligonucleotide comprises: a gap segment consisting of ten linked deoxynucleosides; a 5' wing segment consisting of three linked nucleosides; and a 3' wing segment consisting of three linked nucleosides; wherein the gap segment is positioned between the 5' wing segment and the 3' wing segment; wherein the 5' wing segment and the 3' wing segment comprise cEt sugars; wherein each internucleoside linkage is a phosphorothioate linkage; and wherein each cytosine is a 5-methylcytosine.
4. The oligomeric compound of claim 1, which is a sodium salt or a potassium salt.
5. A pharmaceutical composition comprising the oligomeric compound of claim 1 and a pharmaceutically acceptable diluent.
6. The pharmaceutical composition of claim 5, wherein the pharmaceutically acceptable diluent is water.
7. A pharmaceutical composition comprising the oligomeric compound of claim 2 and a pharmaceutically acceptable diluent.
8. The pharmaceutical composition of claim 7, wherein the pharmaceutically acceptable diluent is water.
9. A pharmaceutical composition comprising the oligomeric compound of claim 3 and a pharmaceutically acceptable diluent.
10. The pharmaceutical composition of claim 9, wherein the pharmaceutically acceptable diluent is water.
11. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 1, thereby treating the disease.
12. The method of claim 11, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.
13. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 2, thereby treating the disease.

14. The method of claim 13, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.

15. A method of treating a disease associated with HSD17B13 in an individual comprising administering to the individual the oligomeric compound of claim 3, thereby treating the disease.

16. The method of claim 15, wherein the individual has a liver disease, NAFLD, NASH, alcoholic steatohepatitis (ASH), alcoholic liver disease, nonalcoholic liver disease, alcoholic cirrhosis, nonalcoholic cirrhosis, steatohepatitis, hepatic steatosis, hepatocellular carcinoma, alcoholic liver disease, HCV hepatitis, chronic hepatitis, hereditary hemochromatosis, or primary sclerosing cholangitis.
